Grammar of Western Armenian

Hossep Dolatian

Draft
of January 31, 2026, 22:28

Comprehensive Grammar Library

language science press

Comprehensive Grammar Library

Editor: Martin Haspelmath

In this series:

1. Jacques, Guillaume. A grammar of Japhug.

This series grew out of the grammars published in *Studies in Diversity Linguistics*, which are proudly mentioned:

4. Berghäll, Liisa. A grammar of Mauwake.

5. Wilbur, Joshua. A grammar of Pite Saami.

7. Schackow, Diana. A grammar of Yakkha.

8. Liljegren, Henrik. A grammar of Palula.

9. Shimelman, Aviva. A grammar of Yauyos Quechua.

11. Kluge, Angela. A grammar of Papuan Malay.

12. Kieviet, Paulus. A grammar of Rapa Nui.

22. Döhler, Christian. A grammar of Komnzo.

23. Yakpo, Kofi. A grammar of Pichi.

Grammar of Western Armenian

Hossep Dolatian

language
science
press

Hossep Dolatian. 2025. *Grammar of Western Armenian* (Comprehensive Grammar Library). Berlin: Language Science Press.

This title can be downloaded at:
redefine\TU\textbackslashlsURL
© 2026, Hossep Dolatian
Published under the Creative Commons Attribution 4.0 Licence (CC BY 4.0):
http://creativecommons.org/licenses/by/4.0/
ISBN: no digital ISBN
no print ISBNs!
no DOI
ID not assigned!

Cover and concept of design: Ulrike Harbort
Fonts: Libertinus, Arimo, DejaVu Sans Mono
Typesetting software: XeLaTeX

redefine \publisherstreetaddress
redefine \publisherurl

Storage and cataloguing done by redefine \storageinstitution

Freie Universität Berlin

Contents

Preface v

Acknowledgments vii

1 citation needs 3

2 Introduction and overview 5
- 2.1 Introduction 5
- 2.2 History of Western Armenian and its people 5
- 2.3 Typological overview of Western Armenian 5
- 2.4 Data sources 5
- 2.5 Transcription and representations 5

I Phonology and orthography 9

3 Orthography 11
- 3.1 Letters and transliteration 11
- 3.2 Writing system and spelling reforms 14
- 3.3 Punctuation symbols 15
- 3.4 Orthography-phonology mismatches 16
  - 3.4.1 Homophony in voiceless letters 17
  - 3.4.2 Mismatches in voicing of clusters 18
  - 3.4.3 Post-rhotic devoicing 19
  - 3.4.4 Deletion of <h> after rhotics 21

4 Segmental phonology 25
- 4.1 Consonants 26
  - 4.1.1 Stops 26
  - 4.1.2 Affricates 29
  - 4.1.3 Fricatives 31
  - 4.1.4 Nasals 34
  - 4.1.5 Rhotic 35

4.1.6 Lateral . . . 37
4.1.7 Glides . . . 37
4.2 Vowels . . . 41
4.2.1 Canonical vowels . . . 41
4.2.2 Phonology of the schwa . . . 44
4.2.3 Front round vowels . . . 49
4.2.4 Turkish-induced centralization . . . 52
4.3 Voicing and aspiration in obstruents . . . 54
4.3.1 Voicing assimilation . . . 56
4.3.2 Aspiration and deaspiration . . . 68
4.4 Place assimilation of nasals . . . 81
4.4.1 Velar assimilation in underived contexts . . . 81
4.4.2 Velar assimilation in derived contexts . . . 83
4.4.3 Other forms of nasal place assimilation . . . 87
4.4.4 Lexical frequency of nasal clusters . . . 89
4.5 Allophonic differences from Eastern Armenian . . . 90
4.5.1 Palatalization . . . 91
4.5.2 Deaspiration and voicing assimilation . . . 92
4.5.3 Sonorant devoicing . . . 93
4.6 Sandhi phenomena or connected speech processes . . . 93
4.6.1 Word-final devoicing . . . 94
4.6.2 /h/ deletion . . . 96
4.6.3 Schwa vowel assimilation . . . 100
4.6.4 Schwa elision . . . 101
4.6.5 Gemination . . . 106

5 Syllable structure **109**
5.1 Overview of syllable structure . . . 109
5.1.1 Possible syllables and their lexical frequency . . . 109
5.1.2 Complex onsets are generally banned . . . 112
5.1.3 Overview of complex codas . . . 112
5.1.4 Appendix or extrasyllabic consonants . . . 113
5.1.5 Maximality of complex codas . . . 115
5.1.6 Onsetless syllables . . . 116
5.1.7 Loanwords . . . 117
5.2 Syllabification of final two-consonant clusters or complex codas 118
5.2.1 Falling-sonority and common complex codas . . . 118
5.2.2 Falling sonority but either rare, extrasyllabic, or uses schwa epenthesis . . . 129

5.2.3 Flat or rising sonority and either rare, extrasyllabic, or uses schwa epenthesis . . . 138
5.2.4 Geminate codas . . . 152
5.3 Syllabification of final three-consonant clusters . . . 153
5.3.1 No native complex codas with three consonants . . . 153
5.3.2 Codas with an appendix . . . 153
5.3.3 Schwa epenthesis in other word-final three-consonant clusters . . . 156
5.4 Other restrictions on complex codas . . . 160
5.4.1 Infrequency and asymmetry of word-internal complex codas . . . 161
5.4.2 Avoidance of certain word-medial complex codas . . . 163
5.4.3 Vowel-based or consonant-based restrictions on codas and complex codas . . . 167
5.4.4 Complex codas created by schwa elision . . . 172
5.5 Complex onset restrictions . . . 172
5.5.1 Complex onsets of consonant + glide . . . 172
5.5.2 Variation in syllabifying word-medial consonant-glide clusters . . . 174
5.5.3 Complex onsets created from schwa elision . . . 177
5.6 Vowel-vowel sequences or vowel hiatus repair . . . 177
5.6.1 Vowel hiatus in underived words and roots . . . 178
5.6.2 Vowel hiatus repair between roots and derivational suffixes . . . 179
5.6.3 Vowel hiatus repair in compounds and prefixoids . . . 191
5.6.4 Vowel hiatus repair before regular inflectional suffixes . . . 198
5.6.5 Vowel hiatus repair in irregular inflection and verbal inflection . . . 204
5.6.6 Vowel hiatus repair before clitics . . . 205
5.6.7 Overapplication of vowel hiatus repair . . . 205
5.6.8 Diachronic problems in glide epenthesis . . . 208

6 note **211**

References **213**

Preface

Acknowledgments

“Our survival is our revenge”

... ...
... ...

TODO

- ctrl-f to find th, lv, neg, inf, def to make sure got the right tags
- make sure suppletive roots get aor tag

1 citation needs

Orthography chapter

- orthography section of WA grammars to talk about unpredictabilities in depth
- Classical loss of final glide
- Classical midvowel change
- Punctuation pedagogy
- Classical voicing change
- Eastern Armenian final stop devoicing espeically after rhotic

Segmental chapter

- place of articulation of dental stops
- Bolsahayeren D-T^h values and Turkish consonant citation
- place for affricates
- place for fricative s z
- place for fricative χ ʁ
- place for nasal
- spirantizing for rhotic
- history of trill in WA
- lateral bolsahayeren
- vowel acoustics
- schwa phonemic status

- reduction of y to u in tsun
- julianne, samuel, and hrayr [y]
- deaspiration othesr
- utjun suffix from vaux
- EA sonoorant devoicing
- bolsahayeren citation for devoicing
- kaisse phrasal allomorphy
- vaux on final Cəɾ
- examples of other words like vagəɾ. i cant remember if 3-syllable words liek ⊠⊠⊠⊠⊠ would do it too

Syllable chapter

- vaux or someone else on final apopcope as cause of final clusters

2 Introduction and overview

2.1 Introduction

TODO: turn the following into prose

Word Order vs. Inflectional vs. Agglutinative (Turkish)

Sneaky presence of Classical Armenian

hasmig: Your words (descriptive, factual, and clear) have brought up another issue that should be well-articulated in the introduction. This grammar is developed according to the principles of the science of language, linguistics. No prescription allowed. Believe it or not, this is a distinction that a lot of people are not aware of. I was looking at a couple of reviews of Dum-Tragut's book and they emphatically point out that it is based on modern linguistics. Perhaps we should have a subtitle to that effect.

origins of WA, and hate against gor and ne

lisa's issues with teaching

Lisa Gulessarian gave me her list of grammatical issues that challenge learners:

1) Word order 2) �ն vs ն 3) ն-ն vs ն-ն 4) the aorist – both the irregularity and when to use it as opposed to the imperfect 5) conditionals 6) instrumental infinitive eg, գրելու ն գրելու

2.2 History of Western Armenian and its people

2.3 Typological overview of Western Armenian

2.4 Data sources

2.5 Transcription and representations

For every example in this grammar, we provide at most the following types of representations in the following order. Those with the asterisk *** are present for every example.

1. Representations used in this grammar:
   a) ***Surface Pronunciation**: Every word or sentence is given a broad phonetic transcription in IPA. This transcription encodes noticeable types of allophony, such as voicing assimilation. This form is demarcated with either brackets [...] or nothing.
   b) **Underlying Pronunciation**: When relevant, we provide the underlying pronunciation (underlying form, underlying representation) of words. This pronunciation is in IPA and it undoes allophony and other morpho-phonological alternations from the surface pronunciation. This representation is not based on the orthography or diachrony, but based on allophony and on changes in a morpheme's pronunciation across its inflectional paradigm. This form is demarcated with slashes /.../.
   c) **Morphological Gloss**: For sentences and for some words, we provide a morpheme segmentation. In the phonology chapters, we tend to minimize morpheme segmentations if they're not relevant. Sentences get full segmentation. Paradigms have full segmentation.
   d) ***Translation**: We translate the word or sentence into English. In most but not all cases, the translation of a word can be found in online dictionaries.
   e) ***Orthographic Representation**: We write the sentence or word based on the traditional orthography of Western Armenian. In rare cases, if a word or sentence is from Eastern Armenian, we use the reformed orthography.
   f) **Transliteration**: When useful, we transliterate Western Armenian examples using our own transliteration scheme. Eastern Armenian data is transcribed with ISO 9985.[1] We generally provide transliterations only when we are discussing the orthography of an example. This form is demarcated with arrows <>. The transliteration system is discussed in §3.1.

We illustrate with the following example (1).

(1) meŋk hos-teˈʁ-e-n gə-skəˈs-i-ŋk
/menk hos-deʁ-e-n g-sks-i-nk/
we this-place-ABL-DEF IND-start-TH-1PL
'We start from here.'

[1] https://www.translitteration.com/transliteration/en/armenian-eastern-classical/iso-9985/

Մենք հոստեղէն կը սկսինք:
<Menk’ hosdeɣēn gə sgsink’.>

In the phonetic surface pronunciation, we generally don’t mark aspiration. The problem is that voiceless stops and affricates have quite variable aspiration based on the speaker’s geographical origin. Aspiration is not meaningful, contrastive, or phonemic. This is discussed in §4.1.1.

When inside a text or paragraph, the orthographic form of words is often rendered with <>.

In some cases, we provide a hypothetical earlier pronunciation or version of a word. We use double slashes for this //...//. We also use double-slahes for intermediate forms of a word, i.e., a form where some but not all phonological rules have applied.

Part I

Phonology and orthography

3 Orthography

This chapter goes over the orthographic system of Western Armenian. We focus on describing the basic elements of Armenian orthography. We do not go in depth in explaining the relationship between orthography and phonology in this chapter. We instead refer readers to specialized sections within different phonology chapters. This is because understanding the linguistic use of various orthographic elements depends on understanding the phonology. More information of Armenian orthography can be found in the references of more specialized sources (Sanjian 1996).

As an overview, Section §3.1 goes over the basic letters of the Armenian alphabet. In Section §3.2, we describe the history of the Armenian script and past spelling reforms.

Section §3.3 goes over punctuation symbols in Western Armenian. Some symbols are placed at the end of phrases and words, while some symbols are 'infixed' or inserted inside of a word. These infixed symbols target the stressed or focused syllable in both declaratives and interrogatives (§??).

Section §3.4 goes over some a set of mismatches between the orthography and phonology. Although Armenian orthography is quite close to the surface pronunciation of words, there are mismatches and homophony in the voicing of consonants. These are due to diachronic sound changes.

3.1 Letters and transliteration

The Armenian script was invented in around the 5th century by saint Mesrop Mashtots, an Armenian clergyman. The script is written left-to-right. The script originally had only 36 letters, but an additional two letters օ <ō>, ֆ <f> were added later. The digraph ու <ow> is often treated as an additional symbol because its primary pronunciation is the vowel /u/. There is also a digraph ﬓ that's found in Eastern Armenian as an abbreviation for ﬓﬓ. In both dialects, it is pronounced as /ev/. We transliterate it as <ew>.

The native name of the Armenian script is in (1). Morphologically, the word 'alphabet' is a compound of the first two letters of the script (ա /ɑjp/, բ /pen/) that are connected with the conjunction /u/ 'and'.

(1) hɑjeren-i ɑjpupen-ə
Armenian.language-GEN alphabet-DEF
‘the Armenian alphabet’
հայերէնի այբուբենը

Throughout this grammar, we utilize the transliteration scheme in Table 3.1. Outside of this orthography chapter and of other orthography-based sections, we generally skip providing transliterations and just provide the IPA transcription.

Note that the transliteration system that we use in this book is not a standardized transliterations. Most existing transliterations of the Armenian script are based on the pronunciation rules of Classical or Eastern Armenian. Thus these transliteration systems are unsuitable to Western Armenian phonemes. As for those systems which were designed for Western Armenian, many of these systems are difficult to use because either a) they don’t have a 1-to-1 symbol association, or b) they represent affricates with difficult-to-remember symbols such as <č> for չ [t͡ʃ].

Armenian has graphemes or letters for every phonemic consonant and vowel. Vowels show some complications however (Table 3.2). For all but the schwa, a pronounced vowel is always written in the orthography in some manner or another. For the schwa, some instances of a pronounced schwa are written with ը <ə>, while some are not written at all.

Table 3.2: Vowel-to-letter associations for non-schwas

Vowel	Letter(s)		Example			
/ɑ/	ա	<a>	**ա**պ	<a**p**’>	[**ɑ**p]	‘palm’
/e/	ե	<e>	**ե**րգ	<**e**rk>	[j**e**ɾk]	‘song’
	է	<ē>	**է**շ	<**ē**ʃ>	[**e**ʃ]	‘donkey’
/o/	ո	<o>	**ո**չ	<**o**t͡ʃ’>	[v**o**t͡ʃ]	‘no’
	օ	<ō>	**օ**ձ	<**ō**t͡s>	[**o**t͡s]	‘snake’
/i/	ի	<i>	**ի**մ	<**i**m>	[**i**m]	‘my’
/u/	ու	<ow>	**ու**ս	<**ow**s>	[**u**s]	‘shoulder’
/ə/	ը	/ə/	**ը**ստ	<**ə**sd>	[**ə**st]	‘according to’
	∅		դրամ	<tram>	[t**ə**ɾɑm]	‘money’

The vast majority of words with a pronounced schwa do not represent the schwa in the orthography. We discuss this asymmetry in Section §4.2.2.2 within the context of the schwa’s phonology. In brief, when a schwa is unwritten, the orthography reflects the origins of the schwa as being epenthetic, a reduced vowel, or a syncopated vowel.

Table 3.1: Letters of Armenian script

Uppercase	Lowercase	Name		Transliteration	Pronunciation
Ա	ա	այբ	ɑjp	<a>	/ɑ/
Բ	բ	բեն	pen	<p>	/p/
Գ	գ	գիմ	kim	<k>	/k/
Դ	դ	դա	tɑ	<t>	/t/
Ե	ե	եչ	jetʃ	<e>	/e/, /je/
Զ	զ	զա	zɑ	<z>	/z/
Է	է	է	e	<ē>	/e/
Ը	ը	ըթ	ət	<ə>	/ə/
Թ	թ	թո	to	<t’>	/t/
Ժ	ժ	ժէ	ʒe	<ʒ>	/ʒ/
Ի	ի	ինի	ini	<i>	/i/
Լ	լ	լիւն	lʏn	<l>	/l/
Խ	խ	խէ	χe	<x>	/χ/
Ծ	ծ	ծա	d͡zɑ	<d͡z>	/d͡z/
Կ	կ	կեն	gen	<g>	/g/
Հ	հ	հո	ho	<h>	/h/
Ձ	ձ	ձա	t͡sɑ	<t͡s>	/t͡s/
Ղ	ղ	ղատ	ʁɑd	<ɣ>	/ʁ/
Ճ	ճ	ճէ	d͡ʒe	<d͡ʒ>	/d͡ʒ/
Մ	մ	մեն	men	<m>	/m/
Յ	յ	յի	hi	<y>	/j/, /h/, silent
Ն	ն	նու	nu	<n>	/n/
Շ	շ	շա	ʃɑ	<ʃ>	/ʃ/
Ո	ո	ո	vo	<o>	/o/, /vo/
Չ	չ	չա	t͡ʃɑ	<t͡ʃ’>	/t͡ʃ/
Պ	պ	պէ	be	<b>	/b/
Ջ	ջ	ջէ	t͡ʃe	<t͡ʃ>	/t͡ʃ/
Ռ	ռ	ռա	ɾɑ	<ṙ>	/ɾ/
Ս	ս	սէ	se	<s>	/s/
Վ	վ	վեւ	vev	<v>	/v/
Տ	տ	տիւն	dʏn	<d>	/d/
Ր	ր	րէ	ɾe	<r>	<ɾ>
Ց	ց	ցո	t͡so	<t͡s’>	/t͡s/
Ւ	ւ	հիւն	hʏn	<w>	/v/
Փ	փ	փիւր	pʏɾ	<p’>	/p/
Ք	քէ	քէ	ke	<k’>	/k/
Օ	օ	օ	o	<ō>	/o/
Ֆ	ֆ	ֆէ	fe	<f>	/f/
ԵՒ	ևՎ	ևՎ	u	<ow>	/u/, /v/

For ease of use, the supplementary materials include a Python script that converts between the transliteration and the orthography.

3.2 Writing system and spelling reforms

Western Armenian differs from Eastern Armenian because Western Armenian uses a more conservative spelling system called the Classical Orthography, Traditional Orthography, or Mesropian Orthography. Eastern Armenian instead uses the Reformed Orthography based on Soviet-era spelling reforms. Examples in Table 3.3 illustrate some of the differences. The Reformed system removed silent letters in words. Depending on the word, word-medial /e/ can be written with either grapheme է or ե; similarly word-medial vowel /o/ can be written with either ո or օ. The Reformed system removed this unpredictability by uniformity using ե, ո for word-medial /e, o/.

Table 3.3: Example of differences across spelling systems

Classical spelling	ծառայ	լեռ	տէր	մոմ	մօտ
Transliteration	<d͡zaṙay>	<leṙ>	<dēr>	<mom>	<mōm>
Reformed spelling	ծառա	լեռ	տեր	մոմ	մոտ
Transliteration	<d͡zaṙa>	<leṙ>	<der>	<mom>	<mom>
Pronunciation	[d͡zɑrɑ]	[ler]	[deɾ]	[mom]	[mom]

The tradition spelling system included many more types of unpredictability between the orthography and pronunciation. We don't survey these unpredictabilities because they have limited connection to the synchronic phonology of Armenian. But they are useful for learners of the Armenian script. These unpredictabilities are amply documented in various teaching grammars of Armenian (bucket).

The above unpredictabilities are ultimately due to various sound changes from Classical Armenian to modern Western Armenian. For example, word-final glides in Classical Armenian were lost in polysyllabic words (cite macak or historical source) (Macak 2016: ch5; 2017: 1041).. This loss created the silent letter յ as in the word <d͡zaṙay> [d͡zɑrɑ] 'servant' above. For midvowels, the graphemes է <ē> and ե <e> were originally pronounced as different midvowels in Classical Armenian (cite macak or historical source). The <ē> form may have been a tense or long version /eː/, while <e> was a plain vowel /e/. Eventually, the two types of midvowels merged into just /e/, thus creating unpredictable spellings in Modern Armenian.

The spelling reform removed essentially all types of unpredictability, surveyed in Dum-Tragut (2009: 12ff). Because Western Armenian is spoken in the Armenian diaspora, Western Armenian publications and literature never adopted the soviet spelling reforms.

3.3 Punctuation symbols

Armenian utilizes a small set of punctuation symbols. One set of symbols is placed at the end of phrases and clauses. One set is placed before or between words. And one set is placed inside words.

This chapter provides a simple overview of the types of symbols. The use of these symbols do not significantly differ from Eastern Armenian. For more in-depth discussion of Armenian punctuation and their orthographic rules, see Dum-Tragut (2009: ch5)

For the word-final symbols in Table 3.4, these symbols are used for ending clauses and sentences. Their uses are largely the same in Armenian as they are in other European languages.

Table 3.4: Phrase-final or clause-final punctuation symbols

Symbol	Name		English analog
՝	ստորակէտ	əstoɾɑɡed	comma
՝	բութ	put	semicolon
.	միջակէտ	mit͡ʃɑɡed	colon
:	վերջակէտ	veɾt͡ʃɑɡed	period

Table 3.5 shows the set of punctuation symbols that are placed at the edges of words. These include the Armenian analog of apostrophes and brackets.

Table 3.5: Punctuation that is placed around words or at edges

Symbol	Name		English analog
« »	չակերտ	t͡ʃɑɡeɾd	brackets
՝	ապաթարց	ɑbɑtɑɾt͡s	apostrophe

Finally, Armenian has punctuation symbols that are infixed or placed inside the word (Table 3.6). These symbols are placed on the stressed vowel of the word which has the strongest prominence in the sentence. In other words, these markers are placed on the syllable that carries nuclear stress or sentential stress. See also Section §?? for discussion on stress and orthography.

Table 3.6: Punctuation symbols that are infixed inside the word

Symbol	Name		English analog
՞	պարոյկ	bɑrɤg	question mark
՛	երկար	jerɡɑr	exclamation mark
՛	շեշտ	ʃeʃt	emphasis mark

The stressed vowel is typically the final non-schwa vowel (2a). But the symbol can be placed further inward if the word has irregular non-final stress (2b). The examples below illustrate with the question mark symbol ՞, which we transliterate as ˀ. The marker is placed on the word which carries the nuclear stress of the sentence. We underline this syllable.

(2) a. mɑrjɑ-n u'ɾɑχ e
Maria-DEF happy is
'Is Maria happy?'
Մարիան ուրա՞խ է:
<Marian owra$^{?}$x ē>

b. 'vorkɑn g-uz-e-s
how.much IND-want-TH-2SG
'How much do you want?
Ո՞րքան կ՚ուզես:
<O$^{?}$rk'an g'owzes>

For general information on the stylistic uses of these markers, refer to teaching grammars and similar material bucket, maybe yeghian, maybe dum, probs just teaching grammar

3.4 Orthography-phonology mismatches

Armenian orthography does not exactly match the surface pronunciations of words, but it is fairly close. As said in Section §3.2, the traditional spelling system creates various types of homophony and unpredictability. This section overviews significant areas of mismatch between orthography and pronunciation. Some of these mismatches are only present in Eastern Armenian, while some are present in both.

Unless otherwise specified, the Eastern examples are from English Wiktionary. The Wiktionary examples are heavily moderated and are reliable for Eastern

Armenian. For transliterations, we adopt ISO 9985 to transliterate the words for Eastern Armenian,[1] while our own transliteration for Western Armenian.

3.4.1 Homophony in voiceless letters

In Western Armenian, stops and affricates have a 2-way laryngeal contrast. Stops and affricates are phonologically either voiced or voiceless.[2] However the orthography displays a 3-way contrast between the graphemes for stops and affricates. Each voiced sound has one corresponding voiced grapheme, but each voiceless sound has two homophones letters. Table 3.7 illustrates for all stops and affricates.

Table 3.7: Homophonous letters for stops and affricates

Letter	Trans.	Pron.	Example			
բ	<p>	/p/	բառ	<paṙ>	[pɑɾ]	‘word’
փ	<p’>	/p/	փառք	<p’aṙk’>	[pɑɾk]	‘glory’
պ	<b>	/b/	պար	<bar>	[bɑɾ]	‘dance’
դ	<t>	/t/	դեր	<ter>	[teɾ]	‘role’
թ	<t’>	/t/	թեւ	<t’ew>	[tev]	‘wing’
տ	<d>	/d/	տեղ	<deɣ>	[deʁ]	‘place’
գ	<k>	/k/	գամ	<kam>	[kɑm]	‘nail’
ք	<k’>	/k/	քար	<k’ar>	[kɑɾ]	‘rock’
կ	<g>	/g/	կար	<gar>	[gɑɾ]	‘string’
ձ	<t͡s>	/t͡s/	ձագ	<t͡sak>	[t͡sɑk]	‘cub’
ց	<t͡s’>	/t͡s/	ցաւ	<t͡s’aw>	[t͡sɑv]	‘pain’
ծ	<d͡z>	/d͡z/	ծակ	<d͡zag>	[d͡zɑg]	‘hole’
ջ	<t͡ʃ>	/t͡ʃ/	ջուր	<t͡ʃowr>	[t͡ʃuɾ]	‘water’
չ	<t͡ʃ’>	/t͡ʃ/	չու	<t͡ʃ’ow>	[t͡ʃu]	‘flight’
ճ	<d͡ʒ>	/d͡ʒ/	ճուտ	<d͡ʒowd>	[d͡ʒud]	‘chick’

.

For a given voiceless sound like [p], there are two homophonous graphemes բ, փ <p, p’>. For a speaker of Western Armenian, the choice of grapheme is unpredictable and has no phonological correlation. The homophony is a cause for common spelling errors for Western Armenian.

[1] https://www.translitteration.com/transliteration/en/armenian-eastern-classical/iso-9985/

[2] Acoustically, the actual correlates of voice vary by geographic region. This is overviewed in Section §4.1.1.

The voiceless homophony is because in Classical Armenian, the different graphemes did reflect different voicing quality (Table 3.8). In modern Western Armenian, the stops/affricates show a two-way contrast between voiced and voiceless. But Classical stops/affricates had a 3-way contrast between voiced, voiceless unaspirated, and voiceless aspirated. The 3-way contrast from Classical Armenian survived into Eastern Armenian, but not Western. The examples below illustrate with the labial stops.

Table 3.8: Labial stops in Western Armenian (WA) vs. Eastern Armenian (EA) and Classical Armenian (CA)

Letter	Example		Pronunciation		
			WA	EA/CA	
բ		<p>	/p/	/b/	
	բառ	<paṙ>	[pɑɾ]	[bɑr]	‘word’
փ		<p’>	/p/	/pʰ/	
	փառք	<p’aṙk’>	[pɑɾk]	[pʰɑrkʰ]	‘glory’
պ		<b>	/b/	/p/	
	պար	<bar>	[bɑɾ]	[pɑɾ]	‘dance’

When the 3-way contrast changed into a 2-way contrast for Western Armenian, this created homophony for the voiceless letters. In terms of the exact sound change from Classical to Western, the Classical voiceless aspirates stayed voiceless with variable aspiration: փ /pʰ/ → /p/. But the voiced and unaspirated sounds switched voiced. The Classical voiceless unaspirates became voiced: պ /p/ → /b/. And the Classical voiced became voiceless: բ /b/ → /p/. This diachronic process is quite complicated and there has bee a lot of work in diachronic phonology to explain such a change CITE Baronian and other folks.

3.4.2 Mismatches in voicing of clusters

The second area of mismatches comes from voicing in clusters. In some monomorphemic words, the orthography shows consonants clusters with non-agreeing voice. In pronunciation though, these clusters are pronounced with identical voicing. Table 3.9 illustrates with clusters of a sibilant and an orthographic voiced stop, and clusters of a voiced fricative and voiceless stop.

Table 3.9: Orthographic mismatches in voicing of consonant clusters

սպասել	<**sb**asel>	[ə**sp**ɑsel]	'to wait'
հաստ	<has**d**>	[hɑ**st**]	'thick'
սկիզբ	<**sg**izb>	[ə**sk**isp]	'beginning'
զբօսանք	<**zp**ōsank'>	[ə**sp**osɑnk]	'recess'
աղբ	<a**ɣp**>	[ɑ**χp**]	'trash'
խեղդել	<xe**ɣt**el>	[χe**χt**el]	'to strangle'
յաղթել	<ya**ɣt**'el>	[hɑ**χt**el]	'to win'
ազգ	<a**zk**>	[ɑ**sk**]	'nation'
աղքատ	<a**ɣk**'ad>	[ɑ**χk**ɑd]	'poor'

For the clusters above, there is no synchronic evidence that the cluster is composed of consonants with different voicing. That is, for a word like ազգ <azk> [ɑsk] 'nation', there is no synchronic evidence that the fricative [s] is derived from an underlying /z/. In all the above words, the cluster is part of a single morpheme, and the consonant never alternates in voicing. That is, for a word like [ɑsk], the fricative is never pronounced as [z] in any morphologically-related word.

Within Armenian philology, there is a lot of work in cataloging words with such mismatches between the spelling and pronunciation, mostly for Eastern Armenian (**Adjarian-1971-LiakatarPhono**; Minassian 1980: 29; Սուքիասյան 2004: 62,74; Ավետիսյան 2007: 21; Ավետիսյան 2011: 30). See Հովհաննիսյան (2014: 59) and Ղարագյուլյան (1974: 185) for a summary and systematic catalog. Teaching grammars likewise provide pedagogical tips on how to spell these clusters (Եզեկյան 2007: 75; Սևակ 2009: 92). But synchronically though, these clusters are just residues of sound changes from Classical to modern Armenian. They do not reflect modern Western morphology or phonology.

For example, for ազգ <azk> [ɑsk], the orthography reflects the fact this word is a reflex of Classical [ɑzg] where the cluster was voiced. The orthography matches the classical pronunciation. Eventual sound changes caused the grapheme գ /k/ to switch from being a voiced segment /g/ in Classical to a voiceless /k/ in Western. Once this change occurred, adjacent fricatives had to assimilate in voicing: CA [ɑzg] → [ɑsk], not [*ɑzk].

3.4.3 Post-rhotic devoicing

In Eastern Armenian, there are words which are orthographically written with a final voiced stops and affricates but which are pronounced with a voiceless

aspirated form (Table 3.10). This orthography mismatch is especially common after the rhotic /ɾ/.

cite

Table 3.10: Eastern Armenian words with orthography-phonology mismatch for voicing after rhotics

Spelling	Transliteration		Pronunciation		Meaning
	EA	WA	EA	WA	
նուրբ	<now**rb**>	<nowrp>	[ˈnu**ɾp**ʰ]	[ˈnuɾp]	‘gentle’
բարդ	<bar**d**>	<part>	[ˈbɑ**ɾt**ʰ]	[ˈpɑɾt]	‘complex’
երգ	<e**rg**>	<erk>	[ˈje**ɾk**ʰ]	[ˈjeɾk]	‘song’
բարձ	<bar**j**>	<par͡ts>	[ˈbɑ**ɾts**ʰ]	[ˈpɑɾ͡ts]	‘pillow’
վերջ	<ver**ǰ**>	<ver͡tʃ>	[ˈve**ɾt͡ʃ**ʰ]	[ˈveɾ͡tʃ]	‘end’

Because of how frequent this mismatch is, many philologists and phonologists have argued that Eastern Armenian has an allophonic rule of devoicing final voiced stops and affricates after rhotics. But this rule is not true allophony in Eastern Armenian (Table 3.11). One can find words where devoicing does not apply. There are likewise no cases of this rule applying in derived environments. The most likely scenario is that this orthography-phonology mismatch in Eastern Armenian is just a diachronic change and not a synchronically active rule.

Table 3.11: Eastern Armenian words where final voiced stops or affricates surface after rhotics

Spelling	Transliteration		Pronunciation		Meaning
	EA	WA	EA	WA	
բորբ	<bo**rb**>	<porp>	[ˈbo**ɾb**]	[ˈboɾp]	‘bright’
թակարդ	<t’aka**rd**>	<t’agart>	[tɑˈkɑ**ɾd**]	[tɑˈgɑɾt]	‘trap’
գորգ	<go**rg**>	<kork>	[ˈgo**ɾg**]	[ˈkoɾk]	‘carpet’
մերձ	<me**rj**>	<mer͡ts>	[ˈme**ɾd͡z**]	[ˈmeɾ͡ts]	‘near’
կամուրջ	<kamow**rǰ**>	<gamowrt͡ʃ>	[kɑˈmu**ɾd͡ʒ**]	[gɑˈmuɾt͡ʃ]	‘bridge’

It is possible that this this rule of devoicing stops/affricates after /ɾ/ is a diachronic sound change that is in progress in Eastern Armenian (Table 3.12) (Ասատրյան 1976, Ավետիսյան 2005). For example, Vahagn Petrosyan informs us that some words are prescriptively pronounced with a final voiced stop, but this voiced sound is often colloquially devoiced.

Table 3.12: Eastern Armenian words where final voiced stops or affricates are variably devoiced after rhotics

Spelling	Transliteration		Pronunciation			Meaning
	EA	WA	Std. EA	Coll. EA	WA	
կախարդ	<kaɣa**rd**>	<gaɣart>	[kɑˈʁɑ**ɾd**]	[kɑˈʁɑ**ɾt**ʰ]	[gɑˈʁɑɾt]	‘witch’
լուրջ	<k’ow**rǰ**>	<k’owrt͡ʃ>	[ˈku**ɾd͡ʒ**]	[ˈku**ɾt͡ʃ**ʰ]	[kluɾt͡ʃ]	‘rag’

read the above EA references again for data for other stuff

No such issue arises in Western Armenian (Table 3.13). Word-final voiced stops and affricates exist after rhotics.

Table 3.13: No post-rhotic devoicing in Western Armenian

Spelling	Transliteration (WA)	Pronunciation (WA)	Meaning
կերպ	<ge**rb**>	[ˈge**ɾb**]	‘form’
աւարտ	<awa**rd**>	[ɑˈvɑ**ɾd**]	‘end’
մերկ	<me**rg**>	[ˈme**ɾg**]	‘naked’
գործ	<ko**rd͡z**>	[ˈko**ɾd͡z**]	‘work’
սուրճ	<sow**rd͡ʒ**>	[ˈsu**ɾd͡ʒ**]	‘coffee’

3.4.4 Deletion of <h> after rhotics

In modern Armenian, the orthography has a letter հ for the sound /h/. But this sound is deleted in some words. This deletion is obligatory and likely due to a diachronic rule.

There are some roots which are orthographically spelled with a rhotic + <h>. Some of these roots pronounce the <h>, and some don’t. Table 3.14 lists such roots where the <h> is not pronounced. For these roots, this <h> was likely pronounced in earlier stages of the language. The sound /h/ is absent both when the word is said in isolation and when suffixes are added.

Table 3.14: Words where the letter <h> is not pronounced after rhotics

Root	աշխարհ	<aʃxar**h**>	[ɑʃˈχɑɾ]	‘world’
→	աշխարհային	<aʃxar**h**ayin>	[ɑʃχɑɾ-ɑˈjin]	‘worldly’
→	աշխարհի	<aʃxar**h**i>	[ɑʃχɑˈɾ-i]	‘world-GEN’
Root	խոնարհ	<xonar**h**>	[χoˈnɑɾ]	‘humble’
→	խոնարհում	<xonar**h**owm>	[χonɑˈɾ-um]	‘conjugation’
→	խոնարհէ	<xonar**h**ē>	[χonɑˈɾ-e]	‘humble-ABL’
Root	ճանապարհ	<d͡ʒanabar**h**>	[d͡ʒɑnɑˈbɑɾ]	‘road’
→	ճանապարհորդ	<d͡ʒanabar**h**ort>	d͡ʒɑnɑbɑˈɾ-oɾt	‘traveller’
→	ճանապարհներ	<d͡ʒanabar**h**ner>	[d͡ʒɑnɑbɑɾ-ˈner]	‘road-PL’
Root	շնորհ	<ʃnor**h**>	ʃəˈnoɾ	‘grace’
→	շնորհակալ	<ʃnor**h**agal>	[ʃənoɾ-ɑˈgɑl]	‘thankful’
→	շնորհով	<ʃnor**h**ov>	[ʃənoˈɾ-ov]	‘grace-INS’

For the above words, the absence of a pronounced /h/ is not a phonological rule but an orthography-phonology mismatch. Such a rule of deleting an /h/ after a rhotic must have been a diachronic rule, and it is not a synchronic rule. Evidence for this is that there are roots where the <rh> sequence is pronounced as /ɾh/ (Table 3.15).

Table 3.15: Words where the letter <h> is pronounced after rhotics

ժպիրհ	<ʒbi**rh**>	[ʒəˈbi**ɾh**]	‘insolent’
նիրհ	<ni**rh**>	[ˈni**ɾh**]	‘light slumber’
արհեստ	<a**rh**esd>	[ɑɾˈ**h**est]	‘handicraft’
արհամարհ	<a**rh**amarh>	[ɑ**ɾh**ɑˈmɑɾ]	‘despicable’
զարհուր	<za**rh**ur>	[zɑɾˈ**h**uɾ]	‘terrifying’

There is likewise the bound root օրհն <ōrhn> where the <h> is typically pronounced as a /t/ in all its derivatives (Table 3.16).

Table 3.16: Words where the letter <h> is pronounced as [t]

օրհնել	<ōr**h**nel>	[o**ɾt**ˈn-e-l]	‘to bless’	√-TH-INF
օրհնեալ	<ōr**h**neal>	[o**ɾt**ˈn-jɑl]	‘blessed’	√-ADJZ
օրհնութիւն	<ōr**h**nut’iwn>	[o**ɾt**n-uˈtʏn]	‘blessing’	√-NMLZ

This deletion can vary by dialect (Table 3.17). For the root <xorh> խորհ, the archaic form pronounces the /h/ while the the modern form does not. For some

derivatives, the lack of an /h/ is the norm. But for derivatives like the verb form, Western Armenian lacks the /h/ (Սաքապետոյեան 2011) while Eastern Armenian keeps the /h/ (Vahagn Petrosyan, p.c. on the Wiktionary entry). Petrosyan reports that the verb form is treated as a learned word in Eastern Armenian, thus preserving the archaic pronunciation with /h/, while we report that the Western cognate is perceived as a common word and thus without an /h/.

Table 3.17: Variation in pronouncing the letter sequence <rh>

		Western	Eastern	
խորհ	<xo**rh**>	[ˈχo**ɾh**]	[ˈχo**ɾh**]	'thought'
խորհուրդ	<xo**rh**owrt>	[χoˈ**ɾ**-uɾt]	[χoˈ**ɾ**-uɾtʰ]	'thought'
խորհել	<xo**rh**el>	[χoˈ**ɾ**-el]	[χo**ɾ**ˈ**h**-el]	'to think'

In sum, the above data is just an orthography-phonology mismatch, and is not a synchronic phonological rule of /h/ deletion.

4 Segmental phonology

This chapter goes over the segmental phonology of Western Armenian (henceforth Armenian or WA). We focus on providing the basic phoneme inventory of Armenian.When possible, we provide frequency data on differnet segments based on the grammar's associated dictionary.

We document the set of attested allophonic processes in Armenian. For consonants, these processes involve changes in the laryngeal or voicing quality of obstruents, i.e., voicing assimilation. For vowels, there is little known about any allophonic alternations. There are some reports of vowel rounding for the underlying sequence /ju/. We briefly list segmental processes that have been reported in previous grammars but which seem to be either unsystematic or obsolete in modern speech.

This section focuses more so on the phonology of segments, and not on their phonetics or acoustics. For an overview of segmental phonetics, see Seyfarth et al. (2024).

For ease of reference, Figures 4.1 and 4.2 provide the consonant inventory and vowel inventory.

	Bilabial	Labio -dental	Dental	Alveolar	Post-alveolar	Palatal	Velar	Uvular	Glottal
Plosive	p b		t d				k g		(ʔ)
Affricate			t͡s d͡z		t͡ʃ d͡ʒ				
Nasal	m			n			(ŋ)		
Tap									
Fricative		f v		s z	ʃ ʒ			χ ʁ	h
Rhotic				ɾ					
Lateral				l					
Glide	. (w)					j			

Figure 4.1: Consonant inventory

The consonant inventory includes [ŋ] in parenthesis. This sound is not a contrastive phoneme in Armenian; it is derived from /n/ when /n/ precedes a velar stop (§4.4). The glide [w] is in parenthesis because it is restricted to non-nativized loanwords (§4.1.7). The glottal stop [ʔ] is epenthesized in vowel hiatus in some morphological constructions (§5.6).

The voiceless stops and affricates are aspirated some geographically-defined groups of WA speakers. We discus aspiration and allophonic deaspiration in write something on aspiration problems.

For vowels, we include the sound /ʏ/ even though this sound is often interchangeable with the sound sequence /uj/. We include the schwa /ə/. For the mid vowels, many past phonological studies of Armenian treated these vowels as lax /ɛ, ɔ/. We treat them as tense /e, o/. We discuss this difference in Section §4.2.1. We include /œ/ as a marginal phoneme.

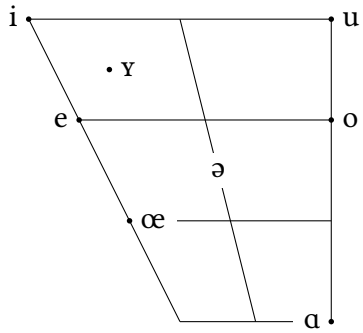


Figure 4.2: Vowel inventory

We go over each type of segment below.

4.1 Consonants

4.1.1 Stops

Western Armenian has a 2-way distinction for stops: phonologically voiced vs. phonologically voiceless (Table 4.1). Stops can have one of 3 places of articulation: bilabial, coronal (dental), and velar. Near-minimal pairs are below for word-initial, intervocalic, and word-final stops.

In terms of articulation, the coronal series /t, d/ is usually pronounced with tongue-tip touching the back of the teeth, i.e. a dental articulation. Dental articulation is previously reported for both Western Armenian and Eastern Armenian CITEpadd stuff. A more narrow phonetic transcription would transcribe these consonants as [t̪, d̪]. We opt for simpler [t, d].

In terms of acoustics, the distinction between the phonologically voiced and voiceless stops varies by geographic region and by influence of other languages. Depending on the region where Western Armenian is spoken, the phonological

Table 4.1: Near-minimal pairs for stops

	Initial		Intervocalic		Final	
/p/	ˈpa͡ts	‘open’	ʃɑˈpɑt	‘week’	ˈt͡ʃɑp	‘size’
		բաց		շաբաթ		չափ
	pɑˈɾi	‘kind’	pɑpɑˈkel	‘to hope’	ɑˈɾɑp	‘Arab’
		բարի		փափաքել		արաբ
/b/	ˈbɑʁ	‘cold’	bɑˈbɑ	‘dad’	ˈgɑb	‘link’
		պաղ		պապա		կապ
	bɑʃtel	‘to worship’	ɑbɑˈgi	‘glass’	bɑˈɾɑb	‘empty’
		պաշտել		ապակի		պարապ
/t/	ˈtɑs	‘lesson’	tɑˈtɑɾ	‘pause’	ˈpɑt	‘duck’
		դաս		դադար		բադ
	tɑˈdel	‘to judge’	ɑtɑˈmɑnt	‘diamond’	ɑɾˈd͡zɑt	‘silver’
		դատել		ադամանդ		արծաթ
/d/	ˈdɑɾ	‘letter’	gɑˈdɑg	‘joke’	ˈmɑd	‘finger’
		տառ		կատակ		մատ
	dɑˈɾi	‘year’	bɑdɑˈni	‘teenager’	ɑˈzɑd	‘free’
		տարի		պատանի		ազատ
/k/	ˈkɑm	‘nail’	ʃɑˈkɑɾ	‘sugar’	ˈtɑk	‘crown’
		գամ		շաքար		թագ
	kɑˈvɑt	‘cup’	ɑkɑˈɾɑg	‘farm’	ɑˈɾɑk	‘fast’
		գաւաթ		ագարակ		արագ
/g/	ˈgɑt	‘milk’	d͡ʒɑˈgɑd	‘forehead’	ˈpɑg	‘closed’
		կաթ		ճակատ		փակ
	gɑˈɾɑb	‘swan’	dɑgɑˈvin	‘still’	ɑˈɾɑg	‘proverb’
		կարապ		տակաւին		առակ

voiced-voiceless distinction /D-T/ is acoustically manifested by either prevoicing vs. unaspiration [D-T], unaspiration vs. aspiration [T-T^{h}], or prevoicing vs. aspiration [D-T^{h}]. Table 4.2 illustrates.

Table 4.2: Acoustic variation of stops

Phonological value		[D-T^h]	[D-T]	[T-T^h]
Voiced	/b/	[b]	[b]	[p]
Voiceless	/p/	[p^h]	[p]	[p^h]
Voiced	/d/	[d]	[d]	[t]
Voiceless	/t/	[t^h]	[t]	[t^h]
Voiced	/g/	[g]	[g]	[k]
Voiceless	/k/	[k^h]	[k]	[k^h]
Region		Turkey	Lebanon	USA

The earliest work on Western Armenian was done by Hrachia Adjarian in the late 19th century (Adjarian 1899; translated in **Adjarian-2024-StopsaffricatesClassicalArmenia** He collected acoustic data on speakers of Armenian across the Ottoman Empire. His work likewise one of the earliest work to utilize what is now called Voice Onset Time (VOT) (Braun 2013). Some of his speakers had a [D-T^h] distinction, whereby phonologically voiced stops were prevoiced while phonologically voiceless stops were voiceless aspirated. Various dialects of Armenian in the Ottoman Empire were reported to have D-T^h distinction as well (see different dialect chapters in Hossep 2024, translated from Աճառեան 1911).

Because of Adjarian's foundational work, nearly all subsequent linguistic discussions on Western Armenian treat the language as having a [D-T^h] distinction. But more recent work has shown the actual acoustic value of stops is subject to extensive geographic variation. This variation is based on the dominant language of the community in which Western Armenian is spoken.

For example, for speakers of Western Armenian in Istanbul, these speakers have a [D-T^h] distinction CITE: bolsahayeren paper, just as previously reported by Adjarian over a century ago CITE: bolsahaten bookturkish. This VOT distinction is likewise found for Turkish, the dominant language of the Istanbul CITE: turkish. But for speakers in Lebanon, these people have a [D-T] distinction where the phonologically voiceless stop is unaspirated (Kelly & Keshishian 2019). This distinction matches that of Lebanese Arabic. As for speakers in the US, they have a [T-T] distinction where the phonologically voiced stops are phonetically voiceless unaspirated, while the phonologically voiceless stops are phonetically voiceless aspirated (Kelly & Keshishian 2021). This matches the situation for North American English. Similar geographic effects are documented for Armenian communities in Canada (Tahtadjian 2021).

For consistency, all phonologically voiced and voiceless stops in this grammar are transcribed as just /D-T/, without explicit aspiration marking, because con-

trast phonetically varies by speaker. For example for HD, he lived in Lebanon up until 2014 at the age of 21, so his voicing system was likely a [D-T] system. But since 2014, he has been in an English-dominant environment for the US so his voicing system is [T-T^{h}] with some occasional prevoicing, as described in Seyfarth et al. (2024).

It is an open question how the voicing distinction is acoustically manifested in other geographic areas where Western Armenian is spoken, including France, Armenia, Syria, Latin America, and elsewhere. It is likely that their voicing system would match with that of the dominant language in their society.

In terms of lexical frequency, we calculated the number of stops in different word-positions from the associated dictionary (Table 4.3). The stops seem well distributed across natural class and position. Labial stops seem less frequent word-finally compared to other stops.

Table 4.3: Lexical frequency of stops

Sound	Total	Initial	Medial	Final
p	12105	3662	7615	828
t	24605	3289	19162	2154
k	17689	3677	11079	2933
b	7061	2374	4085	602
d	15192	2138	10088	2966
g	22487	4195	13694	4598

4.1.2 Affricates

Western Armenian has affricates in two places of articulation: dental /t͡s, d͡z/ and post-alveolar /t͡ʃ, d͡ʒ/. We provide minimal pairs in Table 4.4.

Table 4.4: Near-minimal pairs for affricates

	Initial		Intervocalic		Final	
/t͡s/	ˈt͡sɑχ	‘left’	kɑˈt͡sɑχ	‘vinegar’	ˈtɑt͡s	‘wet’
	ձախ			քացախ		թաց
	t͡sɑˈmɑk	‘continent’	pɑt͡sɑˈgɑ	‘absent’	gɑˈmɑt͡s	‘slow’
	ցամաք			բացակայ		կամաց
/d͡z/	ˈd͡zɑp	‘clap’	ɑˈd͡zɑnt͡s	‘suffix’	t͡sɑd͡z	‘low’
	ծափ			ածանց		ցած
	d͡zɑˈnot	‘familiar’	ɑd͡zɑˈgɑn	‘adjective’	ɑˈɾɑd͡z	‘proverb’
	ծանօթ			ածական		առած
/t͡ʃ/	ˈt͡ʃɑɾ	‘bad’	hɑˈt͡ʃɑd͡z	‘barked’	ˈɑt͡ʃ	‘right’
	չար			հաչած		աջ
	t͡ʃɑˈmit͡ʃ	‘raisin’	ɑt͡ʃɑˈgit͡s	‘assistant’	gɑˈgɑt͡ʃ	‘tulip’
	չամիչ			աջակից		կակաչ
/d͡ʒ/	ˈd͡ʒɑʃ	‘food’	dɑˈd͡ʒɑɾ	‘temple’	ˈhɑd͡ʒ	‘satisfied’
	ճաշ			տաճար		հաճ
	d͡ʒɑmˈpɑ	‘road’	hɑd͡ʒɑˈχel	‘to frequent’	ɑnˈhɑd͡ʒ	‘unsatisfied’
	ճամբայ			յաճախել		անհաճ

In terms of articulation, the series /t͡s, d͡z/ is usually reported to have dental contact. But alveolar contact is reported for some speakers. CITEpcite

As with the stops, there is widespread geographic variation for the acoustics of affricates. This is summarized in Table 4.5 .

Table 4.5: Acoustic variation of affricates

Phonological value		[DZ-TS^{h}]	[DZ-TS]
Voiced	/d͡z/	[d͡z]	[d͡z]
Voiceless	/t͡s/	[t͡s^{h}]	[t͡s]
Voiced	/d͡ʒ/	[d͡ʒ]	[d͡ʒ]
Voiceless	/t͡ʃ/	[t͡ʃ^{h}]	[t͡ʃ]
Region		Turkey	Lebanon & USA

Traditional reports from Adjarian treat the distinction between the phonologically voiced and voiceless affricates as being between prevoicing vs. voiceless aspirated (**Adjarian-2024-StopsaffricatesClassicalArmenianstudiedmoderndialects**, Adjarian 1899). This distinction is maintained in modern Istanbul Armenian cite: bolsahahyeren. For communities in Lebanon and the US, more recent acoustic studies find that the distinction is between prevoiced vs. voiceless unaspirated

(Kelly & Keshishian 2019, 2021, Seyfarth et al. 2024). As with the stops, the variation is due to language contact. Turkish has aspirated affricates, while Lebanese Arabic and North American English do not. CITEpcite, section 1.3.1 of this paper https://scottseyfarth.com/docs/SeyfarthGarellek2020.pdf–

An open question is whether there are subdialects of Western Armenian which acoustically mark the distinction in terms of voiceless unaspirated vs. voiceless aspirated. We expect to find such a distinction for speakers who live in a society where the dominant language has such a distinction.

For consistency, we transcribe all voiceless affricates in this grammar as unaspirated.

In terms of lexical frequency, we calculated the number of affricates in different word-positions from the associated dictionary (Table ??). The affricates seem well distributed across natural class and position. The affricate /d͡ʒ/ seems relatively rare compared to the other affricates.

Table 4.6: Lexical frequency of affricates

Sound	Total	Initial	Medial	Final
t͡s	10527	1569	6840	2118
t͡ʃ	6593	1348	3916	1329
d͡z	5915	1220	3560	1135
d͡ʒ	3016	893	1810	313

4.1.3 Fricatives

Western Armenian has the following set of fricatives: /s, z, ʃ, ʒ, χ, ʁ, h/. Each voiceless fricative has a voiced counterpart, except for /h/. Near-minimal pairs are in Table 4.7.

In terms of lexical frequency, we calculated the number of fricatives in different word-positions from the associated dictionary (Table 4.8).

Table 4.7: Near-minimal pairs for fricatives

	Initial		Intervocalic		Final	
/f/	ˈ**f**ilm	‘film’ ֆիլմ	ʁɑˈ**f**ɑ	‘head’ ղաֆա	ˈu**f**	interjection ուֆ
/v/	ˈ**v**ɑɾt	‘rose’ վարդ	ɑˈ**v**ɑk	‘senior’ աւագ	ˈlɑ**v**	‘good’ լաւ
	vɑˈzel	‘to run’ վազել	ɑ**v**ɑˈzɑn	‘pool’ աւազան	ɑkˈɾɑ**v**	‘crow’ ագռաւ
/s/	ˈ**s**ɑɾ	‘ice’ սառ	hɑˈ**s**ɑg	‘height’ հասակ	ˈmɑ**s**	‘part’ մաս
	sɑˈhun	‘smooth’ սահուն	tɑ**s**ɑˈgɑn	‘classical’ դասական	vəˈnɑ**s**	‘damage’ վնաս
/z/	ˈ**z**ɑd	‘separate’ զատ	kɑˈ**z**ɑn	‘beast’ գազան	ˈmɑ**z**	‘hair’ մազ
	zɑˈdig	‘Easter’ Զատիկ	ʁɑ**z**ɑˈɾoz	masc. name Ղազարոս	ɑˈvɑ**z**	‘sand’ աւազ
/ʃ/	ˈ**ʃ**ɑh	‘gain’ շահ	d͡ʒɑˈ**ʃ**ɑg	‘taste’ ճաշակ	ˈd͡ʒɑ**ʃ**	‘food’ ճաշ
	ʃɑˈbig	‘shirt’ շապիկ	ɑ**ʃ**ɑˈgeɾd	‘student’ աշակերտ	lɑˈvɑ**ʃ**	‘lavash’ լաւաշ
/ʒ/	ˈ**ʒ**ɑm	‘time’ ժամ	pɑˈ**ʒ**ɑg	‘cup’ բաժակ	ˈu**ʒ**	‘strength’ ուժ
	ʒɑˈɾɑŋk	‘heir’ ժառանգ	tɑ**ʒ**ɑˈnil	‘to tire’ տաժանիլ	bɑˈdi**ʒ**	‘punishment’ պատիժ
/χ/	ˈ**χ**ɑʁ	‘game’ խաղ	d͡zɑˈ**χ**ɑd͡z	‘sold’ ծախած	ˈt͡sɑ**χ**	‘left’ ձախ
	χɑˈpel	‘to trick’ խաբել	nɑ**χ**ɑˈkɑh	‘president’ նախագահ	uˈɾɑ**χ**	‘happy’ ուրախ
/ʁ/	ˈ**ʁ**eg	‘helm’ ղեկ	ɑˈ**ʁ**ɑnt	‘sect’ աղանդ	ˈɑ**ʁ**	‘salt’ աղ
	ʁɑˈzɑɾ	masc. name Ղազար	ɑ**ʁ**ɑvˈni	‘pigeon’ աղաւնի	χɑˈ**ʁ**ɑ**ʁ**	‘peaceful’ խաղաղ
/h/	ˈ**h**ɑz	‘cough’ հազ	bɑˈ**h**ɑg	‘guard’ պահակ	ˈkɑ**h**	‘throne’ գահ
	hɑˈzɑɾ	‘thousand’ հազար	ɑ**h**ɑˈkin	‘numerous’ ահագին	səˈɾɑ**h**	‘hall’ սրահ

Table 4.8: Lexical frequency of fricatives

Sound	Total	Initial	Medial	Final
f	832	73	759	0
v	11979	3330	7518	1131
s	12861	2387	8985	1489
z	4362	1096	2900	366
ʃ	6014	1632	4019	363
ʒ	1873	431	1220	222
χ	8862	2641	5717	504
ʁ	8459	81	6553	1825
h	10433	5879	4252	302

The fricatives are attested in all prosodic positions, but some fricatives are less common. The fricative /ʁ/ is rarely found word-initially. The fricative /f/ is rare throughout Armenian. Most occurrences of /f/ come from loanwords that entered Armenian after the Middle Ages. For example, two out of three words with /f/ in Table 4.7 are loanwords: [ˈfilm] from 'film', and [ʁɑˈfɑ] from Turkish "kafa".[1]

In term of articulation, there is some divergence on the place of articulation for the series /s, z/. Some grammars report a dental articulation and some alveolar. For some individuals that we've asked, some report that the tongue touches the upper teeth, while some report that the tongue touches the lower teeth.

cite: fricative s z coronal place

For /χ, ʁ/, the voiced fricative has a typically uvular articulation. But the voiceless fricative can vary between velar and uvular. We suspect that the fricatives /χ, ʁ/ have free variation between velar and uvular place. Part of our suspicion is the fact that the native authors of this grammar cannot easily hear the difference between velar vs. uvular fricatives.

cite: fricative χ ʁ coronal place

For the fricative [v], some argue that this sound is always a surface pre-vocalic allophone of /u/, and that /v/ is not phonemic (Vaux 1998: 13). Evidence for this is that /u/ is sometimes replaced by [v] before vowels, as a type of vowel hiatus repair discussed in Section §5.6. We treat /v/ as a separate phoneme though. This is because of the following reasons:

1. Our native intuitions treat /v/ as a phoneme.
2. There is a dedicated grapheme for /v/ վ that is used in the Reformed orthography.

[1] https://en.wiktionary.org/wiki/kafa#Turkish

3. There are words where [v] is used even though there's no evidence that this [v] is synchronically related to an [u] sound such as in the examples in Table 4.7 like [vɑzel] 'to run'. If we treat [v] as non-phonemic, then we would have to argue that this word is derived from an underlying /uɑzel/ but there's no evidence for this underlying /u/.
4. There are words which show schwa epenthesis breaking up an orthographic <vC> cluster, such as [vənɑs] 'harm' վնաս <vnas>. If we treat [v] as not phonemic, then we would have to argue that such words are either derived from /unɑs/ with un-motivated /u/→[v] change, or derived from /uənɑs/ where the otherwise epenthetic schwa is causing the /u/ to change to [v] (Vaux 1998: 13).

Thus, although there is a synchronic rule of /u/ becoming [v] before vowels, there is evidence that [v] is also a separate phoneme.

4.1.4 Nasals

Western Armenian has nasals /m, n/. Near-minimal pairs are in Table 4.9.

Table 4.9: Near-minimal pairs for nasals

	Initial		Intervocalic		Final	
/n/	ˈ**m**ɑh	'death' մահ	ɑˈ**m**ɑn	'vessel' աման	ˈhɑ**m**	'taste' համ
	ˈ**m**ɑd͡zun	'yogurt' մածուն	ɑ**m**ɑˈnoɾ	'New Year' ամանոր	ɑnˈtɑ**m**	'member' անդամ
/m/	ˈ**n**ɑv	'ship' նաւ	tɑˈ**n**ɑg	'knife' դանակ	ˈtɑ**n**	'ayran' թան
	nɑˈmɑg	'letter' նամակ	ɑ**n**ɑˈbɑd	'desert' անապատ	iʃˈχɑ**n**	'prince' իշխան

In terms of articulation, /m/ is bilabial. The nasal /n/ typically has dental articulation [n̪], but we transcribe this segment as [n] for ease.

There is a velar nasal [ŋ]. But this sound is not a phoneme. It is an allophone of /n/ before velar stops. This is a discussed in Section §4.4.

In terms of lexical frequency, we calculated the number of nasals in different word-positions from the associated dictionary (Table 4.10). The velar nasal is restricted to word-medial position, before velar stops.

Table 4.10: Lexical frequency of nasals

Sound	Total	Initial	Medial	Final
m	15418	4598	8679	2141
n	42734	2186	23907	16641
ŋ	3050	0	3050	0

cite:nasal dental

4.1.5 Rhotic

Western Armenian has only one rhotic phoneme /ɾ/ (Table 4.11).

Table 4.11: Near-minimal pairs for rhotic /ɾ/

	Initial		Intervocalic		Final	
/ɾ/	ɾoˈbe	‘second’	tɑˈɾɑg	‘shelf’	ˈtɑɾ	‘century’
		որպէ		դարակ		դար
	ɾɑˈfi	masc. name	ɑɾɑˈɾɑd͡z	‘creature’	təʒˈvɑɾ	‘difficult’
		Ռաֆֆի		արարած		դժուար

The rhotic /ɾ/ is alveolar. Acoustically, the rhotic is often spirantized (Toparlak 2019). Word-finally, the rhotic is very fricated for some speakers and subdialects, such as in Istanbul Armenian bolsahay paper

In terms of lexical frequency, we calculated the number of rhotics in different word-positions from the associated dictionary (Table 4.12). It is relatively rare to find rhotic-initial words.

Table 4.12: Lexical frequency of rhotics

Sound	Total	Initial	Medial	Final
ɾ	43007	4598	36188	6531

Note that Eastern Armenian has a phonemic flap-trill distinction: /ɾ, r/. Each Eastern phoneme is presented by two graphemes ր, ռ <r, ṙ>. The same trill-flap distinction is reported for Classical Armenian (Macak 2017: 1040; Hossep 2024: 29). For Western Armenian, the two types of rhotics have merged into a single flap /ɾ/, causing the letters ր, ռ to be homophonous. This is illustrated in Table 4.13.

Table 4.13: Trill-flap merger in Western (WA) but not Eastern (EA)

	Initial		Intervocalic		Final	
ր	Ռաֆֆի		ծարաւ	հարազատ	քար	տկար
<r>	<raffi>		<d͡zaraw>	<harazad>	<k’ar>	<dgar>
EA /ɾ/	ɾɑ(f)ˈfi		t͡sɑˈɾɑv	hɑɾɑˈzɑt	kɑɾ	təˈkɑɾ
WA /ɾ/	ɾɑˈfi		d͡zɑˈɾɑv	hɑɾɑˈzɑd	kɑɾ	dəˈgɑɾ
	masc. name		‘thirsty’	‘kindred’	‘rock’	‘weak’
ռ	ռուս	ռազմիկ	առաջ	առարկայ	ծառ	օճառ
<ṙ>	<ṙus>	<ṙazmig>	<aṙat͡ʃ>	<aṙargay>	<d͡zaṙ>	<od͡ʒaṙ>
EA /r/	ˈrus	rɑzˈmik	ɑˈrat͡ʃʰ	ɑrɑrˈkɑ	ˈt͡sɑr	oˈt͡ʃɑr
WA /ɾ/	ˈɾus	ɾɑzˈmik	ɑˈɾat͡ʃ	ɑɾɑɾˈgɑ	ˈd͡zɑɾ	oˈd͡ʒɑɾ
	‘Russian’	‘warrior’	‘before’	‘object’	‘tree’	‘soap’

Historically, earlier stages of Western Armenian did have a phonemic flap-trill distinction adjarian. This is reported in early textbooks of Western Armenian cite.

For modern communities, there is no longer an active flap-trill distinction. Some textbooks acknowledge this and state that the letters ր ռ <r ṙ> are homophonous. cholak

But many schools and textbooks artificially teach a trill. They often qualify the distinction by saying that the pronunciation difference between the letters is “blurry”. These textbooks teach a distinction for two reasons. One reason is diachronic conservatism – they want to teach what was spoken a century or more ago. The other reason is to teach students ways to figure out the right spelling of words. For example, a teacher may exaggerate the pronunciation of a word that has the letter ռ <ṙ> by excessively trilling the letter. But neither students, teachers, nor communities actually use a trill phoneme in real speech. Many teachers in HD’s experience don’t even bother to speak artificially and acknowledge to students that the letters ր ռ <r, ṙ> are homophonous.

There is no active trill-flap distinction for communities in the US (Samuel Chakmadjian), Canada (Talia Tahtadjian), Turkey (Tabita Toparlak), Lebanon (HD, Avedis Samuelian), or Syria (HS, Setrag Hovsepian).

To illustrate this complex set of affairs, consider the Armenian community of Canada (Tahtadjian 2020). Tahtadjian reports us some older speakers have a trill-flap distinction and schools try to teach this distinction. However, students don’t truly acquire this distinction because there is little significant difference in their articulation of the letter ր <r> and ռ <ṙ>.

Note that there are reports that some Western Armenian communities have an allophonic rule of changing /ɾ/ to [r] before nasals CITEpsakabedoyan in jipa. We have not been able to confirm or reproduce such reports.

Some speakers likewise variably trill their flaps in certain environments (Seyfarth et al. 2024). Tabita Toparlak notes that in her impression, the Lebanese rhotic sounds like a trill more often than the Istanbul rhotic.

4.1.6 Lateral

Western Armenian has a lateral /l/ (Table 4.14).

Table 4.14: Near-minimal pairs for lateral /l/

	Initial		Intervocalic		Final	
/l/	ˈlajn լայն	‘wide’	tɑˈlɑɾ դալար	‘green’	ˈkɑl գալ	‘to come’
	lɑˈt͡soʁ լացող	‘crier’	hɑlɑˈd͡zel հալածել	‘to persecute’	səˈχɑl սխալ	‘wrong’

In terms of lexical frequency, we calculated the number of laterals in different word-positions from the associated dictionary (Table 4.15).

Table 4.15: Lexical frequency of laterals

Sound	Total	Initial	Medial	Final
l	14558	1390	5067	8101

The liquid /l/ is palatal. The lateral is a clear lateral and there is no allophonic lateral darkening or lateral velarization for the Lebanese community.

Armenians in Istanbul often velarize the lateral [ɫ] because of influences from Turkish. The general rule is that the lateral is velarized when it is in a syllable with a back vowel. This rule mimics Turkish. cite: bolsahayeren L

4.1.7 Glides

Western Armenian has a phonemic glide /j/ that is used throughout the language. The sound /w/ also exists as a marginal phoneme that’s restricted to a small set of loanwords, mostly recent. This sound is usually nativized with [v]. See Table 4.16.

Table 4.16: Near-minimal pairs for glides

	Initial		Intervocalic		Final	
/j/	ˈjeɾk	‘song’	tiˈjɑg	‘corpse’	ˈχoj	‘ram’
		եօթ		դիակ		խոյ
	jɑvˈrig	‘dear’	mijɑˈnɑl	‘to unite’	ˈtej	‘tea’
		եաւրիկ		միանալ		թէյ
/w/	wikipedˈjɑ	‘Wikipedia’	sɑmˈwel	‘Samuel’		
	vikipedˈjɑ	Ուիքիփեդիա	sɑmˈvel	Սամուէլ		

The glide [j] has a rather complex distribution. Word-initially, it is mostly found before the vowel [e]. Orthographically, the word-initial [je] is written as just ե <e> . The high rate of word-initial [je] sequences is due to a process of glide-epenthesis, discussed in (cite chapter dipthongization). Outside of [je] sequences, word-initial [j] is rarely found. Some common natives word with initial [jV] where V is not /e/ include the words for ‘seven’ and ‘oil’, and their derivations. There are some. loanwords with [jɑ]. These were borrowed and adapted from Turkish ‘yavru’ and its related forms[2].

Table 4.17: Distribution of the glide [j] word-initially

[je]	ˈjez	‘ox’	jeˈɾɑz	‘dream’
	եզ	<ez>	երազ	<eraz>
[jo]	ˈjot	‘seven’	jotnɑmˈjɑ	‘septennial’
	եօթ	<eōt’>	եօթնամեայ	<eōtnameay>
[ju]	ˈjuʁ	‘oil’	juˈʁod	‘oily’
	իւղ	<iwʁ>	իւղոտ	<iwʁod>
[jɑ]	ˈjɑvˈɾəm’	‘my dear’	jɑvˈɾig	‘dear’
	եաւրըմ	<eawrəm>	եաւրիկ	<eawrig>

To illustrate, Table 4.18) shows the distribution of /j/ + vowel substrings at the beginning of words in our associated dictionary.

[2] https://en.wiktionary.org/wiki/yavru

Table 4.18: Lexical frequency of the glide /j/ plus a vowel at the beginning of words

Substring	Initial
#jɑ	1
#jə	0
#je	1952
#jo	47
#ji	0
#jɤ	0
#ju	62

Word-medially, many occurrences of [j] are epenthetic. See Table 4.19. Inserting [j] is a common repair for vowel-vowel sequences, i.e., vowel hiatus (see Section §5.6). There are some morphemes where the surface [j] is not epenthetic and is adjacent to a consonant. These morphemes can be either roots or suffixes. Their syllabification is complicated, see Section §5.5.2.

Table 4.19: Distribution of the glide [j] word-medially

Epenthetic		Root-medial		Suffix-initial	
/V-V/→[VjV]	cf.	/CjV/	/Vj(C)/	/C-jV/	cf.
/kodi + e/	/kodi/	/ɑɾɑkjɑl/	/kɑjl/	/pɑjd + jɑ/	/pɑjd/
kodi'je	ko'di	ɑɾɑk'jɑl	'kɑjl	pɑjd'jɑ	'pɑjd
'belt-ABL'	'belt'	'apostle'	'wolf'	'wooden'	'wood'
գօտիէ	գօտի	առաքեալ	փայտեայ	գայլ	փայտ
<kōdiē>	<kōdi>	<aṙak'eal>	<p'aydeay>	<kayl>	<p'ayd>
/gɑdu + ov/	/gɑdu/	/mɑɾjɑm/	/kujn/	/gɑdɑɾ + jɑl/	/gɑdɑɾ/
gɑdu'jov	gɑ'du	mɑɾ'jɑm	'kujn	gɑdɑɾ'jɑl	gɑ'dɑɾ
'cat-INS'	'cat'	'Mary'	'color'	'perfect'	'end'
կատուով	կատու	Մարիամ	գոյն	կատարեալ	կատար
<gadowov>	<gadow>	<mariam>	<koyn>	<gadareal>	<gadar>

In the phonology of Armenian, some have argued that [j] is a separate phoneme /j/ (Fairbanks 1948: 10). In contrast, Vaux (1998: 12-3) argues that there is no phonemic /j/. He argues that non-epenthetic cases of surface [j] are due to allophony from underlying /i/. For example, the bi-morphemic word [gɑdɑɾ-jɑl] 'perfect' would be analyzed as underlyingly /gɑdɑɾ-iɑl/ (Vaux 1998: 28). And similarly, the name [mɑɾjɑm] 'Mary' would be derived from underlying /mɑɾiɑm/. He would analyze [j] as derived from /i/ via some rule of vowel-hiatus repair. His evidence

is based on diachrony and orthography. The classical orthography represents the word-medial surface [jɑ] via various diagraphs, such as եա <ea> or <ia>.

But in our native intuitions, such cases of non-epenthetic [j] are not due to allophony at all but are from an underlying /j/. The fact that the digraph sequence <ea> or <ia> is pronounced as [jɑ] is just a spelling-pronunciation rule. Furthermore, there are no morpheme-alternations to support treating these non-epenthetic [jɑ] sequences as anything other than /jɑ/.

Word-finally (Table 4.20), the glide [j] is found only in monosyllabic nouns and in compounds where the second member is a monosyllabic noun. For some polysyllabic words and monosyllabic verbs, the orthography has a final <y> letter but this letter is silent. Such a glide was pronounced in Classical Armenian, but has been lost in the modern language (Macak 2016: ch5; 2017: 1041).

Table 4.20: Distribution of the glide [j] word-finally

Monosyllabic noun and derivatives		Monosyllabic non-noun	Polysyllabic
/pɑj/	/mɑg + pɑj/	/gɑ/	/ɑgɾɑ/
ˈpɑj	mɑkˈpɑj	ˈgɑ	ɑgˈɾɑ
‘verb’	‘adverb’	‘exists.PRS.3SG’	‘witness’
բայ	մակբայ	կայ	ակռայ
<pay>	<magpay>	< gay>	<agṙay>
/hɑj/	/iɾɑn + ɑ + hɑj/	/lɑ/	/kulbɑ/
ˈhɑj	iɾɑnɑˈhɑj	ˈlɑ	kulˈbɑ
‘Armenian’	‘Iranian Armenian’	‘cries.PRS.3SG’	‘sock’
հայ	իրանահայ	լայ	գուլպայ
<hay>	<iranahay>	<lay>	<kowlbay>

For the above polysyllabic words, Vaux (1998: 20) treated the silent letter յ <y> as indicating an underlying glide /j/ that got deleted. We disagree with his analysis. Our native intuitions don’t ‘feel’ that there is any such glide. The orthography just has a silent letter. See discussion of the silent glide in Section §3.2, in regards to glide epenthesis and rule reversal.

In terms of lexical frequency, we calculated the number of glides in different word-positions from the associated dictionary (Table 4.21).

Table 4.21: Lexical frequency of glides

Sound	Total	Initial	Medial	Final
j	10981	2062	8860	59

4.2 Vowels

As a descriptive device, we divide the vowel inventory into the following sets:

- core vowels: /ɑ, e, o, i, u/ (§4.2.1)
- schwa: /ə/ (§4.2.2)
- front round vowels: /ʏ, œ/ (§4.2.3)

4.2.1 Canonical vowels

Armenian uses the following core vowels: /ɑ, e, i, o, u/. These vowels can be used in all phonological and prosodic positions. Table 4.22 lists words which have a core vowel in either a stressed or unstressed position. Stress is final.

Table 4.22: Near-minimal pairs for core vowels /ɑ, e, i, o, u/

	σ́		σσ́		σσ́	
/ɑ/	ˈtɑd դատ	‘cause’	d͡zɑˈkum ծագում	‘origin’	tuˈtɑg թութակ	‘parrot’
	ˈbɑk պաք	‘fast’	sɑˈloɾ սալոր	‘plum’	jeˈlɑg ելակ	‘strawberry’
/e/	ˈseχ սեխ	‘melon’	keˈdin գետին	‘ground’	luˈd͡zel լուծել	‘to solve’
	ˈsev սեւ	‘black’	peˈduɾ փետուր	‘feather’	hɑˈmeʁ համեղ	‘tasty’
/i/	ˈkit քիթ	‘nose’	kɨˈdɑg գիտակ	‘adept’	pɑˈɾik բարիք	‘good deed’
	ˈpiʁ փիղ	‘elephant’	tɨˈdel դիտել	‘to watch’	toˈniɾ թոնիր	‘tandoor’
/o/	ˈgoʁ կող	‘rib’	voˈɾɑg որակ	‘quality’	liˈmon լիմոն	‘lemon’
	ˈbot͡ʃ պոչ	‘tail’	koˈmeʃ գոմէշ	‘buffalo’	heˈɾos հերոս	‘hero’
/u/	ˈnuɾ նուռ	‘pomegranate’	suˈlit͡ʃ սուլիչ	‘whistle’	moˈɾuk մորուք	‘beard’
	ˈʃuk շուք	‘shadow’	puˈʒoʁ բուժող	‘healer’	siˈɾun սիրուն	‘lovely’

The vowels can be used in any position (Table 4.23). All core vowels can appear word-initially, word-finally, and word-medially. Though there are some asymmetries. Note that the words with a single vowel like [u] 'and' ու are counted in both 'Word-initial' and 'Word-final' columns.

Table 4.23: Lexical frequency of vowels in different word-positions

	Word-initial #_	Word-medial #..._... #	Word-final _#	Total
ɑ	7017	96427	994	104438
e	52	26710	151	26913
i	667	14388	2596	17651
o	692	19346	190	20228
u	670	18729	460	19859

There are some asymmetries when it comes to vowels at the edges of the word.

The most common initial vowel is /ɑ/, while the rarest is /e/. The other core vowels /i, o, u/ occupy an intermediate spot. The reason for the relative rarity of word-initial /e/ is due to diachronic sound changes. A series of sound changes from Classical Armenian to Modern Armenian caused the initial [e] sound from Classical Armenian to become [je] in Modern Armenian, while a Classical initial [ē] sound became modern [e]. Such sound changes are reflected in the orthography (cite chapter diphthongization. The letter է <ē> is used to mark a word-initial [e], while the letter ե <e> is used to mark a word-initial [je]. We discuss the phonological and morphological effects of these pronunciation rules in (diphtonziation chapter).

Word-finally, /i/ is most common, followed by /ɑ/, while the other vowels /e, o, u/ are relatively low frequency. The high frequency of final /i/ is likely because there are derivational suffixes that end in /i/, such as the plant-naming suffix /-eni/, the ethnicity-marking suffix /-ɑt͡si/, and the deverbal suffix /-i/ that is added onto any infinitive verb to create an adjective.

In terms of stress, the core vowels can appear stressed or unstressed. Table 4.22 listed target vowels in either the first or second (stressed) syllable. Table 4.24 lists words where the vowel is in a word-medial unstressed syllable.

Table 4.24: Core vowels /ɑ, e, i, o, u/ in word-medial unstressed position

/ɑ/	hun**ɑ**ˈɾen	‘Greek language’	jez**ɑ**ˈgi	‘singular’	hoʁ**ɑ**ˈtɑp	‘slipper’
		յունարէն		եզակի		հողաթափ
/e/	nɑɾənt͡ʃ**e**ˈni	‘orange-tree’	ɑv**e**ˈdis	‘good news’	ɑhɾ**e**ˈli	‘horrible’
		նարնջենի		աւետիս		ահռելի
/i/	vost**i**ˈɡɑn	‘police officer’	oɾ**i**ˈnɑg	‘example’	hoɾ**i**ˈnel	‘to fashion’
		ոստիկան		օրինակ		յօրինել
/o/	ɑχ**o**ɾˈʒɑg	‘appetite’	ɑs**o**ˈɾi	‘Assyrian’	ɑŋg**o**ˈʁin	‘bed’
		ախորժակ		ասորի		անկողին
/u/	bɑjt**u**ˈt͡sig	‘firework’	tit͡s**u**ˈhi	‘goddess’	ɑɾk**u**ˈni	‘royal’
		պայթուցիկ		դիցուհի		արքունի

Synchronically, there is no rule of reducing or deleting unstressed word-medial vowels. But diachronically, there have been idiosyncratic cases where a word would lose its medial vowel. We call such a process ‘syncope’. Syncope is not synchronically active but is part of a fossilized set of morphologically-conditioned alternations. We discuss syncope in (syncope chapter).

The core vowels can be used in virtually any type of phonologically-possible syllable. They can be used in a syllable with or without an onset, and with or without coda (Table 4.25).

Table 4.25: Core vowels in different types of syllables

	VC		VCC		CVC		CV	
/ɑ/	ˈɑp	‘palm’	ˈɑχt	‘disease’	ˈbɑb	‘pope’	ɡɑˈɡuʁ	‘soft’
		ափ		ախտ		պապ		կակուղ
/e/	ˈet͡ʃ	‘page’	ˈet͡ʃk	‘descent’	ˈged	‘dot’	seˈɾunt	‘generation’
		էջ		էջք		կէտ		սերունդ
/i/	ˈiʒ	‘viper	ˈint͡ʃ	‘what’	ˈbid͡z	‘stain’	viˈʃɑb	‘dragon’
		իժ		ինչ		բիծ		վիշապ
/o/	ˈot	‘air’	ɔrt.ˈnel	‘to bless’	ˈt͡sɔɾ	‘valley’	boˈɾɑl	‘to yell’
		օդ		օրհնել		ձոր		պօռալ
/u/	ˈuʃ	‘late’	ˈuʁd	‘camel’	ˈt͡sul	‘bull’	t͡ʃuˈtɑg	‘violin’
		ուշ		ուղտ		ցուլ		ջութակ

In terms of acoustics, there have been past studies on the acoustics of Armenian vowels, both Western and Eastern CITEUCKET. One of the largest studies for Western Armenian is Toparlak (2019), which was later interpreted by Seyfarth et al. (2024). We showed the vowel space in Figure 4.2.

For the mid vowels /e, o/, most previous phonological studies of Armenian transcribe the mid vowels as lax /ɛ, ɔ/. These studies include BUCLET. However acoustically, these vowels are quite close to [e, o] and we transcribe them as /e, o/, following Seyfarth et al. (2024). Furthermore, our native intuitions don't hear a clear difference between between [e, o] and [ɛ, ɔ]. This suggests that the Armenian mid vowels can phonetically range from [e, o] and [ɛ, ɔ] as a type of free variation.

4.2.2 Phonology of the schwa

The schwa vowel /ə/ has a complicated treatment in Armenian linguistics and philology. These complications involve a) disagreement over its phonemic status, b) orthographic representations of the schwa, and c) origin of the schwa as being derived or underived.

In terms of the phonological status of the schwa, most occurrences of the schwa are due to morpho-phonological processes. These processes are closely tied to the orthography. All these interactions have caused some researchers to argue that the schwa is not a phoneme. We disagree with this stance and treat the schwa as a phoneme. We first go over asymmetries in the distribution of schwas (§4.2.2.1), the derived role of the unwritten schwa (§4.2.2.2), and the phonemic status of underived and written schwas (§4.2.2.3).

4.2.2.1 Minimal pairs and asymmetries

In terms of phonemic status, some sources treat the schwa as a phoneme in Armenian cite bucket. As a phoneme, the schwa can be used to form minimal or near-minimal pairs with other vowels. We provide such pairs in Table 4.26.

Table 4.26: Near-minimal pairs for the schwa against core vowels

/ə, ɑ/	gəˈɾag	'fire'	gɑˈɾak	'butter'
	կրակ		կարագ	
/ə, e/	həˈɾad	'Mars'	heˈɾu	'far'
	Հրատ		հեռու	
/ə, i/	məˈnal	'to stay'	miˈnag	'alone'
	մնալ		մինակ	
/ə, o/	təˈkal	'spoon'	toˈkaχt	'tuberculosis'
	դգալ		թոքախտ	
/ə, u/	ʃəˈʃug	'whisper'	ʃuˈʃan	'lily'
	շշուկ		շուշան	

A schwa can be found in virtually any type of syllable (Table 4.27).

Table 4.27: Schwas in different types of syllables

VC	**əs**.kɑl	‘to feel’	զգալ
VCC	**əst**.ˈɾug	‘slave’	ստրուկ
CV	ˈmɑɾ.**tə**	‘man-DEF’	մարդը
CVC	**pəɾ**.ˈtel	‘to break’	փրթել
CVCC	**gəɾg**.ˈnel	‘to repeat’	կրկնել

However, the schwa is subject to more restrictions than other vowels. For example, there are virtually no native words where the only vowel is a schwa. The exceptions are a handful of onomatopoeic words, letter names, prepositions, and borrowings (Table 4.28). One common example is the derivational prefix /ənt-/ which is diachronically derived from the archaic preposition of the same form.

Table 4.28: Words where the only vowel is a schwa

ˈfəɾ	‘rustling sound’ ֆըռ	onomatopoeic
ˈət	‘name of letter ը <ə>’ ըթ	letter name
ˈəst	‘according to’ ըստ	preposition
ˈənt	‘to’ ընդ	preposition (archaic)
ənt-hɑgɑˈrɑg	‘on the contrary’ ընդհակառակ	prefixed to /hɑgɑrɑg/ ‘opposite’ հակառակ
fəsˈtəχ	‘pistachio’ ֆստըխ	borrowed from Turkish “fıstık”

The schwa is found in all possible word-positions: initial, medial, and final (Table 4.29).

Table 4.29: Lexical frequency of the schwa in different word-positions

	Word-initial #_	Word-medial #..._... #	Word-final _#	Total
ə	1515	19920	27	21462

Word-finally, the schwa is found only in the definite suffix /-ə/ (Table 4.30). Some words have a special functional meaning when definite, thus making them listed in dictionaries.

Table 4.30: Word-final schwa

ˈuʃ-ə	ութը	literally ‘eight-DEF’	functionally ‘eight’
uˈɾiʃ-ə	ուրիշը	literally ‘other-DEF’	functionally ‘the other’
ˈdɑʃ-ə	տակը	literally ‘under-DEF’	functionally ‘under it other’

4.2.2.2 Phonology of unwritten schwas

Although there is a schwa grapheme <ը> /ə/, most instances of a spoken schwa are unwritten (Table 4.31). These unwritten schwas fall into different morphophonological categories: epenthetic schwas, reduced vowels, and syncopated vowels. The three categories for these unwritten schwas are summarized below.

Table 4.31: Categories of unwritten schwa

Type	Inserted	Reduced	Syncopated
Example:	գրպան <krban> [kəɾˈbɑn] ‘pocket’	գծել <kd͡zel> [kəˈd͡zel] ‘to draw’	հասկնալ <hasgnal> [hɑskəˈnɑl] ‘to understand’
Related:	N/A	գիծ <kid͡z> [ˈkid͡z] ‘line’	հասկանալ <hasganal> [hɑskɑˈnɑl] ‘to understand (archaic)’

The first category of the unwritten schwa is ‘inserted’ or ‘epenthetic’ schwas. Most schwas fall into this category. These are schwas that surface in words like [kəɾbɑn] ‘pocket’. Such words do not have any morphologically-related word where the schwa is replaced by a non-schwa vowel, e.g., there is no such thing as word like **kiɾban*.

Armenian orthography allows rather long clusters of consonants to be written. These clusters are broken up by a schwa in pronunciation. We discuss the phonology of such inserted schwas in (cite chapter schwa epenthesis).

The second category of schwas is reduced schwas. These are schwas which are derived from destressed high vowels /i, u/. Such schwas are created when words are derived from other words that have destressed high vowels. For example, the

word [ˈkid͡z] ‘line’ has a stressed high vowel /i/. The word [kəˈd͡zel] ‘to draw’ is derived from this word by adding the suffix sequence /-el/. Stress shifts from the vowel /i/ to the suffix vowel /e/. This causes the high vowel to be replaced by a schwa in pronunciation. The orthography does not mark this schwa.

The derivation of schwas via high vowel reduction is quite complicated. We discuss the phonology/morphology of high vowel reduction in (cite chapter vowel reduction).

The last category of unwritten schwas is syncopated schwas. This set of words is rather small. These are words which, in earlier stages of the languages, had a word-medial and unstressed non-high vowel like /ɑ/: [hɑskɑˈnɑl] ‘to understand (archaic)’. In more contemporary registers of the language, this word-medial vowel is either deleted or pronounced as a schwa: [hɑskəˈnɑl]. The phonology and morphology of syncope is discussed in (cite syncope chapter).

Note that of the three categories, schwa insertion/epenthesis and high vowel reduction are synchronically productive and wide-spread in the Armenian lexicon. The third category, syncope of non-high vowels, is unproductive and limited to a small set of words. Syncope is more of a sporadic diachronic process than a productive synchronic process. Speakers have to memorize the set of words which have a syncope-derived schwa.

4.2.2.3 Phonology of written schwas

Because of the existence of epenthesis and vowel reduction, some phonological treatments of Armenian treat the schwa as a non-phoneme cite. Such work argues that the schwa is always epenthetic or derived, and thus not part of the phonemic inventory of the language. We disagree with this stance. Although it is true that most occurrences of the schwa are derived from epenthesis and reduction, there are some words or morphemes where the schwa must part be of the underlying form of the morpheme. In such cases, the orthography would represent the underlying schwa with the grapheme ը.

The intuition among speakers is that if a pronounced schwa cannot be predicted from epenthesis and reduction, then it must be written in the orthography. Such cases are few, but they exist. They are found in both functional and non-functional morphemes.

Among functional morphemes (Table 4.32), the most common use of an underlying schwa is the definite suffix. This suffix is /-ə/ after consonants, and /-n/ after vowels. This allomorphy is discussed further in (cite definite alloomprohy). Another common case is the irregular ablative suffix for words of time: /-vəne/. This suffix is explained further in (cite irregular dative -van). In both of these

cases, the schwa is written in the orthography with the grapheme ը <ə>. If the schwa was unwritten, then the word would be pronounced incorrectly.

Table 4.32: Functional morphemes with a written schwa

Definite			Ablative		
/-ə/	ˈpɑɾ	ˈpɑɾ-ə	/-vəne/	ɑjˈsoɾ	ɑjˈsoɾ-vəne
-ը	բառ	բառ**ը**	-ուընէ	այսօր	այսօրուընէ
<-ə>	<pʾaṙ>	<pʾaṙə>	<owəne>	<aysōr>	<aysōrowəne>
	‘word’	‘word-DEF’		‘today’	‘today-ABL’
		‘the word’			‘from today’
If unwritten:		ˈpɑɾ			*ɑjsorune
		‘word’			nonce word

Among non-functional or lexical morphemes (Table 4.33), a written schwa is used when the word starts with a schwa. Such morphemes include verbs, nouns, and adjectives. If the schwa as unwritten, then a schwa would be epenthesized in the wrong slot due to the rules for schwa epenthesis. There is likewise a derivational prefix /ənt-/, described in (cite chapter prefixes). For these lexical morphemes, the initial schwa usually precedes a nasal sound.

Table 4.33: Non-functional or lexical morphemes with a written schwa

				If unwritten
Verb	əˈsel	‘to say’	ըսել	*sel
	əmˈbel	‘to drink’	ըմպել	*məbel
Noun	əχt͡sɑnk	‘desire’	ըղձանք	*χət͡sɑnk
	ənˈdiɾ	‘fine’	ընտիր	*nədiɾ
Adjective	əndɑˈni	‘familiar’	ընտանի	*nədɑni
	əmˈpost	‘stubborn’	ըմբոստ	*məpost

In sum, the schwa is a controversial sound in Armenian. In many cases, a surface schwa is not present in the underlying form of words. Although such derived schwas exists, there are likewise morphemes where the schwa is part of the underlying form of the word. Because of the existence of these underived schwas, we treat the schwa as a phoneme.[3]

[3] As an alternative, Vaux argues that the schwa is non-phonemic and always derived. For cases of written schwas that we argue are underived, like [əndiɾ] ‘fine’ ընտիր <əndir>, Vaux would argue that the underlying form has an empty vocalic slot /Vndiɾ/ (Vaux 1998: 68)). A rule would then fill these empty slots with epenthetic schwas. We don’t entertain this analysis.

4.2.3 Front round vowels

The basic set of vowels in Western Armenian are the core vowels /ɑ, e, i, o, u/ and the schwa /ə/. However, due to contact with Turkish, Western Armenian has developed a sound /ʏ/ which is found across the Armenian lexicon. It like has developed a marginal phoneme /œ/ which is found in a handful of loanwords.

For /œ/ (Table 4.34), this vowel is written with the digraph էօ <ēō> and is optionally nativized with /o/. This vowel developed out of contact with Ottoman Turkish and is found in a handful of loanwords from Ottoman Turkish. Speakers vary in the rate of nativizing such words. For example, our intuition is that in Lebanon, it is more common to nativize these words with /o/ than to use the marginal phoneme /œ/. The rate of nativization likely varies by area and age.

Table 4.34: Words with marginal phoneme /œ/

With /œ/	Nativized		Meaning & origin
t͡ʃœˈɾeɡ	t͡ʃoˈɾeɡ	չէօրէկ	pastry item from Turkish “çörek”
bœˈɾeɡ	boˈɾeɡ	պէօրէկ	pastry item from Turkish “börek”
œʒeˈni	oʒeˈni	Էօժէնի	fem. given name from French “Eugénie”
dœˈʃeɡ	doˈʃeɡ	տէօշէկ	‘mattress’ from Turkish “döşek”
kœˈfte	koˈfte	քէօֆթէ	‘kofta’ from Turkish “köfte”

We transcribe the vowel as /œ/, though we think it’s free to vary with /ø/ without a consistent articulatory target.

More such loanwords are reported in Աճառեան (1902), which is a study on Turkish borrowings in early modern Istanbul Armenian. We have not been able to extensively analyze this dictionary in order to find more such loanwords that survived into the colloquial Western of non-Istanbul Armenians.

For the vowel /ʏ/, its use is more complicated and is closely tied with the orthography. The Armenian script has the digraph իւ <iw> (Table 4.35). This digraph is pronounced as [iv] word-finally and before vowels in both Western and Eastern Armenian. Word-initially, this digraph is pronounced as [ju] or [ʏ]. In all other positions, Eastern Armenian pronounces this digraph as [ju],[4] while Western Armenian generally uses [ʏ]. There are some complications with the nominalizing suffix /-utʏn/ -ութիւն; discussed in Section §4.5.1.

[4]In Eastern Armenian, when the digraph իւ in traditional spelling is pronounced as [ju], it is replaced by յու <yow> in the reformed spelling system.

Table 4.35: Pronunciations of the digraph իւ <iw>

	Western	Eastern			
Initial	ˈ**juʁ** **juɾ**ɑhɑˈdug **ʏɾ**ɑhɑˈdug	ˈ**juʁ** **juɾ**ɑhɑˈtuk	‘oil’ ‘specific’	**իւղ** **իւր**այատուկ	<**iw**ɣ> <**iw**rayadowg>
Medial pre-C	ˈ**d͡ʒʏʁ** ɑŋˈg**ʏn** ɑˈl**ʏɾ**	ˈ**t͡ʃjuʁ** ɑŋˈk**jun** ɑˈl**juɾ**	‘branch’ ‘corner’ ‘flour ’	**ճիւղ** **անկիւն** **ալիւր**	<**d͡ʒiw**ɣ> <ang**iw**n> <al**iw**r>
Medial pre-V	ti**v**ˈvɑn hi**v**ˈvɑnt ti**v**ɑˈgɑn	di**v**ˈvɑn hi**v**ˈvɑnd di**v**ɑˈkɑn	‘divan’ ‘sick’ ‘diabolic’	**դիւան** **հիւանդ** **դիւական**	<ti**w**an> <hi**w**ant> <ti**w**agan>
Final	ˈ**tiv** ɑˈ**niv** gəˈ**ɾiv**	ˈ**tiv** ɑˈ**niv** kəˈ**riv**	‘number’ ‘wheel’ ‘fight’	**թիւ** **անիւ** **կռիւ**	<t’**iw**> <an**iw**> <gr̥**iw**>

Table 4.36 shows the lexical frequency of [ʏ] from the associated dictionary.

Table 4.36: Lexical frequency of the front round vowel /ʏ/ in different word-positions

	Word-initial #_	Word-medial #..._... #	Word-final _#	Total
ʏ	17	9113	0	9130

Word-initally, there is only one morpheme that starts with [ʏ]: [ʏɾ] իւր ‘his (archaic)’. This root is used to form various compounds that involve personhood, such as [ʏɾɑhɑdug] ‘specific’ (Table 4.35). This root can optionally pronounced with [ju] instead: [ʏɾɑhɑˈdug].

There are no cases of word-final [ʏ].

The vowel can be stressed or unstressed, and in different possible syllable shapes (Table 4.37). Within a morpheme or root, the segment [ʏ] is typically found in a final closed syllable CVC(C). Suffixation can make this segment lose a coda. However, there are also roots where the vowel is in a non-final open syllable; such words are typically loanwords like ‘Byzantine’.

Diachronically, the modern [ʏ] sound may have from an earlier [iu̯] sequence from Classical Armenian (Ավետյան 2015; Hossep 2024: 27). This sequence changed to [ʏ], whether via dialect-internal sound changes, contact with other dialects, or via contact with Turkish.

Table 4.37: Use of [ʏ] in different types of syllables

ʏ́C	ˈ**ʏɾ**	‘his (archaic)	իւր
ʏC	**ʏ**.r-ɑˈgɑn	‘particular’	իւրական
Cʏ́	ˈn**ʏ**.t-ɑbes	‘materially’	նիւթապէս
Cʏ	b**ʏ**.zɑnt-ɑˈgɑn	‘Byzantine’	բիւզանդական
	h**ʏ**.ˈsis	‘north’	հիւսիս
Cʏ́C	ˈn**ʏ**t	‘material’	նիւթ
	ɑˈl**ʏɾ**	‘flour’	ալիւր
CʏC	d**ʏ**ɾ.ˈzi	‘Druze’	տիւրզի
CʏC + V	n**ʏ**t-ɑˈgɑn	‘material’	նիւթական
Cʏ́CC	ɑɾ.ˈt**ʏ**ŋk	‘product’	արդիւնք
CʏCC	t**ʏ**ɾk.ˈmen	‘Turkmen’	թիւրքմէն

The acoustic quality of this /ʏ/ can range from [ʏ] to [y]. Tabita Toparlak reports that for Istanbul Armenian is more like [y]. For Syrian Armenians, our impression is that their vowel is more often [ʏ]. Of course, in depth acoustic studies are needed to verify or disconfirm these impressions.

Depending on the word and speaker, the vowel [ʏ] can be replaced with [uj], [ʏj], [jʏ], [jʏ], [ju]. Table 4.38 lists a set of common words that are pronounced with [ʏ], along with possible alternative pronunciations from HD’s speech. Our impression is that this variation is a type of free variation that is closely tied to the speaker’s sociolinguistic origins. For example, HD reports that his family and peers in Lebanon would most often have the [ʏj] or [uj] forms. For Hasmig Seropian from Syria, her own idiolect seems to almost always have [ʏ].

Table 4.38: Words with [ɤ] and alternative pronunciations

[ɤ]	[ɤj]	[uj]	[jɤ]	[ju]		
ˈt͡sɤn	ˈt͡sɤjn	ˈt͡sujn			‘snow’	ձիւն
ˈhɤɾ	ˈhɤjɾ	ˈhujɾ			‘guest’	հիւր
ˈnɤt	ˈnɤjt	ˈnujt			‘topic’	նիւթ
ˈmɤs	ˈmɤjs	ˈmujs			‘other’	միւս
ˈkɤʁ	ˈkɤjʁ		ˈkjɤʁ	ˈkjuʁ	‘village’	գիւղ
sɤˈnɑg	sɤjˈnɑg				‘column’	սիւնակ
hɤˈsis					‘north’	հիւսիս
ɑˈɾɤn	ɑˈɾɤjn	ɑˈɾujn			‘blood’	արիւն
ɑˈʁɤs	ɑˈʁɤjs	ɑˈʁujs			‘brick’	աղիւս
ɑˈɾɤd͡z	ɑˈɾɤjd͡z	ɑˈɾujd͡z			‘lion’	առիւծ
məɾˈt͡ʃɤn	məɾˈt͡ʃɤjn	məɾˈt͡ʃujn			‘ant’	մրջիւն
hənˈt͡ʃɤn	hənˈt͡ʃɤjn	hənˈt͡ʃujn			‘sound’	հնչիւն
hɑˈɾɤɾ	hɑˈɾɤjɾ	hɑˈɾujɾ			‘hundred’	հարիւր
zeˈpɤɾ	zeˈpɤjɾ	zeˈpujɾ			‘zephyr’	զեփիւռ
ɑχˈpɤɾ	ɑχˈpɤjɾ	ɑχˈpujɾ			‘fountain’	աղբիւր
əsˈpɤɾk					‘diaspora’	սփիւռք
ɑɾˈtɤŋk	ɑɾˈtɤjŋk	ɑɾˈtujŋk			‘result’	արդիւնք

Furthermore, there are reports that colloquial speech can reduce the Western [ɤ] vowel (Eastern [ju]) to a [u] in some high-frequency words cite. For example, in HD’s experience, some attested reduced words are [ˈt͡sun] ‘snow’ ձուն instead of [ˈt͡sɤn] ձիւն.

Another rare pronunciation of [ɤ] is as [əju] or [əjɤ]. In HD’s judgments, such a division happens sometimes for word-initial [Cɤ] sequences like [kəjuʁ, kəjɤʁ] ‘village’ instead of [kɤʁ] ‘village’ գիւղ. We suspect that such a divided pronunciation is restricted to emphatic speech.

There is some phonetic work on the [ɤ] sound. cite and expand. cite julianne and samuel

4.2.4 Turkish-induced centralization

In modern Western Armenian, it is unknown if there are any allophonic rules that affect vowels. The closest example that we know is vowel laxing in Turkish-speaking Armenians.

For speakers of Western Armenian who don't know Turkish, such as HD and the Lebanese community, the vowels /i, u, e, o/ are pronounced as just [i, u, e, o]. But it is reported that Western Armenian speakers who are Turkish-speaking apply an allophonic rule of changing /i, u, e, o/ to [ɪ, ʊ, ɛ, ɔ] in different contexts (Fairbanks 1948: 3-4).

The main source for this process is Fairbanks, who documents extensive allophonic laxing for his informants who were from Istanbul (Fairbanks 1948: 1). Fairbanks reports that in his Istanbuli speakers, the informants would lower a vowel (from close to half-close, or half-close to half-open) in different non-identical contexts. Table 4.39 reorganizes some of his data based on word position. We keep Fairbanks's original transcription.[5] For example, his informants would change /i/ to [ɪ] before a coda like [gɪn], but keep the /i/ as [i] word-finally like [giɻagi]. We do not share any of Fairbanks's judgments. In HD's speech, all these vowels stay tense or high: [e, o, u, i] and not [ɛ, ɔ, ʊ, ɪ].

Table 4.39: Vowel centralization data from Fairbanks (1948: 3-4)

	before vowel	before glide	word-final	before coda	before onset	word-initial
e	eí	t'eĭ				
ɛ			guzɛ	dɛsnɛl	p'aɻɛgam	
o		jɛɻɛgoɪn	asdudzo			
ɔ				vɔtš	jɔt'ə	
u		k'uĭɻ	jɛɻgu			udɛl
ʊ				dʊn		
i	hianali		gɪɻagi			
ɪ				gɪn	ɪɻɪgʊn	

This process is specific to Turkish-speaking Armenians. Hasmig Seropian (p.c.) seems to show this allophonic process as well in her own speech. Although she was raised in Syria, her grandparents were Turkish-speaking and she was raised as a Turkish-Armenian bilingual. In a study on modern Istanbul Armenian, self cite report that centralization is still present. Though the exact allophonic contexts for centralization are unclear. They argue that such centralization is due to contact with Turkish, which has similar vowel lowering rules.

[5]HD's pronunciation, translation, and orthography are as follows. For /e/: [eji] 'I was' էի, [tej] 'tea' թէյ, [guze] 'he wants' կ'ուզէ, [desnel] 'to see' տեսնել, [pɑɾegɑm] 'friend' բարեկամ. For /o/: [jeɾegojn] 'of evening' երեկոյն, [ɑstud͡zo] 'of God' Աստուծոյ, [vo͡tʃ] 'no' ոչ, [jotə] 'seven' եօթը. For /u/: [kujɾ] 'sister' քոյր, [jeɾgu] 'two' երկու, [dun] 'house' տուն, [udel] 'to eat' ուտել. For /i/: [hijɑnɑli] 'magnificient' հիանալի, [giɾɑgi] 'Sunday' կիրակի, [gin] 'lady' կին, [iɾigun] 'evening' իրիկուն.

4.3 Voicing and aspiration in obstruents

In terms of its phonemes, Western Armenian has voiced stops and voiceless obstruents (stops, affricates, and fricatives). The obstruents however can change their voicing quality or aspiration quality depending on the phonological (phonotactic) context that they're used in. Two such processes are voicing assimilation and deaspiration. Both processes tend to occur simultaneously.

Voicing assimilation is when in a cluster of obstruents, the obstruents have to agree in voicing: either both voiced or both voiceless (Rule 1). When words are derived or morphemes are combined, we can get sequences of obstruents that underlyingly have different voicing qualities. But when pronounced, voiced obstruents become voiceless when next to a voiceless sound, regardless if the underlyingly voiced obstruent is before or after the voiceless obstruent. An exception is the sound /v/ which is devoiced before voiceless obsteuents, but can variably maintain its voicing after voiceless obstruents (§4.3.1.5).

When the underlyingly voiced sound is after the voiceless sound, we find progressive assimilation. When the underlyingly voiced sound is before the voiceless sound, we find regressive assimilation.

Rule 1. Voicing assimilation in obstruent clusters

Progressive assimilation:				
After another voiceless obstruent, a voiced obstruent becomes voiceless.				
. /tɑntɑʁ-**g**od/	→	[tɑntɑʁ-ˈ**g**od]	'slowish'	դանդաղկոտ
/vɑχ-**g**od/	→	[vɑχ-ˈ**k**od]	'cowardly'	վախկոտ
Regressive assimilation:				
Before another voiceless obstruent, a voiced obstruent becomes voiceless.				
/hɑleb/	→	[hɑleb]	'Aleppo'	Հալէպ
/hɑle**b-t͡s**i/	→	[hɑle**p-**ˈ**t͡s**i]	'Aleppoite'	հալէպցի

Deaspiration is more complex. As discussed in §4.1.1, Armenian speakers vary in the exact acoustic quality of voiced vs. voiceless stops. For example, voiceless stops are unaspirated for speakers from Lebanon, but they're aspirated for speakers from Turkey and the USA. HD is a speaker who aspirates his stops by default. For speakers like HD who have aspiration as the default value for voiceless stops, such speakers also have deaspiration rules where a voiceless stop loses its aspiration (Rule 2). If a voiceless stop is part of an obstruent cluster, then it generally loses its aspiration. Deaspiration can apply in different types of obstruent clusters, such as fricative-stop, stop-fricative, affricate-stop, and stop-affricate. Stop-stop clusters have a single aspiration after the cluster. Word-final clusters /VCC#/

can variably aspirate.[6,7]

Rule 2. Deaspiration in obstruent clusters for speaker HD with default aspiration for stops

Before or after a voiceless obstruent, voiceless stops are deaspirated.

Post-fricative deaspiration: After a fricative, deaspirate a voiceless stop

/hɑs**t**ʰ/	→	[ˈhɑ**st**]	'thick'	հաստ

Pre-fricative deaspiration: Before a fricative, deaspirate a voiceless stop

/ɑ**p**ʰse/	→	[ɑ**p**ˈse]	'tray'	ափսէ

Post-affricate deaspiration: After an affricate, deaspirate a voiceless stop

/lu**t͡sk**ʰ**i**/	→	[lu**t͡s**ˈ**k**i]	'match (fire)'	լուցկի

Pre-affricate deaspiration: Before an affricate, deaspirate a voiceless stop

/ɑ**k**ʰ**t͡s**ɑn/	→	[ɑ**k**ˈ**t͡s**ɑn]	'switch'	աքցան

Stop-stop deaspiration: In a stop-stop cluster, deaspirate a voiceless stop

/t͡sɑ**t**ʰ**k**ʰel/	→	[t͡sɑ**t**ˈ**k**ʰel]	'to jump'	ցատքել

For speakers like HD that have aspiration, voicing assimilation and stop deaspiration often apply in the same words, meaning they interact together. In terms of rule interaction, voicing assimilation feeds deaspiration: /hɑle**b**-t͡si/ → //hɑle**p**ʰ-t͡si// → [hɑle**p**-t͡si]. The end result is transparent application of both rules

The two processes can be found in both underived and derived contexts. A context is underived when the segments involved are part of the same root or morpheme, and they are always pronounced the way they are. A context is derived when it's created via combining morphemes or by applying word-internal changes. Because these processes apply in derived contexts, the same morpheme like 'Aleppo' [hɑle**b**] can change its pronunciation depending on the following morpheme: [hɑle**p**-t͡si] 'Aleppoite'.

The following subsections provide examples for both of these processes, both in derived and underived contexts. Data is organized by the subcategory of the process, cluster size, and word position.

Note that some possible morphological exceptions to voicing assimilation reduplication. Rduplication is discussed in (cite chapter, include vazvezl types and gab-gabujd types, tepteʁin).

[6] We predict that affricates also undergo deaspiration after fricatives. However, the authors' subdialects (Syrio-Lebanese/USA) do not aspirate affricates in general. So we cannot test this hypothesis for now. Such a hypothesis can be tested with speakers from Turkey who do have aspirated affricates.

[7] Fairbanks (1948: 7) also reports that /kʰn/ clusters deaspirate to [kn] but we couldn't replicate this finding.

4.3.1 Voicing assimilation

Voicing assimilation is when in a consonant cluster C_1C_2, the two consonants have identical voicing quality. Such a cluster consists of two obstruents (stop, affricate, fricative). Either both consonants are voiced or both are voiceless. Voicing assimilation is found in underived and derived contexts.

4.3.1.1 Voicing assimilation in underived contexts

In an underived context, the orthography may suggest that within a morpheme or root, the two consonants of a root have different voicing. But the consonants are always pronounced with the same voicing quality. In this case, it's more accurate to say that this is a phonology-orthography mismatch (§3.4.2), rather than a synchronic phonological rule that changes the voicing quality of root-internal clusters (Table 4.40).[8]

Table 4.40: Voicing assimilation in underived contexts

/jekib**bd**os/	/bɑ**tk**ɑm/	/χe**χ**tel/	/ˈɑs**tχ**/	/ˈɑs**tv**ɑd͡z/
[jekib**ˈd**os]	[bɑ**tˈk**om]	[χe**χˈt**el]	[ˈɑs**tχ**]	[ɑs**tˈv̥**ɑd͡z]
<ek**ibd**os>	<ba**dk**am>	<xe**ɣd**el>	<a**sdɣ**>	<asdowad͡z>
'Egypt'	'message'	'to strangle'	'star'	'God'
Եգիպտոս	պատգամ	խեղդել	աստղ	Աստուած

The issue of ambiguously devoiced /v/ as in 'God' is discussed later in Section §4.3.1.5.

4.3.1.2 Progressive assimilation in intervocalic VC(C)CV contexts

Voicing assimilation also occurs in derived contexts. terms of suffixation, there are few productive suffixes that start with a voiced obstruent. One such suffix is /-bɑn/ which is used to derive nouns that loosely have the meaning of 'guardian or keeper of X'. This suffix starts with a voiced stop /-b/ after voiced segments.[9] The orthography marks this as a voiced stop պ <b> as well. But after a voiceless obstruent, this stop devoices to /-p/ (Table 4.41).

[8] Some speakers (Tabita Toparlak) can pronounce the word 'star' as [ɑstəʁ] instead of [ɑstχ]. For such speakers, the final fricative is underlyingly /ɑstʁ/ with schwa epenthesis. But for speakers like HD who never have the fricative non-adjacent from the stop, there is never any alternations so the word is underlyingly /ɑstχ/.

[9] The vowel /-ɑ-/ in these examples is the vowel used to connect stems to form compounds. Some suffixes also use this vowel.

Table 4.41: Progressive voicing assimilation for the derivational suffix /-ban/ in VC(C)CV contexts

		bɑɾˈdez	‘garden’	պարտէզ
→	/bɑɾdez-ban/	bɑɾdiz-ˈban	‘gardener’	պարտիզպան
		jegeʁeˈt͡si	‘church’	եկեղեցի
→	/jegeʁet͡si-ɑ-ban/	jegeʁet͡s-ɑ-ˈban	‘church warden’	եկեղեցապան
		ˈhɑt͡s	‘bread’	հաց
→	/hɑt͡s-ban/	hɑt͡s-ˈpan	‘baker’	հացպան
		bɑʃˈt-el	‘to worship’	պաշտել
→	/bɑʃt-ban/	bɑʃt-ˈpan	‘protector’	պաշտպան
		bɑˈh-el	‘to keep’	պահել
→	/bɑh-ban/	bɑh-ˈpan	‘guardian’	պահպան

Another suffix with a voiced stop is /-god/. This suffix surfaces with [g] after voiced sounds. It is devoiced after voiceless obstruents (Table 4.42).[10]

Table 4.42: Progressive voicing assimilation for the derivational suffix /-god/ in VCCV contexts

Base		d͡zɑˈɾɑv	‘thirsty’	ծարաւ
→	/d͡zɑɾɑv-god/	d͡zɑɾɑv-ˈgod	‘very thirsty’	ծարաւկոտ
Base		məˈs-il	‘to shiver’	մսիլ
→	/məs-god/	məs-ˈkod	‘chilly’	մսկոտ
Base		pɑmpɑˈs-el	‘to gossip’	բամբասել
→	/pɑmpɑs-god/	pɑmpɑs-ˈkod	‘gossipy’	բամբասկոտ
Aorist stem		jeɾɑˈt͡s-ɑd͡z	‘bubbled’	եռացած
→	/jeɾɑt͡s-god/	jeɾɑt͡s-ˈkod	‘effervescent’	եռացկոտ
Aorist stem		pɑɾgɑˈt͡s-ɑd͡z	‘angry’	բարկացած
→	/pɑɾgɑt͡s-god/	pɑɾgɑt͡s-ˈkod	‘irritable’	բարկացկոտ
Aorist stem		t͡sɑnt͡sɾɑt͡s-ɑd͡z	‘bored’	ձանձրացած
→	/t͡sɑnt͡sɾɑt͡s-god/	t͡sɑnt͡sɾɑt͡s-ˈkod	‘easily bored’	ձանձրացկոտ
Aorist stem		zɑɾmɑt͡s-ɑd͡z	‘surprised’	զարմացած
→	/zɑɾmɑt͡s-god/	zɑɾmɑt͡s-ˈkod	‘easily surprised’	զարմացկոտ

This suffix is likewise found after CC-final roots (Table 4.43). Here, we see progressive assimilation happening in VCC-CV contexts.

[10] This suffix is often used after aorists stems. Most aorist stems end with /t͡s/, and this explains why Table 4.42 is overpopulated with /t͡s/ examples.

Table 4.43: Progressive voicing assimilation for the derivational suffix /-god/ in VCCCV contexts

		bɑˈhɑ**ntʃ**	‘demand’	պահանջ
→	/bɑhɑ**ntʃ**-**g**od/	bɑhɑ**ntʃ**-ˈ**k**od	‘demanding’	պահանջկոտ
		pɑ**χ**ˈ**tʃ**-il	‘to flee’	փախչիլ
→	/pɑ**χtʃ**-**g**od/	pɑ**χtʃ**-ˈ**k**od	‘fugitive’	փախչկոտ
		nɑˈχɑ**nts**	‘jealousy’	նախանձ
→	/nɑχɑ**nts**-**g**od/	nɑχɑ**nts**-ˈ**k**od	‘jealous’	նախանձկոտ
		dəɾˈdu**ntʃ**	‘grunt’	տրտունջ
→	/dəɾdu**ntʃ**-**g**od/	dəɾdə**ntʃ**-ˈ**k**od	‘grunting’	տրտնջկոտ

Progressive assimilation is also found in compounds (Table 4.44). Most compounds are formed via a linking vowel /-ɑ-/. But there are cases of vowel-less or unlinked compounds where a) there is no such vowel, and b) the lack of a vowel causes the two stems to be adjacent. Here, we can find voicing assimilation.

Table 4.44: Progressive voicing assimilation for the compounds without a linking vowel /VC-CV/

Bases	ˈχɑtʃ + ˈgɑb	‘cross’ + ‘link’	խաչ, կապ
→	χɑ**tʃ**-ˈ**k**ɑb	‘cross piece’	խաչկապ
Bases	ˈhɑts + ˈgeɾ	‘cross’ + ‘eat!’	հաց, կեր
→	hɑ**ts**-**k**eˈɾ-ujt	‘banquet’	հացկերույթ

Another context for progressive assimilation is vowel reduction of /i,u/. For the forms in Table 4.45, the root has a final (C)CVC sequence where the final onset and coda are separated. When a suffix is added, the root’s vowel is deleted, causing the consonants to assimilate.

Table 4.45: Progressive voicing assimilation from vowel reduction in /VCCV/

	tɑˈ**hid͡ʒ**	‘executioner’	դահիճ
→	tɑ**ht͡ʃ**-ɑˈbed	‘chief executioner’	դահճապետ
	sɑsˈ**tig**	‘intense’	սաստիկ
→	sɑ**st.k**-ɑˈgɑn	‘intensive’	սաստկական
	pəsˈ**tuʁ**	‘pistachio’	փստուղ
→	pə**st.χ**-eˈni	‘pistachio tree’	փստղենի
	hənˈ**tig**	‘Indian’	հնդիկ
→	hə**nt.k**-ɑsˈtɑn	‘India’	հնդկաստան
	mɑɾˈ**tig**	‘mankind’	մարդիկ
→	mɑ**ɾt.k**-ɑˈjin	‘human (adj)’	մարդկային
	t͡sənˈ**t͡suʁ**	‘bronchus’	ցնցուղ
→	t͡sə**nt͡s.χ**-ɑˈjin	‘bronchial’	ցնցղային
	t͡ʃəχˈ**t͡ʃig** + nəˈmɑn	‘bat’ + ‘like’	չղջիկ, նման
→	t͡ʃə**χt͡ʃ.k**-ɑ-nəˈmɑn	‘bat-like’	չղջկանման

4.3.1.3 Regressive assimilation in intervocalic VC(C)CV contexts

In [V(C)C_1-C_2V] we likewise find cases of regressive assimilation where a voiceless obstruent C_2 causes C_1 to devoice.

A common morphological construction that causes regressive devoicing is forming aorist stems. The roots in Table 4.46 can form inchoative verbs by adding the suffix sequence /-n-ɑ-l/. Their aorist stems are formed by replacing the suffix /n/ with the suffix /t͡s/. This stem is used to from the simple past (past perfective).[11] The /t͡s/ causes the preceding obstruent to devoice.[12]

[11]The full glosses are /-n-ɑ-l/ [INCH-TH-INF], /t͡s-ɑ-v/ [AOR-PST-3SG.

[12]The sequence of [Vt͡s-t͡sV] as a single segment, i.e., a geminate [t͡s:].

Table 4.46: Regressive voicing assimilation before aorist suffix /-t͡s-/ in VCCV contexts

Base		ˈʃɑ**d**	‘many’	շատ
Verb		ʃɑ**d**-ˈn-ɑ-l	‘to multiply (intr.)’	շատնալ
Aorist	/ʃɑ**d-t͡s**-ɑ-v/	ʃɑ**t**-ˈ**t͡s**-ɑ-v	‘it multiplied’	շատցաւ
Base		bəzˈdi**g**	‘small’	պզտիկ
Verb		bəzdi**g**-ˈn-ɑ-l	‘to get small’	պզտիկնալ
Aorist	/bəzdi**g-t͡s**-ɑ-v/	bəzdi**k**-ˈ**t͡s**-ɑ-v	‘it got small’	պզտիկցաւ
Base		ˈme**d͡z**	‘big’	մեծ
Verb		me**d͡z**-ˈn-ɑ-l	‘to grow up’	մեծնալ
Aorist	/me**d͡z-t͡s**-ɑ-v/	me**t͡s**-ˈ**t͡s**-ɑ-v	‘he grew up’	մեծցաւ
Base		dəˈke**ʁ**	‘ugly’	տգեղ
Verb		dəke**ʁ**-ˈn-ɑ-l	‘to get ugly’	տգեղնալ
Aorist	/dəke**ʁ-t͡s**-ɑ-v/	dəke**χ**-ˈ**t͡s**-ɑ-v	‘it got ugly’	տգեղցաւ

We find the same regressive pattern before the derivational suffix /-t͡si/ (Table 4.47). This suffix is used to create names for people from cities, countries, and ethnic groups (ethnonyms). This suffix is underlyingly just a voiceless affricate. It causes regressive devoicing on obstruents.

Table 4.47: Regressive voicing assimilation for the derivational suffix /-t͡si/ in VCCV contexts

		pɑˈɾi**z**	‘Paris’	Պարիզ
→	/pɑɾi**z-t͡s**i/	pɑɾi**s**-ˈ**t͡s**i	‘Parisian’	պարիզցի
		ʁɑɾɑˈbɑ**ʁ**	‘Karabakh’	Ղարաբաղ
→	/ʁɑɾɑbɑ**ʁ-t͡s**i/	ʁɑɾɑbɑ**χ**-ˈ**t͡s**i	‘Karabaghian’	ղարաբաղցի
		hɑˈle**b**	‘Aleppo’	Հալէպ
→	/hɑlɑ**b-t͡s**i/	hɑle**p**-ˈ**t͡s**i	‘Aleppoite’	հալէպցի

We again find regressive assimilation before the causative suffix [-t͡sən-] (Table 4.48).[13]

[13]The gloss for /t͡sən-e-l/ is [CAUS-TH-INF].

Table 4.48: Regressive voicing assimilation before the causative suffix /-t͡sən-/ in VCCV contexts

		ˈɡɑ**b**	‘link’	կապ
→	/ɡɑ**b-t͡s**ən-e-l/	ɡɑ**p-t͡s**əˈn-e-l	‘to connect (trns.)’	կապցնել
		ɑʃχɑˈ**d**-i-l	‘to work’	աշխատիլ
→	/ɑʃχɑ**d-t͡s**ən-e-l/	ɑʃχɑ**t-t͡s**əˈn-e-l	‘to employ’	աշխատցնել
		pɑˈɾɑ**g**	‘thin’	բարակ
→	/pɑɾɑ**g-t͡s**ən-e-l/	pɑɾɑ**k-t͡s**əˈn-e-l	‘to make thin’	բարակցնել
		t͡sɑd͡z	‘low’	ցած
→	/**t͡sɑd͡z-t͡s**ən-e-l/	**t͡sɑt͡s-t͡s**əˈn-e-l	‘to lower’	ցածցնել
		vɑ**z**-e-l	‘to run’	վազել
→	/vɑ**z-t͡s**ən-e-l/	vɑ**s-t͡s**əˈn-e-l	‘to hasten’	վազցնել
		ˈχɑ**ʁ**	‘game’	խաղ
→	/χɑ**ʁ-t͡s**ən-e-l/	χɑ**χ-t͡s**əˈn-e-l	‘to make to play’	խաղցնել

We’ve also found another C-initial suffix /-kin/ (Table 4.49). This derivational suffix is rare but it can cause regressive assimilation. There are also some cases where compounds create contexts for regressive assimilation.

Table 4.49: Regressive voicing assimilation before the derivational suffix /-kin-/ and in compounds in VCCV contexts

		ˈu**ʒ**	‘strength’	ուժ
→	/u**ʒ-k**in/	u**ʃ**-ˈ**k**in	‘strong’	ուժգին
		ˈpujʒ + ˈ**k**ujɾ	‘healing’ + ‘sister’	բոյժ, քոյր
→		pu**ʃ**-ˈ**k**ujr	‘strong’	բուժքոյր

There are some rare derivational prefixes that end in a voiced obstruent: synonymous [təʒ-] դժ and [dəʒ-] ժդ (Table 4.50). The fricative surfaces as voiced before voiced segments, and devoices before voiceless obstruents. We report on just [təʒ-] becomes its more common in speech.[14]

[14]Note that the schwa in this prefix is epenthetic, but we transcribe the schwa here for illustration.

Table 4.50: Regressive voicing assimilation after the derivational prefix [təʒ-] in VCCV contexts

		ˈ**g**eɾb	‘form’	կերպ
→	/təʒ-**g**eɾb/	təʃ-ˈ**g**eɾb	‘ugly’	դժկերպ
		ˈ**p**ɑχt	‘luck’	բախտ
→	/təʒ-**p**ɑχt/	təʃ-ˈ**p**ɑχt	‘unlucky’	դժբախտ
		ˈ**k**oh	‘satisfied’	գոհ
→	/təʒ-**k**oh/	təʃ-ˈ**k**oh	‘dissatisfied’	դժգոհ
		ˈ**k**ujn	‘color’	գույն
→	/təʒ-**k**ujn/	təʃ-ˈ**k**ujn	‘discolored’	դժգույն

Vowel reduction can likewise create contexts for regressive assimilation. For the word in Table 4.51, the root has a final CVC syllable. When a derivational suffix is added, the root’s vowel is deleted and this causes the consonants to become adjacent.

Table 4.51: Regressive voicing assimilation from vowel reduction or syncope in /VCCV/

	gəˈ**dut͡s** + ˈ**t͡s**ev	‘beak’ + ‘shape’	կտուց, ձեւ
→	gə**tt͡s**-ɑ-ˈ**t͡s**ev	‘beak-shaped’	կտցաձեւ
	seˈ**buh**	‘gentleman’	սեպուհ
→	se**ph**-ɑˈgɑn	‘appropriate’	սեպհական
	bɑˈ**gɑs**	‘missing’	պակաս
→	bɑ**k**ˈ**s**-il	‘to lessen’	պակսիլ
	ɑt͡ʃɑˈ**git͡s**	‘assistant’	աջակից
→	ɑt͡ʃɑ**kt͡s**-uˈtɤn	‘assistance’	աջակցութիւն
	ɑˈ**dz͡uχ**	‘coal’	ածուխ
→	ɑ**t͡sχ**-ɑˈjin	‘carbonic’	ածխային
	hɑmoˈ**z-it͡ʃ**	‘persuasive’	համոզիչ
→	hɑmo**s-t͡ʃ**-uˈtɤn	‘persuasion’	համոզչութիւն

The above regressive assimilation patterns are also attested for VCCCV clusters with 3 consonants (Table 4.52). Such sequences are relatively rare but attested. We find regressive devoicing in these clusters, such as before the aorist suffix /-t͡s-/ or causative /-t͡sən-/.

Table 4.52: Regressive voicing assimilation in VCCCV contexts before aorist suffix /-t͡s-/ or causative /-t͡sən-/

Base		ˈɡɑɾd͡ʒ	‘short’	կարճ
Verb		ɡɑɾd͡ʒ-ˈn-ɑ-l	‘to get short’	կարճնալ
Aorist	/ɡɑɾd͡ʒ-t͡s-ɑ-v/	ɡɑɾt͡ʃ-ˈt͡s-ɑ-v	‘it got short’	կարճցաւ
Caus.	/ɡɑɾd͡ʒ-t͡sən-e-l/	ɡɑɾt͡ʃ-t͡səˈn-e-l/	‘to make short’	կարճցնել
Base		ɡɑˈbujd	‘blue’	կապոյտ
Verb		ɡɑbujd-ˈn-ɑ-l	‘to become blue’	կապոյտնալ
Aorist	/ɡɑbujd-t͡s-ɑ-v/	ɡɑbujt-ˈt͡s-ɑ-v	‘it became blue’	կապոյտցաւ
Caus.	/ɡɑbuɾd-t͡sən-e-l/	ɡɑbujt-t͡səˈn-e-l/	‘to make blue’	կապոյտցնել
Base		χoˈɾuŋɡ	‘deep’	խորունկ
Verb		χoɾuŋɡ-ˈn-ɑ-l	‘to become deep’	խորունկնալ
Aorist	/χoɾuŋɡ-t͡s-ɑ-v/	χoɾuŋk-ˈt͡s-ɑ-v	‘it became deep’	խորունկցաւ
Caus.	/χoɾuŋɡ-t͡sən-e-l/	χoɾuŋk-t͡səˈn-e-l/	‘to make deep’	խորունկցնել

Vowel reduction can likewise create VCCCV clusters with regressive assimilation (Table 4.53). So far, the only examples that we've found involved the derivational suffix /-it͡ʃ/, whose vowel can delete and thus cause regressive assimilation.

Table 4.53: Regressive voicing assimilation from vowel reduction in /VCCCV/

	ɡɑzmɑɡeɾˈb-it͡ʃ	‘organizer’	կազմակերպիչ
→	ɡɑzmɑɡeɾp-t͡ʃ-ɑˈɡɑn	‘organizational’	կազմակերպչական
	məɡəɾˈd-it͡ʃ	‘baptist’	մկրտիչմ
→	məɡəɾt-t͡ʃ-ɑˈɡɑn	‘Baptist (religion)’	մկրտչական
	neɾˈɡ-it͡ʃ	‘painter’	ներկիչ
→	neɾk-t͡ʃ-uˈhi	‘fem. painter’	ներկչուհի
	pəɾˈɡ-it͡ʃ	‘savior’	փրկիչ
→	pəɾk-t͡ʃ-ɑˈɡɑn	‘pertaining to Christ’	փրկչական
	mɑɾˈz-it͡ʃ	‘trainer’	մարզիչ
→	mɑɾs-t͡ʃ-ɑˈɡɑn	‘pertaining to trainers’	մարզչական

4.3.1.4 Regressive assimilation in appendix contexts: /VC(C)-k/

Regressive assimilation happens in a word-final context for the suffix /-k/. This suffix can be added after any consonant cluster and cause devoicing. We go over this devoicing process as a type of regressive assimilation.

The suffix /-k/ can be after virtually any type of consonant, even if the consonant + *k* cannot form a complex coda based on their sonority (Table 4.54). Because of this behavior, this suffix is often phonologically analyzed as an appendix (§??). In terms of segmental phonology, this voiceless suffix triggers regressive voicing assimilation (devoicing) on any preceding obstruent.[15]

Table 4.54: Regressive voicing assimilation from appendix /-k/ reduction in /VC-k/ sequences

	ˈte**b**	'towards (archaic)'	դէպ
→	ˈte**p-k**	'event'	դէպք
	hɑˈvɑ**d**	'belief (archaic)'	հաւատ
→	hɑˈvɑ**t-k**	'belief'	հաւատք
	dɑˈɾɑ**d͡z**	'spread out'	տարած
→	dɑˈɾɑ**t͡s-k**	'extent'	տարածք
	vɑˈ**z**-el	'race'	վազել
→	ˈvɑ**s-k**	'race'	վազք
	ˈgo**ʁ**	'side'	կող
→	ˈgo**χ-k**	'book cover'	կողք

After a consonant cluster as well (VCC-*k*), the suffix triggers devoicing (Table 4.55).

Table 4.55: Regressive voicing assimilation from appendix /-k/ reduction in /VCC-k/ sequences

	ˈbɑ**ɾd**	'debt (archaic)'	պարտ
→	ˈbɑ**ɾt-k**	'debt'	պարտք
	ˈgu**ɾd͡z**	'core of pumpkin'	կուրծ
→	ˈgu**ɾt͡s-k**	'chest'	կուրծք
	ˈʃɑ**ɾʒ**	'motion'	շարժ
→	ˈʃɑ**ɾʃ-k**	'motion'	շարժք

4.3.1.5 Variable voicing assimilation for /v/

The segment /v/ has complicated patterns of voicing assimilation. When /v/ is before a voiceless obstruent, it devoices. But when the /v/ is after a voiceless ob-

[15] As discussed in Section §4.3.2.6, the level of aspiration on the final *k* seems variable. When word-final and sentence-final, the suffix tends to resist deaspiration: [vɑs-k] or [vɑs-kʰ] 'race'. The problem is that, by being word-final, it's difficult to measure the degree of aspiration.

struent, it can variably devoice. Thus, /v/ undergoes complete regressive assimilation (complete devoicing) but incomplete progressive assimilation (incomplete devoicing).

First consider regressive assimilation. When the /v/ is before a voiceless obstruent in a derived context, it devoices. Possible morphological contexts are before suffixes and in vowel reduction.

Table 4.56: Regressive voicing assimilation of /v/ before voiceless obstruents

	ˈhɑ**v** + ˈpujn	‘chicken’ + ‘house’	հաւ, բոյն
→	hɑ**f**-ˈ**p**ujn	‘praise’	հաւբոյն
	kɑˈvi**t**	‘courtyard’	գաւիթ
→	kɑ**ft**-ɑ-ˈbɑh	‘gatekeeper’	գաւթապահ
	ˈko**v**	‘praise (archaic)’	գով
→	ˈko**f**-**k**	‘praise’	գովք
	ˈse**v**	‘black’	սեւ
	se**v**-ˈn-ɑ-l	‘to blacken’	սեւնալ
→	ˈse**f**-ˈ**t͡s**-ɑ-v	‘it blackened’	սեւցաւ
	kərɑˈ**v**-**it͡ʃ**	‘attractive’	գրաւիչ
→	kərɑ**f**-**t͡ʃ**-uˈtɤn	‘attractiveness’	գրաւչութիւն
	teˈte**v**	‘light’	թեթեւ
→	tete**f**-**s**oˈlig	‘lightheaded’	թեթեւսոլիկ
	ɑˈ**viʃ**	‘lymph’	աւիշ
→	ɑ**f**-**ʃ**ɑˈjin	‘lymphatic’	աւշային

Now consider progressive assimilation. In some morphological contexts, a root-final /u/ becomes [v] before a vowel suffix. This [v] can then devoice after voiceless consonants (Table 4.57).

Table 4.57: Progressive voicing assimilation from /u/ frication: /VCu-V/ → //VCv-V// → [VCf-C]

	gɑˈ**du**	‘cat’	կատու
→	gɑ**dv**-ɑˈgɑn	‘feline’	կատուական
	dɪɾɑˈ**t͡su**	‘clerk’	տիրացու
→	dɪɾɑ**t͡sf**-uˈtɤn	‘clerkship’	տիրացութիւն
	təˈ**tu**	‘sour’	թթու
→	tə**tf**-eˈʁen	‘pickles’	թթուեղէն

After a voiceless sound, we transcribe /v/ as either fully devoiced [f] or as partially devoiced [ɣ̥]. When the fricative is part of the stressed syllable, we strongly perceive hearing a [v] sound even though the spectrogram shows little voicing on the fricative. We transcribe this situation as [ɣ̥]. But when the fricative is not part of the stressed syllable, the perception of a [f] is more salient to our ears.

We found similar ambiguity from vowel reduction and the irregular inflectional suffix /-vɑn/. (Table 4.58). The devoicing of /v/ seems more salient when unstressed.

Table 4.58: Variable or incomplete progressive voicing assimilation for /v/ in /VCCV/

	hɑˈ**ʃiv**	'account'	հաշիւ
→	hɑʃˈ**ɣ̥**-el	'to count'	հաշուել
→	hɑʃˈ**f**-ɑbɑh	'accountant'	հաշուապահ
	ˈje**ɾp**	'when'	երբ
→	je**ɾp**-ˈ**ɣ̥**ɑn	'when-GEN'	երբուան
→	je**ɾp**-**f**əˈn-e	'when-GEN-ABL'	երբուընէ

We find similar ambiguity for post-voiceless [v] from the passive suffix *-v-* (Table 4.59). This suffix surfaces as voiced after voiced obstruents and sonorants. It is likewise written with the digraph ու that is 'supposed' to be pronounced as [v] before vowels.[16] When stressed and after a voiceless segment, we see little to no voicing on the fricative (on the spectogram), but we strongly hear a [v] sound. When unstressed, the perception of [v] is less strong.[17]

[16] The glosses for the words in Table 4.59 is as follows. The active verbs consist of a root, theme vowel (-e-), and then an infinitive suffix *-l*. The passives have the passive suffix *-v-*, theme vowel *-i-*, and infinitive *-l*. Infinitives can get further nominal inflection like with case markers; adding case causes the theme vowel to change to *-e-*. See (cite chapter theme i neutralization).

[17] After CC clusters, the passive /-v-/ triggers schwa epenthesis: [ɑsˈt-e-l] 'to influence' vs. [ɑstəˈv-i-l] 'to be influenced'. Thus voicing assimilation is blocked. See (cite chapter passive epenthesis).

Table 4.59: Progressive voicing assimilation for passive /-v-/

	gɑˈ**b**-e-l	‘to connect’	կապել
→	gɑ**b**ˈ-**v**-i-l	‘to be removed’	կապուիլ
	hɑˈ**n**-e-l	‘to remove’	հանել
→	hɑ**n**ˈ-**v**-i-l	‘to be removed’	հանուիլ
	t͡ʃɑˈ**p**-e-l	‘to measure’	չափել
→	t͡ʃɑ**p**ˈ-**v̥**-i-l	‘to be measured’	չափուիլ
→	t͡ʃɑ**p**-**f**-e-ˈl-ov	‘to be removed (INS)’	չափուելով
	χoˈ**t**-e-l	‘to push in’	խոթել
→	χo**t**ˈ-**v̥**-i-l	‘to be pushed in’	խոթուիլ
→	χo**t**-**f**-e-ˈl-e	‘to be pushed in (ABL)’	խոթուելէ
	t͡səˈ**k**-e-l	‘to leave s.o.’	ձգել
→	t͡sə**k**ˈ-**v̥**-i-l	‘to be left’	ձգուիլ
→	t͡sə**k**-**f**-e-ˈl-u	‘to be left (GEN)’	ձգուելէ
	əˈ**s**-e-l	‘to say’	ըսել
→	ə**s**ˈ-**v̥**-i-l	‘to be said’	ըսուիլ
→	ə**s**-**f**-e-ˈl-ov	‘to be said (INS)’	ըսուելով
	kɑˈ**ʃ**-e-l	‘to say’	քաշել
→	kɑ**ʃ**ˈ-**v̥**-i-l	‘to be said’	քաշուիլ
→	kɑ**ʃ**-**f**-e-ˈl-e	‘to be said (ABL)’	քաշուելէ
	t͡sɑˈ**χ**-e-l	‘to sell’	ծախել
→	t͡sɑ**χ**ˈ-**v̥**-i-l	‘to be sold’	ծախուիլ
→	t͡sɑ**χ**-**f**-e-ˈl-u	‘to be sold (GEN)’	ծախուելու
	koˈ**t͡s**-e-l	‘to close’	գոցել
→	ko**t͡s**ˈ-**v̥**-i-l	‘to be closed’	գոցուիլ
→	ko**t͡s**-**f**-e-ˈl-ov	‘to be closed (INS)’	գոցուելով
	goˈ**t͡ʃ**-e-l	‘to close’	կոչել
→	go**t͡ʃ**ˈ-**v̥**-i-l	‘to be closed’	կոչուիլ
→	go**t͡ʃ**-**f**-e-ˈl-e	‘to be closed (ABL)’	կոչուելէ

It’s possible that the the sound /v/ is gesturally pronounced at the same time as the following vowel, thus making it difficult to determine when its voicing (or lack of voicing) starts. Ideally future phonetic research can look into why we have such difficulties in determining the voicing of /v/.

It is possible that the behavior of /v/ is complicated by its articulatory behavior. Cross-linguistically, it is known that some langauges have the segment /v/ undergoing voicing assimilation rules when in a pre-consontal (vC) or word-final po-

sition (v#). The segment however resists undergoing or applying voicing assimilation in post-consontal (Cv) contexts. Such languages include Russian, Czech, Hungarian, among others (Padgett 2002, Hall 2004, Barkaï & Horvath 1978). A surprising similarity is that the Armenian /v/ obligatorily undergoes regressive devoicing assimilation, while progressive devoicing is variable.

4.3.2 Aspiration and deaspiration

This section discusses aspiration and deaspiration. In this section, we mark aspiration on voiceless stops in the phonemic and phonetic forms, because HD tends to aspirate his stops. We mark the locations where the aspiration is lost.

Note that data for this section applies only to speakers who do use aspiration in the first place. These include Armenians from Turkey and the US. Armenians in Arab countries do not aspirate.

Data here is mainly from HD who has aspiration because he lives in the US.

4.3.2.1 Post-fricative deaspiration (intervocalic)

Word-initially, the orthography has roots that start with a /s/-stop and /ʃ/-stop cluster. In Western Armenian, this cluster undergoes schwa prothesis: /stʰoɾ/ → [əstoɾ] ‘blind’. The sibilant causes the following voiceless stop to deaspirate (Table 4.60).

Table 4.60: Post-fricative deaspiration in root-initial sibilant-stop clusters (underived contexts) for HD

/#ʃtʰV/ [əʃtV]	/**ʃt**ʰɑb/ [ə**ʃ**ˈ**t**ɑb] <**ʃd**ab>	‘haste’ շտապ	cf. /#tʰV/ [tʰV]	/**t**ʰɑpʰoɾ/ [**t**ʰɑˈpʰoɾ] <**t**’ap’or>	‘procession’ թափօր
/#spʰV/ [əspV]	/**sp**idɑg/ [ə**sp**iˈdɑg] <**sb**idag>	‘white’ սպիտակ	cf. /#pʰV/ [pʰV]	/**p**ʰipʰ/ [ˈ**p**ʰipʰ] <**p**ip>	‘eye pupil’ բիբ
/#stʰV/ [əstV]	/**st**ʰeʁ**d͡z**el/ [ə**st**eʁˈ**d͡z**el] <**sd**eɣ**d͡z**el>	‘to create’ ստեղծել	/#tʰV/ [tʰV]	/**t**ʰeʁ/ [ˈ**t**ʰeʁ] <**t**eɣ>	‘medicine’ դեղ
/#skʰV/ [əskV]	/**sk**ʰuʃ/ [əsˈ**k**uʃ] <**zk**oyʃ>	‘careful’ զգոյշ	cf. /#kʰV/ [kʰV]	/**k**ʰuʃɑg/ [**k**ʰuˈʃɑg] <**k**owʃag>	‘diviner’ գուշակ

HD has default aspiration for his stops, but not in this context. Fairbanks documents that her Istanbuli speakers have aspirated stops in general, but that these stops are unaspirated after sibilants (Fairbanks 1948: 5-7).

There is evidence that the cluster lacks a schwa in the underlying or lexical representation. For discussion of schwa epenthesis and prothesis, see (cite schwa epenthesis chapter). Furthermore, many of these clusters orthographically have voiced stops like տ <d> in ստոր <sdor> [əstoɾ] 'blind'. This voiced stop is just an orthographic residue of diachronic sound changes. There is no synchronic evidence that the post-fricative stops in Table 4.60 are underlyingly voiced. For discussion of voicing mismatches with the orthography, see Section §3.4.2.

We likewise find deaspiration in word-medial intervocalic contexts (Table 4.61).

Table 4.61: Post-fricative deaspiration in word-medial intervocalic sibilant-stop clusters (underived contexts)

/Vsp^{h}V/ [VspV]	/te**sp**ɑrez/ [te**s**ˈ**p**ɑn] <te**sb**an>	'ambassador' դեսպան	cf./Vp^{h}V/ [VpV]	/ɑjk^{h}ep^{h}ɑn/ [ɑjk^{h}eˈ**p**^{h}ɑn] <ayke**p**an>	'viticulturist' այգեբան
/Vst^{h}V/ [VstV]	/ɑ**st**iʒɑn/ [ɑ**s**ˈ**t**iʒɑn] <a**sd**id͡ʒan>	'degree' աստիճան	cf. /VtV/ [Vt^{h}V]	/gɑ**t**^{h}ig/ [gɑˈ**t**^{h}ig] <ga**t**'ig>	'small drop' կաթիկ
/Vsk^{h}V/ [VskV]	/gɑ**sk**^{h}ɑd͡z/ [gɑ**s**ˈ**k**ɑd͡z] <ga**sg**ad͡z>	'doubt' կասկած	cf. /Vk^{h}V/ [Vk^{h}V]	/d͡ʒɑɾɑ**k**^{h}ajt^h/ [d͡ʒɑɾɑˈ**k**^{h}ɑjt^h] <d͡ʒaɾa**k**ayd>	'ray' ճառագայթ
/Vʃt^{h}V/ [VʃtV]	/ʃe**ʃt**^{h}oʁ/ [ʃe**ʃ**ˈ**t**oʁ] <ʃe**ʃd**oɣ>	'stressing' շեշտող	cf. /Vt^{h}V/ [Vt^{h}V]	/me**t**^{h}od/ [meˈ**t**^{h}od] <me**t**'od>	'method' մեթոտ
/Vχt^{h}V/ [VχtV]	/χe**χt**^{h}el/ [χe**χ**ˈ**t**el] <xe**ɣd**el>	'to strangle' խեղդել	cf. /Vt^{h}V/ [Vt^{h}V]	/ɑnhe**t**^{h}et^h/ [ɑnheˈ**t**^{h}et^h] <anhe**t**'et'>	'fantastical' անհեթեթ

cite fairbanks and other folks on post-fricative deaspiration

Post-fricative aspiration is also blocked in derived contexts, such as vowel reduction (Table 4.62). The root consists of an underlying fricative-vowel-stop sequence. In compounds, the vowel is deleted,[18] and thus causing the fricative to precede the stop. The fricative deaspirates the stop, while the stop devoices the fricative.

[18]The deletion is due to destressed high vowel reduction. See (cite chapter reduction).

Table 4.62: Post-fricative deaspiration from vowel reduction in intervocalic (VCCV) derived contexts

	[kɑˈvi**t**ʰ]	‘courtyard’	գաւիթ
→	/kɑ**vit**ʰ-ɑ-bɑh/	with compound linker /-ɑ-/	
	[kɑ**ft**-ɑ-ˈbɑh]	‘gatekeeper’	գաւթապահ

There are some derivational prefixes that end in a voiced fricative: synonymous [təʒ-] or [dəʒ-] (Table 4.63). The fricative causes deaspiration of the following stop, and the fricative then gets devoiced.

Table 4.63: Post-fricative deaspiration after the derivational prefixes in VCCV contexts

		ˈ**p**ʰɑχt	‘luck’	բախտ
→	/təʒ-**p**ʰɑχtʰ/	tə**ʃ**-ˈ**p**ɑχt	‘unlucky’	դժբախտ
→	/dəʒ-**p**ʰɑχtʰ/	də**ʃ**-ˈ**p**ɑχt	‘unlucky’	տժբախտ
		ˈ**k**ʰoh	‘satisfied’	գոհ
→	/təʒ-**k**ʰoh/	tə**ʃ**-ˈ**k**oh	‘dissatisfied’	դժգոհ
→	/dəʒ-**k**ʰoh/	də**ʃ**-ˈ**k**oh	‘dissatisfied’	տժգոհ
		ˈ**k**ʰujn	‘color’	գույն
→	/təʒ-**k**ʰujn/	tə**ʃ**-ˈ**k**ujn	‘discolored’	դժգույն
→	/dəʒ-**k**ʰujn/	də**ʃ**-ˈ**k**ujn	‘discolored’	տժգույն

The above data concerned VCCV clusters where the cluster consists of only two consonants. We can also find a few cases where the cluster has 3 consonants $VC_1C_2.C_3V$ such that the C_1C_2 form a complex coda (Table 4.64). Here again, we find deaspiration of C_3. If C_2 is a voiceless obstruent and C_3 is voiced obstruent, we also find devoicing.

Table 4.64: Post-fricative deaspiration in intervocalic VCCCV derived contexts

	bɑɾˈ**sig**	‘Persian person’	պարսիկ
→	bɑ**ɾs.k**-eˈɾen	‘Persian language’	պարսկերէն
	ˈɑ**js** + ˈ**de**ʁ	‘this’ + ‘place’	այս, տեղ
→	ɑ**js**-ˈ**te**ʁ	‘here, this place’	այստեղ
	ˈɑ**js** + ˈ**k**ʰɑn	‘this’ + ‘much’	այս, քան
→	ɑ**js**-ˈ**k**ɑn	‘this much’	այսքան

4.3.2.2 Pre-fricative deaspiration (intervocalic)

Compared to post-fricative stops, it is relatively rarer to find pre-fricative stops. In the few types of examples that we find, we also find pre-fricative deaspiration (also reported in Fairbanks 1948: 7).

It is relatively rare to find roots with stop-fricative clusters (Table 4.65). Such clusters can be voiced or voiceless. When voiceless, the stop is unaspirated.

Table 4.65: Pre-fricative deaspiration in VCCV clusters in underived contexts

/ɑp^{h}se/	ɑpˈse	‘tray’	ափսէ
/ɑp^{h}ʃ-um/	ɑpˈʃum	‘surprise’	ապշում
/ɑp^{h}sos/	ɑpˈsos	‘alas’	ափսոս
/ɑk^{h}sor/	ɑkˈsoɾ	‘exile’	աքսոր
/tɑt^{h}χ-el/	tɑtˈχel	‘to wet’	թաթխել

Many contexts for pre-fricative deaspiration come from derived contexts, such as come from vowel reduction of destressed /i,u/ (Table 4.66). Some cases come from reduction or syncope of /ɑ/ in some roots.

Table 4.66: Pre-fricative deaspiration from vowel reduction and syncope in VC(C)CV derived contexts

	ʃɑmˈp^{h}uʃ	‘foolish’	շամբուշ
→	ʃɑmp.ʃ-ɑˈɡɑn	‘foolish’	շամբշական
	əmˈp^{h}iʃ	‘athlete’	ըմբիշ
→	əmp.ʃ-ɑˈɡɑn	‘athletic’	ըմբշական
	kəŋˈk^{h}uʃ	‘delicate’	քնքուշ
→	kəŋk.ˈʃ-ɑŋk	‘delicacy’	քնքշանք
	ɡəŋˈk^{h}uʁ	‘hood’	կնգուղ
→	ɡəŋk.χ-ɑˈvoɾ	‘Capuchin friar’	կնգղաւոր

Another context is the names for the days of the week (Table 4.67). These names are compounds without the linking vowel /-ɑ-/. Note that the second root in these compounds changes its form, likely a type of allomorphy.

Table 4.67: Pre-fricative deaspiration in compounds for days of the week

	je'ɾe**k**^{h} + **ʃ**ɑ'**p**^{h}**ɑt**^{h}	'two' + 'week'	երեք, շաբաթ
→	je'ɾe**k-ʃ**ɑ**p**'**t**^{h}i	'Tuesday'	երեքշաբթի
	/t͡ʃoɾe**k**^{h}-/ + **ʃ**ɑ'**p**^{h}**ɑt**^{h}	'quatro-' + 'week'	
→	t͡ʃoɾe**k-ʃ**ɑ**p**'**t**^{h}i	'Wednesday'	չորեքշաբթի
	'hiŋ**k**^{h} + **ʃ**ɑ'**p**^{h}**ɑt**^{h}	'five' + 'week'	հինգ, շաբաթ
→	hiŋ**k-ʃ**ɑ**p**'**t**^{h}i	'Thursday'	հինգշաբթի

Some numerals likewise show pre-fricative deaspiration (Table 4.68). These are formed from a root and decade suffix /-sun/.

Table 4.68: Pre-fricative deaspiration in some numerals

	'u**t**^{h} + /-sun/	'eight' + '-th'	ութ
→	u**t**-'sun	'eighty'	ութսուն
Base	/vɑ**t**^{h}/ + /-sun/	'hexa-' + '-th'	
→	vɑ**t**-'sun	'sixty'	վաթսուն

Another context is passivization (Table 4.69). The passive suffix /-v-/ is devoiced after voiceless stops and triggers deaspiration. Note that the devoicing of passive /-v-/ is complicated. See Section §4.3.1.5. The gloss for active verbs is √-TH-INF, while for passives it's √-PASS-TH-INF.

Table 4.69: Pre-fricative deaspiration from passivization in VCCV derived contexts

Base	t^{h}ɑ'**p**^{h}-e-l	'to throw away'	թափել
→	t^{h}ɑ**p**-'v̥-i-l	'to be thrown away'	թափուիլ
Base	χɑ'**p**^{h}-e-l	'to trick'	խաբել
→	χɑ**p**-'v̥-i-l	'to be tricked'	խաբուիլ
	noɾo'**k**^{h}-e-l	'to restore'	նորոգել
→	noɾo**k**-'v̥-i-l	'to be restored'	նորոգուիլ
	hɑvɑ'**k**^{h}-e-l	'to gather'	հաւաքել
→	hɑvɑ**k**-'v̥-i-l	'to be gathered'	հաւաքուիլ

4.3.2.3 Post-affricate deaspiration (intervocalic)

Similar to post-fricative deaspiration, we also have post-affricate aspiration. There are fewer examples of this process though.

In derived contexts (Table 4.70), most but not all attested examples of post-affricate deaspiration simultaneously involve devoicing of the stop because the stop was underlyingly voiced.

Table 4.70: Post-affricate deaspiration in intervocalic VCCV derived contexts

Aorist stem:	mɔɾɑ'**t͡s**-ad͡z	'forgotten'	մոռացած
→ /mɔɾɑ**t͡s-g**od/	mɔɾɑ**t͡s**-'**k**od	'forgetful'	մոռացկոտ
	'pʰɑ**t͡s** + **p**ʰe'ɾɑn	'open' + 'mouth'	բաց, բերան
→ /pʰɑ**t͡s-p**ʰeɾɑn]	pʰɑ**t͡s-pe**'ɾɑn	'babbler'	բացբերան
	'χɑ**t͡ʃ** + '**k**ʰɑɾ	'cross' + 'stone'	խաչ, քար
→ /χɑ**t͡ʃ-k**ʰɑɾ/	χɑ**t͡ʃ**-'**k**ɑɾ	'cross-stone'	խաչքար
	jeˈɾe**t͡s** + 'gin	'elder' + 'woman'	երէց, կին
→ /jeɾe**t͡s-g**in/	jeɾe**t͡s**-'**k**in	'pastor's wife'	երէցկին
	'me**t͡ʃ** + 'deʁ	'in' + 'place'	մէջ, տեղ
→ /me**t͡ʃ** -deʁ/	me**t͡ʃ**-'teʁ	'middle'	մէջտեղ
	mi'**t͡ʃ**ug	'nucleus'	միջուկ
→ /mi**t͡ʃ**ug-ɑjin/	mi**t͡ʃ**k-ɑ'jin	'nuclear'	միջկային
	kʰeʁe'**t͡s**ig	'pretty'	գեղեցիկ
→ /kʰeʁe**t͡s**ig-ɑnɑl/	keʁe**t͡s**k-ɑ'nɑl	'to become pretty'	գեղեցկանալ

The above data concerned VCCV clusters where the cluster consists of only two consonants. We can also find cases where the cluster has 3 consonants $VC_1C_2.C_3V$ such that the C_1C_2 form a complex coda (Table 4.71). Here again, we find deaspiration of C_3. If C_2 is a voiceless obstruent and C_3 is voiced obstruent, we also find devoicing.

Table 4.71: Post-affricate deaspiration in intervocalic VCCCV derived contexts

	ɑχ'**t͡ʃ**ig	'girl'	աղջիկ
→ /ɑ**χt͡ʃ**ig-ɑgɑn/	ɑ**χt͡ʃ**.k-ɑ'gɑn	'feminine'	աղջկական
Base	tʰɑpʰɑn'**t͡s**ig	'transparent'	թափանցիկ
→ /tʰɑpʰɑ**nt͡s**ig-ɑnɑl/	tʰɑpʰɑ**nt͡s**k-ɑ'nɑl	'to become transparent'	թափանցկանալ
	hoˈɾɑ**nt͡ʃ**	'yawn'	յօրանչ
→ /hoɾɑ**nt͡ʃ-g**od/	hoɾɑ**nt͡ʃ**-'**k**od	'yawning'	յօրանչկոտ

It should be noted that in these intervocalic VCCCV contexts, HD perceives a smaller degree of deaspiration than in VC.CV clusters. But based on observing

the spectrograms, such clusters do show deaspiration and devoicing. It is possible that what HD perceives is incomplete neutralization, or that this is just a perceptual illusion.

4.3.2.4 Pre-affricate deaspiration (intervocalic)

Deaspiration can also apply before affricates.

So far, we've only found very few cases of deaspiration in underived contexts (Table 4.72).

Table 4.72: Pre-affricate deaspiration in underived contexts

/ɑk^{h}**t͡s**ɑn/	ɑkˈ**t͡s**ɑn	'switch'	աքցան

In derived contexts, stops deaspirate before affricates. A common morphological construction that shows this process is causativization. For the roots in Table 4.73, the root ends in either voiceless aspirated stop or a voiced stop. Aspiration is clearer when the stop is intervocalic. To form a causative verb, the suffix sequence /-t͡sən-e-l/ is added. The affricate causes devoicing and deaspiration.[19] This affects both VCCV and VCCCV clusters

Table 4.73: Pre-affricate deaspiration before the causative suffix /-t͡sən-/

		ˈmu**t**^{h}	'dark'	մութ
		mu**t**^{h}-e-n	'dark (ABL-DEF)'	մութէն
→	/mu**t**^{h}-**t͡s**ən-e-l/	mu**t-t͡s**əˈn-e-l	'to darken'	մութցնել
		k^{h}i**d**-e-m	'I know (√-TH-1SG)'	գիտեմ
→	/k^{h}i**d-t͡s**ən-e-l/	ki**t-t͡s**əˈn-e-l	'to notify'	գիտցնել
		ˈdɑ**k**^{h}	'hot'	տաք
		dɑˈ**k**^{h}-e-n	'hot (ABL-DEF)'	տաքէն
→	/dɑ**k**^{h}-**t͡s**ən-e-l/	dɑ**k-t͡s**əˈn-e-l	'to heat'	տաքցնել
		lo**k**^{h}-ɑŋk^{h}	'bath (√-NMLZ)'	լոգանք
→	/lo**k**^{h}-**t͡s**ən-e-l/	lo**k-t͡s**əˈn-e-l	'to notify'	լոգցնել
		χoˈɾu**ŋg**	'deep'	խորունկ
→	/χoɾu**ng-t͡s**ən-e-l/	χoˈɾu**ŋk-t͡s**əˈn-e-l	'to deepen'	խորունկցնել

Another case involves irregular verbs with an affricate infix. The roots in Table 4.74 have an underlying voiced stop. This voiced stop surfaces in some inflected

[19] The gloss for /t͡sən-e-l/ is [CAUS-TH-INF].

forms when the stop is intervocalic. In their citation or infinitive form, the root takes a meaningless affricate suffix /-t͡ʃ-/ suffix that triggers devoicing and deaspiration on the root's consonant.

Table 4.74: Pre-affricate deaspiration before the meaningless infix /-t͡ʃ-/

		tʰəˈ**b**-ɑ-v	'he touched'	√-PST-3SG	դպաւ
→	/tʰə**b**-**t͡ʃ**-i-l/	tə**p**-ˈ**t͡ʃ**-i-l	'to touch'	√TH-INF	դպչիլ
		pʰɑˈ**g**-ɑ-v	'he stuck to'	√-PST-3SG	փակաւ
→	/pʰɑ**g**-**t͡ʃ**-i-l/	pɑ**k**-ˈ**t͡ʃ**-i-l	'to stick to'	√TH-INF	փակչիլ

Vowel reduction can likewise create contexts for pre-affricate devoicing. All the examples in Table 4.75 involve roots that take a derivational suffix, either /-it͡ʃ/ or /-git͡s/. When another derivational suffix is added, the vowel of /-it͡ʃ/ is deleted, causing the consonants to become adjacent and trigger deaspiration.

Table 4.75: Pre-affricate deaspiration from vowel reduction in VC(C)CV contexts

	noɾoˈ**k**ʰ-**it͡ʃ**	'reformer'	նորոգիչ
→	noɾo**k**-**t͡ʃ**-ɑˈgɑn	'of reforms (adj.)'	նորոգչական
	nəvɑˈ**k**ʰ-**it͡ʃ**	'musician'	նուագիչ
→	nəvɑ**k**-**t͡ʃ**-uˈhi	'fem. musician'	նուագչուհի
	mij-ɑˈ**git͡s**	'united'	միակից
→	mij-ɑ**kt͡s**-uˈtjɤn	'junction'	միակցութիւն
	χənta-ˈ**git͡s**	'united'	խնդակից
→	χənta-**kt͡s**-ɑˈgɑn	'junction'	խնդակցական
	jeɾˈ**k**ʰ-**it͡ʃ**	'singer'	երգիչ
→	jeɾ**k**-**t͡ʃ**-uˈhi	'fem. singer'	երգչուհի

4.3.2.5 Deaspiration in stop-stop clusters (intervocalic)

In stop-stop clusters, initial voiceless stops are deaspirated in both underived and derived contexts. The second stop is aspirated.

In underived contexts (Table 4.76), it is relatively difficult to find intervocalic stop-stop clusters. There are some roots and bound roots which have such clusters. Here, we find that the two stops are either both voiced stops or both voiceless stops. The second stop is aspirated if voiceless.

Table 4.76: Intervocalic stop-stop restrictions in underived contexts

Voiced + Voiced	jeki**b**ˈ**d**os	‘Egypt’	Եգիպտոս
	ɑ**b**ˈ**d**ɑg	‘slap’	ապտակ
	pe**g**ˈ**d**-el	‘to break’	բեկտել
Voiceless + Voiceless	bɑ**t**ˈ**k**ʰam	‘message’	պատգամ
	t͡sɑ**t**ˈ**k**ʰ-el	‘to jump’	ցատքել
* Aspirated + Stop	*bɑ**t**ʰ**g**ɑm		
	*bɑ**t**ʰ**k**ɑm		
* Stop + Aspirated	*bɑ**gt**ʰɑm		

More cases are found in derived contexts, when a voiceless aspirated stop becomes adjacent to another stop. For the derived contexts in Table 4.77, the voiceless stop precedes an underlyingly voiced stop. The voiced stop devoices due to voicing assimilation. Both stops become voiceless, and second stop is aspirated.

Table 4.77: Stop-stop deaspiration in derived /VCCV/ contexts where the first is voiceless and second is voiced

Bases	ˈ**t**ʰɑ**k**ʰ + ˈ**g**ɑb	‘crown’ + ‘link’	թագ, կապ
→	tʰɑ**k-k**ʰab-uˈtʰɤn	‘coronation’	թագկապութիւն
	χɑˈnu**t**ʰ	‘store’	խանութ
→	χɑnu**t**-ˈ**p**ʰɑn	‘shopkeeper’	խանութպան
	oˈ**k**ʰ**ud**	‘benefit’	օգուտ
→	o**kt**ʰ-ɑˈgɑɾ	‘useful’	օգտակար

This constraint is likewise active in /VCCCV/ contexts. For the words in Table 4.78, the voiceless stop is underlyingly aspirated and part of a complex coda. In derivation, the voiceless stop precedes a voiced stop. Both stops become voiceless. Aspiration is on the second stop.

Table 4.78: Stop-stop deaspiration in derived /VCCCV/ contexts where the second is voiceless and third is voiced

	pʰɑˈ**pʰug**	‘delicate’	փափուկ
→	pʰɑ**p.k**ʰ-ɑˈgɑn	‘delicate’	փափկական
	tʰə**m**ˈ**pʰug** + hɑˈɾ-el	‘drum’ + ‘to beat’	թմբուկ, հարել
→	tʰə**mp.k**ʰ-ɑ-ˈhɑɾ	‘drummer’	թմբկահար
	kʰə**n**ˈ**tʰig** + ˈkʰɑɾ	‘marble’ + ‘rock’	գնդիկ, քար
→	kʰə**nt.k**ʰ-ɑ-ˈkʰɑɾ	‘globulite’	գնդկաքար
	sə**n**ˈ**tʰig**	‘mercury’	սնդիկ
→	sə**nt.k**ʰ-ɑˈjin	‘mercurial’	սնդկային

4.3.2.6 Deaspiration in final clusters

The previous sections focused on deaspiration in word-medial contexts. Word-finally, we think there is deaspiration.

Among underived contexts, deaspiration can apply in word-final clusters of a fricative+stop (Table 4.79).

Table 4.79: Post-fricative deaspiration in word-final sibilant-stop clusters (underived contexts)

/Vspʰ#/ [Vsp#]	/bɑɾi**sp**ʰ/ [bɑˈɾi**sp**]	‘fortress’	պարիսպ <bari**sb**>	/zu**sp**ʰ/ [ˈzu**sp**]	‘restrained’	զուսպ <zow**sb**>
/Vstʰ#/ [Vst#]	/bɑdɾɑ**st**ʰ/ [bɑdˈɾɑ**st**]	‘ready’	պատրաստ <badra**sd**>	/χi**st**ʰ/ [ˈχi**st**]	‘strict’	խիստ <xi**sd**>
/Vskʰ#/ [Vsk#]	/ɑ**sk**ʰ/ [ˈɑ**sk**]	‘nation’	ազգ <a**zk**>	/i**sk**ʰ/ [ˈi**sk**]	‘but’	իսկ <xi**sg**>
/Vʃtʰ#/ [Vʃt#]	/d͡ʒi**ʃt**ʰ/ [ˈd͡ʒi**ʃt**]	‘correct’	ճիշդ <d͡ʒi**ʃt**>	/gu**ʃt**ʰ/ [ˈgu**ʃt**]	‘sated’	կուշտ <gow**ʃd**>
/Vχp#/ [Vχp#]	/ɑ**χp**ʰ/ [ˈɑ**χp**]	‘trash’	աղբ <a**ɣp**>	/vo**χp**ʰ/ [ˈvo**χp**]	‘lament’	ողբ <xi**sg**>
/Vχt#/ [Vχt#]	/ɑ**χt**ʰ/ [ˈɑ**χt**]	‘disease’	ախտ <a**xd**>	/kɑ**χt**ʰ/ [ˈkɑ**χt**]	‘emigration’	գաղթ <ka**ɣt**’>

Note that although we transcribe these stops as deaspirated, it is possible that there is some degree of aspiration.[20] When a word-final like [ɑsk] is said in isolation, the stop is often produced with a perceivable release. It’s difficult to tease apart aspiration and just audible releases: [ɑsk(ʰ)] ‘nation’. If the word is however sentence-medial, then aspiration is fully absent (1).

[20] Fairbanks (1948: 5) transcribes these final clusters as aspirated.

(1) ɑsk əs-ɑ-v ‘He said “nation”.’
nation say-PST-3SG
«Ազգ» ըսաւ:

Word-finally in derived contexts, a VCC or VCCC cluster is formed by adding the nominalizer suffix /-kʰ/.

The suffix can derive nouns from roots and from other words. Its syllable structure is quite complicated, and it is often analyzed as an extrasyllabic appendix (§??). When this suffix is word-final after a fricative, it has some degree of aspiration in our recordings when uttered in isolation. But when another inflectional suffix is added, this /-kʰ/ becomes word-medial and fully deaspirates.[21] Furthermore, if the word is sentence-medial, then aspiration is again fully lost. We first show data from VC-kʰ contexts (Table 4.80).

Table 4.80: Variable post-fricative deaspiration for the suffix /-kʰ/ in VC-kʰ contexts

	tʰəˈʒoχ	‘hard’	դժոխ
Derived	tʰəˈʒoχ-**k(ʰ)**	‘hell’	դժոխք
Inflected	tʰəʒoχ-ˈ**k**-ov	‘hell-INS’	դժոխքով
	tʰəʒoχ-**k**-ˈneɾ	‘hell-PL’	դժոխքներ
Root	/hɾɑʃ-/	bound root as in...	
	həɾɑʃ-ɑˈli	‘marvelous’	հրաշալի
Derived	həˈɾɑʃ-**k(ʰ)**	‘miracle’	հրաշք
Inflected	həɾɑʃ-ˈ**k**-i	‘miracle-GEN’	հրաշքի
	həɾɑʃ-**k**-ˈneɾ	‘miracle-PL’	հրաշքներ
Root	/des-/	bound root as in...	
	deˈs-ɑd͡z	‘seen’	տեսած
Derived	ˈdes-**k(ʰ)**	‘sight’	հրաշք
Inflected	des-ˈ**k**-i	‘sight-ABL’	տեսքէ
Root	/χos-/	bound root as in...	
	χoˈs-il	‘to speak’	խօսիլ
Derived	ˈχos-**k(ʰ)**	‘speech’	խօսք
Inflected	χos-ˈ**k**-eɾ	‘speech-PL’	խօսքեր

If the fricative is part of a complex coda, we also find that the /-kʰ/ seems to resist deaspiration in isolation (Table 4.81). Deaspiration is more visible when this

[21] Between a fricative and consonant (/VC-kʰ-CV/), the suffix /-kʰ/ tends to significantly overlap with the preceding fricative. This suggests some type of gestural overlap.

segment becomes word-medial by adding inflectional suffixes, or even sentence-medial.

Table 4.81: Variable post-fricative deaspiration for the suffix /-kʰ/ in VC-k contexts

Base	ˈbɑɾs	‘Persian (archaic)	պարս
Derived	ˈbɑɾs-**k**(ʰ)	‘Persia	Պարսք
Inflected	bɑɾs-ˈ**k**-e-n	‘Persia-ABL-DEF’	Պարսքէն
Base	ˈvɑɾs	‘hair (archaic)	վարս
Derived	ˈvɑɾs-**k**(ʰ)	‘hair of head	վարսք
Inflected	vɑɾs-ˈ**k**-ov	‘hair-INS’	վարսքով

The same patterns are found when the preceding consonant is a voiceless affricate (Table 4.82).

Table 4.82: Variable post-affricate deaspiration for the suffix /-kʰ/ in VC-k contexts

Base	ˈmet͡ʃ	‘in’	մէջ
Derived	ˈmet͡ʃ-**k**(ʰ)	‘waist’	մէջք
Inflected	met͡ʃ-ˈ**k**-e	‘waste-ABL	մէջքէ
Base	tʰəˈɾit͡ʃ	‘flight’	թռիչ
Derived	tʰəˈɾit͡ʃ-**k**(ʰ)	‘flight’	թռիչք
Inflected	tʰəɾit͡ʃ-ˈ**k**-i	‘flight-GEN	թռիչքի
	təɾit͡ʃ-**k**-ˈneɾ	‘flight-GEN	թռիչքներ
Aorist stem	uɾeˈt͡s-ɑd͡z	‘swollen’	ուռեցած
Derived	uˈɾet͡s-**k**(ʰ)	‘swelling (n)’	ուռեցք
Inflected	uɾet͡s-ˈ**k**-ov	‘swelling-INS	ուռեցքով
	uɾet͡s-**k**-ˈneɾ	‘swelling-PL	ուռեցքներ
Aorist stem	ləvɑt͡s-oʁ	‘washer’	լուացող
Derived	ləˈvɑt͡s-**k**(ʰ)	‘laundry’	լուացք
Inflected	ləvɑt͡s-ˈ**k**-ov	‘laundry-GEN	լուացքի
	ləvɑt͡s-**k**-ˈneɾ	‘laundry-PL	լուացքներ

We find the same variation when the affricate is part of a complex coda (Table 4.83).

Table 4.83: Variable post-affricate deaspiration for the suffix /-k/ in VCC-k contexts

Base	ˈvɑɾt͡s	‘reward’	վարձ
Derived	ˈvɑɾt͡s-k(ʰ)	‘wage’	վարձք
Inflected	vɑɾt͡s-ˈk-er	‘wage-PL’	վարձքեր
Base	ˈveɾt͡ʃ	‘end’	վերջ
Derived	ˈveɾt͡ʃ-k(ʰ)	‘edge’	վերջք
Inflected	veɾt͡ʃ-ˈk-i-n	‘edge-PL’	վերջքին
Base	zəˈɾujt͡s	‘tale’	զրույց
Derived	zəˈɾujt͡s-k(ʰ)	‘conversation’	զրույցք
Inflected	zəɾujt͡ʃ-ˈk-e	‘tale-ABL’	զրույցքէ
	zəɾuɾt͡ʃ-k-ˈneɾ	‘tale-PL’	զրույցքներ
Base	bɑˈhɑnt͡ʃ	‘demand’	պահանջ
Derived	bɑˈhɑnt͡ʃ-k(ʰ)	‘credit’	պահանջք
Inflected	bɑhɑnt͡ʃ-ˈk-ov	‘credit-INS’	պահանջքով
	bɑhɑnt͡ʃ-k-ˈneɾ	‘credit-PL’	պահանջքներ

After a stop, the suffix /-kʰ/ causes devoicing and deaspiration of the stop. The suffix has the same variable deaspiration as before. Note that many of the roots in Table 4.84 are bound roots. Many also involve vowel reduction.

Table 4.84: Stop-stop deaspiration in derived /VC-k/ contexts

	ˈhed	‘with’	հետ
→	ˈhet-k(ʰ)	‘trace’	հետք
	/ɑʁotʰ-/	bound root for praying	
	ɑʁoˈtʰ-el	‘to pray’	աղօթել
→	ɑˈʁot-k(ʰ)	‘trace’	աղօթք
	/vod-/	bound root for feet	
	voˈd-ig	‘tiny foot’	ոտիկ
→	ˈvot-k(ʰ)	‘foot’	ոտք
	/mid-/	bound root for mind	
	məd-ɑˈjin	‘mental’	մտային
→	ˈmit-k(ʰ)	‘mind’	միտք

We acknowledge though that the lack of deaspiration of /-k/ is confounded with how this segment is word-final and said in isolation. It’s possible that this segment is truly deaspirated after all fricatives/affricates even in isolation. But

by being word-final in isolation, the stop gets some degree of intonational prominence, noisy enhancement, a stronger release, or breathiness.

As of writing this grammar, we have not been able to do a systematic acoustic study of this suffix's deaspiration across multiple speakers. That seems like a worthwhile future research question.

4.4 Place assimilation of nasals

Before the velar stops /k, g/, the nasal /n/ becomes velar [ŋ] (Rule 3). Orthographically, the nasal is still written as a coronal nasal Ն <n>. This process applies both in underived (§4.4.1) and derived contexts (§4.4.2). In derived contexts, we can find cases where a surface [n] alternates with [ŋ]. Complications arise though from cases where /n/ has stress and can resist velarization.

Rule 3. Nasal place assimilation before velars

Before a velar stop /k, g/, the nasal /n/ becomes [ŋ].				
/ɑ**nk**ɑm/	→	[ɑŋˈ**k**ɑm]	'time'	անգամ
/t͡sɑ**ng**/	→	[t͡sɑ**ŋg**]	'list'	ցանկ

Nasal place assimilation does not occur before other dorsal sounds, such as before /χ, ʁ/ (§4.4.3). There is likewise no productive rule of assimilation before labials. However, labial assimilation was a widespread diachronic rule. There is likewise a degree of labial assimilation in connected speech for high-frequency words.

4.4.1 Velar assimilation in underived contexts

In VCCV clusters, the nasal /n/ becomes [ŋ] before the velar stops /k, g/. Table 4.85 shows the application of this rule in roots (underived contexts).

Table 4.85: Nasal place assimilation before velar stops in VCCV contexts

/kɑ**nk**ɑd/	'complaint'	/kɑ**nk**uɾ/	'curly'	/hɑ**nk**ist/	'comfortable'
[kɑŋˈ**k**ɑd]	գանգատ	[kɑŋˈ**k**uɾ]	գանգուր	[hɑŋˈ**k**ist]	հանգիստ
/ə**ng**ujz/	'walnut'	/hɑ**ng**ɑɾd͡z/	'suddenly'	/ɑ**ng**oʁin/	'bed'
[əŋˈ**g**ujz]	ընկոյզ	[hɑŋˈ**g**ɑɾd͡z]	յանկարծ	[ɑŋ**g**oˈʁin]	անկողին

Velar place assimilation also applies in VCCCV clusters where the first two consonants are the nasal and velar (Table 4.86). Such clusters tend to be found in bound roots.

Table 4.86: Nasal place assimilation before velar stops in VCCCV (VNCCV) contexts

/ɑ**nk**l-ijɑ/	‘English’ (√-NMLZ)	/gɑ**nk**n-i-l/	‘to stand’ (√-TH-INF)
[ɑ**ŋk**liˈjɑ]	Անգլիա	[gɑ**ŋk**ˈnil]	կանգնիլ
/ə**ng**d͡ʒ-i-l/	‘to succumb’ (√-TH-INF)	/hɑ**nk**-t͡ʃ-i-l/	‘to rest’ (√-VX-TH-INF)
[ə**ŋg**ˈd͡ʒil]	ընկճիլ	[hɑ**ŋk**ˈt͡ʃil]	յանգչիլ

Place assimilation also applies word-finally in VCC clusters (Table 4.87).

Table 4.87: Nasal place assimilation before velar stops in final VCC contexts

/ʒɑɾɑ**nk**/	‘heir’	/nɑhɑ**nk**/	‘state’	/vɑɾu**nk**/	‘cucumber’
[ʒɑˈɾɑ**ŋk**]	ժառանգ	[nɑˈhɑ**ŋk**]	նահանգ	[vɑˈɾu**ŋk**]	վարունգ
/χu**ng**/	‘incense’	/vɑ**ng**/	‘syllable’	/ɑnɑ**ng**/	‘that way’
[ˈχu**ŋg**]	խունկ	[ˈvɑ**ŋg**]	վանկ	[ɑˈnɑ**ŋg**]	անանկ

Although rare, there is one word which shows velar assimilation in a final VCCC cluster (Table 4.88).

Table 4.88: Nasal place assimilation before velar stops in final VCCC contexts

/ɑ**nk**χ/	[ˈɑ**ŋk**χ]	‘vulture’	անգղ

The above examples concerned place assimilation in roots. There are likewise derivational suffixes that contain a sequence of a nasal /n/ and a velar stop: /-ɑnk, -unk/. These suffixes show place assimilation: [-ɑŋk, -uŋk]. Note that many of the suffixed forms in Table 4.89 are derived from bound roots, that are often used in verbs.

Table 4.89: Nasal place assimilation before velar stops in derivational suffixes /-ɑnk, -unk/

ɑbˈɾ-i-l	‘to live’	√-TH-INF	ապրիլ
ɑbˈɾ-ɑŋ**k**	‘goods’	√-NMLZ	ապրանք
d͡zɑʁˈɾ-e-l	‘to mock’	√-TH-INF	ծաղրել
d͡zɑʁˈɾ-ɑŋ**k**	‘mockery’	√-NMLZ	ծաղրանք
mɑɾˈz-e-l	‘to exercise’	√-TH-INF	մարզել
mɑɾˈz-ɑŋ**k**	‘exercise’	√-NMLZ	մարզանք
hɑɾˈk-e-l	‘to respect’	√-TH-INF	յարգել
hɑɾˈk-ɑŋ**k**	‘respect’	√-NMLZ	յարգանք
pəsˈχ-e-l	‘to vomit’	√-TH-INF	փսխել
pəsˈχ-uŋ**k**	‘vomit’	√-NMLZ	փսխունք
iˈɾɑv	‘truly’	√	իրաւ
iɾɑˈv-uŋ**k**	‘right (n)’	√-NMLZ	իրաւունք

Thus nasal place assimilation is productive before velar stops in underived contexts.

4.4.2 Velar assimilation in derived contexts

In derived contexts, we can see roots which on their own surface with a coronal nasal [n]. When new words are derived from these roots, the nasal /n/ becomes adjacent to a velar stop /k, g/ and it becomes [ŋ].

In terms of suffixation, consider the derivational suffix /-ɡod/ (Table 4.90). This suffix is relatively rare. We’ve found roots that end in /n/ and that take this suffix. We see nasal place assimilation.

Table 4.90: Nasal place assimilation before velar stops in suffixation

	bɑɾd͡zeˈ**n**-ɑ-l	‘to boast’	√-TH-INF	պարծենալ
→	bɑɾd͡zeŋ-ˈ**g**od	‘boastful’	√-NMLZ	պարծենկոտ
	ˈku**n**	‘sleep’	√	քուն
→	kəŋ-**g**od	‘sleepy’	√-NMLZ	քնկոտ

Vowel reduction creates more contexts for nasal place assimilation. For the words in Table 4.91, the base ends in a NVC sequence. Some of these bases are suffixed forms themselves. In the derived forms, the vowel is deleted, the nasal and velar stop become adjacent, and the nasal assimilates.

Table 4.91: Nasal place assimilation before velar stops from vowel reduction

	ˈpu**n**	‘original’	√	բուն
	pəˈ**n-ig**	‘native’	√-ADJZ	բնիկ
→	pə**ŋ-g**-ɑˈt͡sum	‘nativization’	√-ADJZ-NMLZ	բնկացում
	ˈʃu**n**	‘dog’	√	շուն
	ʃəˈ**n-ig** + ˈt͡sug	‘puppy’ + ‘fish’	√-DIM	շնիկ, ձուկ
→	ʃə**ŋ-g**-ɑˈt͡sug	‘dog-fish’	√-DIM-LV-√	շնկաձուկ
	mɑˈ**nug**	‘child’	√	մանուկ
→	mɑ**ŋg**-ɑˈgɑn	‘childish’	√-ADJZ	մանկական
	jeɾt͡ʃɑˈ**nig**	‘happy’	√	երջանիկ
→	jeɾt͡ʃɑ**ŋg**-ɑˈved	‘happy’	√-ADJZ	երջանկաւէտ

Place assimilation likewise applies word-finally in derived contexts. For the words in Table 4.92, the suffix /-k/ is added after the nasal, causing the nasal to assimilate.

Table 4.92: Nasal place assimilation before velar stops from the suffix /-k/ in VC-C contexts

	t͡ʃɑˈ**n**-ɑ-l	‘to try’	√-TH-INF	ջանալ
→	ˈt͡ʃɑ**ŋ-k**	‘effort’	√-NMLZ	ջանք
	gəɾo**n**-ɑgɑn	‘religious’	√-ADJZ	կրօնական
→	gəˈɾo**ŋ-k**	‘religion’	√-NMLZ	կրօնք
	məˈgɑ**n**	‘muscle’	√	մկան
→	məˈgɑ**ŋ-k**	‘muscles’	√-NMLZ	մկանք
	ɑɾˈʒɑ**n**	‘cheap’	√	արժան
→	ɑɾˈʒɑ**ŋ-k**	‘worthiness’	√-NMLZ	արժանք
	oˈɾe**n**	‘law (archaic)’	√	օրէնք
→	oˈɾe**ŋ-k**	‘law’	√-NMLZ	օրէնք

Assimilation likewise occurs when /-k/ is added after a VCN sequence (Table 4.93).

Table 4.93: Nasal place assimilation before velar stops from the suffix /-k/ in VCC-C contexts

	ˈlɑ**jn**	'wide'	√	լայն
→	ˈlɑ**jŋ-k**	'width'	√-NMLZ	լայնք
	jeɾˈgɑ**jn**	'long'	√	երկայն
→	jeɾˈgɑ**jŋ-k**	'length'	√-NMLZ	երկայնք
	hɑˈmɑ**jn**	'whole'	√	համայն
→	hɑˈmɑ**jŋ-k**	'community'	√-NMLZ	համայնք

Complications arise when place assimilation interacts with stress (Table 4.94). Armenian has primary stress on the final non-schwa vowel. There is reports of initial secondary stress but such a stress is very weak and not really perceptible. The exception is the negative prefix /ɑn-/. This prefix takes very perceptible secondary stress. In casual speech, the nasal becomes [ŋ] before velar stops. But in careful speech, the secondary stress on /ɑn-/ can cause the nasal to optionally surface as [ɑn-].

Table 4.94: Variable nasal place assimilation for the prefix /ɑn-/

	ˈhɑm	‘taste’	համ
→	ˈɑn-ˈhɑm	‘tasteless’	անհամ
	ɑɾˈtɑɾ	‘just’	արդար
→	ˈɑn-ɑɑɾˈtɑɾ	‘unjust’	անարդար
	ˈ**k**oɾd͡z	‘work’	գործ
→	ˌɑŋ-ˈ**k**oɾd͡z	‘unemployed’	անգործ
→	ˌɑ**n**-ˈ**k**oɾd͡z		(careful)
	ˈ**g**eʁd͡z	‘false’	կեղծ
→	ˌɑŋ-ˈ**g**eʁd͡z	‘sincere’	անկեղծ
→	ˌɑ**n**-ˈ**g**eʁd͡z		(careful)
	kiˈdɑg	‘wise’	գիտակ
→	ˌɑŋ-**k**iˈdɑg	‘unknowing’	անգիտակ
→	ˌɑ**n**-**k**iˈdɑg		(careful)
	gɑsˈkɑd͡z	‘doubt’	կասկած
→	ˌɑŋ-**g**ɑsˈkɑd͡z	‘doubtless’	անկասկած
→	ˌɑ**n**-**g**ɑsˈkɑd͡z		(careful)
	keʁeˈt͡sig	‘beautiful’	գեղեցիկ
→	ˌɑŋ-**k**eʁeˈt͡sig	‘ugly’	անգեղեցիկ
→	ˌɑ**n**-**k**eʁeˈt͡sig		(careful)
	gɑɾeˈli	‘possible’	կարելի
→	ˌɑŋ-**g**ɑɾeˈli	‘impossible’	անկարելի
→	ˌɑ**n**-**g**ɑɾeˈli	(careful)	

The above is based on HD’s perception though. Experimental data is needed to accurately know the degree of velarization (or lack of velarization) of the prefix /ɑn-/ in natural and controlled speech.

The effect of stress is stronger the words in Table 4.95. These words are compounds where the first stem ends in a nasal, and the second stem starts with a velar stop. These words are quantifiers and they have irregular primary stress on the first stem. In HD’s perception, the nasal is preferably [n] instead of [ŋ].

Table 4.95: Variable nasal place assimilation words with irregular stress

	ˈɑ**jn** + ˈ**k**ɑn		‘that’ + ‘than’	այն, քան
→	ˈɑ**jn**-ˈ**k**ɑn	ˈɑ**jŋ**-ˈ**k**ɑn	‘that much’	այնքան
	ˈnu**jn** + ˈ**k**ɑn		‘same’ + ‘than’	նոյն, քան
→	ˈnu**jn**-ˈ**k**ɑn	ˈnu**jŋ**-ˈ**k**ɑn	‘as much’	նոյնքան

In sum, nasal place assimilation is productive before velar stops in derived contexts. There is some complications from stress. Stressed nasals seem to resist velarization, but more acoustic data is needed.

4.4.3 Other forms of nasal place assimilation

Nasal place assimilation is productive before the velar stops /k, g/. But there is little to no evidence of productive place assimilation before other types of places, i.e., there is no assimilation before uvulars or labials.[22] . The dorsal fricatives /χ, ʁ/ do not trigger any place assimilation (Table 4.96). The nasal /n/ surfaces as [n] before them.[23] We haven't found relevant examples from underived contexts, but there are some cases from derived contexts.

Table 4.96: No nasal place assimilation before dorsal uvular fricatives

	gɑˈ**nuχ**	'early'	√	կանուխ
→	gɑ**n**ˈ**χ**-e-l	'to anticipate'	√-TH-INF	կանխել
	ˈ**χ**iʁd͡ʒ	'conscience'	√	խիղճ
→	ˌɑ**n**-ˈ**χ**iʁd͡ʒ	'unscrupulous'	NG-√	անխիղճ
	ʁeg-ɑ-ˈvɑɾ	'leader'	√-LV-√	ղեկավար
→	ˌɑ**n**-**ʁ** eg-ɑ-ˈvɑɾ	'undirected'	NG-√-LV-√	անղեկավար

Before labial stops /p, b/, the nasal /n/ surfaces as [n] (Table 4.97). However, it is rare to find roots or underived contexts which have an underlying /np/ or /nb/ sequence. In derived contexts, compounding and prefixation can create [np] and [nb] sequences, without any assimilation.

[22] Vaux (1998: 19) reports a palatal nasal [ɲ] before coronal affricates, like [ɲd͡ʒ]. We think such an assimilation is likely a low-level phonetic process that's not easily perceptible.

[23] Fairbanks (1948: 8) reports a velar [ŋ] before a dorsal fricative [x] but we couldn't replicate this finding.

Table 4.97: No synchronic nasal place assimilation before bilabial stops

	ˈɑj**n** + ˈ**b**es	‘that’ + ‘way’	այն, պէս
→	ɑj**n**-ˈ**b**es	‘like that’	այնպէս
	ˈpɑ**n** + ˈ**p**eɾ	‘word (archaic)’ + ’bring!’	բան, բեր
→	pɑ**n**-ˈ**p**eɾ	‘messenger’	բանբեր
	bəˈduχ +	‘fruit’	պտուղ
→	ɑ**n**-**b**əˈduχ	‘unfruitful’	անպտուղ
	pənɑˈgɑn +	‘natural’	բնական
→	ɑ**n**-**p**ənɑˈgɑn	‘unnatural’	անբնական

Diachronically however, there is a process of nasal place assimilation before labials (Table 4.98). For example, there are modern words which have a [mp] or [mb] cluster, but these cluster diachronically derived from an /n(V)p/ or /n(V)b/ sequence.

Table 4.98: Diachronic nasal place assimilation before bilabial stops

Modern form:	d͡ʒɑ**m**ˈ**p**ɑ	‘road’	ճամբայ	
Historical source:	d͡ʒɑ**nɑ**ˈ**p**ɑɾ	‘road’	ճանապարհ	(Աճառեան 1926: 182-3)
Modern form:	ɑ**mb**ɑˈɾiʃt	‘wicked’	ամբարիշտ	
Historical source:	<a**nb**ariʃd>	reconstructed	անպարիշտ	(Աճառեան 1926: 149)

Before labial nasal /m/, the nasal /n/ does not assimilate. We have not found this sequence in any underived contexts, but it is abundant in prefixation.

Table 4.99: Lack of synchronic nasal place assimilation before nasal /m/

	ˈ**m**ɑh	‘death’	մահ
→	ˌɑ**n**-ˈ**m**ɑh	‘deathless’	անմահ
	ˈ**m**ɑjɾ	‘mother’	մայր
→	ˌɑ**n**-ˈ**m**ɑjɾ	‘motherless’	անմայր
	mɑˈkuɾ	‘clean’	մաքուր
→	ˌɑ**n**-**m**ɑˈkuɾ	‘unclean’	անմաքուր
	meɾʒeˈli	‘rejectable’	մերժելի
→	ˌɑ**n**-**m**eɾʒeˈli	‘irrecusable’	անմերժելի

However, there are some high-frequency words where we do find optional assimilation of /n/ before labials (Table 4.100).

Table 4.100: Exceptional assimilation of /n/ before labials

	pɑ**n-m**ə	'a thing'	բան մը
~	pɑ**m-m**ə	(thing-INDF)	(casual speech)
	ˌɑ**n-b**ɑjˈmɑn	'necessarily'	անպայման
~	ˌɑ**m-b**ɑjˈmɑn	(NG-√)	(casual speech)

Within the Armenian lexicon, it is rare to find /n/+labial sequences in roots, but it is quite common to find /m/+labial sequences (Table 4.101).

Table 4.101: Roots with /mp/ or /mb/

ɑ**m**ˈpoχt͡ʃ	'entire'	ամբողջ	ɑ**m**ˈ**p**op	'tight'	ամփոփ
ˈɑ**mb**	'cloud'	ամպ	ˈχu**mp**	'group'	խումբ

It is unclear if there is a synchronically active constraint against having /n/+labial sequences in roots in modern Armenian. Such a constraint is likely just diachronic, not synchronic. The only case where do see a synchronic alternation is in high-frequency collocation.

4.4.4 Lexical frequency of nasal clusters

To give a fuller picture of nasal place assimilation, Table 4.102 provides the lexical frequency of different nasal + sound clusters. We focus on clusters where the second sound is a stop, affricate, or nasal. Data is from the associated dictionary.

Table 4.102: Lexical frequency of nasal + sound clusters where the second sound is a stop, affricate, or nasal

	m	n	ŋ
p	mp = 742	np = 151	ŋp = 0
b	mb = 101	nb = 127	ŋb = 0
t	mt = 12	nt = 4448	ŋt = 0
d	md = 30	nd = 811	ŋd = 0
t͡s	mt͡s = 4	nt͡s = 1043	ŋt͡s = 0
d͡z	md͡z = 4	nd͡z = 171	ŋd͡z = 0
t͡ʃ	mt͡ʃ = 5	nt͡ʃ = 968	ŋt͡ʃ = 0
d͡ʒ	md͡ʒ = 4	nd͡ʒ = 171	ŋd͡ʒ = 0
mk	mk = 25	nk = 0	ŋk = 1893
mg	mg = 36	ng = 0	ŋg = 1157
mm	mm = 4	nm = 118	ŋm = 0
mn	mn = 383	nn = 205	ŋn = 0
mŋ	mŋ = 0	nŋ = 0	ŋŋ = 0

We derive various generalizations:

1. **Labial stops**: The labial nasal [m] is well attested before labial stops [p, b]. The coronal nasal [n] is also found before labial stops, but in smaller numbers. Many of these n+labial sequences are from the prefix /ɑn-/.

2. **Coronal stops and affricates**: The labial nasal [m] is attested before coronal stops and affricates [t, d, t͡s, d͡z, t͡ʃ, d͡ʒ], but quite rarely. The coronal nasal [n] is well-attested before coronal stops and affricates

3. **Velar stops**: The velar nasal [ŋ] is restricted to before velar stops [k, g]. These nasals are common. The coronal nasal [n] and labial nasal [m] are rarely before a velar stop.

4. **Nasal clusters**: The nasal clusters [mn, nn, nm] are well-attested, but [mm] is rare.

4.5 Allophonic differences from Eastern Armenian

Eastern and Western Armenian are different varieties of Armenian. The two have similar but non-identical phoneme inventories. The dialects however share a large proportion of their allophony in common. In this section, we overview

allophonic processes that have been reported in Eastern Armenian and which either exist or don't exist in Western Armenian.

We likewise overview some processes which seem to apply in some varieties of Western Armenian but not others. We also note phonological processes that are optional and restricted to connected speech.

Unless otherwise specified, the Eastern examples are from English Wiktionary. The Wiktionary examples are heavily moderated and are reliable for Eastern Armenian. For transliterations, we adopt ISO 9985 to transliterate the words for Eastern Armenian,[24] while we use our own transliteration for Western Armenian.

4.5.1 Palatalization

Because of contact with Russian, Eastern Armenian has been slowly developing a rule of palatalizing dental stops to affricates before /j/: /tj/ → [t͡sj]. This rule is particularly common in the nominalizing suffix /-utʰjun/ -ություն which is almost always pronounced as [-ut͡sʰjun] in modern Eastern Armenian.

cite vaux, and cite examples

For Western Armenian, there is no Russian contact so no such palatalization rule exists. The closest analog is affrication of the suffix /-utɤn/ -ութիւն. Whereas this suffix is often pronounced as [-ut͡sʰjun] in Eastern Armenian, this suffix is often pronounced as [-t͡ʃɤn] in Western Armenian.

Note that the suffix /-utɤn/ shows a lot of speaker and register variation (Table 4.103). The most formal pronunciation is [-utɤn]. But in casual speech, this suffix can variably be pronounced as [-utjun], [-utjɤn], [-ut͡ʃjun], [-ut͡ʃɤn], among other options. We do not know the probability or the frequency of the different pronunciations. We do not know what social factors correlate with any of these choices.

[24] https://www.translitteration.com/transliteration/en/armenian-eastern-classical/iso-9985/

Table 4.103: Variation in the pronunciation of the nominalizer /-utjun/ suffix

Adjective:	uˈɾɑχ	'happy'	ուրախ
Nominalized	uɾɑχ-uˈtjun	'happiness'	ուրախութիւն
	uɾɑχ-uˈtjɤn		
	uɾɑχ-uˈtɤn		
	uɾɑχ-uˈt͡ʃɤn		
	uɾɑχ-uˈt͡ʃjɤn		
	uɾɑχ-uˈt͡ʃjun		
	uɾɑχ-uˈt͡ʃun		

For the vowel, the vowel /ɤ/ is often broken up into a glide+vowel sequence /ju/ or some other type of round vowel.[25] This is a common process in Armenian (§4.2.3). The stop /t/ often becomes [t͡ʃ] in this context. The change from /t/ to [t͡ʃ] is unique to this morpheme and is not a language-general rule, i.e., it is a morpheme-specific rule.

4.5.2 Deaspiration and voicing assimilation

Western Armenian has deaspiration of stops when adjacent to a fricative, affricate, or another stop. This was surveyed in Section §4.3. But to our knowledge, such deaspiration does not exist in Eastern Armenian.

To illustrate, the Table 4.104 provides Western and Eastern forms. The Western forms show deaspiration of the stop, while the Eastern form does not. Note that the initial schwa in Eastern is optional. WA data is from HD who has a general rule of stop aspiration, and a rule of post-fricative deaspiration.

Table 4.104: Post-fricative deaspiration in Western but not Eastern Armenian

Spelling	Western	Eastern	Meaning
սփոփել	**ə**s**p**oˈp^{h}el	(ə)**sp**^{h}oˈ.pel	'to comfort'
սթափ	**ə**s**t**ˈɑp^h	(ə)**s**ˈ**t**^{h}ɑp^h	'sober'
սքանչելի	**ə**s**k**ɑnt͡ʃeˈli	(ə)**sk**^{h}ɑnt͡ʃ^{h}eˈli	'wonderful'

One possible reason as to why the dialects differ in this respect is phonemicity. Aspiration is phonemic in Eastern, but it is not in Western. Thus, there is no loss in phonemic contrasts when a Western stop is deaspirated after a fricative.[26]

[25] See Ավետյան (2015) for discussion on the diachronic changes in this suffix's pronunciation.
[26] We thank Scott Seyfarth for discussion.

Another apparent area of difference is voicing assimilation (Table 4.105). In obstruent clusters, both Western and Eastern Armenian are reported to have regressive assimilation in voiced+voiceless clusters (Խաչատրյան 1988: 35,100-107). But for voiceless+voiced clusters, Western Armenian has progressive assimilation (devoicing) while Eastern Armenian can keep the cluster unchanged.

Table 4.105: Dialectal differences in voicing assimilation

	Western		Eastern	
Regressive	ʁɑrɑˈpɑʁ	'Karabagh' Ղարաբաղ	ʁɑrɑˈbɑʁ	'Karabagh' Ղարաբաղ
→	ʁɑrɑpɑχ-ˈ**t͡s**i	'Karabaghian' ղարաբաղցի	ʁɑrɑbɑχ-ˈ**t͡s**ʰi	'Karabaghian' Ղարաբաղցի
Progressive	tɑntɑ**ʁ**-ˈ**g**od	'slowish' դանդաղկոտ	pəˈ**tuʁ**	'fruit' պտուղ
→	vɑχ-ˈ**k**od	'coward' վախկոտ	pə**tʁ**-ɑ-beɾ	'fruit-bearing' պտղաբեր

Unfortunately to our knowledge, there isn't a systematic study on productive voicing assimilation processes in Eastern Armenian. So we cannot say if Eastern Armenian truly lacks progressive assimilation.

4.5.3 Sonorant devoicing

For Eastern Armenian, it is reported that sonorants can devoice when word-final. We have not been able to verify whether this process applies in Western or not. Our impression is that this process is a rather low-level phonetic rule, and thus not perceptible to speakers.

cite

Although there is voicing assimilation of obstruents in an obstruent cluster (§4.3), we also don't know if sonorants get devoiced when adjacent to a voiceless obstruent.

4.6 Sandhi phenomena or connected speech processes

Sandhi or connected speech phonology is about how the way that are words are pronounced can change when used in isolation vs. inside a casual conversation. This section goes over segmental processes that occur in connected speech.

4.6.1 Word-final devoicing

In both Western and Eastern Armenian, voicing contrasts can be found word-finally for stops and affricates. We illustrate below with the labial series (Table 4.106).

Table 4.106: Phonemic voicing for final labials in Eastern and Western

	Eastern	Western	
թագ	ˈtɑ**g**	ˈtɑ**k**	‘crown’
թակ	ˈtɑ**k**	ˈtɑ**g**	‘mallet’
թաք	ˈtɑ**k**ʰ	ˈtɑ**k**	‘hiding’

However in both Eastern and Western Armenian, there is evidence that there is some sort of gradient devoicing process. For Eastern Armenian, there is likewise a diachronic devoicing process.

In Eastern Armenian, there are some words which are spelled with a final voiced stop/affricate, but this sound is pronounced as voiceless. We provide Western forms for completeness (Table 4.107).

Table 4.107: Words in Eastern Armenian that are spelled with a final voiced stop/affricate but are pronounced as voiceless

Letter	Word	Transliteration		Pronunciation		
		EA	WA	EA	WA	Meaning
բ	Հակոբ	<Hako**b**>	<Hagop>	hɑˈko**p**ʰ	hɑˈgop	masc. name
գ	ձագ	<ja**g**>	<t͡sak>	ˈd͡zɑ**k**ʰ	ˈt͡sɑk	‘cub’
դ	օդ	<ò**d**>	<ōt>	ˈo**t**ʰ	ˈot	‘air’
ձ	օձ	<ò**j**>	<ōt͡s>	ˈo**ts**ʰ	ˈot͡s	‘snake’
ջ	աջ	<a**ǰ**>	<at͡ʃ>	ˈɑ**tʃ**ʰ	ˈɑt͡ʃ	‘right’

However, it is unlikely that this devoicing is due to a synchronic phonological rule (Table 4.108). For example, voiced stops and affricates can surface word-finally in some words.

Table 4.108: Words in Eastern Armenian that are spelled with a final voiced stop/affricate and are voiced in pronunciation

Letter	Word	Transliteration		Pronunciation		
		EA	WA	EA	WA	
բ	արաբ	<ara**b**>	<arap>	ɑˈɾɑ**b**	ɑˈɾɑp	‘Arab’
գ	արագ	<ara**g**>	<arak>	ɑˈɾɑ**g**	ɑˈɾɑk	‘fast’
դ	բադ	<ba**d**>	<pat>	ˈbɑ**d**	ˈpɑt	‘duck’
ձ	նախանձ	<naxan**j**>	<naxan**ts**>	nɑˈχɑn**d͡z**	nɑˈχɑnt͡s	‘jealousy’
ջ	քաջ	<k’a**ǰ**>	<k’at͡ʃ>	ˈkɑ**d͡ʒ**	ˈkɑt͡ʃ	‘brave’

Furthermore, for those words which have this devoiced stop or affricate in their citation form, the sound is still pronounced as devoiced in other derived or inflected forms (Table 4.109).

Table 4.109: Inflected form of words in Eastern Armenian that are spelled with a final voiced stop/affricate but are pronounced as voiceless

Letter	Word	Transliteration		Pronunciation		Meaning
		EA	WA	EA	WA	
բ	Հակոբը	<Hako**b**ë>	<Hago**p**ə>	hɑˈko**p**ʰ-ə	hɑˈgo**p**-ə	masc. name (DEF)
գ	ձագեր	<ja**g**er>	<t͡sa**k**er>	d͡zɑˈ**k**ʰ-eɾ	t͡sɑˈ**k**-eɾ	‘cub-PL’
դ	օդի	<ò**d**i>	<ō**t**i>	oˈ**t**ʰ-i	oˈ**t**-i	‘air-GEN’
ձ	օձս	<ò**j**s>	<ō**ts**s>	ˈo**ts**ʰ-**əs**	ˈo**ts**-əs	‘snake-POSS.1SG’
ջ	աջով	<a**ǰ**ov>	<a**t͡ʃ**ov>	ɑˈ**t͡ʃ**ʰ-ov	ɑˈ**t͡ʃ**-ov	‘right-INS’

If Eastern Armenian had true final devoicing, we would expect to see morpheme alternations where some morpheme is pronounced with a voiceless stop when said in isolation [hɑkop], but then pronounced with a voiced stop when suffixes are added [*hɑkob-ə]. This does not happen.

The most likely scenario is that, again, this final devoicing rule is just an orthography-phonology mismatch which applied as a diachronic rule, not an active synchronic rule.

For Western Armenian, we don’t see such an orthography-phonology mismatch. Words that are spelled with a final voiced stop are pronounced as such. However, there seems to be a gradient rule of final devoicing that varies by word, speaker, region, and by register.

For example, in HD’s ideolect, certain words are prescriptively pronounced with a final voiced stop (Table 4.110). But in causal speech, the stop is optionally

devoiced word-finally. Such devoicing doesn't occur when suffixes are added, making the stop intervocalic. HD self-reports that the "devoicing" can also manifest as just un-releasing the final voiced stop. We transcribe this "devoiced" or unreleased form as just a voiceless unaspirate.

Table 4.110: Words with variable final devoicing in HD's Western Armenian pronunciation

		Final voicing	Final devoicing	
Root	կապ	ˈɡɑ**b**	ˈɡɑ**p**	'connection'
→	կապեր	ɡɑˈ**b**-er		'connection-PL'
Root	կապիկ	ɡɑˈbi**g**	ɡɑˈbi**k**	'monkey'
→	կապիկս	ɡɑˈbi**g**-əs		'monkey-POSS.1SG'
Root	ազատ	ɑˈzɑ**d**	ɑˈzɑ**t**	'free'
→	ազատը	ɑˈzɑ**d**-ə		'free-DEF'
Root	տաբատ	dɑˈpɑ**d**	dɑˈpɑ**t**	'pants'
→	տաբատով	dɑpɑˈ**d**-ov		'pants-INS'
Root	շատ	ˈʃɑ**d**	ˈʃɑ**t**	'many'
→	շատեր	ʃɑˈ**d**-eɾ		'many-PL'
Root	ծակ	ˈd͡zɑ**g**	ˈd͡zɑ**k**	'hole'
→	ծակի	d͡zɑˈ**g**-i		'hole-GEN'
Root	մէկ	ˈme**g**	ˈme**k**	'one'
→	մէկը	ˈme**gg**əi		'one-DEF'

For the Lebanese community, this devoicing process is optional and limited to a handful of high-frequency words in connected speech. For Turkish-speaking communities such as in Istanbul, word-final devoicing is much more common because Turkish has final devoicing. For the Istanbul community, final devoicing is so common that it's likely an allophonic rule bolsahayen Anaid Donabedian self-reports devoicing in her French community as well.

We cannot study in depth the rate of final devoicing. It seems that such a process is highly variable by speaker, geographic region, and by register. An ideal future research question is to examine the rate of devoicing in an oral corpus of natural speech. We speculate that devoice will vary not only by speaker, but may also show signs of incomplete devoicing or incomplete neutralization.

4.6.2 /h/ deletion

In modern Armenian, the orthography has a letter հ for the sound /h/. There are many words which are spelled with an <h> either word-initially or word-finally.

The /h/ is pronounced in careful speech. But in casual connected speech, this /h/ is optionally deleted in some words (Ղարագյուլյան 1974: 162; Մարգարյան 1997: 64).

We illustrate below with some common words which start with /h/ (2). This /h/ is pronounced in careful speech, but can be optionally dropped in casual speech after a consonant or vowel. A frequent target of deletion is the classifier [hɑd].

(2) Words which show optional /h/ deletion in connected speech

a. jeɾgu **h**ɑzɑɾ (Careful)
jeɾgu ɑˈzɑɾ (Casual)
two thousand
'two thousand.'
երկու հազար

b. t͡ʃoɾoɾt ˈ**h**ɑɾg-ə (Careful)
t͡ʃoɾoɾt ˈɑɾg-ə (Casual)
forth floor-DEF
'the fourth floor'
չորրորդ յարկը

c. meg hɑd ˈkiɾk (Careful)
meg ɑd kiɾk (Casual)
one CLF book
'one book.'
մէկ հատ գիրք

A frequent target of /h/ deletion is the classifier [hɑd] (3). It follows numerals and precedes nouns. The /h/ deletes in casual speech after either a vowel or consonant.

(3) /h/ deletion for the classifier /hɑd/

a. jeɾgu **h**ɑd ˈkirk (Careful)
jeɾgu ɑd ˈkiɾk (Casual)
two CLF book
'two books.'
երկու հատ գիրք

b. jeɾek **h**ɑd ˈkirk (Careful)
jeɾek ɑd ˈkirk (Casual)
three CLF book
'three books.'
երեք հատ գիրք

For words with an initial <h>, the deletion seems especially common after a /ɾ/-final word, such as after a possessive pronoun (4). It is likewise frequent after the word [ʃɑd] ‘very’ or ‘much’.

(4) Words which show optional /h/ deletion in connected speech, especially after /ɾ/ or dentals

a. meɾ **h**ɑkust-ˈneɾ-ə (Careful)
meɾ ɑkust-ˈneɾ-ə (Casual)
our clothing-PL-DEF
‘our clothes’
մեր հագուստներ

b. ɑsoɾ **h**ɑˈmɑɾ (Careful)
ɑsoɾ ɑˈmɑɾ (Casual)
this.GEN reason
‘for this reason’
ասոր համար:

c. ʃɑd **h**ɑˈɾust e (Careful)
ʃɑd ɑˈrust e (Casual)
very rich is
‘He is very rich.’
Շատ հարուստ է:

HD feels though that there are some /h/-initial words which resist deletion (5). These words seem to all be monosyllabic so there might be some prosodic constraint involved.

(5) Words which resist /h/ deletion in connected speech

a. meɾ ˈ**h**ɑjɾ-ə (Careful, casual)
our father-DEF
‘our father’
մեր հայրը

b. im ˈ**h**ɑz-əs (Careful, casual)
my cough-POSS.1SG
‘my cough’
իմ հազս

For verbs with an initial /h/, the deletion seems common after verbal proclitics or prefixes, such as the future particle [bidi], indicative [gə-], and negative [t͡ʃə-] (6). .

(6) Inflected verbs which show /h/ deletion in casual speech

a. bidi **h**ɑχˈt-e-ŋk (Careful)
bid ɑχˈt-e-ŋk (Casual)
FUT win-TH-1PL
'We will win.'
Պիտի յաղթենք:

b. gə-**h**iˈʃ-e-m goɾ (Careful)
g-iˈʃ-e-m goɾ (Casual)
IND-remember-TH-1SG PROG
'I remember.'
Եթէ չհասկնամ:

c. jete t͡ʃə-**h**ɑskəˈ-n-ɑ-m (Careful)
jete t͡ʃ-ɑskəˈ-n-ɑ-m (Casual)
if NEG-understand-INCH-TH-1SG
'If I don't understand.'
Եթէ չհասկնամ:

For words with a final /h/, it is much rarer to find such words (7). In HD's judgments, most of these words don't show deletion in connected speech, whether intervocalically or sentence-finally.

(7) Words that don't delete final /h/ in connected speech or sentence-finally

a. ˈmɑh. mɑˈh-eɾ. (Careful, casual)
death death-PL
'Death. The deaths.'
Մահ: Մահեր:

b. vəsˈtɑh. vəsˈtɑh e-m. (Careful, casual)
sure Sure is-1SG
'For sure. I am sure.'
Վստահ: Վստահ եմ:

For the 'president', the final /h/ can delete and its deleted form can affect allomorphy (8). The definite suffix is /-n/ after vowels, and /-ə/ after consonants. The deletion of the /h/ affects the choice of allomorph. The deletion and subsequent allomorphy is represented in the orthography.

(8) Words where /h/ deletion affects allomorphy

a. nɑχɑˈkɑh. nɑχɑˈkɑh-ə. (Careful)
nɑχɑˈkɑ. nɑχɑˈkɑ-n. (Casual)
president president-DEF
‘President. The president’
Նախագահ: Նախագահը:
Նախագա: Նախագան:

Because of how /h/ deletion applies in only some words and because of how it can feed other morphophonological rules, it is possible that /h/ deletion is actually a type of allomorphy in connected speech find suitable kaisse citation, im thinking of an articel in the inkelas-zec book 1990.

4.6.3 Schwa vowel assimilation

The schwa /ə/ is present in many Armenian words. Many of its occurrences are epenthetic. As discussed in cite chapter schwa epenth, consonant clusters in the orthography are broken up by schwas in pronunciation.

In careful speech, a pronounced schwa is pronounced simply as [ə]. But in casual speech, there are some words where the schwa is assimilates to the vowel quality of the following vowel, to create an identical or near-identical vowel (Table 4.111).[27]

Table 4.111: Words where the schwa assimilates to the following vowel

		Careful speech	Casual speech	
գլուխ	/kluχ/	kəˈluχ	kuˈluχ	/ ‘head’
պտուղ	/bduʁ/	bəˈduʁ	buˈduʁ	‘fruit’
դգալ	/tkɑl/	təˈkɑl	tɑˈkɑl	‘spoon’
լռէ՛	/lɾe/	ləˈɾe	leˈɾə	‘be quiet!’
ձգէ՛	/t͡ske/	t͡səˈke	t͡siˈke	‘leave it!’

One common occurrence of schwa assimilation is from the indicative prefix (Table 4.112). This prefix is /g-/ before vowels, and /gə-/ before consonants. The schwa is epenthetic (cite chapter schwa epenthesis). Before some /hi/-initial

[27] The word ‘easy’ դիւրին <tiwrin> is prescriptively pronounced as [tʏrin] in careful speech, but it’s much more common to say [təɾʏn] or [tʏɾʏn] in careful speech. It’s possible that the connected form derives from a similar UR /trʏn/ with schwa epenthesis and assimilation. For [ləɾe] ‘be quiet’ լռէ and [t͡səke] ‘leave it’ ձգէ, the underlying forms are simplified. The schwa is caused by desressed high vowel reduction from the morphological roots [luɾ] ‘quiet’ լուռ and [t͡sik] ‘projectile’ ձիգ.

words, the prefix is optionally pronounced as [gi-]. The /h/ can optionally delete as well, causing the two vowels to then fuse into one vowel.

Table 4.112: Words where indicative prefix and /hi/-initial words fuse

Careful	gə-**hi**ˈʃ-e-n	gə-**hi**vɑnt-ɑ-ˈn-ɑ-m
Casual	gi-**hi**ˈʃ-e-n	gi-**hi**vɑnt-ɑ-ˈn-ɑ-m
Casual	g-**i**ˈʃ-e-n	g-**i**vɑnt-ɑ-ˈn-ɑ-m
	IND-remember-TH-3PL	IND-sick-LV-INCH-TH-1PL
	'They remember.'	'I become sick.'
	Կը յիշեն:	Կը հիւանդանամ:

This process of schwa vowel assimilation (vowel harmony) is limited to a handful of high-frequency words. We suspect that these alternations are just grammaticalized from some type of vowel-vowel coarticulation that occurs in casual speech. Data from oral corpora is needed in order to find out how much schwas can alternate in their vowel quality. For words other than the ones listed above, we suspect that the schwa can have its vowel quality be *gradiently* affected by neighboring vowels, not categorically.

4.6.4 Schwa elision

Armenian has a process of inserting schwas in consonant clusters to avoid creating complex onsets. This is discussed in cite schwa chapter. The inserted schwas are found in isolation and careful speech, and they're the typical way to list a word's pronunciation in a dictionary. However, in casual or connected speech, an inserted schwa can be optionally deleted. We call this process 'schwa elision'.

Consider the words in Table 4.113 that start with a stop. In isolation, the initial consonant cluster gets an epenthetic schwa. The schwa is used to prevent different complex onsets. However in casual speech, some of these schwas can be optionally deleted.

Table 4.113: Stop-initial clusters that undergo schwa epenthesis, with or without optional schwa elision

Cluster	UR	SR	Can elide?		
stop + stop	/**tk**ɑl/	**tək**ɑl	no	‘spoon’	դգալ
stop + sibilant	/**kʃ**el/	**kəʃ**el	yes: **kʃ**el	‘to drive’	քշել
stop + non-sibilant	/**tv**il/	**təv**il	no	‘to appear’	թուիլ
stop-nasal	/**tn**el/	**tən**el	no	‘to put’	դնել
stop + rhotic	/**kr**istos/	**kər**istos	yes: **kr**istos	‘Christ’	Քրիստոսս
stop + lateral	/**kl**uχ/	**kəl**uχ	no	‘head’	գլուխ

Deletion is possible in stop-sibilant and stop-rhotic clusters. But it is odd when the schwa is between a stop and another type of consonant.

For fricative-initial clusters, schwa elision has similar but not identical patterns (Table 4.114). For sibilant-stop clusters, such clusters tend to get prothesis (initial schwa epenthesis). Such clusters can undergo elision: [(ə)stoɾ]. But when the fricative-stop cluster has a schwa inserted inside the cluster, then elision is odd: [ʃəpotil]. For fricative-rhotic clusters, schwa elision is possible for a light fricative like [h], but not for other fricatives.

Table 4.114: Fricative-initial clusters that undergo schwa epenthesis, with or without optional schwa elision

	Careful	Casual	Can elide?		
sibilant + stop	/**st**oɾ/	**əst**oɾ	yes: **st**oɾ	‘curtain’	ստոր
	/**ʃp**otil/	**ʃəp**otil	no	‘to be confused’	շփոթիլ
non-sibilant + stop	/**vg**ɑ/	**vəg**ɑ	no	‘witness’	վկայ
fricative + fricative	/**ʃʃ**ug/	**ʃəʃ**ug	no	‘whisper’	շշուկ
	/**sχ**ɑl/	**səχ**ɑl	no	‘wrong’	սխալ
fricative + nasal	/**χm**el/	**χəm**el	no	‘to drink’	խմել
fricative + rhotic	/**sɾ**ɑh/	**sər**ɑh	no	‘hall’	սրահ
	/**hɾ**ɑʃk/	**hər**ɑʃk	yes: **hɾ**ɑʃk	‘miracle’	հրաշք
fricative + lateral	/**sl**ɑk/	**səl**ɑk	no	‘arrow’	սլաք

Rhotics seem to especially trigger elision after a stop or fricative, but not after sonorants. Other sonants like nasals, glides, and laterals do not typically allow elision before or after them.

The closest systematic study to schwa elision is Gharagycite ghara who listed different phonotactic contexts which can cause elision in Eastern Armenian. For Eastern Armenian, hovakimyan also reports that schwa elision is possible when

the relevant consonants have a high degree of voiceless frication, such as between an aspirated stop and a sibilant.

For a word that shows elision in the initial syllable, there is evidence that the schwas were inserted and then deleted, rather than never inserted in the first place. Consider plural allomorphy (Table 4.115). The plural suffix is /-eɾ/ after monosyllables but /-neɾ/ elsewhere: [kɑm-eɾ, ɑntɑm-neɾ]. Epenthetic schwas are counted by the plural, thus triggering the polysyllabic-selecting suffix /-neɾ/: [təkɑl-neɾ, kəʃoʁ-neɾ]. Schwa elision can later apply to delete epenthetic schwas: [kʃoʁ-neɾ]. For the elided cases, the schwa must have been present in the first place; otherwise we would get the incorrect plural suffix /-eɾ/: **kʃoʁ-eɾ*. cite schwa paper

Table 4.115: Plural allomorphy and schwa elision

	Singular	Plural	Elision		
/kɑm/	kɑm	kɑm-eɾ		‘nail(s)’	գամ(եր)
/ɑntɑm/	ɑntɑm	ɑntɑm-neɾ		‘member(s)’	անդամ(ներ)
/**tk**ɑl/	**tək**ɑl	**tək**ɑl-neɾ		‘spoon(s)’	դգալ(ներ)
/**kʃ**oʁ/	**kəʃ**oʁ	**kəʃ**oʁ-neɾ	**kʃ**oʁ-neɾ	‘driver(s)’	քշող(ներ)

The above concerns word-initial clusters. For word-medial clusters, it is difficult to get clear data because most cases of schwa epenthesis occur word-initially (or root-initially in compounds). There are some compounding structures though where the second root shows possible schwa elision (Table 4.116). The root [kiɾ] is used to mean ‘writing’, and its derived forms replace the high vowel with a schwa: [kəɾ-el] ‘to write’. This root is used to form compounds analogously to how English has compounds with the second stem *-graph*. Various derivational suffixes can be then productively added.

Table 4.116: Schwa elision in compounds where the second stem is the root /kiɾ/

Less frequent root+root combination		More frequent root+root combination	
ɑsk ‘nation’ ազգ	ɑsk-ɑ-kəɾ-ɑgɑn ‘ethnographical’ ազգագրական	kid͡z ‘line’ գիծ	kəd͡z-ɑ-kɾ-ɑgɑn ‘descriptive’ գծագրական
luɾ ‘news’ լուր	ləɾ-ɑ-kəɾ-oʁ ‘journalist’ լրագրող	əstoɾ ‘below’ ստոր	əstoɾ-ɑ-kɾ-oʁ ‘undersigner’ ստորագրող
nɑχ ‘first’ նախ	nɑχ-ɑ-kəɾ-e-l ‘to prescribe’ նախագրել	nəgɑɾ ‘picture’ նկար	nəgɑɾ-ɑ-kɾ-e-l ‘to depict’ նկարագրել
lusin ‘moon’ լուսին	lusn-ɑ-kəɾ-utɤn ‘selenography’ լուսնագրութիւն	ɑɾt͡sɑn ‘statue’ արձան	ɑɾt͡sɑn-ɑ-kɾ-utɤn ‘registration’ արձանագրութիւն
gujɾ ‘blind’ կոյր	guɾ-ɑ-kəɾ-it͡ʃ ‘typhlograph’ խումբ	χump ‘group’ կուրագրիչ	χəmp-ɑ-kɾ-it͡ʃ ‘editor’ խմբագրիչ
oɾen ‘law’ օրէն	oɾin-ɑ-kəɾ-ɑjin ‘statutory’ օրինագրային	d͡ziɾ ‘circle’ ծիր	d͡zəɾ-ɑ-kɾ-ɑjin ‘programmatic’ ծրագրային

For some of these compounds, the second root’s schwa is typically pronounced. This is the case for compounds where the root+root combination is rather uncommon or a highly technical word: [ɑsk-ɑ-kəɾ-ɑgɑn] ‘ethnographical’ with optional elision in casua speech [ɑsk-ɑ-kɾ-ɑgɑn]. But in more frequent compounds like ‘descriptive’, the default pronunciation doesn’t use the schwa: [kəd͡z-ɑ-kɾ-ɑgɑn]. Using a schwa here sounds odd: ?*kəd͡z-ɑ-kəɾ-ɑgɑn*. For these frequent compounds, schwa elision is morphologized for these words as the typical pronunciation rule. That means, these words diachronically were pronounced by default with a schwa in an earlier stage of the language, with optional elision or deletion. But because these compounds were high-frequency, the schwa was pronounced less and less often, to the point that these compounds likely don’t insert insert the schwa in the first place.[28]

[28]We say ‘compound’ here, but the generalization is for the family of compounds that uses this root+root combination. For example, all words that contain the root+root pair of [kid͡z] ‘line’ and [kiɾ] ‘writing’ use a schwaless form: [kəd͡z-ɑ-kɾ-ɑgɑn] ‘descriptive’, [kəd͡z-ɑ-kɾ-it͡ʃ] ‘designer’ գծագրիչ, and so on.

A similar example is the root [tiɾ] ‘position’ դիր that has a schwa in its derived forms like [təɾ-utɤn] ‘situation’ դրութիւն. When a compound has this root as its second stem (Table 4.117), it is very common to not use a schwa. Using a schwa generally suggests that the word is a highly technical word, or that the first root is a learned prefix like /hɑm-/ ‘homo-’. It’s also arguable that the surface sequence [-ɑ+tɾ] (with two morphemes: linking vowel plus root) has grammaticalized into a single suffix [-ɑtɾ]. Whereas the schwa form [-ɑ+təɾ] is treated as containing a root.

Table 4.117: Schwa elision in compounds where the second stem is the root /tiɾ/

Less frequent root+root combination			More frequent root+root combination		
sɑhmɑn	sɑhmɑn-ɑ-tiɾ	sɑhmɑn-ɑ-təɾ-ɑgɑn	kor͡dz	kor͡dz-ɑ-tiɾ	kor͡dz-ɑ-tɾ-e-l
‘border’	‘constituent’	‘constitutionalist’	‘work’	‘executive’	‘to execute’
սահման	սահմանադիր	սահմանադրական	գործ	գործադիր	գործադրել
hɑm-	hɑm-ɑ-tiɾ	hɑm-ɑ-təɾ-ujt	uʃ	uʃ-ɑ-tiɾ	uʃ-ɑ-tɾ-utɤn
‘homo-’	‘homologous’	‘synthesis’	‘late’	‘careful’	‘to carefulness’
համ	համադիր	համադրոյթ	գործ	ուշադիր	ուշադրութիւն

Thus for the [ɑtɾ] sequence in these words, the diachronic origin is schwa elision but it likely grammaticalized in the modern language.

For word-final clusters, there are many words that end in an obstruent+rhotic cluster (§5.2.3.4, §5.2.3.12). The prescriptive rule is that these words are pronounced with an epenthetic schwa (Table 4.118). But in casual speech, the schwa can be absent. Though it’s unclear if the schwa was inserted and then deleted, vs. not inserted in the first place. For example, some speakers in our experience almost always skip the schwa in these clusters. vaux

Table 4.118: Optional schwas in final clusters

/vɑkɾ/	vɑkəɾ, vɑkɾ	‘tiger’	վագր
/pokɾ/	pokəɾ, pokɾ	‘small’	փոքր
/meʁɾ/	meʁəɾ, meʁf	‘honey’	մեղր

find other exampels from publised sources

To summarize, the surface appearance and disappearance of schwas can vary by type of cluster, location, and morpheme.

4.6.5 Gemination

Armenian does not have phonemic gemination or phonemic length for consonants. However, there are contexts where geminates are created. We mark gemination here with doubled letters like [mm] and not with a gemination symbol [mː].

There are some high frequency words which do not have a geminate in careful speech. But in casual connected speech, there is a geminated consonant in these words.

Table 4.119: Words that have gemination in casual speech

Careful	Casual		
d͡zɑmel	d͡zɑmmel	'to chew'	ծամել
ɑmen	ɑmmen	'all'	ամեն
ɑseʁ	ɑsseʁ	'needle'	ասեղ
hisun	hissun	'fifty'	յիսուն
kəsɑn	kəssɑn	'twenty'	քսան
təzuɡ	təzzuɡ	'dwarf'	թզուկ
kənel	kənnel	'to buy'	գնել

There is no obvious phonological reason why these words geminate their middle consonant while most other words don't.

5 Syllable structure

This chapter discusses the syllable in Armenian. The first section give a basic overview of possible syllable types (§5.1). The remaining sections go in depth on the range of possible complex codas (§5.2.1-5.4), complex onsets (§5.5), and vowel hiatus repair or vowel-vowel sequences (§5.6).

5.1 Overview of syllable structure

We first discuss the possible types of syllables in the language (§5.1.1), with statistical data on their frequency. Armenian has a maximal syllable template of CjVCCC. But in general, complex onsets are banned and complex codas are at most 2 consonants. So although the maximal syllable is CjVCCC, the typical syllable is CVC or CVCC.

We then go over how complex onsets are generally banned (§5.1.2), and possible complex codas (§5.1.3). For consonant clusters like an underlying /VCC/, the final CC can be pronounced as a complex coda within the same syllable[VCC] or pronounced with schwa epenthesis [V.CəC]. In some cases like [betk] 'need' պէտք, the cluster doesn't have a schwa, but it is as analyzed as consisting of a coda plus an extrasyllabic appendix (§5.1.4). Complex codas are generally at most 2 consonants, but 3 consonants can occur, mostly because of appendixes (§5.1.5).

We also discuss onsetless syllables (§5.1.6) and special syllable structures found in loanwords (§5.1.7).

5.1.1 Possible syllables and their lexical frequency

In terms of syllable structure, Armenian utilizes various templates. The most common template is CVC or CVCC, while the most maximal template is CjVCCC.

For most words, the syllable is at most CVCC. This means that a syllable can consist of a simple onset (CV) or no onset (VC). The syllable can have a coda (CVC), a complex coda (CVCC), or no coda (CV). A complex coda typically has two consonants, but three is possible (CVCCC). The onset is typically just a single consonant (CVC) but it is possible to have a complex onset made up of a consonant plus glide (CGVC). Table 5.1 illustrates the basic types of syllables.

Table 5.1: Basic syllable types

Onset	CV	pu	‘owl’	բու
	CVC	pɑn	‘thing’	բան
	CVCC	pɑɾt	‘complex’	բարդ
	CVCCC	bɑɾtk	‘debt’	պարտք
Complex onset	CGV	sjɑ.vɑ.ˈgɑn	‘black’	սեաւական
	CGVC	njɑɾ.tig	‘fibril’	նեարդիկ
	CGVCC	ljɑɾt	‘liver’	լեարդ
No Onset	V	u	‘and’	ու
	VC	ɑʁ	‘salt’	աղ
	VCC	ɑʁd	‘stain’	աղտ
	VCCC	ɑn͡tsk	‘passage’	անցք

In terms of word size, there is no minimal syllable size for a word. That is, a word can be just a single syllable vowel *V* or a single open syllable *CV* (Table 5.4). There is no limit on how a big a word can be. See Section §**??** for examples of stress shift applying in very large words. Words can get larger whether by compounding, adding derivational suffixes, or by adding inflectional suffixes.

Virtually any consonant can act as an onset or coda. To illustrate, the tables in Section §4.1 showed how each consonant can be found word-initially or word-finally. Virtually any vowel can be found in any type of syllable. For the core vowels /ɑ, e, i, o, u/, see Table 4.25; for the schwa see Table 4.27. For /ɤ/, it can be found with or without a coda, but it is very rare to find a /ɤ/ without an onset (Section §4.2.3).

Complex onsets are generally banned (§5.1.2), while complex codas are generally at most two consonants with falling sonority (§5.1.3). Flat sonority clusters can be created via extrasyllabic appendixes (§5.1.4). Final clusters of 3 consonants are exceedingly rare (§5.1.5). There are some restrictions on complex codas based on location or vowel (§5.4). Onsetless syllables are quite rare word-medially, but they exist (§5.1.6).

In terms of lexical frequency, the different types of syllables are possible across the word, including in monosyllabic words, word-initial polysyllabic words, and so on (Table 5.2).

Table 5.2: Lexical frequency of syllable shapes

	Monosyllabic #σ#	Polysyllabic Initial #σσ…	Polysyllabic Medial …σσσ…	Polysyllabic Final …σσ#	Total
CGV	0	4	104	402	510
CGVC	5	34	7	1055	1101
CGVCC	6	0	0	180	186
CV	34	30139	83011	3971	117155
CVC	1083	18331	11878	48034	79326
CVCC	766	1998	540	7245	10549
CVCCC	30	2	0	119	151
V	6	4156	16	1	4179
VC	51	5412	3	5	5471
VCC	68	937	1	1	1007
VCCC	8	0	0	0	8

When examining the lexical frequency of possible syllables shapes, there are some noticeable asymmetries. Monosyllabic words can virtually have any possible syllable shape. But, the most common is to have an onset and coda (CVC: 1083, CVCC: 766). It is rare for a monosyllabic word to have a /Cj/ complex onset (CGV: 0, CGVC: 5, CGVCC: 6).

Within polysyllabic words, virtually any type of syllable can be found in any position. We list the most common types in Table 5.3. That is, an open CV, closed CVC, or closed CVCC syllable can be found word-initially, word-medially, or word-finally.

Table 5.3: Different syllable shapes in different word positions

	Word-initial		Word-medial		Word-final	
CV	ˈ**pɑ**.ˈɾi	‘good’ բարի	hɑχ.ˈ**tɑ**.ˈɡɑn	‘triumphal’ յաղթական	t͡sə.ˈ**ri**	‘free’ ձրի
CVC	ˈ**kul**.ˈbɑ	‘sock’ գուլպայ	d͡ʒəʃ.**mɑɾ**.ˈdel	‘to verify’ ճշմարտել	hɑjd.ˈ**nel**	‘to reveal’ յայտնել
CVCC	ˈ**keχt͡ʃ**.kɑ.ˈjin	‘boorish’ գեղջկային	əs.**kəsp**.nɑ.ˈɡɑn	‘original’ սկզբնական	mɑχ.ˈ**tɑŋk**	‘wish’ մաղթանք
CVCCC	**kuɾt͡sk**.ˈkɑl	‘apron’ կուրծքկալ			ɑ.ve.ˈ**lɔɾtk**	‘apocenosis’ աւելորդք

5.1.2 Complex onsets are generally banned

Within the onset, the norm is that the syllable consists of at most a single consonant. However, complex onsets are possible in special cases. The most common example is when the complex onset consists of a consonant plus a glide /j/ (Table 5.4). Word-initially, the /Cj/ sequence is pronounced as a complex onset within the same syllable. Word-medially, the /Cj/ can be pronounced into separate syllables. These and other possible complex onsets are discussed in §5.5.

Table 5.4: Basic syllable types with a /Cj/ complex onset

	Monosyllabic	Word-initial	Word-medial	Word-final
CGV		**mjo**.kɑ.ˈɡeɾb ‘similar to us’ մեօքակերպ	mi.ˈ**ljo**.nɑ.ˈdeɾ ~ mil.ˈ**jo**.nɑ.ˈdeɾ ‘millionaire’ միլիոնատէր	ɑ.tɑ.mɑn.ˈ**tjɑ** ~ ɑ.tɑ.mɑnt.ˈ**jɑ** ‘insured’ ադամանդեայ
CGVC	ˈ**kjɑɾ** ‘necklace’ քեառ	**ljɑɾ**.tɑ.ˈjin ‘hepatic’ լեարդային	ɑŋ.**gjɑn**.tɑ.ˈɾɑn ~ ɑŋg.**jɑn**.tɑ.ˈɾɑn ‘corner’ անկեանդարան	ɑ.bɑ.ho.ˈ**vjɑl** ~ ɑ.bɑ.hov.ˈ**jɑl** ‘diamond-encrusted’ ապահովեալ pɑ.ɾi.ˈ**zjɑn** ‘Parisian’ փարիզեան
CGVCC	ˈ**gjɑŋk** ‘life’ կեանք			o.ˈ**ɾjoɾt** ~ oɾ.ˈ**joɾt** ‘miss’ օրիորդ

For the gaps in Table 5.4, see the lexical frequency data (Table 5.2) in §5.1.1. The high frequency of word-final CGV(C)(C) is because of common derivational suffixes that have a glide: *-ja, -jan, -jal*. Cf. the roots [ɑtɑmɑnt] ‘diamond’ ադամանդ, [ɑbɑhov] ‘safe’ ապահով, [pɑɾiz] ‘Paris’ Փարիզ with the words in Table 5.4.

For other restrictions on complex onsets, see Section §5.5. §5.1.7 has some words with atypical complex onsets in loanwords.

5.1.3 Overview of complex codas

Orthographically, Armenian has many words that are written with two or three final consonants. To illustrate, Table 5.5 is an overview of the syllabification of words that are written with a word-final 2-consonant cluster. We include the

frequency of these word-final clusters. Their syllabification is surveyed in depth in Section §5.2. Larger clusters of 3 consonants are discussed in §5.1.5 and §5.3.

Table 5.5: Pronunciation of word-final 2 consonant clusters

Sonority	Underlying	Surface shape					Final #
Falling	/VCC/	[VCC]	<ba**rz**>	[bɑ**ɾz**]	'simple'	պարզ	7738
Falling	/VCC/	[VCəC]	<xa**ṙn**>	[χɑ**ɾən**]	'mixed'	խառն	193
Geminate	/VCG/	[VC:]	<da**rr**>	[dɑ**ɾɾ**]	'element'	տարր	93
Flat	/VCC/	[VCəC]	<e**ws**>	[je**vəs**]	'morever'	եւս	360
Flat	/VCC/	[VCC]	<o**tk**'>	[vo**tk**]	'foot'	ոտք	286
Rising	/VCC/	[VCəC]	<d͡za**nr**>	[d͡zɑ**nər**]	'heavy'	ծանր	445
Rising	/VCC/	[VCC]	<go**ɣm**>	[go**ʁm**]	'side'	կողմ	149

Some of these clusters are pronounced as just a complex coda. These clusters tend to have falling-sonority: [bɑɾz] 'simple'. Obstruent complex codas are always homogeneous in voicing (§4.3.1).

But there are orthographic clusters with falling-sonority clusters, but which usually take schwa epenthesis: [χɑɾən] 'mixed'.

Very rarely, we find words that end in two identical consonants, and this cluster is pronounced as a geminate or single long consonant: [dɑɾɾ] 'element'.

As for clusters with flat or rising sonority, some take schwa epenthesis, while some can form consonant clusters without epenthesis. The clusters without a schwa are often analyzed with an extrasyllabic appendix (§5.1.4).

Besides the above categories of complex codas, there are some arbitrary restrictions on word-medial complex codas and some vowel-coda dependencies. These miscellaneous restrictions are covered in Section §5.4. Schwa epenthesis is likewise a quite complicated morphophonological process, briefly overviewed in Section §4.2.2 and discussed in depth in cite epenthesis chapter.

5.1.4 Appendix or extrasyllabic consonants

For those flat or rising-sonority clusters which don't take schwa epenthesis, the final consonant is often analyzed as some type of extrasyllabic appendix (Table 5.6). The final segment is one of the following segments /k, m, χ, s, ʃ/. Lexical frequency data is taken from the associated dictionary.

Table 5.6: Syllables that end in a flat or rising sonority cluster of 2 or more consonants

	Sonority	Shape				Final #	Total #
-k	Flat	[VCk]	li**t͡sk**	'stuffing'	լիցք	218	220
	Falling-Flat	[VCCk]	ɑntu**ntk**	'abyss'	անդունդք	106	107
-m	Rising	[VCm]	go**ʁm**	'side'	կողմ	127	198
-χ	Flat	[VCχ]	vɑ**ʃχ**	'usury'	վաշխ	57	59
	Rising	[VCχ]	pi**t͡ʃχ**	'common ivy'	փիչխ	2	2
	Falling-Rising	[VCCχ]	ɑ**stχ**	'star'	աստղ	9	9
-s	Flat	[VCs]	**d͡zɑχs**	'expense'	ծախս	7	8
	Rising	[VCs]	medɑ**ks**	'silk'	մետակս	20	20
	Falling-Rising	[VCs]	əspi**ŋks**	'sphinx'	սփինքս	3	3
-ʃ	Flat	[VCʃ]	**d͡ʒɑfʃ**	'breast-plate'	ճաւշ	1	1

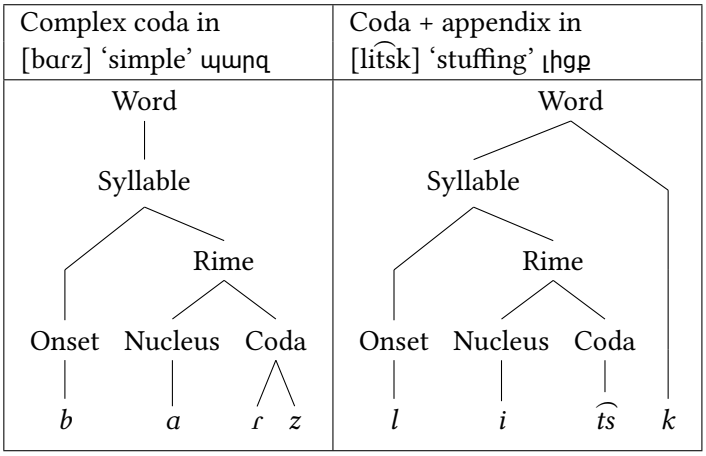
For syllables that end in an appendix, most of these syllables are word-final syllables. For example, there are in total 220 syllables in the associated dictionary, which contain an appendix /k/. Most of these syllables (218) appear at the end of the word. Usually, a word-medial appendix exists because of adding derivational suffixes or compounding.

The appendix /m/ is quite common after fricatives (§5.2.3.10), while the fricative appendixes are mostly found after other fricatives (§5.2.3.9). Some of these fricatives can also follow stops (§5.2.3.2) and complex codas (§5.3.2).

Among these appendixes, the nominalizer suffix *-k* ք is special in how its consistently violates all syllable rules. It can follow any type of consonant or consonant cluster, including laterals (§5.2.2.8), stops, (§5.2.3.1), affricates (§5.2.3.6), and complex codas (§5.3.2.1). Diachronically, this suffix *-k* was a plural suffix in Classical Armenian, and thus it was freely added after words. In the modern language, this inflectional suffix was reanalyzed as a derivational suffix, and it developed special behaviors in terms of syllabification. It can form a complex coda that is otherwise found in the language, such as *ɾ-k*, but it can also form consonant clusters that are otherwise absent. Though there are some restrictions on word-medial appendixes (§5.4.2.5).

Because of this special behavior, the suffix *-k* is often analyzed as not actually being part of the syllable. It is instead an extrasyllabic appendix, i.e., a segment that is added after any syllable (Vaux 1998: 83-4; Vaux & Wolfe 2009). Representation 1 illustrates.

Representation 1. Syllable structure of a complex coda vs. coda + appendix



Note that in Representation 1, we show the *-k* as attached directly to the word, and not to the Coda node. But it's possible that the appendix is actually added to the Syllable node instead. Crucially, the *-k* must be present somewhere within the phonological structure, so that it can trigger allophonic processes such as voicing assimilation (§4.3.1.4, 4.3.1.4), cf. voicing assimilation in Polish appendixes: Rubach & Booij 1990, Rubach 1996, 1997).[1]

The extrasyllabic or appendix representation is assumed for the other possible segments in Table 5.6. Besides these segments, one rarely comes across the other stops /p, b, t, d, g/ as an appendix in a flat sonority cluster. These are discussed in §5.2.3.1 and §5.2.3.6.

5.1.5 Maximality of complex codas

For complex codas, these are usually at most 2 consonants. If the orthography has a final 3-consonant cluster (Table 5.7; §5.3), this cluster is pronounced with either schwa epenthesis or with an appendix such as /-k, χ, -s/. Kouyoumdjian (1970) lists only one word [vɛɾst] (a loanword from Russian) with a final 3-consonant cluster which a) we pronounce without epenthesis and which b) doesn't have an appendix. The appendix /k/ is generally a separate suffix.

[1] Vaux (1998: 83-4) argues that *-k* is an appendix at the end of cyclic domains, and illustrates it as connected to a prosodoc word (p. 32). Dolatian (2021: 870) argues that this segment is attached to a prosodic constituent that's that is below the prosodic word, specifically the prosodic stem (Downing 1999). The main evidence for that is because the appendix can precede inflectional suffixes which are argued to be outside the prosodic stem. However, one could argue that the appendix can variably attach either to a syllable node, prosodic stem node, or prosodic word node without any empirical predictions or consequences.

Table 5.7: Words with final 3-consonant clusters

Sonority	Shape					#
Falling-Falling	VCəCC	<xaɾ̌nk’>	[χɑɾənk]	‘copulation’	խառնք	1
Rising-Falling	VCəCC	<gozɾ̌n>	[gozərn]	‘young camel’	կոզռն	1
Falling-Falling	VCCəC	<ewayln>	[jevɑjlən]	‘et cetera’	եւայլն	13
Falling-Rising	VCCəC	<par͡tsr>	[pɑɾ͡tsər]	‘high’	բարձր	201
Falling-Flat	VCCəC	<t͡ʃermn>	[t͡ʃermən]	‘fever’	ջերմն	6
Flat-Rising	VCCəC	<magdr>	[mɑgdər]	‘mactra’	մակտր	2
Rising-Flat	VCCəC	<saɣmn>	[sɑʁmən]	‘embryon’	սաղմն	9
Falling-Falling	VCCk	<d͡zajrk’>	[d͡zajɾk]	‘extremity’	ծայրք	28
	cf.	<d͡zajr>	[d͡zajɾ]	‘side’	ծայր	
Falling-Geminate	VCCk	<ergk>	[jeɾkk]	‘work of art’	երկք	7
	cf.	<erg>	[jeɾg]	‘literary work’	երկ	
Flat-Geminate	VCCk	<xozanad͡zad͡zgk>	[χozɑnad͡zɑt͡skk]	‘small thatched cottage’	խոզանածածկք	1
	cf.	<d͡zad͡zg>	[d͡zad͡zg]	‘cover’	ծածկ	
Geminate-Falling	VCCk	<uɣxk’>	[uχχk]	‘torrent’	ուղխք	2
	cf.	<uɣx>	[uχχ]	‘torrent’	ուղխ	
Falling-Flat	VCCk	<t͡sajtk’>	[t͡sajtk]	‘splash’	ցայտք	106
	cf.	<t͡sajt>	[t͡sajt]	‘gush’	ցայտ	
Falling-Rising	VCCs	<garank’s>	[gɑɾɑŋks]	‘cavalla’	կարանքս	3
Falling-Rising	VCCχ	<asdɣ>	[ɑstχ]	‘star’	աստղ	9
Falling-Falling	VCCC	<versd>	[veɾst]	‘verst’	վերստ	1

For those clusters that use schwa epenthesis, the final consonant is almost always a sonorant or fricative. The preceding cluster almost always have falling sonority. These restrictions are because of diachrony (Vaux 1998: 26; **Beekes-2003-HistoricalPh**
It has been postulated that in earlier stages of the language (Classical Armenian and Proto-Armenian), the ancestor of these <V_1CCC> [V_1CCəC] words would have an extra final syllable (perhaps <V_1CCCV_2> or <V_1CCV_2C>). Over time, the final syllable was lost, and the loss of a syllable required schwa epenthesis.

5.1.6 Onsetless syllables

For onsetless syllables like V(C)(C), such syllables are generally restricted to the word-initial position (Table 5.8). Word-medially, such onsetless syllables are quite restricted. They can be found across a compound boundary or a prefixoid boundary /ɑ/ in some words, and often in loanwords. But in this case, there is a slight glottal stop before the V(C)(C) syllable. The glottal stop isn’t marked in the traditional transcriptions in Armenian philology or dialectology.

Table 5.8: Onsetless syllables in different word positions

	Word-initial	Word-medial	Word-final
V	**u**.ˈʒeʁ ‘strong’ ուժեղ	ɑ.me.n-ɑ-**ʔu**.ˈɾɑχ ‘happiest’ ամենաուրախ	bo.ˈ**ʔɑ** ‘boa’ պօա
VC	**iʃ**.ˈχun ‘rhubarb’ իշխուն	kɑɾ.di.s-ɑ-**ʔən**.teɾ.ˈnum ‘map-reading’ քարտիսաքնթեռնում	tɑ.te.**ʔos** ‘Thaddeus’ Թադէոս
VCC	**ənt**.lɑj.ˈnel ‘to enlarge’ ընդլայնել	ɑ.me.n-ɑ-**ʔoɾt**.ˈnjɑl ‘most blessed’ ամենաօրհնեալ	i.mɑs.t-ɑ-**ʔiχ͡ts** ‘sensible’ իմաստաիղձ

Such words are quite few. The associated dictionary has less than 30 words that have vowel hiatus. Most of them are compounds or loanwords. The only common words found were numerals like [dɑsn-ə-utə] ‘eighteen’ տասնրութը, where the schwa is used as a meaningless connector between the numbers ‘ten’ and ‘eight’ (1).

(1) dɑs-n-ə-ut-ə, dɑs-ə, ut-ə ‘eighteen, ten, eight’
ten-DEF-CON-eight-DEF, ten-DEF, eight-DEF,
տասնրութը, տասը, ութը

Further description of word-medial onset-less syllables is discussed in the section on vowel hiatus repair (§5.6), especially for loanword roots (§5.6.1) and prefixoids (§5.6.3.2).

5.1.7 Loanwords

We do not discuss loanwords at length, but there are some loanwords which have atypical syllable structures, kept from the source language. The atypicality is either a complex onset that’s not /Cj/ or a complex coda that’s rare in the native lexicon. Some examples are in Table 5.9. For the word-medial consonant clusters, they can be variably pronounced with a complex onset.

Table 5.9: Atypical syllable structures in loanwords

‘Sanskrit’	sɑnsk.ˈɾit ~ sɑns.ˈkɾit	սանսկրիտ
‘bread-pan’	mɑgd.ˈɾɑ ~ mɑg.ˈdɾɑ	մակտրայ
‘electron’	je.legd.ˈɾon ~ je.leg.ˈdɾon	ելեկտրոն
‘sphinx’	əs.pinks	սփինքս

5.2 Syllabification of final two-consonant clusters or complex codas

This section goes through all attested and un-attested complex codas in Western Armenian. This section expands on section §5.1.3. To find the attested clusters, we went through the Kouyoumdjian dictionary and kept track of all words that were written with two final consonants. We cataloged the consonants in terms of their sonority and pronunciation. We then double-checked against word-internal complex codas.

For sonority, we use the conventional sonority scale of *stop/affricate < fricative < nasal < liquid < glide < vowel.*

In Section §5.2.1, we go through word-final consonant clusters that had falling sonority, formed a complex coda in pronunciation, and were very common in the dictionary, such as fricative-stop clusters like [ɑχp] 'trash' աղբ. In contrast, Section §5.2.2 goes through clusters that had falling sonority but had some exceptional behavior. Such exceptional behavior is one of the following:

- The orthographic cluster is pronounced as a complex coda but is very rare or restricted to loanwords like the lateral-fricative cluster in [vɑls] 'waltz' վալս. This category includes clusters that are simply unattested in either Kouyoumdjian or other sources like Wiktionary.
- The orthographic cluster requires an intervening schwa, either always or optionally like rhotic-/n/ clusters in [χɑɾ(ə)n] 'mixed' խառն.
- The orthographic cluster is pronounced without a schwa but the second consonant was almost always a certain segment, suggesting that this segment is an extrasyllabic appendix, such as lateral-/k/ clusters like [χelk] 'mind' խելք.

The dictionary likewise listed many words that end in a two-consonant cluster with either flat or rising sonority (§5.2.3). Here, we find the same types of exceptional behavior: rarity vs. epenthesis vs. appendixes. Gemination was vanishingly rare as well (§5.2.4).

Because the data is quite complicated, we've had difficulty provided succinct summaries over the possible complex codas. Instead, each subsection has a list of what natural classes of clusters pattern together in terms of their syllabification.

5.2.1 Falling-sonority and common complex codas

The majority of common complex codas were falling sonority and belonged to one of the following groups based on the identity of the first and second consonant (C1, C2):

- Fricative /s,ʃ/ + stop (§5.2.2.1)
- Fricative /χ,ʁ/ + stop or affricate (§5.2.1.2)
- Nasal /m/ + labial stop (§5.2.1.3)
- Nasal /n/ + stop or affricate (§5.2.1.4)
- Rhotic /ɾ/ + obstruent (§5.2.1.5)
- Glide /j/ + consonant (§5.2.1.7)

5.2.1.1 Fricative /s,ʃ/ + stop

The fricatives /s,ʃ/ can form complex codas with voiceless stops [p, t, k] with stop deaspiration. The most common stop is coronal [t]. The fricatives however cannot form complex codas with voiced obstruents, in order to avoid a voicing mismatch (§3.4.2, §4.3.1). The fricatives /s,ʃ/ also avoid combining with affricates (§5.2.2.2).

The fricative /s/ is a pretty common segment. It can form complex codas with any type of voiceless stop: [sp, st, sk] (Table 5.10). The [k] can be part of either the root (written as գ,կ) or part of the nominalizer suffix *-k* (written as ք).

Table 5.10: Complex codas where C1 is fricative /s/, and C2 is a voiceless stop

				Final #	Total #
[sp]	ˈvo**sp**	'lentil'	ոսպ	22	70
	bɑ.ˈɾi**sp**	'fortress'	պարիսպ		
	ɑ**sp**.ˈɾig	'mugwet'	ասպրիկ		
[st]	ˈpu**st**	'coral'	բուստ	284	611
	nə.ˈbɑ**st**	'subsidy'	նպաստ		
	ən.to**st**.ˈnum	'rebound'	ընդոստնում		
[sk]	ˈgɑ**sk**	'malt'	կասկ	43	120
	bə.ˈɾi**sk**	'drias plant'	պրիսկ		
	vo**sk**.ɾɑ.ˈpɑn	'osteologist'	ոսկրաբան		
/s-k/	ˈke**s-k**	'head of hair' (√-NMLZ)	գէսք		
	cf. ˈkes	'long hanging hair'	գէս		

Similarly, the fricative /ʃ/ can form a complex coda with a voiceless stop [ʃp, ʃt, ʃk] (Table 5.11). The [k] can be part of the root (written as գ,կ) or part of the nominalizer suffix *-k* (written as ք).

Table 5.11: Complex codas where C1 is fricative /ʃ/, and C2 is a voiceless stop

				Final #	Total #
[ʃp]	ˈku**ʃp**	‘crevice’	գուշպ	1	1
[ʃt]	ˈge**ʃt**	‘sect’	կեշտ	174	207
	peˈhe**ʃt**	‘paradise’	բեհեշտ		
	bɑ**ʃt**.pɑˈnel	‘to protect’	պաշտպանել		
[ʃk]	ˈmɑ**ʃk**	‘cuticle’	մաշկ	51	53
	təˈmi**ʃk**	‘Damascus blade’	դմիշկ		
	d͡ʒə**ʃk**.ˈɾid	‘exact’	ճշգրիտ		
/ʃ-k/	ˈd͡ʒo**ʃ-k**	‘defamation’ (√-NMLZ)	ճօշք		
	cf. d͡ʒoˈʃ-ɑ-l	‘to defame’ (√-TH-INF)	ճօշալ		

5.2.1.2 Fricative /χ, ʁ/ + stop or affricate

The fricative /χ/ can form a complex coda with any voiceless stop or affricate: [χp, χt, χk, χt͡s, χt͡ʃ] (Table 5.12). The [k] can be part of either the root (written as գ) or part of the nominalizer suffix *-k* (written as ք).

Table 5.12: Complex codas where C1 is fricative /χ/, and C2 is a voiceless stop or affricate

				Final #	Total #
[χp]	ˈʃe**χp**	‘blade’	շեղբ	19	20
	ʃɑχˈpɑ**χp**	‘bragger’	շաղփաղփ		
	ɑ**χp**.kuˈdot͡s	‘manure’	աղբկուտոց		
[χt]	ˈsɑ**χt**	‘saddle’	սախտ	220	238
	təˈɾɑ**χt**	‘paradise’	դրախտ		
	kɑ**χt**.ˈnɑd	‘stammering’	քաղթնատ		
[χk]	ˈhe**χk**	‘lazy’	հեղգ	45	46
/χ-k/	hɑˈd͡ʒɑ**χ-k**	‘frequency’ (√-NMLZ)	յաճախք		
	cf. hɑd͡ʒɑˈχ-e-l	‘to frequent’ (√-TH-INF)			
	me**χ-k**-ˈn-ɑ-l	‘to pity’ (√-NMLZ-INCH-TH-INF)	մեղքնալ		
	cf. me**ʁ**-ɑvor	‘sinner’ (√-ADJZ)	մեղաւոր		
[χt͡s]	ˈte**χt͡s**	‘peach’	դեղձ	53	147
	ɑˈdɑ**χt͡s**	‘timber’	ատաղձ		
	kɑ**χt͡s**.ˈke**ʁ**	‘cancer’	քաղցկեղ		
[χt͡ʃ]	ˈze**χt͡ʃ**	‘discount’	զեղչ	25	34
	t͡ʃəɾ.ˈʃe**χt͡ʃ**	‘basin’	ջրշեղջ		
	ke**χt͡ʃ**.kɑ.ˈjin	‘boorish’	գեղջկային		

The fricative /ʁ/ is generally infrequent as a segment. But it can form complex codas with voiced stops and affricates (Table 5.13). It cannot co-occur with voiceless obstruents. Any orthographic sequences of <ɣ> and a voiceless obstruent are pronounced as voiceless (§3.4.2).

Table 5.13: Complex codas where C1 is fricative /ʁ/, and C2 is a voiced stop or affricate

				Final #	Total #
[ʁb]	ˈd͡ʒɑ**ʁb**	‘coffin’	ջաղպ	1	1
[ʁd]	ˈge**ʁd**	‘stain’	կեղտ	20	34
	ən.ˈt͡su**ʁd**	‘giraffe’	ընծուղտ		
	dɑ**ʁd**.ˈgɑl	‘to be disgusted’	տաղտկալ		
[ʁg]	ˈme**ʁg**	‘soft’	մեղկ	21	22
	mə'ʒu**ʁg**	‘gnat’	մժուղկ		
	no**ʁg**.ˈdɑl	‘to dislike’	նողկտալ		
[ʁd͡z]	ˈme**ʁd͡z**	‘soot’	մեղծ	37	40
	ɑ.ˈne**ʁd͡z**	‘incorruptible’	անեղծ		
	ke**ʁd͡z**.ˈmɑmp	‘abusively’	զեղծմամբ		
[ʁd͡ʒ]	ˈχe**ʁd͡ʒ**	‘wretched’	խեղճ	4	13
	pɑ.ɾe.ˈχi**ʁd͡ʒ**	‘conscientious’	բարեխիղճ		
	χɑ**ʁd͡ʒ**.ˈmuŋk	‘tonsil’	խաղճմունք		

5.2.1.3 Nasal /m/ + labial stop

The nasal /m/ can precede labial stops /p, b/ (Table 5.14). The cluster [mp] is significantly more common than [mb]. However, /m/ seems to avoid forming a complex coda with other types of stop. A spurious exception is *mk* sequences, but these utilize an appendix. See Section §5.2.2.5.

Table 5.14: Complex codas where C1 is nasal /m/, and C2 is a labial stop

				Final #	Total #
[mp]	d͡zəˈd͡zu**mp**	‘sulfur’	ծծումբ	182	232
	gɑˈʁɑ**mp**	‘cabbage’	կաղամբ		
[mb]	ˈɑ**mb**	‘cloud’	ամպ	6	8
	ˈu**mb**	‘gulp’	ումպ		

The preponderance of [mp] over [mb] is typologically surprising (Pater 1999) but it makes diachronic sense (Beguš 2019). Most surface [mp] clusters are writ-

ten with final <mp> մբ. This sequence corresponds to Classical/Eastern Armenian [mb] clusters. In contrast, Western [mb] clusters are orthographically <mb> մպ, and they correspond to Classical/Eastern [mp]. The sound changes *p*→*b* and *b*→*p* caused Western Armenian to end up having [mp] be more common than [mb].

5.2.1.4 Nasal /n/ + stop or affricate

The nasal /n/ can form a complex coda with coronal /t/ or /d/ (Table 5.15).

Table 5.15: Complex codas where C1 is nasal /n/, and C2 is a coronal stop

				Final #	Total #
[nt]	dɑˈʁɑ**nt** ɑtɑˈmɑ**nt**	‘talent’ ‘diamond’	տաղանդ ադամանդ	295	673
[nd]	ˈχɑ**nd** ˈʒɑ**nd**	‘lewd’ ‘pestilent’	խանտ ժանտ	32	113

There are no examples of /n/ + a labial stop /p, b/. It’s unclear if this is an accidental gap, or if there’s an active constraint against having /np, nb/ complex codas. Such clusters can arise across different syllables however, like [pɑn.peɾ] ‘messenger’ բանբեր (§4.4.3).

The nasal /n/ can appear before velar /k, g/ (Table 5.16). In this situation, the nasal becomes a velar [ŋ] (§4.4). Note that the /k/ can be part of the same root as the nasal (often written as գ, but sometimes ք). The nasal+stop can also be part of common nominalizer suffixes /-ɑnk, -unk, -(ɑ)munk/. The stop can also be part of a separate nominalizer suffix /-k/. For all these latter cases, the stop is written as ք. We do not provide nuanced frequencies for these different morphological constructions.

Table 5.16: Complex codas where C1 is nasal /n/, and C2 is a velar stop

				Final #	Total #
/ng/	ˈt͡sɑŋg	‘list’	ցանկ	201	245
	gəˈɾuŋg	‘heel’	կրունկ		
/nk/ root	ˈɾuŋk	‘nostril’	ռունգ	870	1162
	ˈt͡sɑŋg	‘list’	ցանկ		
	vəˈdɑŋk	‘danger’	վտանգ		
/n-k/	ˈvɑŋ-k	‘convent’ (√-NMLZ)	վանք		
	cf. vɑn-ɑˈgɑn	‘monastic’ (√-ADJZ)	վանական		
/-ɑnk/	ɑʃχɑˈd-ɑŋk	‘work’ (√-NMLZ)	աշխատանք		
	cf. ɑʃχɑˈd-i-l	‘to work’ (√-TH-INF)	աշխատիլ		
/-unk/	həɾɑˈʃ-uŋk	‘miracle’ (√-NMLZ)	հրաշունք		
	cf. həˈɾɑʃ-k	‘miracle’ (√-NMLZ)	հրաշք		
/-(ɑ)munk/	bɑʃt-ɑmuŋk	‘ceremony’ (√-NMLZ)	պաշտամունք		
	cf. bɑʃˈt-e-l	‘to worship’ (√-TH-INF)	պաշտել		

The nasal /n/ can precede any affricate /t͡s, d͡z, t͡ʃ, d͡z/ (Table 5.17).

Table 5.17: Complex codas where C1 is nasal /n/, and C2 is an affricate

				Final #	Total #
[nt͡s]	ˈdɑnt͡s	‘pear’	տանձ	134	415
	bəˈʁint͡s	‘copper’	պղինձ		
[nd͡z]	ˈχɑnd͡z	‘bait’	խանձ	24	34
	goˈʁund͡z	‘hard crust’	կողունձ		
[nt͡ʃ]	ˈmɑnt͡ʃ	‘lad’	մանչ	222	270
	ɑˈgɑnt͡ʃ	‘ear’	ականջ		
[nd͡ʒ]	ˈd͡ʒɑnd͡ʒ	‘fly’	ճանճ	56	78
	jeˈʁind͡ʒ	‘large nettle’	եղինճ		

5.2.1.5 Rhotic /ɾ/ + obstruent

The rhotic /ɾ/ can form a complex coda with a) any obstruent, and b) with /m/. The nasal /n/ however has complications in syllabification; postponed to Section §5.2.2.7.

The rhotic /ɾ/ can form a complex coda with any type of stop: /p, b, t, d, k, g/ (Table 5.18). For /k/, the stop can either be part of the root (written as ɢ) or arguably be the nominalizer suffix *-k* (written as -ᵽ).

Table 5.18: Complex codas where C1 is rhotic /ɾ/, and C2 is a stop

				Final #	Total #
[ɾp]	ˈɑɾ**p**	‘sunlight’	արփ	41	79
	ˈsuɾ**p**	‘holy’	սուրբ		
[ɾb]	ˈd͡ʒɑɾ**b**	‘grease’	ճարպ	216	216
	ˈd͡zeɾ**b**	‘crevice’	ծերպ		
[ɾt]	ˈvɑɾ**t**	‘rose’	վարդ	571	752
	zəˈvɑɾ**t**	‘joyous’	զուարթ		
[ɾd]	ˈkoɾ**d**	‘frog’	գորտ	297	388
	həˈbɑɾ**d**	‘proud’	հպարտ		
[ɾk]	ˈhɑɾ**k**	‘esteem’	յարգ	379	592
	ˈgɑɾ**k**	‘order’	կարգ		
	ˈpɑɾ-**k**	‘glory’ (√-NMLZ)	փառք		
	cf. pɑɾ-ɑ-zɑɾt	‘glorious’ (√-LV-√)	փառազարդ		
[ɾg]	ˈneɾ**g**	‘paint’	ներկ	184	534
	ɑˈd͡zɑɾ**g**	‘switch’	ածարկ		

The rhotic /ɾ/ can be appear before any affricate /t͡s, d͡z, t͡ʃ, d͡ʒ/ (Table 5.19).

Table 5.19: Complex codas where C1 is rhotic /ɾ/, and C2 is an affricate

				Final #	Total #
[ɾt͡s]	ˈhɑɾ**t͡s**	‘issue’	հարց	187	284
	χoˈloɾ**t͡s**	‘orchid’	խոլորձ		
[ɾd͡z]	ˈvoɾ**d͡z**	‘belch’	ործ	340	387
	ləˈbiɾ**d͡z**	‘slippery’	լպիրծ		
[ɾt͡ʃ]	ˈɑɾ**t͡ʃ**	‘bear’	արջ	41	111
	hɑˈʁɑɾ**t͡ʃ**	‘currant’	հաղարջ		
[ɾd͡ʒ]	ˈgoɾ**d͡ʒ**	‘griffin’	կորճ	37	49
	zəˈvɑɾ**d͡ʒ**	‘joyfully’	զուարճ		

The rhotic can appear before the fricatives /s, z, ʃ, ʒ, χ/ (Table 5.20). Complications arise for the other fricatives.

Table 5.20: Complex codas where C1 is rhotic /ɾ/, and C2 is a fricative

				Final #	Total #
[ɾs]	ˈhɑɾs	‘bride’	հարս	36	88
	ɑˈʁɛɾs	‘supplication’	աղերս		
[ɾz]	ˈmɑɾz	‘confine’	մարզ	14	29
	ˈbɑɾz	‘simple’	պարզ		
[ɾʃ]	ˈkoɾʃ	‘gray’	գորշ	44	52
	ʃəˈʁɑɾʃ	‘gauze’	շղարշ		
[ɾʒ]	ˈvɑɾʒ	‘accustomed’	վարժ	84	85
	ɑˈχoɾʒ	‘pleasant’	ախորժ		
[ɾχ]	ˈtɑɾχ	‘sketch’	թարխ	4	4
	ˈχoɾχ	‘hide’	խորխ		

The Kouyoumdjian dictionary does have any words with final /ɾf, ɾv, ɾʁ/. For /ɾf, ɾv/, this is likely because these fricatives are quite rare in the first place. On Armenian Wiktionary, we've found a handful of words with final /ɾf, ɾv/. These all seem to be loanwords: [ɑlomoɾf] ‘allomorph’ ալոմորֆ, [neɾv] ‘nerve’ ներվ. For /ɾʁ/, the handful of Wiktionary examples seem to be dialectal words that are absent from Western Armenian.

Orthographically, the rhotic can form a cluster with the fricative /h/, whether as ռh or րh (Table 5.21). For example, as of date, we counted code:46 words in the dictionary that end in an orthography րh <rh>. However, most words that have this final cluster don't pronounce the /h/, such that the [ɾh] pronunciation is archaic or obsolete. Only a subset of these words have the final /h/ still pronounced, thus creating a [ɾh] complex coda. See Section §3.4.4 for general data on this orthography-phonology mismatch.

Table 5.21: Complex codas where C1 is rhotic /ɾ/, and C2 is a fricative /h/

				Final #
Silent <h>	ɑʃˈχɑɾ	‘world’	աշխարհ	
	χoˈnɑɾ	‘humble’	խոնարհ	46-8=38
Pronounced <h>	ˈχoɾh	‘thought’	խորհ	8
	ʒəˈbiɾh	‘insolent’	ժպիրհ	

5.2.1.6 Rhotic /ɾ/ + nasal /m/

Orthographically, there are many words that end in a rhotic /ɾ/ + nasal /m/. These clusters are pronounced as [ɾm] without schwa epenthesis (Table 5.22).

Table 5.22: Complex codas where C1 is rhotic /ɾ/, and C2 is a nasal /m/

				Final #	Total #
[ɾm]	ˈzɑ**ɾm**	'tribe'	զարմ	99	251
	’t͡ʃe**ɾm**	'warm'	ջերմ		
	’ɑ**ɾm**	'stamp'	արմ		
	voˈʁo**ɾm**	'pity'	ողորմ		

Word-medially however, the [ɾm] complex coda shows some idiosyncrasies (§5.4.2.1).

5.2.1.7 Glide /j/ + consonant

The glide /j/ can form a complex coda with virtually any type of consonant. Though there are some accidental gaps in the Kouyoumdjian dictionary.

As a C1, the glide /j/ can precede virtually any type of stop, whether voiced or voiceless (Table 5.23). For final [jk], the final /k/ can either be part of the same morpheme as the glide (written as յգ), or part of a separate nominalizer suffix *-k* (written as յք) .

Table 5.23: Complex codas where C1 is glide /j/, and C2 is a stop

				Final #	Total #
[jt]	ˈχɑ**jt**	‘sting’	խայթ	94	108
	məʃɑˈgu**jt**	‘culture’	մշակոյթ		
[jd]	ˈɑ**jd**	‘cheek’	այտ	150	181
	bəˈdu**jd**	‘tour’	պտոյտ		
[jk]	ˈɑ**jk**	‘dawn’	այգ	182	183
	ˈzu**jk**	‘twin’	զոյգ		
/j-k/	ˈgɑ**jk**	‘station’	կայք		
	cf. ˈgɑj	‘station’	կայ		
	həˈmɑ**jk**	‘charm’	հմայք		
	cf. həmɑˈj-e-l	‘to charm’	հմայել		
[jg]	ˈhɑ**jg**	masc. name	Հայկ	20	20
	bɑˈɾu**jg**	question mark	պարոյկ		
[jŋ]	lɑ**jŋ**.ˈkin	’broadwise’	լայնքին	0	10
	hɑ**jŋ**.ˈgujs	‘on the other side’	յայնկոյս		

But as an accidental gap, the Kouyoumdjian dictionary doesn’t list any final /jp/ or /jb/ words. Such clusters however are not impossible, but they may be restricted to loanwords. For example, the name of the first Armenian letter is [ɑjp] այբ, possibly a loanword of ‘alpha’. For /jb/, Armenian Wiktionary lists some such words (written with final յպ) but these seem to all be loanwords: այպ for Eastern [ɑjp] or Western [ɑjb] from Arabic ‘shame’. Our dictionary lists only the following words with a (non-final) complex coda /jp/ or /jb/ (Table 5.24).

Table 5.24: Complex codas where C1 is glide /j/, and C2 is a bilabial stop

				Final #	Total #
[jp]	ɑ**jp**.pe.nɑ.ˈɾɑn	‘alphabet’	այբբենարան	0	1
[jb]	ɑ**jb**.nɑ.gɑ.ˈdɑg	‘mimic’	այպնակատակ	0	1

The glide can precede an affricate (Table 5.25). The Kouyoumdjian dictionary lists word with a /j/ + /t͡s, d͡z, d͡ʒ/. The dictionary lacks /jt͡ʃ/. This seems to be an accidental gap. Armenian Wiktionary likewise lacks words which would get pronounced with /jt͡ʃ/ in Western Armenian.

Table 5.25: Complex codas where C1 is glide /j/, and C2 is an affricate

				Final #	Total #
[jt͡s]	ˈɑj**t͡s**	'visit'	այց	221	228
	sɑˈɾu**jt͡s**	'frost'	սառոյց		
[jd͡z]	ˈɑj**d͡z**	'goat'	այծ	71	82
	ɑɾˈgɑ**jd͡z**	'wavering'	առկայծ		
[jd͡ʒ]	bɑˈd͡ʒu**jd͡ʒ**	'adorned'	պաճոյճ	22	22

The glide can also precede a voiced or voiceless fricative (Table 5.26). The Kouyoumdjian dictionary lists word with a /j/ + /s, z, ʃ, ʒ/.

Table 5.26: Complex codas where C1 is glide /j/, and C2 is a fricative

				Final #	Total #
[js]	ˈhɑ**js**	'paste'	հայս	220	233
	ˈlu**js**	'light'	լոյս		
[jz]	ˈhu**jz**	'sap'	հոյզ	83	86
	əŋˈgu**jz**	'walnut'	ընկոյզ		
[jʃ]	kəˈmu**jʃ**	'imagination'	քմոյշ	5	6
[jʒ]	ˈdu**jʒ**	'damage'	տոյժ	41	47

For final [ujʃ] and [ujʒ] sequences, such pronunciations are rather archaic for most roots. The modern language tends to turn such [ujʃ] sequences to [uʃ], such as archaic [ɑnujʃ] անոյշ but modern [ɑnuʃ] անուշ 'sweet'. Similarly, most words with final [ujʒ] are pronounced with final [uʒ], such as archaic [ujʒ] ոյժ vs. modern [uʒ] ուժ 'strength'.

The fricatives /f, v, h/ are generally rare so their absence after /j/ is not surprising. Armenian Wiktionary listed a handful of words with final /jf, jv/; all of these are loanwords such as [sejv] 'save' սեյվ or [sejf] 'safe (n)' սեյֆ.

For the uvular /χ, ʁ/, these sounds aren't generally rare. The absence of /jχ/ or /jʁ/ may be an accidental gap. For example on Armenian Wiktionary, we found only two words that end in /jχ/, both of these are loanwords such as [ʃejχ] 'sheikh' շեյխ. For /jʁ/, Armenian Wiktionary only had one word [ʃujʁ] շույղ which was listed as a dialectal word, and thus wouldn't be in Western Armenian.

Finally, the glide can be precede any other sonorant: /m, n, ɾ, l/ (Table 5.27).

Table 5.27: Complex codas where C1 is glide /j/, and C2 is a sonorant

				Final #	Total #
[jm]	ˈɡɑ**jm** mɑnɾɑˈɡɑ**jn**	‘mast’ ‘small mast’	կայմ մանրակայմ	10	10
[jn]	ˈ͡tsɑ**jn** sɑˈɡɑ**jn**	‘voice’ ‘but’	ձայն սակայն	404	424
[jɾ]	ˈʒɑ**jɾ** hɑmˈpu**jɾ**	‘rock’ ‘kiss’	ժայռ համբույր	280	287
[jl]	ˈkɑ**jl** ʃəˈɾɑ**jl**	‘wolf’ ‘prodigal’	գայլ շռայլ	137	143

Note that for [jm], although this coda is possible, it seems very rare. For example, all of Kouyoumdjian’s examples were for compounds with the final root [ɡɑjm] ‘mast’. As we discuss elsewhere in Section §5.2.3.10), final [Cm] codas have quite complicated behaviors. Furthermore, word-medial [jm] codas seem even rarer (§5.4.2.1).

5.2.2 Falling sonority but either rare, extrasyllabic, or uses schwa epenthesis

The previous section looked at final consonant clusters that were a) falling sonority, and b) were commonly syllabified as complex codas. This section goes through cases of falling sonority clusters that for some reason or another are either a) rare complex codas, b) potentially fake complex codas made up a coda and an appendix, or c) get an epenthetic schwa. Such clusters and their behavior are the following:

- Fricative /f,v/ + stop or affricate: rare, likely just accidental gaps (§5.2.2.1)
- Fricative /s,ʃ/ + affricate: unattested, either accidental gaps or banned (§5.2.2.1)
- Fricative /z,ʒ/ + stop or affricate: unattested, likely just accidental (§5.2.2.1)
- Fricative /h/ + stop or affricate: rare, likely just coda + appendix (§5.2.2.4)
- Nasal /m/ + non-labial stop or affricate: rare, either generally banned or coda + appendix (§5.2.2.4)
- Nasal /m,n/ + fricative: rare word-finally; unclear if rarity is because of a ban or just accidental gaps. Somewhat avoided word-medially (§5.2.2.6)
- Rhotic /ɾ/ + nasal /n/: rare word-finally, and often avoided with schwa epenthesis (§5.2.2.7)
- Lateral /l/ + obstruent: rare and most are analyzable as coda + appendix (§5.2.2.8)

- Lateral /l/: nasal /m/: unattested outside of loanwords (§5.2.2.9)
- Lateral /l/ + nasal /n/: triggers schwa epenthesis (§5.2.2.10)

Data in this section expands on §5.1.4.

5.2.2.1 Fricative /f,v/ + stop or affricate

In general, the fricatives /f, v/ seem to avoid being the first consonant in a complex coda. In the Kouyoumdjian (1970) dictionary, we found very few words with such clusters. The words which existed are also low-frequency words.

When C1 is a fricative /f/, the C2 can be a stop /p, t, k/ (Table 5.28). /fp/ and /ft/ seem restricted to loanwords, especially Biblical loanwords as in the table below or other Semitic loanwords like [nɑft] 'oil' Նավթ. /fk/ seems restricted to cases where the *k* is the nominalizer suffix -*k* -ք. Thus the /fk/ cluster could arguably be treated as being a false complex coda, where *f* is a coda but *k* is an appendix.

Table 5.28: Complex codas where C1 is fricative /f/ and C2 is a stop

				Final #	Total #
[fp]	ˈho**fp**	'Job'	Յովբ	1	1
[ft]	ˈnɑ**ft** behˈmo**ft**	'naphta' 'behemoth'	Նաւթ բեհմովթ	12	12
[fk]	χoˈɾo**fk** cf. χoɾoˈv-e-l	'roasting' 'to roast'	խորովք խորովել	12	12

When C1 is /v/, C2 can be /b, d, g/ (Table 5.29). Again, these words are few and rare. Of the words in Kouyoumdjian (1970), a lot of these words are names of flora and fauna; these may possibly be old loanwords.

Table 5.29: Complex codas where C1 is fricative /v/ and C2 is a stop

				Final #	Total #
[vb]	ʒiˈlɑ**vb**	'white broom (plant)'	ժիլաւպ	1	1
[vd]	ɑɾɑˈbo**vd** heɾesiˈjo**vd**	'dried fig' 'heretic'	արաբովտ հերեսիովտ	5	5
[vg]	kəˈʁɑ**vg** nɑ**vg**.ˈnjɑɾ	'small river fish' 'boats'	գղաւկ նաւկնեար	2	3

We found no cases of Western words with a labial fricative and then an affricate. One possible case is a non-Western dialectal word թովջ on Wiktionary,

which seems to have something to do with taxes. We can at best guess that it's pronounced as [toft͡ʃ]. The rarity of such cases suggests that Western Armenian just doesn't have such clusters.

5.2.2.2 Fricative /s, ʃ/ + affricate

It seems that /s,ʃ/ cannot form a complex coda with affricates. We found no /s,ʃ/ + affricate complex codas in either the Kouyoumdjian dictionary or Armenian Wiktionary. This may just be an accidental gap. But this could be also due to some constraint against having an /s,ʃ/ + affricate cluster because both the fricatives and the affricate would have their own type of frication. To illustrate, the voiceless affricates are /t͡s, t͡ʃ/, and they both have end in a fricative-like element. Thus, it is possible that Armenian bans words like **ast͡s* in order to avoid a complex coda that both starts and ends with an *s*-like element.[2]

5.2.2.3 Fricative /z, ʒ/ + stop or affricate

There seems to be an accidental gap such that there are no syllables that end in a [z] or [ʒ] plus a stop or affricate.

For [zC], the absence of final [zC] words may just be an accidental gap since a) Eastern Armenian can end in such sequences like [skizb] 'beginning' <skizp> սկիզբ, and b) other fricatives like /s/ don't have such gaps. It is possible that this accidental gap arose via diachrony. Orthographically, a Western cluster [zb] would be written as զպ <zb>. But such an orthographic cluster would have to be pronounced as [zp] in Classical Armenian and Eastern Armenian; such a cluster is unattested for Eastern Armenian (= 0 hits on Wiktionary). In contrast, a [zb] cluster in Classical/Eastern would correspond to a [sp] cluster in Western: [əskisp] 'beginning'. Thus, in the development of Classical to modern Western Armenian, [zb] sequences switched to being [sp], but no such original cluster could have changed to [zb].

Note that although there are many words that end in an orthographic cluster of <z> plus a voiceless sound, such clusters are pronounced as voiceless: <azk> [ɑsk] 'people' ազգ. See Section §3.4.2.

Similar for [ʒC] codas, it is unclear if Western Armenian either a) bans [ʒC] complex codas as a language-general rule, or if b) the absence of such clusters is an accidental gap. It is possible that the absence of such clusters is an accidental

[2] Wiktionary did have some Russian loanwords with a final /ʃt͡ʃ/ clusters, but these don't exist in Western Armenian. And even if they did exist, loanwords often violate a language's phonological constraints.

gap that's caused by diachrony, for the same reasons as for the absence of [zC] clusters.

5.2.2.4 Fricative /h/ + stop or affricate

The fricative /h/ seems to avoid being in a complex coda (Table 5.30). In the Kouyoumdjian dictionary, we found only 5 words that have a /hC/ cluster inside a syllable. For C2, all these words involved the suffix *-k* -ք, suggesting that these words may instead be parsed as ending in a coda+appendix rather than a complex coda.

Table 5.30: Tautosyllabic CC clusters where C1 is fricative /h/ (total n=5; word-final n=5)

ˈχɑ**h-k**	'kitchen'	խահք	cf. ˈχɑh	'dish'	խահ
ˈbɑ**h-k**	'fasting'	պահք	cf. ˈbɑh	'preservation'	պահ
pɛɾt-ɑ-bɑ**h-k**	'garrison'	բերդապահք	cf. ˈpɛɾt	'fortress'	բերդ
ʃɑpɑt-ɑ-bɑ**h-k**	'Sabbatarians'	շաբաթապահք	cf. ʃɑpɑt	'Saturday'	շաբաթ
ɑt͡ʃ-ɑ-bɑ**h-k**	'leather screens'	աչապահք	cf. ɑt͡ʃ-k	'eye'	աչք

On Armenian Wiktionary, we found a handful more words with a final [hC] cluster, but these were all limited to specific non-standard dialects like Karabagh Armenian, thus their pronunciations cannot be extended to Western Armenian.

5.2.2.5 Nasal /m/ + non-labial stop or affricate

The nasal /m/ seems to start complex codas only with labial stops, and it avoids all other consonants.

For example, the Kouyoumdjian dictionary lists zero final /mt, md/ clusters. It has only one word with an /mg/ coda: [dɑmg] 'damp' տամկ. The dictionary states that this word is derived from a synonymous [dɑmug] տամուկ. This suggests that the word [dɑmg] is just a grammaticalized weak form of the larger word [dɑmug]. Armenian Wiktionary provides a handful of examples of final orthographic <mt, mt', md>, but these are all either loanwords or obscure dialectal words that aren't found in Western Armenian.

Before velar /k/, the nasal /m/ is found (Table 5.31). Kouyoumdjian lists 18 words with an /mk/ cluster. All of these words have the cluster in word-final position. But, all of these examples involve the nominalizer suffix *-k* -ք. So these examples could arguably be syllabified as a coda + appendix *-k*.

Table 5.31: Tautosyllabic CC clusters where C1 is nasal /m/ and C2 is /k/ (total n=18; word-final n=18)

ˈte**m-k**	'face'	դէմք	cf. ˈtem	'facing'	դէմ
gɑ**m-k**	'will'	կամք	cf. gɑˈm-i-l	'to will' (√-TH-INF)	կամիլ
χəˈnɑ**m-k**	'care'	խնամք	cf. χəˈnɑm	'care'	խնամ
ʃoˈʁo**m-k**	'flattery'	շողոմք	cf. ʃoˈʁom	'flattery'	շողոմ
hɑˈme**m-k**	'aromatics'	համեմք	cf. hɑˈmem	'aroma'	համեմ

Neither Kouyoumdjian nor Wiktionary provided any examples of final /m/ + affricate clusters.

Word-internally, we find evidence that such clusters are avoided. The bound root /ɑmt͡ʃ/ always surfaces with an epenthetic schwa after /m/ (Table 5.32). See schwa epenthesis chapter for elaboration.

Table 5.32: Avoidance of complex coda /mt͡ʃ/

/ɑmt͡ʃ-god/	[ɑ.mət͡ʃ.ˈkod]	'shy'	ամչկոտ
√-ADJZ	*ɑmt͡ʃ.kod		
/ɑmt͡ʃ-n-ɑ-l/	[ɑ.mət͡ʃ.ˈnɑl]	'to be ashamd'	ամչնալ
√-INCH-TH-INF	*ɑmt͡ʃ.nɑl		

5.2.2.6 Nasal /m,n/ + fricative

Nasals can form complex codas with stops and affricates. But it seems that nasals rarely form complex codas with fricatives. In the Kouyoumdjian dictionary, we found only few words that have a syllable /VNC/ cluster a) where N is a nasal and b) C is a fricative (Table 5.33). All of these examples had the cluster /ms/.

Table 5.33: Tautosyllabic CC clusters where C1 is nasal /m/ and C2 is fricative /s/ (total n=18; word-final n=18)

ˈko**ms**	'count'	կոմս
ˈdo**ms**	'ticket'	տոմս
təˈɑm + ˈdo**ms**	'money + ticket'	դրամ, տոմս
→ təɾɑm-ɑ-ˈdo**ms**	'banknote'	դրամատոմս

There are two generalization. First, it seems that the nasal /m/ can form a cluster with only the fricative /s/, and with no other fricative. Based on this, we argue that the /s/ in these words is actually an extrasyllabic appendix /-s/, not part of a complex coda. See Section §5.2.3 for more data on this appendix.

As counter-examples, we only found a few cases of orthographic <m>+fricative clusters on Armenian Wiktionary. All these were either obvious loanwords like [təɾijumf] 'triumph' տրիումֆ, dialectal words, or words that can exist in Eastern Armenian but not Western.

Second, the nasal /n/ seems to avoid forming word-final complex codas with fricatives. We found zero such examples in Kouyoumdjian. On Armenian Wiktionary, the attested examples look primarily as either loanwords like [ɑlijɑns] 'alliance' ալիանս or [oɾɑnʒ] 'orange' օրանժ, or obscure dialectal words from non-Western Armenian.

Word-finally, orthographic /ns/ is always pronounced as [nəs] with schwa epenthesis. The /s/ here is a suffix. The /-s/ typically marks first person possession like in /im-s/ [iməs] 'mine' իմս, but it also is used as an archaic obsolete derivational suffix from Classical Armenian: /voɾgɑn-s/ [voɾ.gɑ.nəs] 'the present' որկանս.

However, it seems easier to find some types of /n/+fricative complex codas inside words than at the end of words. See §5.4.2.3 for word-medial data. But note that word-medial nasal-fricative sequences are still atypical.

5.2.2.7 Rhotic /ɾ/ + nasal /n/

The rhotic /ɾ/ can form a complex coda with any obstruent and with the nasal /m/. But with the nasal /n/, we find the following complications:

- with monosyllabic roots, the /ɾn/ cluster preferably gets epenthesis [ɾən], but schwa-less [ɾn] is possible in casual speech for some words
- with polysyllabic roots, the /ɾn/ cluster avoids epenthesis to get [ɾn]
- with compounds that have a monosyllabic final root, the /ɾn/ again prefers epenthesis [ɾən]

More restrictions are found when the /ɾn/ sequence is word-medial, discussed in Section §5.4.2.2.

For the first group of 'monosyllabic roots' (Table 5.34), the root orthographically has one vowel followed by a cluster <rn> րն or <ṙn> ռն: <xaṙn> 'mixed'. Prescriptively, the final rhotic-nasal sequence is pronounced with an intervening epenthetic schwa: [χɑɾən]. This is the 'prescriptive rule' to pronounce these words. In casual speech, the norm is to also use the schwa, but it is possible to delete the schwa: [χɑɾn].

Table 5.34: Final CC clusters where C1 is rhotic /ɾ/ and C2 is nasal /n/ – ‘monosyllabic’ root

<xaṙn>	[ˈχɑɾən]	‘mixed’	խառն
<t͡seṙn>	[ˈt͡seɾən]	‘hand’	ձեռն
<taṙn>	[ˈtɑɾən]	‘bitter’	դառն
<saṙn>	[ˈsɑɾən]	‘mountain’	սառն
<puṙn>	[ˈpuɾən]	‘fist’	բուռն
<tʾoṙn>	[ˈtoɾən]	‘grandson’	թոռն

In terms of type frequency in the dictionary, we found 177 words with word-final [ɾən] syllables.

It is an open question on how often the schwa-less forms are used in natural speech, and it’s unknown what linguistic or extra-linguistic factors would condition the optional use of the schwa-less form.

Alongside the above orthographically monosyllabic roots, there are also orthographically polysyllabic roots (Table 5.35). These roots contain two orthographic vowels, and end in a rhotic-nasal sequence: <eɣeṙn> ‘crime’. However for pronunciation, the norm is to **not** add a schwa: [jeʁeɾn]. For some of these words though, the use of schwa is possible: [titeɾ(ə)n] ‘butterfly’.

Table 5.35: Final CC clusters where C1 is rhotic /ɾ/ and C2 is nasal /n/ – ‘polysyllabic’ root

<eɣeṙn>	[jeˈʁeɾn]	‘crime’	եղեռն
<tʾitʾeṙn>	[tiˈteɾn]	‘butterfly’	թիթեռն
<agaṙn>	[ɑˈgɑɾn]	‘citadel’	ակառն
<gawaṙn>	[gɑˈvɑɾn]	‘trench’	կաւառն
<ʃowaṙn>	[ʃəˈvɑɾn]	‘lance’	շուառն
<liseṙn>	[liˈseɾn]	‘axle-tree’	լիսեռն

In terms of type frequency in the dictionary, we found 35 words with word-final [CVɾn] syllables (cf. 35 cases throughout the word).

In terms of token frequency, the two groups have impressionistically different usages. The monosyllabic roots are very common words, while the polysyllabic roots are all very low-frequency. It is possible that the difference in prosodic behavior is tied with this frequency difference.

It is possible that the over-arching generalization is that word-final [ɾn] complex codas prefer being in minimally bisyllabic words, thus triggering epenthesis in a word like [χɑɾən] ‘mixed’. However, when compounds are formed from

these /ɾn/-final words (Table 5.36), we find that the compound inherits the schwa behavior of its component stems: [ɑvɑz-ɑ-ˈχɑɾən] ‘sand mixed’.

Table 5.36: Final CC clusters where C1 is rhotic /ɾ/ and C2 is nasal /n/ – derived compounds

ɑˈvɑz + ˈχɑ**ɾən**	‘sand + mixed’	աւազ, խառն
→ ɑvɑz-ɑ-ˈχɑ**ɾən**	‘sand mixed’	ազատախառն
sɑˈgɑv + ˈt͡se**ɾən**	‘few + hand’	սակաւ, ձեռն
→ sɑgɑv-ɑ-ˈt͡se**ɾən**	‘sand mixed’	սակաւաձեռն
teˈtev + ˈpe**ɾən**	‘light + load’	թեթեւ, բեռն
→ tetev-ɑ-ˈpe**ɾən**	‘sand mixed’	թեթեւաբեռն
hɑˈɾʏɾ + ˈtu**ɾən**	‘hundred + door’	հարիւր, դուռն
→ hɑɾʏɾ-ɑ-ˈtu**ɾən**	‘having a hundred doors’	հարիւրադուռն

Polysyllabic roots like [titeɾ(ə)n] butterfly also percolate their lack of a schwa to compounds. But such compounds are even rarer than their component root: [ɑt͡ʃ-ɑ-tiˈte**ɾ(ə)n**] ‘pavonian butterfly’.

5.2.2.8 Lateral /l/ + obstruent

For the lateral /l/, this sound seems to avoid starting a complex coda in native Armenian words.

When C1 is /l/ and C2 is anything but the nominalizer /k/ (Table 5.37), the Kouyoumdjian dictionary lists only the following words. Most look like obvious loanwords.

Table 5.37: Final CC clusters where C1 is lateral /l/ and C2 is not nominalizer /k/

ˈɑ**lp**	‘alpha’	ալփ	ge**lb**	‘kelp’	կելպ
ˈvo**lt**	‘volt’	վոլթ	vo**lt**.ˈt͡ʃɑp	‘voltmeter’	վոլթչափ
ɑsˈpɑ**ld**	‘asphalt’	ասփալտ	go ˈpɑ**ld**	‘cobalt’	կոբալտ
əsˈpɑ**ld**	‘spalt’	սպալտ	bɑˈsɑ**ld**	‘basalt’	պասալտ
bɑˈzɑ**ld**	‘basalt’	պազալտ	ˈtɑ**lg**	‘talc’	թալկ
ˈtɑ**lk**	‘talc’	թալք	go ˈbɑ**lg**	‘copalche’	կոպալկ
he**lg**.ɾɑ.ˈmid	‘hellgramite’	հելկրամիտ	ˈvɑ**ls**	‘waltz’	վալս
ˈkɑ**lχ**	‘horned cumin’	քալխ	ˈʁu**lb**	‘common gromwell’	ղուլպ
sɑˈpɑ**ld**	‘raw fruit’	սաբալտ	χe**ld**.ˈɾug	‘trefoil’	խելտրուկ

Many such clusters are found on Armenian Wiktionary. Again, it seems that many of Wiktionary’s examples are loanwords, obscure flora/fauna, or obscure

dialectal words. Whether or not there are some Armenian dialects that allow /lC/ doesn't say anything about the avoidance of such clusters in Western Armenian.

The above data show that /lC/ clusters are generally restricted to non-native words. Spurious counter-examples are words with a final /lk/ sequence (Table 5.38). But here, the /k/ is part of the nominalizer suffix *-k* -ք; so all these words could be syllabified as ending in a coda /l/ + appendix /k/.

Table 5.38: Tautosyllabic CC clusters where C1 is lateral /l/ and C2 is nominalizer /k/ (total n=33; word-final n=32)

ˈje**l-k**	'ascent'	ելք	cf. jeˈl-ad͡z	'risen' (√-RPTCP)	ելած
ɑɾeˈv+e**l-k**	'east'	արեւելք	cf. ɑˈɾev	'sun'	արեւ
ɑɾev+e**l-k**-t͡si	'eastern'	արեւելքցի			
ˈχe**l-k**	'mind'	խելք	cf. χel-ɑˈt͡si	'smart' (√-ADJZ)	խելացի
t͡soˈ**l-k**	'flash'	ցոլք	cf. t͡soˈl-ɑ-l	'to flash' (√-TH-INF)	ցոլալ
ɑɾˈke**l-k**	'obstacle'	արգելք	cf. ɑɾkeˈl-it͡ʃ	'preventative' (√-ADJZ)	արգելիչ
deˈsi**l-k**	'apparition'	տեսիլք	cf. deˈsil	'sight'	տեսիլ

Outside of the appendix /l-k/, the sequence /lk/ undergoes schwa epenthesis word-internally instead of forming a complex coda. For example, /slktɑl/ → [sə.lək.tɑl] and not **sə̀lk.tal* 'to lounge' սլքտալ.

There may be a diachronic reason for the rarity of [lC] complex codas. The modern Armenian /l/ <լ> and /ʁ/ <ղ> descend from Classical Armenian /l/ and /ɫ/. Classical /l/ and /ɫ/ were separate phonemes but they had a partial complementary distribution in the native lexicon of Classical Armenian, such that /l/ is word-initial while /ɫ/ is pre-consonantal (Macak 2016: 70ff). The distribution is partial because there are many exceptions. But it is possible that this old distribution is the reason why so few words exist where the modern /l/ is part of a [lC] complex coda.

5.2.2.9 Lateral /l/ + nasal /m/

We could not find any native words with a complex coda /lm/ in our dictionary. The only case we found was the obvious loanword [film] 'film' ֆիլմ. Wiktionary likewise only had loanwords.

Schwa epenthesis also targets /lm/ word-internally to prevent complex codas: /plmpoʃ/ →[pə.ləm.poʃ] and not **pə̀lm.poʃ* 'awkward' փլմփոշ.

5.2.2.10 Lateral /l/ + nasal /n/

The lateral /l/ generally avoids forming a complex coda with obstruents, with such clusters largely restricted to loanwords (§5.2.2.8). When a word ends in an

orthographic <ln> sequence (Table 5.39), we find obligatory schwa epenthesis.

Table 5.39: Schwa epenthesis in final CC clusters where C1 is lateral /l/ and C2 is nasal /n/

<ow**ln**>	[ˈu**lən**]	'neck'	ուլն
<anow**ln**>	[ɑˈnu**lən**]	'Spanish spider'	անուլն
<part͡sea**ln**>	[pɑɾtˈɾjɑ**lən**]	'the Most High (God)'	բարձրեալն

We found 12 words with word-final syllable [lən]. Add to this, the word /jevɑjln/ → [je.vɑj.lən] 'etc' եւայլն with word-final [Clən] .

5.2.3 Flat or rising sonority and either rare, extrasyllabic, or uses schwa epenthesis

Armenian orthography has many words that end in a consonant cluster that has flat or rising sonority. Some of these form rare complex codas, some are likely a coda + appendix, and some undergo schwa epenthesis.

1. When C1 is a stop:
   - + stop or affricate: rare and most are either loanwords or coda + appendix /-k/ (§5.2.3.1)
   - + fricative: rare, some are likely a coda + appendix, and some take schwa epenthesis (§5.2.3.2)
   - + nasal: either loanword or takes schwa epenthesis (§5.2.3.3)
   - + rhotic /ɾ/: usually schwa epenthesis, but schwa-less forms are possible (§5.2.3.4)
   - + lateral /l/: loanwords and take schwa epenthesis (§5.2.3.5)
2. When C1 is an affricate:
   - + stop: rare and most are either loanwords or coda + appendix /-k/ (§5.2.3.6)
   - + fricative: rare and most undergo schwa epenthesis (§5.2.3.7)
   - + sonorant: rare and undergoes schwa epenthesis (§5.2.3.8)
3. When C1 is a fricative:
   - Fricative + fricative: most use a fricative appendix, and some have schwa epenthesis (§5.2.3.9)
   - + nasal /m/: relatively common and likely just coda + appendix (§5.2.3.10)

- + nasal /n/: relatively common and undergoes schwa epenthesis (§5.2.3.11)
- + rhotic /ɾ/: rare and undergoes schwa epenthesis, with possible schwa elision (§5.2.3.12)
- + lateral /l/: rare and undergoes schwa epenthesis (§5.2.3.13)

4. When C1 is a nasal:
   - + nasal /m/: rare and undergoes schwa epenthesis (§5.2.3.14)
   - + nasal /n/: common and undergoes schwa epenthesis (§5.2.3.15)
   - + rhotic /ɾ/: rare and undergoes schwa epenthesis (§5.2.3.16)
   - + lateral /l/: rare and undergoes schwa epenthesis (§5.2.3.17)

Data in this section expands on §5.1.4.

5.2.3.1 Stop + stop or affricate

Orthographically, syllable-final stop + affricate clusters are exceedingly rare. We found no cases in Kouyoumdjian. As for Armenian Wiktionary, they seem restricted to loanwords like [əskottʃ͡~əskotʃ͡] 'scotch tape' սքոթչ.

We did find cases of final stop-stop clusters. Such final stop-stop clusters are limited to three categories: some roots, some loanwords, and an open class of words where the nominalizer *-k* is added after root-final stop.

For the first category, we found some native roots on Wiktionary, such as [t͡sɑdg] 'jump' ցատկ and [dɑgd] 'musical bar' տակտ on Wiktionary. Kouyoumdjian did not have any such roots. Word-medially, a handful more cases are found like [d͡zə**b**d.jɑl] 'incognito' ծպտեալ, but HD fees that the avoidance of this complex coda feels more common: [d͡zə**b.dj**ɑl].

For the second category, the Kouyoumdjian dictionary listed only the following words which had stop-stop complex coda (Table 5.40). These words seem to be loanwords.

Table 5.40: Syllable- CC clusters where C1 and C2 are stops and the word is a loanword

t͡ʃɑlˈʁu**bd**	'true jalap'	ջալղուպտ
kɑˈɾi**pt**	'charpybdis'	քարիբդ
mɑ**gd**.ˈɾɑ	'bread-pan'	մակտրայ
je.le**gd**.ɾo.gɑn	'electrical'	ելեկտրական

More loanwords are found on Wiktionary, such as [fɑgd] 'fact' ֆակտ. As for the third category, The segment *k* is special in how it can follow any type of consonant or consonant-cluster. This is because Armenian has a nominalizer suffix

-*k* -ք that is used to form many nouns. This suffix can be added after any consonant even if it creates a flat-sonority cluster. When this suffix is added, it triggers the devoicing of preceding obstruents (§4.3.1.4).

In Kouyoumdjian's dictionary, the nominalizer -*k* is quite common after any stop: [d͡zu**p-k**] 'fluctuation' (Table 5.41). It triggers devoicing of preceding stops: [jɛɾɑχʃe**p-k**] 'torment'. It can also follow a velar stop /k, g/ to create a long (geminate) version of itself: [kɑ**k-k**] 'separation'. Almost all of these cases involve word-final complex codas, and not word-medial. See §5.4.2.5 on possible word-medial restrictions against /Ck/ clusters.

Table 5.41: Syllable-final CC clusters where C1 is a stop and C2 is an appendix /k/.

				Final #	Total #
/p-k/→[p-k]	ˈd͡zu**p**	'fluctuation' (√)	ծուփ	n=10	n=10
	ˈd͡zu**p-k**	'fluctuation' (√-NMLZ)	ծուփք		
/b-k/→[p-k]	jɛɾɑχˈʃe**b**	'scar' (√)	երաշխէպ		
	jɛɾɑχˈʃe**p-k**	'torment' (√-NMLZ)	երաշխէպք		
/t-k/→[t-k]	ɑʁoˈ**t**-e-l	'to pray' (√-TH-INF)	աղօթել	n=57	n=58
	ɑˈʁo**t-k**	'prayer' (√-NMLZ)	աղօթք		
/d-k/→[t-k]	ɑˈʁe**d**	'misfortune' (√)	աղէտ		
	ɑˈʁe**t-k**	'calamity' (√-NMLZ)	աղէտք		
/k-k/→[k-k]	ˈho**k**	'concern' (√)	հոգ	n=44	n=44
	ˈho**k-k**	'hindrance' (√-NMLZ)	հոգք		
/g-k/→[k-k]	ˈkɑ**k**	'to untie' (√-TH-INF)	քակել		
	ˈkɑ**k-k**	'separation' (√-NMLZ)	քակք		

5.2.3.2 Stop + fricative

It is relatively rare to find word-final orthographic clusters that end in a stop + fricative. In the Kouyoumdjian dictionary, we found the following categories: stop + /s/, stop + /χ/, and stop + /ʁ/. For /Cs/ and /Cχ/, the stop-fricative cluster is pronounced together without schwa epenthesis. The final fricative acts as an appendix. But for /Cʁ/, we find schwa epenthesis. However word-medially, the clusters seem to prefer schwa epenthesis.

For stop + /s/, we found many examples (Table 5.42). The stop can be a voiceless [p, t, k].

Table 5.42: Sylalble-final CC clusters where C1 is a stop and C2 is a fricative /s/

				Final #	Total #
[ps]	ˈti**ps**	‘sirup’	տիբս	n=3	n=3
	jeˈɾe**ps**	‘priest’	երեփս		
[ks]	meˈdɑ**ks**	‘silk’	մետաքս	n=17	n=17
	əstɑˈmo**ks**	‘stomach’	ստամոքս		

For the stop + /s/ clusters, it’s difficult to tell how many of these are purely native words vs. old nativized borrowings, such as the word for [əstɑmoks] ‘stomach’ which has an old Greek origin (Աճառեան 1926: 269), or the obvious loanword [gegɾops] ‘Cecrops’ Կեկրոպս.

For some words, the final /s/ is often parsed as a possessive suffix /-s/. This suffix is pronounced with a schwa after a consonant: /pɑɾ-s/ [pɑɾ-əs] ‘my finger’ բառս. We discard thes words.

For /t/+/s/, we seee schwa epenthesis. This is beacuse of two factors. The first factor is that /s/ is typically a suffix. For an archaic low frequency word like <yeds> յետս ‘behind’, it is parsed as complex word with a root /hed/ ‘with’ and possessive suffix /s/: [hedəs]. Second, the /ts/ as a complex coda would sound the same as an affricate /t͡s/. So such a word is instead pronounced with schwa epenthesis. Consider archaic monomorphemic: /hɑnbeds/ → [hɑnbedəs] ‘vainly’ յանպէտս

Word-medially, stop + /s/ is avoided as a complex coda. For the bound root /bɑdsbɑɾ-/, its derivatives use schwa epenthesis: [bɑdəsbɑɾ-il] ‘to shelter’ պատըսպարիլ

For stop + /χ/, we found only one example in Kouyoumdjian: [dipχ] ‘common mistletoe’ տիպխ.

For stop + /ʁ/, these are relatively few (Table 5.43). They take schwa epenthesis.

Table 5.43: Schwa epenthesis in word-final CC clusters where C1 is a stop and C2 is a fricative /ʁ/

<bɣ>→[bəʁ]	<gowb**ɣ**>	ˈgub**əʁ**	‘padlock’	կուպղ	n=2
<dɣ>→[dəʁ]	<sid**ɣ**>	ˈsid**əʁ**	‘waterpot’	սիտղ	n=8
<kɣ>→[kəʁ]	<ʃik**ɣ**>	ˈʃik**əʁ**	‘hock’	շիգղ	n=n=3
<gɣ>→[gəʁ]	<sig**ɣ**>	ˈsig**əʁ**	‘shekel’	սիկղ	n=1

[redacted]

Based on comparing the Kouyoumdjian dictionary against Wiktionary, it seems that all other possible stop + fricative clusters are either unattested, found in non-Western dialects, or are loanwords: [vakf] ‘waqf’ վակֆ.

5.2.3.3 Stop + nasal

There are words which end in an orthographic stop + nasal cluster. In native words, the nasal is always /n/. We could not find native words with /m/.

For final stop + /m/ clusters, Kouyoumdjian had no such examples. On Wiktionary, we found some examples but they all seem like loanwords: [ɾitm~ɾitəm] ‘rhythm’ ռիթմ with unclear syllabification.

As for stop + /n/, we always find schwa epenthesis in pronunciation (Table 5.44). Many of these words though have an archaic connotation, or are restricted to high-level formal registers.

Table 5.44: Schwa epenthesis in final CC clusters where C1 is a stop and C2 is a nasal /n/

<p(’)n>→[pən]	<a**p’n**>	ˈɑ**pən**	‘shore (archaic)’	ափն	n=8
<t(’)n>→[tən]	<ka**t’n**>	ˈɡɑ**tən**	‘milk (archaic)’	կաթն	n=3
<dn>→[dən]	<o**dn**> <ma**dn**>	voˈ**dən** mɑˈ**dən**	‘foot (archaic)’ ‘finger (archaic)’	ոտն մատն	n=103
<k(’)n>→[kən]	<yantow**kn**> <yo**kn**>	hɑnˈtu**kən** ˈho**kən**	‘audacious’ ‘numerous’	յանդուգն յոքն	n=18
<g(’)n>→[gən]	<t͡s’a**gn**> <arekaɡ**n**>	ˈt͡sɑ**gən** ɑɾeˈkɑ**gən**	‘misery’ ‘sun’	ցակն	n=76

We could not find final <bn>→[bən] clusters on either Kouyoumdjian or Wiktionary, but this is likely an accidental gap.

For final <dn>-[dən], Kouyoumdjian listed 103 such words. The vast majority of them (n=99) were derived from the words ‘foot’ or finger’: [vɑɾt-ɑ-mɑdən] ‘rose-fingered’ վարդամատն.

The reason why many of these words have an archaic connotation is because of word-final nasal deletion in final <Cn> cluster. Such a nasal deletion happened in the development of Classical Armenian to Modern Armenian. Thus the Classical word for ‘finger’ is traditionally pronounced as [mɑdən] մատն, while the modern common word is just [mɑd] մատ. The nasal is retained in archaic-sounding words and their derivatives. See cite nasal liaison for more discussion on such nasals.

5.2.3.4 Stop + rhotic /ɾ/

Word-finally, stop + /ɾ/ usually undergo schwa epenthesis (Table 5.45). This is the prescriptive norm, but there is a degree of variation. The rhotic /ɾ/ can be orthographically either <r> ր or <ṙ> ռ.

Table 5.45: Schwa epenthesis in final CC clusters where C1 is a stop and C2 is a rhotic /ɾ/

<pr>→[pəɾ]	<i**pr**>	ˈi**pəɾ**	'as'	իբր	n=1
<br,bṙ>→[bəɾ]	goi**br**>	ˈgu**bəɾ**	'tar'	կուպր	n=4
<t(')ṙ>→[təɾ]	<k'ara**tr**> <mē**t'r**>	kɑˈɾɑ**təɾ** ˈme**təɾ**	'plover' 'meter'	քարադր մէթր	n=14
<dr>→[dəɾ]	<nō**dr**> <a**dr**>	ˈno**dəɾ** ˈɑ**dəɾ**	'cursive' 'flame'	նօտր ատր	n=33
<k'r>→[kəɾ]	<da**kr**>	ˈdɑ**kəɾ**	'brother-in-law'	տագր	n=6
<gr,gṙ>→[gəɾ]	<sa**gr**>	ˈsɑ**gəɾ**	'axe'	սակր	n=3

Prescriptively, the norm is to have a schwa in a word-final stop-rhotic cluster: <p'ok'r> → [pokəɾ] 'small' փոքր. But in casual speech, some of these words can be pronounced without a schwa: [pokɾ]. The absence of this schwa is possibly tied with schwa elision (§4.6.4). The absence of this schwa does however vary across the types of stops. For example, [kəɾ] sequences can easily simplify to [kɾ] in speech: [vɑkəɾ, vɑkɾ] 'tiger' վագր. As can [təɾ] sequences and [dəɾ] sequences: [metəɾ, metɾ] 'meter' մէթր, [dedəɾ, dedɾ] 'tract' տետր. But [pəɾ] clusters don't: [ipəɾ] 'as' իբր.

5.2.3.5 Stop + lateral /l/

Falling or rising sonority clusters with a final /l/ are quite rare. In the Kouyoumdjian dictionary, we found only 2 words that ended in an orthographic stop + <l> cluster. One is an obvious borrowing and gets schwa epenthesis: <kowntsdabl>→[kuntəstɑbəl] 'constaple' գունդստապլ. The other is a possible native word. We're unfamiliar with this word but we think it should be pronounced with a schwa as well: <mowgl>→[mugəl] 'false myrrh' մուկլ.

Outside of dictionary, we have come across other loanwords with final orthographic <Cl> cluster. Such clusters get epenthesis again: <t͡s'igl>→[t͡sigəl] 'cycle' ցիկլ.

5.2.3.6 Affricate + stop

Affricates and stops both have low sonority. They usually cannot form a complex coda together. There are two classes of exceptions: a handful of roots , a handful of loanwords, and a large set of appendix-final words. This third category consists of words that end in the nominalizer *-k*.

For the first class, the Kouyoumdjian dictionary listed only one word which has an affricate-stop coda. That word is a compound [sɑl-ɑ-d͡zɑd͡zg] ‘paved with stones’ that used the root [d͡zɑd͡zg] ‘cover’ ծածկ.

For the second category of loanwords, Armenian Wiktionary list some examples such as [tone͡tsk] ‘Donetsk’ Դոնեցկ.

The third category are words with the nominalizer *-k*, which violates all the syllable structure rules of Armenian (Table 5.46). This suffix can follow any consonant, including affricates, and it triggers devoicing. This suffix is analyzed as being an extrasyllabic segment simply because it consistently acts in a bizarre fashion.[3]

Table 5.46: Final CC clusters where C1 is an affricate and C2 is an appendix /k/

/t͡s-k/→[t͡s-k]	t͡ʃɑˈχɑ**t͡s**	‘mill’ (√)	ջաղաց	n=55
	t͡ʃɑˈχɑ**t͡s-k**	‘mill’ (√-NMLZ)	ջաղացք	
/d͡z-k/→[t͡s-k]	dɑɾɑˈ**d͡z**-e-l	‘to spread’ (√-TH-INF)	տարածել	n=71
	dɑɾɑˈ**t͡s-k**	‘spread’ (√-NMLZ)	տարածք	
/t͡ʃ-k/→[t͡ʃ-k]	χɑχɑˈ**t͡ʃ**-e-l	‘to gargle’ (√-TH-INF)	խախաջել	n=23
	χɑˈχɑ**t͡ʃ-k**	‘gargle’ (√-NMLZ)	խախաջք	
/d͡ʒ-k/→[t͡ʃ-k]	ˈd͡ʒo**d͡ʒ**	‘oscillation’ (√)	ճօճ	n=4
	ˈd͡ʒo**t͡ʃ-k**	‘swing’ (√-NMLZ)	ճօճք	

Note that the sequence [t͡sk] is quite common because the nominalizer *-k* can be added after the resultative participle suffix *-ad͡z*, and cause devoicing: [ɑjɾ-ɑ**d͡z**] ‘burnt’ (√-RPTCP այրած and [ɑjr-ɑ**t͡s-k**] ‘scald’ (√-RPTCP-NMLZ) այրածք.

5.2.3.7 Affricate + fricative

It is relatively rare to find final orthographic clusters of an affricate + fricative. In the Kouyoumdjian dictionary, we found only one word [pit͡ʃχ] ‘common ivy’

[3] Armenian Wiktionary also lists stop-stop clusters that are were dialectal non-Western words which seemed to use a cognate of the nominalizer *-k*. For example, we speculate that the entry <t’inamet͡ʃ’g> թինամեչկ is one such entry, and Wiktionary just defines it as a possible cognate to some word թիկնամեջք <t’iknamet͡ʃ’k’> which ends in the nominalizer *-k*.

փիչխ where the affricate-fricative cluster is pronounced together without a schwa. The final /χ/ is one of the possible fricative appendixes that can be added at the end of syllables.

Besides this word, we found clusters which undergo schwa epenthesis: affricate + /s/, affricate + /ʁ/. Epenthesis is the norm (Table 5.47).

Table 5.47: Schwa epenthesis in final CC clusters where C1 is an affricate and C2 is a fricative

<d͡ʒɣ>→[dʒəʁ]	<go**d͡ʒɣ**>	ˈgo**d͡ʒəʁ**	‘stump’ կոճղ	n=2
<t͡s’ɣ>→[t͡səʁ]	<p’o**t͡s’ɣ**>	ˈpo**t͡səʁ**	‘rake’ փոցղ	n=1
<t͡s’s>→[t͡səs]	<tʃ’ori**t͡s’s**>	tʃoˈɾi**t͡səs**	‘four times’ չորիցս	n=15

For affricate + /ʁ/, some of these words allow optional deletion of the schwa: [go**d͡ʒəʁ**, go**d͡ʒʁ**] ‘stump’. This is part of schwa elision (§4.6.4).

For <t͡s’s>→[t͡səs], this sequence seems to be restricted to numeral-related words, e.g., <vet͡sit͡ss>→[vet͡sit͡səs] ‘six times’ վեցիցս.

5.2.3.8 Affricate + sonorant

It is extremely rare to find any words which end in an orthographic affricate+sonorant cluster, such as hypothetical /t͡s+n/, /d͡ʒ+ɾ/, and so on. We found zero such cases in the Kouyoumdjian dictionary. Armenian Wiktionary also has few clear cases of such clusters. However, the data that we find shows that affricate + sonorant clusters pattern like stop+sonorant clusters in undergoing schwa epenthesis.

Although affricate + nasal clusters are un-attested, HD’s intuition is that their syllabification is the same as stop+sonorant cluster. That is, because word-final /dn/ clusters undergo schwa epenthesis, then so does word-final /t͡sn/. For example, a nonce word մացն <mat͡s’n> is syllabified as [mɑt͡sən]. But of course, because these words don’t exist, we can’t say much about them. We suspect that their absence is more of an accidental gap.

For affricate + rhotic clusters, the Kouyoumdjian dictionary doesn’t list any words. But Wiktionary lists a handful. Here we find schwa epenthesis: <t͡s’ad͡zr>→[t͡sɑd͡zəɾ] ‘low’ ցածր.

5.2.3.9 Fricative + fricative

Fricatives can take any stop to form a complex coda. Root-final fricative-fricative clusters do exist. But these are severely limited to a handful of fricative combinations, and to an apparent finite number of roots.

There are largely two categories of root-final fricative-fricative clusters. The first category are orthographic clusters that are pronounced as they are spelled, creating a consonant cluster in pronunciation: <d͡zaxs> [d͡zɑχs] ‘cost’ ծախս. We analyze this category of clusters as containing an extrasyllabic final fricative. The second group is words that are spelled with a fricative-fricative cluster, but this cluster undergoes schwa epenthesis in speech: <ews> [jevəs] ‘moreover’ եւս.

We discuss the epenthetic group first because it is an extremely small class (Table 5.48). In Kouyoumdjian’s dictionary, we only found 3 words that end in an orthographic fricative-fricative cluster, and that get schwa epenthesis. These 3 words all use /vs/ and they are function words.

Table 5.48: Schwa epenthesis in final CC clusters where C1 and C2 are fricatives

<e**ws**>	/(j)e**vs**/	[ˈje**vəs**]	‘moreover’	եւս
<aj**lews**>	/ɑj**levs**/	[ɑjˈle**vəs**]	‘anymore’	այլեւս
<t’e**rews**>	/te**revs**/	[teˈɾe**vəs**]	‘perhaps’	թերեւս

For the category of pronounced clusters, there are a handful of roots which end in a fricative + /χ,ʃ,s/ (Table 5.49). The most frequent C2 is /χ/. Each type of /Cχ/ cluster is found in a handful roots: [χɑɾisχ] ‘anchor’ խարիսխ. Such roots can be productively used to form new words via compounding and prefixation: [sɑɾ-ɑ-χɑɾisχ] ‘ice anchor’ սառախարիսխ

Table 5.49: Final CC clusters where C1 is fricative and C2 is /χ/

				# of roots	# of derivatives
/sχ/	χɑˈɾi**sχ** χoˈɾi**sχ**	‘anchor’ ‘honey-comb’	խարիսխ խորիսխ	n=2	n=11
/ʃχ/	ˈvɑ**ʃχ** jeˈɾɑ**ʃχ**	‘usury’ ‘surety’	վաշխ երաշխ	n=7	n=44
/χχ/	ˈze**χχ** ˈse**χχ**	‘lewd’ ‘melon’	զեղխ սեղխ	n=8	n=18

For the /χχ/ clusters, the two segments are spelled differently <ɣx> ղխ, suggesting that they diachronically arose from two separate segments that eventually became an identical long (geminate) segment. In fact, one of the examples from Kouyoumdjian is the word ‘melon’ [seχχ] սեղխ. The more common rendition of this word is however [seχ] սեխ where there is no gemination.

The other pronounced fricative-fricative clusters are /fs, fʃ, χs/ (Table 5.50) which are all restricted to a handful of roots in Kouyoumdjian's dictionary. From this set, only word [d͡zɑχs] 'cost' and its derivatives are frequent. The word 'star' can vary in pronunciation: <asdɣ> [ɑstχ, ɑsχ] 'star' աստղ (§5.3.2.2).

Table 5.50: Final CC clusters where C1 is fricative and C2 is /s,ʃ/

				# of roots	# of derivatives
/fʃ/	ˈd͡ʒɑ**fʃ**	'breast-plate'	ճաւշ	n=1	n=0
/fs/	ˈze**fs**	'Zeus'	Զեւս	n=1	n=0
/χs/	ˈd͡zɑ**χs**	'cost'	ծախս	n=2	n=4
	ˈtu**χs**	'incubation'	թուխս		

On Armenian Wiktionary, we also found a handful of instances in non-dialectal words: [nɑ**ʃχ**] 'pattern' նաշխ, [kedən-so**χs**] 'earth-crawling' գետնսողս.

Based on the above data, it is obvious that fricative-fricative clusters are highly restricted in terms of a) what combinations are attested, and b) how many roots have these clusters. Because both of these factors are small, these post-fricative /ʃ, χ, s/ fricatives have been analyzed as being extrasyllabic appendixes cite vaux. Meaning that a word like [d͡zɑχs] 'cost' doesn't end in a genuine complex coda [χs], but that the syllable is [d͡zɑχ] and the /s/ is added outside the syllable.

5.2.3.10 Fricative + nasal /m/

The nasal /m/ has quite bizarre syllabification sometimes. As part of a complex coda, /m/+ fricative clusters are generally rare (§5.2.2.6). They are rare in that a) few words exist with an /m/+fricative cluster, and b) few of these existing words are high-frequency. In contrast, it is quite common to find words with a fricative + /m/ cluster (Table 5.51). And many of these words are high-frequency.

Table 5.51: Final CC clusters where C1 is fricative and C2 is a nasal /m/

				# of roots	# of derivatives
[zm]	ˈgɑ**zm**	‘construction’	կազմ	n=6	n = 39
	ˈχɑ**zm**	‘quarrel’	խազմ		
[hm]	ˈdo**hm**	‘family’	տոհմ	n=1	n=15
[ʁm]	ˈho**ʁm**	‘wind’	հողմ	n=12	n=33
	ˈgo**ʁm**	‘side’	կողմ		
	ˈme**ʁm**	‘soft’	մեղմ		
[ʃm]	təˈɾo**ʃm**	‘stamp’	դրոշմ	n=5	n=11
	[ɑ**ʃm**]	‘jade’	աշմ		

For /zm/, the Kouyoumdjian dictionary lists 6 roots that end in [zm]. Of these roots, the root [gɑzm] ‘construction’ was used to derive other words, specifically 39 compounds and prefixed words like [noɾ-ɑ-gɑzm] ‘newly-formed’ նորակազմ (literally ‘new+form’). Then there is a handful (n=5) of borrowings with the foreign suffix *-izm* like [fɑʃi**zm**] ‘fascism’ ֆաշիզմ.

Note that for /zm/ clusters, some sources report schwa epenthesis cite vaux. For example, the word [gɑzm] ‘construction’, the typical pronunciation is to not use a schwa. But Vaux reports a schwa form [gɑzəm]. It is possible that such a schwa is a transient vowel that’s created by going from [z] to [m].

For /hm/, only one root in the dictionary had this cluster: [dohm] ‘family. All other words were compounds from this root like [ɑzɑd-ɑ-dohm] ‘noble’ ազատակողմ, literally ‘free+family’.

For /ʁm/, we found 12 roots. Of that 12, [hoʁm] and [goʁm] form all the derived words that have the /ʁm/ cluster: [vet͡s-ɑ-goʁm] ‘hexagonal’ վեցակողմ, literally ‘six+side’.

For /ʃm/, we found 5 roots. Of these roots, [təɾoʃm] formed all the derived forms with this cluster: [nɑmɑg-ɑ-təɾoʃm] ‘postage-stamp’ նամակադրոշմ, literally ‘letter+stamp’.

For /Vʒm/ words, we found no such cases. The closest was a <ayʒm> [ɑjʒəm] ‘now’ այժմ where the <ʒm> cluster is after a consonant. The fact that this word is pronounced as [ɑjʒəm] and not [ɑjəʒm] suggests either that a) [ʒm] can’t be a complex coda or coda+appendix cluster, or b) schwa epenthesis avoids creating [jə] sequences if possible. Data is obviously too limited to know.

Besides the above words, we found only one case of orthographic final <sm> in the Kouyoumdjian dictionary: <t͡ʃasm> ‘chimera’ ջասմ. This word is likely an old loanword. We’re not if a schwa is needed here: [tʃɑs(ə?)m]. Similarly we found one case of <vm>: <hrovm> ‘Rome’ հրովմ. We think the pronunciation is likely

without a schwa before the nasal: [hərovm]. Part of the ambiguity is that these words seem like obvious loanwords and HD never heard of such clusters before before. And finally, one case of <xm> is an obvious loanword without a schwa: <traxm> [tərɑχm] 'Greek currency' դրախմ.

In sum, fricative + /m/ clusters are pretty common. But we're not sure why. The fact that only /m/ can form these clusters but not /n/ suggests that there is something special about the nasal /m/. mention vaux. We treat this /m/ as an extrasyllabic appendix in these words, though we're not sure how Armenian came to develop this strange behavior. Furthermore, such fricative-nasal clusters show more idiosyncrasies word-medially (§5.4.2.1).

5.2.3.11 Fricative + nasal /n/

Although word-final fricative + /m/ clusters are pronounced without an intervening schwa, word-final clusters of fricative + /n/ regularly undergo epenthesis (Table 5.52).

Table 5.52: Schwa epenthesis in final CC clusters where C1 is a fricative and C2 is a nasal /n/

<vn>→[vən]	<goralio**vn**>	gorɑliˈjo**vən**	'coral'	կորալիովն	n=3
<sn>→[sən]	<how**sn**>	ˈhu**sən**	'brink'	հուսն	n=27
	<orbē**sn**>	vorˈbe**sən**	'the wherefore'	որպէսն	n=26
<zn>→[zən]	<a**zn**>	ˈɑ**zən**	'nation'	ազն	n=36
	<t'akaz**n**>	tɑˈkɑ**zən**	'prince'	թագազն	
<ʃn>→[ʃən]	<ta**ʃn**>	ˈtɑ**ʃən**	'contract'	դաշն	n=7
	<yankow**ʃn**>	hɑŋˈku**ʃən**	'closely'	յանգուշն	
<ʒn>→[ʒən]	<a**ʒn**>	ˈɑ**ʒən**	'crack'	աժն	n=1
<xn>→[χən]	<to**xn**>	ˈto**χən**	'funnel'	դոխն	n=1
<ɣn>→[ʁən]	<sde**ɣn**>	əsˈte**ʁən**	'dactyl'	ստեղն	n=17
	<e**ɣn**>	je**ʁən**	'hind'	եղն	

As with stop + /n/ clusters (§5.2.3.3), a lot of these fricative + /n/ words have an archaic connotation.

For <vn>→[vən] clusters, these seem to be restricted to loanwords. While <fn>→[fən] and <hn>→[hən] clusters seem to be accidental gaps.

5.2.3.12 Fricative + rhotic /ɾ/

Word-finally, an orthographic cluster of a fricative + rhotic /ɾ/ undergoes schwa epenthesis (Table 5.53).

Table 5.53: Schwa epenthesis in final CC clusters where C1 is is a fricative and C2 is a rhotic /ɾ/

<wr>→[vəɾ]	<a**wr**>→ˈɑ**vəɾ**	‘bile’	աւր	n=1
<sr>→[səɾ]	<nō**sr**>→ˈno**səɾ**	‘coarse’	նօսր	n=6
<zr>→[zəɾ]	<e**zr**>→ˈje**zəɾ**	‘shore’	եզր	n=11
<ʒ>→[ʒəɾ]	<ʒa**hr**>→ˈʒɑ**həɾ**	‘virus’	ժահր	n=4
<xr>→[χəɾ]	<d͡ʒa**xr**>→ˈd͡ʒɑ**χəɾ**	‘flight’	ճախր	n=3
<ɣr>→[ʁəɾ]	<d͡za**ɣr**>→ˈd͡zɑ**ʁəɾ**	‘mocking’	ծաղր	n=9

For final <zr>→[zəɾ] sequences, all of Kouyoumdjian’s examples are either [jezəɾ] ‘shore’ or its derivatives: [ked-ezəɾ] ‘river-bank’ գետեզր where [kəd] is ‘river’.

For final <wr,vr>→[vəɾ], more cases come from loanwords on Wiktionary: <manewr> → [mɑˈnevəɾ] ‘maneuver’ մանևր.

These fricative-rhotic clusters are generally few. Thus in Kouyoumdjian, there are accidental gaps for the fricatives /f, ʃ/. On Wiktionary, we did find a loanword with [fəɾ]: <ʃifr>→[ʃifəɾ] ‘cipher’ շիֆր.

Prescriptively, all fricative-rhotic clusters are pronounced with a schwa: <meɣr>→[meʁəɾ] ‘honey’ մեղր. But in casual speech, some of these words allow the deletion of the schwa: [meʁɾ]. This is a case of schwa elision. In our experience, [ʁəɾ] sequences often reduce to just [ʁəɾ] in natural speech, while it is less common to see such reduction or schwa elision after the other fricatives.

5.2.3.13 Fricative + lateral /l/

The Kouyoumdjian dictionary lists only one word that end in an orthographic cluster of a fricative + /l/. This word was <p’ahl> ‘stallion’ փահլ. This word is pronounced with a schwa: [pɑhəl].

On Wiktionary, we found some other cases from loanwords. These again take schwa epenthesis: <p’azl> [pɑzəl] ‘puzzle’ փազլ.

5.2.3.14 Nasal + nasal /m/

We did not find such final clusters in Kouyoumdjian. Nor did we find any such words on Wiktionary. However, there is a colloquial word that has this orthographic cluster and get a schwa: <d͡ʒanm> → [d͡ʒɑnəm] ‘my dear’ ճանմ. This word is borrowed from Turkish ‘canim’ [d͡ʒanim ‘my dear’.

5.2.3.15 Nasal + nasal /n/

There are many words which end in an orthographic nasal-nasal sequence: <mn> and <nn>. In pronunciation, a schwa is added between the nasals (Table 5.54).

Table 5.54: Schwa epenthesis in final CC clusters where C1 is and C2 are nasals

Archaic suffix *-umən*		Other *-mən*		*-nən*
<xajt'ow**mn**>	<derewow**mn**>	<ada**mn**>	<owre**mn**>	<**inn**>
/χɑjt-u**mn**/	/dɛrɛv-u**mn**/	/ɑdɑ**mn**/	/uɾu**mn**/	/inn/
[χɑjˈtu**mən**]	[dɛrɛˈvu**mən**]	[ɑˈdɑ**mən**]	[uˈɾe**mən**]	[i**nən**]
խայթումն	տերեւումն	ատամն	ուրեմն	ինն
'pricking'	'foliation'	'tooth'	'thus'	'nine'
n=325		n=36		n=13

Words with a final <mn> [mən] cluster are generally archaic. One common situation are words that end in the nominalizer suffix [-um]. This suffix is pronounced [-um] in the modern language, with an orthographic final <m> ում. But in more archaic stages of Western Armenian, the form of this suffix [-umən] with an orthographic nasal cluster <mn> -ումն. Kouyoumdjian's dictionary uses these archaic forms, thus he has a lot of words that would be pronounced with a final [-umən].

5.2.3.16 Nasal + rhotic /ɾ/

Word-finally, nasal + /ɾ/ clusters undergo schwa epenthesis. The nasal can be /m/ or /n/ (Table 5.55).

Table 5.55: Schwa epenthesis in final CC clusters where C1 is a nasal /m,n/ and C2 is a rhotic /ɾ/

<mr>→[məɾ]	<ha**mr**>	ˈhɑ**məɾ**	'speechless'	համր	n=3
<nr>→[nəɾ]	<ma**nr**>	ˈmɑ**nəɾ**	'small'	մանր	n=5

I am not aware of cases where, in spoken speech, the [nəɾ] cluster is reduced to *[nɾ]: [mɑnəɾ] 'small', not [*mɑnɾ].

5.2.3.17 Nasal + lateral /l/

Nasals cannot start a complex coda with laterals. We found no orthographic cases of final <ml> or <nl> clusters in Kouyoumdjian. On Wiktionary, the few

examples that we found were all loanwords. And these get schwa epenthesis: <pŕanl> [pəɾɑnəl] բռանլ from French ‘branle’, and <greml> [gəɾeməl] ‘Kremlin’ Կրեմլ from Russian.

5.2.4 Geminate codas

There are few roots end in a geminate, i.e. roots where the final consonant is repeated. In Kouyoumdjian’s dictionary, we have found the following types of word-final repeated segments: [kk], [χχ], and [ɾɾ]. Medial geminates are variably avoided (§5.4.2.6).

For [kk], this gemination is derived from adding the nominalizer /-k/ after a velar stop: [hok-k] ‘hindrance’ հոգք. This cluster is discussed in Section §5.2.3.1. This cluster is arguably syllabified as a coda + appendix, such as [.hok.{k}]. Similar geminates are created in final [VCk-k] clusters via this appendix (§5.3.2.1).

For [χχ], this cluster is limited to a handful of roots, some of which sound archaic: [seχχ] ‘melon’ սեղխ. This cluster is part of a pattern where a final [χ] can be found in different fricative-fricative clusters, again as a type of extrasyllabic consonant. See Section §5.2.3.9.

Finally, the [ɾɾ] pattern is found in a handful of roots. This is orthographically represented with a final rhotic getting repeated: <rr> ր՝ր՝. The [ɾɾ] sequence is pronounced as a single long rhotic. In HD’s impression, this orthographic <rr> sequence is pronounced as [ɾɾ] more often in Eastern Armenian than in Western Armenian. In his Western judgments, it feels more typical to degeminate this final cluster into just a singleton [ɾ] (§??).

In Kouyoumdjian’s dictionary, we found 23 words which end in an <rr> [ɾɾ] sequence (Table 5.56). Of these 23 words, 4 are roots. The other words were all derived from the root [dɑɾɾ] ‘element’ (18 words) or [ɑntoɾɾ] ‘peaceful’ (1 word).

Table 5.56: Final CC clusters where C1-C2 are geminate [ɾɾ]

ˈdɑɾɾ	‘element’	տարր
ɑnˈtoɾɾ	‘peaceful’	անդորր
ˈjeɾɾ	‘mustiness’	երր
ˈdoɾɾ	‘turion’	տորր
ˈmed͡z + ˈdɑɾɾ	‘big + element’	մեծ տարր
→ med͡z-ɑ-dɑɾɾ	‘spacious’	մեծատարր

Besides the above sequences of identical sounds, we’ve come across [zz] sequences in onomatopoeic words like [bəzz] ‘buzz’ պըզզ, [uff] ‘wow’ ուֆֆ. We’ve also found loanwords like [miss] ‘Miss’ միսս and [finn] ‘Finn’ ֆինն.

5.3 Syllabification of final three-consonant clusters

Armenian allows at most two consonants in a complex coda. Exceptions are rare and seem limited to non-nativized loanwords (§5.3.1). Among native words, if the orthography ends in a sequence of 3 consonants, then we see one of two outcomes:

- The cluster is a (complex) coda plus one or two appendixes (§5.3.2).
- The cluster is pronounced with schwa epenthesis (§5.3.3).

Data in this section expands on §5.1.5.

5.3.1 No native complex codas with three consonants

It seems that Armenian generally avoids words that have a complex coda larger than two consonants. The only example that we found in Kouyoumdjian was a loanword [verɾst] 'Verst' վերստ, as the name for a Russian unit of length.

Armenian Wiktionary provides much more examples of words with final 3-consonant complex codas. Again, these are loanwords, especially proper nouns like [lisit͡ʃɑnsk] 'Lysychansk' Լիսիչանսկ or[tɑjms] 'The Times' թայմս . Some Wiktionary loanwords even have flat-sonority: [ɑtjungd] 'adjunct' ադյունկտ.

5.3.2 Codas with an appendix

A consonant is an appendix if it can be pronounced after a consonant cluster, despite having flat sonority (§5.1.4). Such appendixes are common in final two-consonant sequences (§5.2.3), but can also occur in final 3-consonant sequences. The set of possible final appendixes that we found in in 3-consonant clusters is /-k, χ, s/. Very rarely, we find a cluster with two final appendixes /χ,k/ (§5.3.2.4).

5.3.2.1 Complex coda + appendix /-k/

The nominalizer suffix *-k* can be added after virtually any attested complex coda (Table 5.57). In some cases, the sequence of three consonants has continuous falling sonority. This occurs when the second consonant is a rhotic, nasal, or fricative.

Table 5.57: Complex codas + appendix /-k/ with continuous falling sonority

[VCl-k]	ʃɑˈɾɑ**jl**	‘ray’	շառայլ	n=4
	ʃɑˈɾɑ**jl-k**	‘glimmer’	շառայլք	
[VCɾ-k]	həˈɾɑ**jɾ**	‘burned’	հրայր	n=12
	həˈɾɑ**jɾ-k**	‘conflagration’	հրայրք	
[VCŋ-k]	pənɑˈt͡sɑ**jn**	‘onomatopoeia’	բնաձայն	n=4
	pənɑˈt͡sɑ**jŋ-k**	‘onomatopoeia’	բնաձայնք	
[VCs-k]	ˈbɑ**ɾs**	‘Persian’	պարս	n=6
	ˈbɑ**ɾs-k**	‘Persia’	Պարսք	
[VCʃ-k]	ˈdu**jʒ**	‘damage’	տոյժ	n=2
	ˈdu**jʃ-k**	‘damage’	տոյժք	

The suffix devoices preceding fricatives: see the [VCʃ-k] row above (§4.3.1.4). The suffix tends to deaspirate after obstruents (§??).

Among falling-sonority clusters, the suffix can likewise follow the lateral /l/: see row [VCl-k] in Table 5.57. As explained in Section §5.2.2.8, the lateral generally resists preceding a consonant in the same syllable, except for the suffix *-k*.

Although the appendix can create falling-sonority clusters, it is much more common to find flat-sonority clusters (Table 5.58). The suffix *-k* can be easily added after a complex coda that ends in a stop or affricate of any place of articulation.

Table 5.58: Complex codas + appendix /-k/ with flat sonority

[VCp-k]	o͡tsɑˈge**ɾb**	‘snake-like’	օձակերպ	n=9
	o͡tsɑˈge**ɾp-k**	‘Ophidia’	օձակերպք	
[VCt-k]	ˈgi**ɾt**	‘well-educated’	կիրթ	n=39
	ˈgi**ɾt-k**	‘instruction’	կիրթք	
[VCk-k]	ˈje**ɾg**	‘work’	երկ	n=6
	ˈje**ɾk-k**	‘work of art’	երկք	
[VCt͡s-k]	ˈhu**nt͡s**	‘harvest’	հունձ	n=34
	ˈhu**nt͡s-k**	‘harvest’	հունձք	
[VCt͡ʃ-k]	ɑˈnu**ɾt͡ʃ**	‘dream’	անուրջ	n=12
	ɑˈnu**ɾt͡ʃ-k**	‘dream’	անուրջք	

Among these final three-consonant clusters, there are cases where the suffix

-k follows another velar stop: see row [VCk-k] in Table 5.58. The two are pronounced as one long geminate.

On a last note, in Kouyoumdjian's dictionary, we only found one case of a /VCCk/ word that's pronounced with schwa epenthesis: <xa**ṙnk**'> [χɑɾə**nk**] 'copulation' խառնք. Here, the schwa is necessary because [ɾn] is difficult to pronounce as a complex coda in general. See the other sections for variations on how [ɾn] clusters are produced, both word-finally (§5.2.2.7) and word-medially (§5.4.2.2).

5.3.2.2 Complex coda + appendix /χ/

Among appendixes, the suffix *-k* is the most common. Another common appendix is /χ/. This fricative can follow obstruents such as stops (§5.2.3.2) and fricatives (§5.2.3.9). It can likewise follow complex codas.

The only relevant example we came across in Kouyoumdjian is the word for 'star'. Orthographically, this word ends in a consonant cluster: <asdɣ> աստղ. In Eastern Armenian, this cluster undergoes schwa epenthesis: [ɑstəʁ]. In some sub-dialects of Western Armenian such as in Istanbul, Istanbuli speakers tell us that this word is also pronounced with epenthesis: [ɑstəʁ]. But in the Lebanese sub-dialect of Western Armenian, there is no schwa epenthesis. The cluster is pronounced either as [ɑstχ] or with stop deletion [ɑsχ].

For the [ɑstχ] case, we analyze the [st] as forming a complex coda. The [χ] is then an appendix.

The word for 'star' has many derivatives (8 in Kouyoumdjian). Compounds that end with this root show the same patterns of pronunciation: [d͡zoˈv-ɑstχ] 'starfish' ծովաստղ, literally 'sea-star'.

5.3.2.3 Complex coda + appendix /s/

Besides /k,χ/, another common appendix is /s/. This segment can follow stops (§5.2.3.2). It can likewise follow a complex coda that ends in a stop.

In Kouyoumdjian's dictionary, we found three such cases. Two of them are alternate spellings of the loanword [əspiŋks] 'sphinx' սպինքս, սփինքս. The other word is the name of some fish species [gɑɾɑŋks] 'Cavalla' կարանքս.

Because we only found the above three examples, it's possible that cases of complex codas + /s/ without schwa epenthesis are limited to either a) loanwords or b) words where the /s/ follows a velar stop [k]. Other clusters of consonants + /s/ undergo schwa epenthesis (§5.2.3.2,5.2.3.7).

5.3.2.4 Coda + two appendixes (χ, k)

Interestingly, there are some words which have a sequence of a coda *χ*, plus an appendix *χ*, and then an appendix *k*. Orthographically, the two *χ* sounds are spelled differently, suggesting their different origins (Table 5.59).

Table 5.59: Sequence of appendixes in final /χχk/ clusters

<aɣx>	ˈɑχχ	'baggage'	ɯɳխ
<aɣxk'>	ˈɑχχ-k	'closing'	ɯɳխք
<owɣx>	ˈuχχ	'torrent'	ուɳխ
<owɣxk'>	ˈuχχ-k	'torrent'	ուɳխք

We only found two such words in Kouyoumdjian that had this final cluster.

5.3.3 Schwa epenthesis in other word-final three-consonant clusters

Schwa epenthesis occurs for final clusters of two consonants that cannot be pronounced together (§5.2.25.2.3). We likewise see schwa epenthesis in certain final clusters of three consonants /VCCC#/ Such clusters are almost always syllabified with an epenthetic before the last consonant [VCCəC] and rarely with a schwa after the first consonant [VCəCC]. The final consonant is almost always a fricative or sonorant, and the preceding two-consonant cluster often has falling sonority. The reason for this is discussed in Section §5.1.5.

- /VCCs/ → [VCCəs] (§5.3.3.1
- /VCCʁ/ → [VCCəʁ] (§5.3.3.2
- /VCCm/ → [VCCəm] (§5.3.3.3
- /VCCn/ → [VCCən], very rarely [VCəCn] (§5.3.3.4
- /VCCɾ/ → [VCCəɾ] (§5.3.3.5
- /VCCl/ → [VCCəl] (§5.3.3.6

5.3.3.1 Epenthesis before final /s/ in /VCCs/ clusters

There are very few words that end in an orthographic cluster of two consonants + /s/. Of the few words we found, these clusters undergo schwa epenthesis: <VCCs>→[VCCəs] (Table 5.60).

Table 5.60: Schwa epenthesis in final 3-consonant clusters with final /s/

<yns>→[jns]	<hzorako**yns**> հզօրագոյնս	həzoɾɑˈku**jnəs** ‘powerfully’	n=12
<rk’s>→[ɾkəs]	<ne**rk’s**> ‘inside’	ˈne**ɾkəs** Ներքս	n=1

In the Kouyoumdjian dictionary, all the cases for the <yns>→[jns] cluster involved the final root [-kujnəs] -գոյնս which is used to create a type of adverbial superlative meaning.

Many more cases are found when the final /s/ is the 1SG possessive suffix. This suffixes always triggers a schwa after a consonant: <mart> [mɑɾt] ‘man’ մարդ vs. <marts> [mɑɾt-əs] ‘my man’ մարդս. This is discussed in posessive schwa

5.3.3.2 Epenthesis before final /ʁ/ in /VCCʁ/ cluster

There are very few words that end in an orthographic cluster of two consonants + /ʁ/. For most words, this cluster is pronounced with schwa epenthesis before the final sound: <VCCɣ>→[VCCəʁ]. The main exception is the word ‘star’ [ɑstχ] աստղ discussed in Section §5.3.2.2.

In the Kouyoumdjian dictionary, we found relatively few cases of such clusters (Table 5.61). Most of them display schwa epenthesis.

Table 5.61: Schwa epenthesis in final 3-consonant clusters with final /ʁ/

<sdɣ>→[stəʁ]	<o**sdɣ**>	ˈvo**stəʁ**	‘lime twig’	ոստղ	n=1
<nkɣ>→[ŋkəʁ]	<ʃi**nkɣ**>	ˈʃi**ŋkəʁ**	‘snake cucumber’	շինգղ	n=4
<ngɣ>→[ŋgəʁ]	<a**ngɣ**>	ˈɑ**ŋgəʁ**	‘angle’	անկղ	n=1
<rgɣ>→[ɾgəʁ]	<a**rgɣ**>	ˈɑ**ɾgəʁ**	‘box’	արկղ	n=7

Besides the above clusters, Armenian Wiktionary lists some non-dialectal words like <gowntsɣ> [gunt͡səʁ] ‘clod’ կունձղ.

5.3.3.3 Epenthesis before final /m/ in /VCCm/ clusters

Word-final orthographic clusters of two consonants + /m/ are rare. In Kouyoumdjian, we found only 3 examples (Table 5.62). These three examples all look morphologically or diachronically related to each other. Here we find schwa epenthesis.

Table 5.62: Schwa epenthesis in final 3-consonant clusters with final /m/

<ayʒm>	ˈɑjʒəm	‘now’	այժմ
<aɾayʒm>	ɑˈɾɑjʒəm	‘presently’	առայժմ
<t͡s’ayʒm>	ˈt͡sɑjʒəm	‘till now’	ցայժմ

Based on the above limited data, it seems that final /ʒm/ clusters just generally prefer epenthesis (cf. Section §5.2.3.10).

5.3.3.4 Epenthesis before final /n/ in /VCCn/ clusters

There are many words that end in an orthographic cluster of two consonants plus the nasal /n/. In such words, schwa epenthesis is needed. In the most typical case, the schwa is added immediately before the nasal: <VCCn>→[VCCən]. For the two-consonant cluster, this cluster is almost always a cluster would have formed a complex coda, if only the nasal /n/ was absent. It is quite rare to find cases of epenthesis as <VCCn>→[VCəCn]

To illustrate, Table 5.63 organizes the set of final /CCn/ clusters that we found in Kouyoumdjian, based on the natural class of the CC cluster.

Table 5.63: Schwa epenthesis in final 3-consonant clusters with final /n/ such that the preceding consonants could have formed a complex coda

fric. + affr. + /n/	<ta**ɣt͡sn**>	ˈtɑ**χt͡sən**	‘horsemint’	դաղձն	n=2
fric. + stop + /n/	<a**sgn**>	ˈɑ**skən**	‘garnet’	ասկն	n=4
nasal + stop + /n/	<pa**ngn**>	ˈpɑ**ŋgən**	‘tale’	բանկն	n=14
nasal + affr. + /n/	<gow**nt͡sn**>	ˈgu**nt͡sən**	‘clod’	կունձն	n=13
rhotic + obstr. + /n/	<a**rtn**>	ˈɑ**ɾtən**	‘lance’	արդն	n=38
rhotic + /m/ + /n/	<se**rmn**>	ˈse**ɾmən**	‘seed’	սերմն	n=6
glide + cons. + /n/	<to**yzn**>	ˈtu**jzən**	‘frivolous’	դոյզն	n=5

For the words above, the pre-nasal consonant cluster would have formed a complex coda if the nasal was absent. For example, <towrkn> [tuɾkən] ‘potter’s wheel’ դուրգն vs. <t’owrk’> [tuɾk] ‘Turk’ թուրք.

Besides the above words, there are also words where the pre-nasal cluster is flat or rising-sonority (Table 5.64). But in these clusters, the middle consonant would have formed an appendix if the nasal was absent. For example, word-final [ʁm] can form a syllabifiable cluster, whether we analyze the [m] as a coda or appendix. Adding a nasal /n/ after this cluster leads to schwa epenthesis.

Table 5.64: Schwa epenthesis in final 3-consonant clusters with final /n/ such that the preceding consonants could have formed a coda+appendix

<ɣmn>→[ʁmən]	<sa**ɣmn**> cf. <go**ɣm**>	ˈsɑ**ʁmən** ˈgo**ʁm**	‘embryo’ ‘side’	սաղմն կողմ	n=9
<ʃxn>→[ʃχən]	<sda**ʃxn**> cf. <pte**ʃx**>	əsˈtɑ**ʃχən** pəˈte**ʃχ**	‘styrax tree’ ‘mayor’	ստաշխն բդեշխ	n=9

It is much more common for the pre-nasal cluster to have a syllabifiable cluster like <Vrtn> (cf. <Vrt>), then to have this cluster be some unsyllabifiable <Vktn> (cf. <Vkt>). I found virtually no such clusters in Kouyoumdjian. The closest example I found was the word <go**zrn**> [ˈgo**zərn**] ‘young came’ կոզռն where the <VCCn> cluster is syllabified with a schwa after the first consonant [VCəCn] instead of before the nasal [VCCən]. The earlier location of the schwa is because [ɾn] is a licit complex coda, and schwa epenthesis tries to maximize the size of complex codas cite schwa epenthesis.

The relatively high number of falling-sonority clusters in before final /n/ is likely because of the diachronic origins of these clusters (§5.1.5).

5.3.3.5 Epenthesis before final /ɾ/ in /VCCɾ/ clusters

There are many words which end in an orthographic cluster of two consonants plus the rhotic /ɾ/. Here, the rule is that these clusters are syllabified with schwa epenthesis before the rhotic: <VCCr>→[VCCəɾ]. There are some attested cases of variable schwa elision in these clusters.

In Kouyoumdjian’s dictionary, there were many cases of <VCCr> words (Table 5.65). Here, the preceding consonants almost always have falling sonority; if the rhotic was absent, then the cluster would have been a complex coda. But because the rhotic is present, we have schwa epenthesis.

Table 5.65: Schwa epenthesis in final 3-consonant clusters with final /ɾ/ such that the preceding consonants could have formed a complex coda

fric. + stop + /ɾ/	<ko**ɣdr**>	ˈko**ʁdəɾ**	‘tender’	գողտր	n=60
fric. + affr. + /ɾ/	<k’a**xts’r**>	ˈkɑ**χ͡tsəɾ**	‘sweet’	քաղցր	n=2
nasal + stop + /ɾ/	<sa**ndr**>	ˈsɑ**ndəɾ**	‘comb’	սանտր	n=23
nasal + affr. + /ɾ/	<t’a**ntsr**>	ˈtɑ**nt͡səɾ**	‘thick’	թանձր	n=2
rhotic + obstr. + /ɾ/	<t’a**ntsr**>	ˈtɑ**nt͡səɾ**	‘tough’	կարծր	n=10
glide + cons. + /ɾ/	<ga**ysr**>	ˈgɑ**jsəɾ**	‘emperor’	կարծր	n=2

As with <VCCn> clusters, the final rhotic is almost always found after a consonant cluster that could have been a complex coda. I have found only one exception in the Kouyoumdjian dictionary: an obvious loanword <magdr> [mɑgdəɾ] ‘mactra’ մակտր.

5.3.3.6 Epenthesis before final /l/ in /VCCl/ clusters

There are vanishingly few words that end in an orthographic cluster of three consonants such that the final consonant is /l/: <VCCl>.

In Kouyoumdjian, we only found one example, and this example is a loanword with schwa epenthesis: <mankl> [mɑŋkəl] ‘mangle’ մանգլ.

On Armenian Wiktionary, we found more cases of this final consonant, and they behaved the same in terms of schwa epenthesis. Many of them were loanwords like <ansampl> [ɑnsɑmpəl] ‘ensemble’ անսամբլ. But some seemed like native words: <pangl> [pɑŋgəl] ‘riddle’ բանկլ.

5.3.3.7 Epenthesis in other types of VCCC clusters

In word-final 3-consonant clusters, the final consonant is almost always either an appendix /k, s, χ/ or a consonant that triggers schwa epenthesis. For this second group of schwa-inducing consonants, these consonants come from a restricted set /s, ʁ, m, n, ɾ, l/. The reason is diachronic, as explained before in Section §5.1.5.

The Kouyoumdjian dictionary did not have any other final consonants that triggered schwa epenthesis in 3-consonant clusters. On Armenian Wiktionary, we found a handle of other possible final consonants that trigger epenthesis: <kovnt> [kovənt] ‘type of circle-dance’ գովնդ, and <eɣrt> [jeʁəɾt] ‘type of willow’ եղրդ. For these words, the schwa creates a complex coda because schwa epenthesis generally prefers creating complex codas over simple codas **jeʁɾət*.

Some are words from other dialects like ‘still’ աշտխ. We exclude them because they’re not Western Armenian, so we don’t know how they ‘should’ be pronounced anyway.

5.4 Other restrictions on complex codas

Besides the restrictions on falling sonority and appendixes, there are some minor restrictions on possible complex codas. These restrictions are the following:

1. Complex codas are generally infrequent word-medially, but still attested (§5.4.1).

2. Some complex codas are relatively easy to form word-finally, but difficult to form word-medially, or variably avoided word-medially (§5.4.2).
3. Some vowel-consonant combinations are generally avoided within the same syllable, specifically when the vowel is a high vowel or schwa (§5.4.3).
4. Some consonant clusters receive schwa epenthesis in careful speech, but they can be pronounced as clusters without schwas casual speech because of schwa elision (§5.4.3.2).

5.4.1 Infrequency and asymmetry of word-internal complex codas

In general, Western Armenian allows word-internal complex codas. Thus, if some complex coda is attested word-finally, then it is in principle also possible word-initially and word-medially. For example, the plural suffix *-ner* can be added after any polysyllabic word, creating a complex coda: [mɑχ.tɑ**ŋk**] 'wish' and [mɑχ.tɑ**ŋk**-ner] 'wishes' մաղթանքներ. And for monosyllabic roots, we can add the sequence [n=e] 'DEF=is' on them: [mɑ**ɾt**] 'man' and [mɑ**ɾt**-n=e] 'is the man' մարդն է.

Furthermore, if some complex coda is banned or atypical word-finally then it also banned or atypical word-medially as well. For example, stop-stop complex codas like [gd] are generally avoided word-finally. Thus also word-medially, the complex coda [gd] is avoided in native words. Loanwords unsurprisingly provide cases of [gd]: [e.le**gd**.ɾon] 'electron' էլեկտրոն which can be resyllabified as [e.leg.dɾon], thus creating a complex onset and further signaling the non-native status of this word.

However, once we look at only uninflected words, it can be difficult to find such word-internal complex codas simply because of the following issues:

1. Word-medial complex codas are relatively rare in the lexicon.
2. Complex codas in roots tend to be root-final instead of root-initial.
3. Compounds tend to insert a vowel between the roots.
4. C-initial suffixes and CC-final prefixes are few.

For the first point on the general rarity of word-internal complex codas, Armenian Wiktionary (Jan 2021) has around 184,862 lexemes that have a listed syllabification. But of these 184K words, only 8,095 words started with a (C)VCC syllable, 2,834 had a listed word-medial CVCC, and 296 were reported with a medial VCC syllable.[4] These low numbers don't mean that word-internal complex codas are impossible, just relatively rarer.

[4] These numbers can't be fully trusted though because Armenian Wiktionary is rampant with syllabification errors.

For the second point on roots, Although complex codas are allowed in the language, there are relatively few roots which contain a word-medial complex coda. Thus, it is difficult to find a root example for every possible complex coda. Some of the few examples are [hɑ**jd**.ni] ‘clear’ յայտնի, [tɑ**ɾk**.mɑn] ‘translator’ թարգման, and [bɑ**ʃt**.pɑn] ‘protector’ պաշտպան.

For the third point on compounds, when new words are created by adding derivational suffixes or by compounding, word-medial complex codas can be created: the compound [ɑjs-t͡ʃɑp] ‘this much’ այսչափ made from the roots [ɑjs, t͡ʃɑp] ‘this, much.’ But it is much more common to include a linking vowel /-ɑ-/: [mɑ**ɾt**, seɾ] ‘man, love’ մարդ,սէր create a compound [mɑ**ɾ**.t-ɑ-seɾ] ‘philanthropist’ մարդասէր without a medial complex-coda.

For the fourth point on derivational morphology, there are a handful of derivational prefixoids that have complex codas: the prefixoid *antɾ-* անդր- is analogous to the English prefix *trans-*, such as [ɑ**nt**.ɾ-ɑt.land.jɑn] ‘trans-Atlantic’ անդրադլանտեան. Another common example is the prefixoid [jeɾg-] ‘bi-,di-’ երկ- such as [je**ɾg**-d͡ʒeχk] ‘bifurcation’ երկճեղք derived from the word [d͡ʒəχk] ‘crack’ ճեղք.

However, even when we use derivational suffixes, it is relatively common for the complex coda to precede one of the following consonants: a rhotic /ɾ/, a nasal /n/, glide /j/, a velar stop /g/ (often devoiced [k]). Each type of consonant often correlates with some morphophonological idiosyncrasy.

For the rhotic, consider [vo**sk**.ɾ-ɑjin] ‘osseous’ ոսկրային, derived from the root [voskoɾ] ‘bone’ ոսկոր with syncope of /o/ (see syncope chapter.

For the nasal, consider the word [əs.kə**sp**.n-ɑ.ˈgɑn] ‘original’ սկզբնական, derived from [əski.**sp**] ‘beginning’ սկիզբ. The nasal is not part of the suffix /-ɑgɑn/, but is a morphologically-epenthetic nasal due to a historic relic of nasal weakening (see nasal liaison).

For the glide /j/, there are three common derivational suffixes that start with a glide /j/: /jɑn, jɑg, jɑl, jɑ/ եան, եակ, եալ, եայ. For all these suffixes, we can optionally syllabify the glide with the preceding consonant: [zəm.ɾu**χt**.jɑ, [zəm.ɾu**χ.tj**ɑ] ‘made of emeralds’ զմրուխտեայ from [zəm.ɾu**χt**] ‘emerald’ զմրուխտ. See Section §5.5.2.

And finally for the stop /g/, a common derivational suffixal is *-ig*: [tɑ.pɑ**n.t͡s**-ig] ‘transparent’ թափանցիկ. This suffix can undergo further derivation and undergo vowel reduction, creating a complex coda: [tɑ.pɑ**nt͡s**.k-u.tjun] ‘transparency’ թափանցկութիւն. This suffix *-ig* is also used to form an irregular plural suffix for the word [mɑɾt] ‘person’ մարդ as [mɑ**ɾt**-ig] ‘people’ մարդիկ. This irregular plural can undergo derivational morphology, creating a complex coda: [mɑ**ɾt**.-k-ɑ.gɑn] ‘humane’ մարդկական.

To demonstrate these tendencies, we took the 2,834 Armenian Wiktionary which were listed as having a word-medial CVCC syllable. From this set, the majority (n=848, 29.92%) preceded the fricative /v/. This fricative was the passive suffix /v/ (spelled ու in the traditional orthography, վ in reformed). This suffix can follow complex codas in Eastern Armenian [ʃərɑjl-v-e-l] 'to be wasted' շռայլվել, but it triggers a schwa after CC clusters in Western Armenian [ʃərɑjlə-v-i-l]. Thus this group is inadmissible in Western Armenian. After /v/, the other most common post-coda sounds were /ɾ/ ր (n=520, 18.35%), /n/ ն (n=341, 12.03%), /j/ յ (n=254, 8.96%), and /g/ կ (n=131, 4.62%). Besides these frequent consonants, we also found <b> պ (n=77, 2.72%) but these were mostly due to the root [**bɑʃt**.pɑn] 'protector' պաշտպան.

Thus for medial CVCC syllables on Armenian Wiktionary, the above consonants accounted for over 75% of reported consonants that follow a complex coda in Eastern Armenian. The other 25% seemed like an arbitrary distribution of other consonants without any clear generalizations.

In sum, Armenian allows complex codas anywhere in the word. But there are some accidental asymmetries due to the structure of the Armenian lexicon and morphology.

5.4.2 Avoidance of certain word-medial complex codas

The previous section explained that essentially every possible word-final complex coda is also possible word-medially. But there are certain consonant clusters that are permitted word-finally, but seem to be avoided word-medially. These avoided clusters are grouped as the following. We go through each case below. cite tese sections into the previous sections

- /Cm/: /jm, ɾm, ʁm, zm, ʃm, hm/ (§5.4.2.1)
- /ɾn/ (§5.4.2.2)
- nasal-fricative: /ns, ms/ (§5.4.2.3)
- fricative-fricative: /χs, χʃ, ʃχ/ (§5.4.2.4)
- appendix /C-k/ (§5.4.2.5)
- geminates (§5.4.2.6)

5.4.2.1 Avoiding word-medial /Cm/ complex codas

Word-finally, /jm/ is a very rare cluster: [gɑ**jm**] 'mast' (§5.2.1.7). Inflection with the definite suffix /-n/ can create a word-medial complex coda: [gɑ**jm**-n=e] 'must-DEF=is' կայմն է. But we could not find any other cases word-medial [jm] complex coda.

Word-finally, /ɾm/ can form a complex coda as in [t͡ʃe**ɾm**] ‘warm’ ջերմ (§5.2.1.6). But word-medially, it seems that this [ɾm] is almost always before a nasal /n/ or glide /j/: [t͡ʃe**ɾm**.n-ɑ.gɑn] ‘feverish’ ջերմնական. Here the nasal is epenthetic because of an obscure morphological rule (see nasal liaison).[5] Exceptions are few, such as [ɑ**ɾm**.dik] ‘cereal’ արմտիք.

Further evidence for the above restriction comes from vowel reduction. There are a few roots which can create a [ɾm] complex coda when undergoing reduction: from [gɑɾ.miɾ] ‘red’ կարմիր to [gɑ**ɾm**.ɾil] ‘to grow red’ կարմրիլ. But, for this root, it is impressionistically more common to have a schwa between the two consonants: [gɑɾ.mə.ɾil]. Similar reduction patterns are found for the root [mɑɾmin] ‘body’ մարմին but [mɑ**ɾm**.n-ɑgɑn, mɑɾ.mə.n-ɑgɑn] ‘corporeal’ մարմնական. In HD’s impression, the schwa form is more common, while the schwaless form sounds higher register.

Word-finally, fricative + /m/ clusters are attested (§5.2.3.10), but each displays some idiosyncrasies word-medially.

The final cluster /ʁm/ can form a complex coda even though it has rising sonority: [gɔ**ʁm**] ‘side’ կողմ. It is possible to analyze this cluster as comprised of a coda and appendix. Word-medially, this coda can be created, but it seems to almost always precede either a nasal /n/ or glide /j/: [gɔ**ʁm**.n-ɑgɑn] ‘lateral’ կողմնական due to an epenthetic nasal (), and [kɑɾ-ɑ-gɔ**ʁm**-jɑn] quadrilateral քառակողմեան with possible resyllabification [kɑɾ-ɑ-gɔ**ʁ.mj**ɑn].

As before, this suggests that the medial complex coda [ʁm] has to before /n/ or /j/. Further evidence comes from vowel reduction. The root [ɑʁmug] ‘noise’ աղմուկ reduces to [ɑʁ**mə**g-e-l] ‘to disturb’ աղմկել. The high vowel is replaced by a schwa, and we can’t just delete the schwa to form **ɑʁm.gel*. The avoidance of a schwa here is evidence that [ʁm] is an undesired complex coda before an obstruent like /g/.

For /ʃm/, this rising-sonority cluster is attested word-finally [tə.ɾo**ʃm**] ‘stamp’ դրոշմ. However, word-medially, the only cases we could find involved adding /n/-initial inflection: [tə.ɾɔ**ʃm**-ner] ‘stamps’ դրոշմեր. We couldn’t find clear cases of such clusters before derivational suffixes. Further, Kouyoumdjian only listed one word with an intervocalic /VʃmCV/ cluster, and this cluster gets schwa epenthesis in HD’s judgments: <ga**ʃm**powṙn> [gɑ**ʃəm**puɾən] ‘robust’ կաշմբուռն, instead of **gɑ**ʃm**.puɾən*.

For /zm/, this rising-sonority cluster can form a word-final complex coda: [bɑ.de.ɾɑ**zm**]# ‘war’ պատերազմ. It can become word-medial by adding the plural suffix *-neɾ*:

[5] In very few cases, we found root-final [ɾm] precede a glide, with possible resyllabification: [χɑ**ɾm**.jɑ, χɑɾ.mjɑ] ‘fake jewel’ խարմեայ where /-ja/ is a derivational suffix.

[bɑ.de.ɾɑ**zm**-neɾ] ‘wars’ պատերազմներ. Derivational suffixation again shows a tendency for a subsequent /j/: [bɑ.de.ɾɑ**zm**.jɑn] ‘of war’ պատերազմեան, with possible resyllabification [bɑ.de.ɾɑ**z**.mjɑn] (§5.5.2). In contrast, vowel reduction avoids creating this cluster: [χɑz.**muz**] ‘must (n)’ խազմուզ derives [χɑz.**mə.z**-ɑt͡ʃɑp] ‘gleucometer’ խազմզաչափ instead of **χɑ**zm**.zɑ.t͡ʃɑp*.

The other fricative-/m/ clusters also seem absent word-medially except before /n/-initial inflection. For example, medial /hm/ can be formed in [do**hm**-n=e] ‘family-DEF=is’ տոհմն է. But we could not find other word-medial case. And /sm/ is too rare even word-finally.

5.4.2.2 Avoiding word-medial /ɾn/ complex codas

For /ɾn/, these cluster can be pronounced word-finally as a complex coda [t͡se**ɾn**] ‘hand’ ձեռն. But schwa epenthesis is much more typical [t͡se**ɾən**] (§5.2.2.7. Word-medially before a consonant, epenthesis is the norm and (in HD’s judgment) the only possible pronunciation, such as in the derived word [t͡se**ɾən**-peg] ‘one-handed’ ձեռնբեկ. Schwa epenthesis in bound roots likewise avoids creating [ɾn] complex codas: <v**ṙn**del>→[və**ɾən**del] ‘to expel’ վռնտել instead of **və**ɾn**.del*.

find rən. word medial syllables

5.4.2.3 Avoiding word-medial nasal-fricative complex codas

For /ns/, this cluster is extremely rare word-finally. As surveyed in Section §5.2.2.6, final [ns] complex codas seem to be primarily loanwords like [finɑ**ns**] ‘finance’ ֆինանս. Word-medially, possible /VnsCV/ clusters arise via making compounds of the bound root /oɾens-/. Before C-initial suffixes or roots, this /ns/ cluster be pronounced with or without a schwa: [oɾe**ns**-tiɾ, soɾe**nəs**-tiɾ] ‘legislator’ օրէնսդիր. In HD’s judgment, the schwa form is much more typical in Western Armenian. VP informs us that the schwa-less form is however more common in Eastern Armenian. Thus, it seems that for Western Armenian at least, medial [ns] complex codas are dispreferred.

For /ms/, this cluster is relatively rare word-finally and it can form complex codas: [do**ms**] ‘ticket’ տոմս. Inflection can also create a medial case: [do**ms**-n=e] ‘ticket-DEF=is’ տոմսն է. This cluster is found word-medially for words that are derived from irregular root [ɑmis] ‘month’ ամիս. Such as the genitive [ɑ**ms**-vɑ] ‘month-GEN’ ամսուայ with an optional schwa [ɑ**məs**-vɑ]. Other cases seem too always involve adding a glide /j/ after the [ms] cluster: [ɑmen-ɑ**ms**-jɑ] ‘monthly’ ամենամսեայ, and the /VmsjV/ cluster can syllabify as either [Vms.jɑ] or [Vm.sjɑ].

Consider /mʃ/. Avoided in լմշտել verb active To trample on, to tread, to crush.

Consider /nʃ/. The native lexicon doesn't have any word-final [nʃ] complex codas. For example, Wiktionary lists a handful of such words and they're obvious loanwords: [ɾomɑnʃ] 'Romansch' ռոմանշ. But it seems possible to create such clusters word-medially – but in derived environments. Consider the word 'pressure' [d͡ʒənʃ-um] ճնշման. The suffix /-um/ is a special nominalizer suffix. For words with this nominalizer suffix, the standard genitive form is created by replacing [-um] with [-m-ɑn]: [dʒənʃ-m-ɑn]. However in casual speech, HD observes that his speech almost always turns this [nʃ] sequence into [nt͡ʃ]. It's an open question if such behavior means that [nʃ] complex codas are truly absent from Armenian, or if the affrication is a type of low-level phonetic change.

Consider /nχ/. Although we couldn't find a word with a final [nχ] cluster, we did find a few cases with word-medial [nχ]. For the latter category, the passive suffix /v/ can follow complex codas in Eastern Armenian: [kɑnχ.vel] 'to be anticipated' կանխվել, but not in Western Armenian [gɑn.χə.vil] կանխուիլ. This suggests that word-medial [nχ] are possible in principle, just rare, and they are subject to dialectal variation.

5.4.2.4 Avoiding word-medial fricative-fricative clusters

For the fricative-fricative clusters like /ʃχ, χʃ, χs/. These clusters are rare word-finally, thus even rarer word-medially (§5.2.3.9). So it's difficult to know whether their word-medial rarity is due to statistical change or something more. Some of the examples that we found are [əmpoʃχnel] 'to savor' ըմբոշխնել, [bɑχʃkil] 'to be refreshed' պաղշկիլ, [təχs.majɾ] 'brooding hen' թխսմայր.

5.4.2.5 Avoiding word-medial /C-k/ appendixes

For /C-k/, as said many times, the nominalizer suffix *-k* can be added after any consonant cluster.

For example, consider a polysyllabic word like the verb [dɑɾɑd͡z-el] 'to spread' տարածել. We can derive the noun [dɑɾɑt͡s-k] 'spread' տարածք. The appendix *k* can be made word-medial by adding an inflectional suffix: [dɑɾɑt͡s-k-neɾ] 'spread-PL' տարածքներ.

cite vaux dolation

However, the morphology generally avoids placing this suffix between a consonant-final and a consonant-initial derivational suffix morphology. For example, the root [mud] 'entry (archaic') մուտ has a more common form as [mut-k] 'entry' մուտք with the nominalizer. It can be preceded by C-initial inflection like [mut-k-n=e] 'entry-DEF-is' մուտքն է. The root 'entry' [mut-k] can also be used in compounds like [ɑɾev-mut-k] 'West' արեւմուտք where [ɑɾev] means 'sun'.

The adjective 'Western' however is formed by deleting the *k*, adding *-jan*, and reducing the root vowel: [ɑɾev-məd-jɑn] 'Western' արեւմտեան. Because of this apparent avoidance stem-medial *-k*, most analyses of the appendix *-k* argue that this suffix must be at the end of the word before inflection. I have only found a handful of counter-examples: [ɑɾev-mut-k-t͡si] 'Westerner' արեւմուտքցի.

5.4.2.6 Avoiding word-medial geminate complex codas

For geminates, because geminate codas are extremely rare word-finally, it's not surprising that they are rarer word-medially. The main example that we found is in the derivatives of the word [məɾ.ɾig] 'tempest' մրրիկ which form geminate codas: [mə**ɾɾ**.gɑ.jin] 'turbulent' մրրկային. However, HD reports that it's also common to remove this medial geminate by either degemination or by using a schwa: [mə**ɾ**.gɑ.jin, mə.**ɾə**ɾ.gɑ.jin]. This suggests that word-medial geminates are somewhat avoided word-medially.

5.4.3 Vowel-based or consonant-based restrictions on codas and complex codas

For virtually any pair of vowels and consonants, we can create a coda or complex coda. But, there are some restrictions for glides and /i,u/ (§5.4.3.1). There are more restrictions for the schwa /ə/ (§5.4.3.2). And the front vowel /ɤ/ simply always takes a coda when in the final syllable (§4.2.3).

5.4.3.1 Restrictions on /j/ and /v/ codas

The consonants /j/ and /v/ are relatively common codas in Armenian. But they have various restrictions on what types of vowels they can follow.

First, in word-final complex codas like /VjC/, the glide only follows /ɑ/ or /u/ in native words. For example, in Kouyoumdjian (1970)'s dictionary, we found 703 words with final /ɑjC/ like [lɑjn] 'wide' լայն, and 1272 words with final /ujC/ like d [kujn] 'color' գոյն; see more examples in Section §5.2.1.7.

The only other vowel that we found in Kouyoumdjian (1970) was /e/ in a single loanword: [kempejn] 'campaign' քէմփէյն. Wiktionary had more cases of loanwords with /ej/. Wiktionary also had some examples of final /əj,oj/, but these were all non-Western words or loanwords like [kɑzojl] 'gas oil' գազոյլ.

Second, word-final simplex codas like /Vj/ are generally rare (Table 5.66). In the Kouyoumdjian (1970) dictionary, there are many words (n=958) that are spelled with a final glide <y> յ; but most such cases involve a silent unpronounced glide

like <d͡zaɾ̓ay> [d͡zɑɾɑ] ‘servant’ ծառայ. After factoring out silent glides, we found only 62 words with a final glide coda. These have either /ɑ, e, o/.

Table 5.66: Restriction on word-final /j/ codas

/ɑj/	ˈh**ɑj**	‘Armenian’	հայ	n=41
	ɾusɑˈh**ɑj**	‘Russian-Armenian’	ռուսահայ	
	ˈp**ɑj**	‘verb’ ’	բայ	
/ej/	ˈt**ej**	‘tea’	թէյ	n=2
	ˈb**ej**	‘bey’	պէյ	
/oj/	ˈk**oj**	‘existent’	գոյ	n=19
	ˈχ**oj**	‘ram’	խոյ	
	iŋkⁿnɑˈk**oj**	‘self-existent’	ինքնագոյ	

For /uj/, this vowel+coda sequence is unattested word-finally in Kouyoumdjian (1970). On Wiktionary, we found a handful of /uj/-final words in loanwords and interjetions, such [uj] ‘ouch’ ույ. Word-medially, /uj/ can be easily derived via resyllabification. For example, /u/ can be used with /j/ as in [k**uj**n] ‘color’ գույն but [k**uj**.neɾ] ‘colors’ գույներ.

Third, /ij/ seems to be rare even as a word-medial syllable. For example in the Kouyoumdjian dictionary, there is only one such word that has a syllable ending in /ij/: [**ij**.nɑl] ‘to fall’ իյնալ. This is a high-frequency irregular verb. However, in HD’s judgment, it’s also common to pronounce this word as just [i.nɑl] without a glide.

Wiktionary listed only a handful of other examples of [(C)ij] syllables, but these were all either obscure dialectal words, or loanwords like [novoɾos**ij**sk] ‘Novorossiysk’ Նովոռոսիյսկ.

Fourth, alongside /ij/, a similar restriction seems to be that /uv/ codas are rare or banned (Table 5.67). We couldn’t find any such cases in Kouyoumdjian. On Wiktionary, the handful of examples we found were either obscure dialectal words, or loanwords like [ve.z**uv**] ‘Vesuvius’ վեզուվ. In contrast, Kouyoumdjian (1970) had many word-final /v/ codas for other vowels.

Table 5.67: Restriction on word-final /v/ codas

/ɑv/	ˈh**ɑv** pəˈn**ɑv**	‘chicken’ ‘never’	հաւ բնաւ	n= 410
/ev/	t͡s**ev** ɑˈɾ**ev**	‘form’ ‘sun’	ձեւ արեւ	n=426
/ov/	ˈd͡z**ov** ʒoˈʁ**ov**	‘sea’ ‘meeting’	ծով ժողով	n=144
/iv/	ˈt**iv** bɑˈd**iv**	‘number’ ‘honor’	պատիւ պատիւ	n=155

It is likely that the reason why /ij, uv/ codas are generally banned is because the vowel and coda would be too similar. The glide /j/ is phonologically a consonant-form of /i/. As for /v/, Armenian does not have a productive /w/ phoneme, so /v/ is the most similar consonant form of /u/. Such similarities for /i/-/j/ and /u/-/v/ are also found in vowel hiatus repair (§5.6).

Although [ij] and [uv] codas are generally banned, it is possible to have [i.jV] and [u.vV] sequences. That is, the sequence /ij/ can be formed as long as the glide is part of the next syllable: [kini-ji] ‘wine-GEN’ գինիի. Such a sequence is easily created because the regular genitive suffix is *-i*.

Similarly for /uv/, such sequences are attested when the /v/ is part of the next syllable: [ve.ɾ**u.v**ɑɾ] ‘up’ վերուվեր. Though it seems that even [u.v] sequences are rather infrequent.

And although /ij/ and /uv/ are generally avoided as codas, it is quite easy to form /ji/ and /vu/ syllables. For /ji/, this syllable is easily created when the genitive suffix /i/ triggers glide epenthesis after a vowel: [d͡zɑɾɑ-ji] ‘servant-GEN’ ծառայի. The /vu/ sequence is easily made when the productive nominalizer /-um/ is added after a /v/-final root: [χoɾov-um] ‘roasting’ խորովում. Furthermore, the sequences /iv/ and /uj/ are also easily found codas: [t**iv**] ‘number’ թիւ, [z**uj**k] ‘twin’ զոյգ.

5.4.3.2 Restrictions on codas for schwas

The schwa is a common vowel in Armenian and it can have virtually any type of coda or complex coda. But there are some restrictions involving a) glide codas (§5.4.3.2.1, b) rhotic-fricative complex codas (§5.4.3.2.2), and c) rhotic-nasal complex codas (§5.4.3.2.3).

5.4.3.2.1 Schwas avoid glide codas

First, it seems that the schwa cannot have a glide /j/ coda. We could not find any examples /(C)əj/ syllables in Kouyoumdjian. As for Wiktionary, we did find a handful of examples but these were all obscure dialectal words that don't exist in Western Armenian.

Furthermore, it seems that even the sequence /əj/ is banned regardless of syllable structure. That is, it is very rare to find a word where a schwa precedes a glide /j/. The closest examples we could find were atypical colloquial pronunciations of word-initial [CʏC] (§4.2.3). Very rarely, a [CʏC] word like like [kʏʁ] 'village' գիւղ can be pronounced as [CəjuC] or [CəjʏC]: [kə.juʁ, kə.jʏʁ].

Note that although /əj/ is a rare sequence, the sequence /jə/ is quite common. This is because the definite suffix is *-ə* after glide-final roots: [pɑj-ə] 'verb-DEF' բայը.

5.4.3.2.2 Schwas avoid rhotic-fricative codas codas

Second, it seems that the schwa avoids having a rhotic-fricative complex coda like [əɾs], [əɾʃ], and so on. The evidence comes mainly from schwa epenthesis. Data is summarized in Table 5.68.

Table 5.68: Avoidance of schwa-rhotic-fricative syllables

<CrsC>	<b**ɾs**del>	bə.**ɾəs**.tel	*b**əɾs**.tel	'to wrinkle'	պռստել
→[Cə.ɾəs.C]	<t'**ɾs**nel>	tə.**ɾəs**.ˈnel	*t**əɾs**.nel	'to soften'	թռսկել
<CrʃC>	<p'**ɾʃ**dal>	pə.**ɾəʃ**.tɑl	*p**əɾʃ**.tɑl	'to sneeze'	փռշտալ
→[Cə.ɾəʃ.C]	<t'**ɾʃ**nel>	tə.**ɾəʃ**.ˈnel	*t**əɾʃ**.nel	'to wither'	թռշնել
<CrχC>	<t͡ʃ'**rχ**gal>	t͡ʃə.**ɾəχ**.ˈkɑl	*t͡ʃ**əɾχ**.kɑl	'to clatter'	չրխկալ
→[Cə.ɾəχ.C]	<t'**rχ**gal>	tə.**ɾəχ**.ˈkɑl	*t**əɾχ**.kɑl	'to rattle'	թրխկալ

For [əɾs], the orthography provides few words where we have a /CɾsC/ cluster that would need schwa epenthesis. In these few cases, the schwa is added between the /ɾs/ cluster: <s**rs**gel> [sə.**ɾəs**.kel] 'to sprinkle' սրսկել. If an [əɾs] complex coda was easily allowed, then we would incorrectly expect **sə*ɾs*.kel*, cf. [h**ɑɾs**.nik] 'wedding' հարսնիք.

Similar behavior is found for [əɾʃ] : <p'**ɾʃ**ni> [pə.**ɾəʃ**.ni] 'type of tree', cf. [g**ɑɾʃ**.neʁ] 'sinewy' կարշնեղ. And for [əɾχ]: <t**rχ**got͡s'> [tə.**ɾəχ**.kot͡s] 'gunshot' դրխկոց, cf. [m**ɑɾχ**] 'resinous pine' մարխ.

For the other orthographic combinations of consonant + rhotic + fricative combinations, these clusters were either absent or astonishingly rare in dictionaries.

We could find suitable examples to see how they would be pronounced in Western Armenian.

Note that although the general behavior is for a schwa /ə/ to avoid getting a rhotic-fricative complex codas, such syllables do occur in special morphologically-induced circumstances. Consider the word [tu**ɾs**] 'outside' դուրս. Before a vowel-initial derivational suffix, the root's high vowel is reduced to a schwa: [tə**ɾ.s**-e.t͡si] 'foreigner' դրսեցի. In very rare cases, a nasal is inserted after this root: [tə**ɾs**.n-ɑgɑn] 'exterior' դրսնական. In such derivatives, the morphology allows a [əɾs] syllable to maintain similarity with the root [tuɾs], and to avoid a non-similar output like **tə.**ɾəs**.na.gan.*[6]

If we look beyond Western Armenian and into Eastern Armenian, we find similar analogical effects happen in passivization passive chapter. In both dialects, the passive stem tends to be phonologically identical to the active stem. In Western Armenian, the passive suffix /v/ cannot follow any complex coda, so it triggers a schwa whether after any rhotic-fricative cluster: [tə**ɾ.ʒ**-e-l] 'to infringe' դրժել but passive [tə**ɾ.ʒ**ə-v-i-l] 'to be infringed' դրժուիլ, cf. [m**ɑɾz**-e-l] 'to exercise' մարզել and passive [m**ɑɾ.z**ə-v-i-l] 'to be exercised' մարզուիլ.

But in Eastern Armenian, the passive /v/ can follow complex codas: [m**ɑɾz**.-v-e-l] մարզվել, and it can follow [əɾF] syllables: [d**əɾʒ**.-v-e-l] դրժվել. The passive data thus shows that schwas generally avoid having a rhotic-fricative complex coda, unless there is a morphological pressure to maintain identity of roots.

5.4.3.2.3 Schwas avoid word-medial rhotic-nasal codas

Because of how the morphology of Armenian works, the orthography creates consonant + rhotic + fricative clusters word-medially, not word-finally. Thus, although word-medial [əɾs] complex codas are banned, it is unknown whether these clusters are fine word-finally. This ambiguity is absent for word-medial rhotic-nasal complex codas, summarized in Table 5.69.

Table 5.69: Avoidance of word-medial schwa-rhotic-nasal syllables

<CrmC>	<k**rm**p'al>	kə.**ɾəm**.ˈpɑl	*k**əɾm**.pɑl	'to thump'	գրմփալ
→[Cə.ɾəm.C]	<x**ṙm**p'et͡s'>	χə.**ɾəm**.ˈpot͡s	*χ**əɾm**.pot͡s	'snore'	խռմփոց
<CrnC>	<z**rn**kal>	zə.**ɾəŋ**.kɑl	*z**əɾŋ**.kɑl	'to tinkle'	զրնգալ
→[Cə.ɾən.C]	<h**ṙn**t͡ʃ'el>	hə.**ɾən**.t͡ʃel	*h**əɾn**.t͡ʃel	'to snort'	հռնչել

[6] This need for similarity of roots does show some variation. For example [m**əɾ.s**-i-l] 'to feel cold' մրսիլ and its derivative [m**əɾs**.-kod, mə.**ɾəs**.-kod] 'chilly' մրսկոտ.

Interestingly, although the schwa can't have a rhotic-fricative complex coda, it can have a word-final /ɾn/ complex coda : [gɔ.z**əɾn**] 'young camel' կոզն. But word-medially, such complex codas are generally avoided for all types of vowels: <a**ɾ̈n**t͡ʃ'agan> [ɑ.**ɾən**.t͡ʃɑ.ˈgɑn] 'relative' առնչական (§5.4.2.2). Schwas also avoid this complex coda word-medially: <p**ɾ̈n**gil> [pə.**ɾəŋ**.gil] 'to be inflamed' բռնկիլ.

For /ɾm/, this cluster can be formed word-medially for non-schwa vowels, but with various restrictive tendencies (§5.4.2.1). For the schwa, it seems that it cannot have a /ɾm/ complex: <x**ɾ̈m**p'al> [χə.**ɾəm**.pɑl] 'to snore' խռմփալ.

5.4.4 Complex codas created by schwa elision

write after schwa elision abstamp

5.5 Complex onset restrictions

The typical syllable in Armenian can be described as CVC or CVCC without a complex onset (§5.1.2). However, in principle, Armenian does creating complex onsets. But the possible types of complex onsets are significantly restricted and they can categorized as one of the following:

1. consonant + glide sequences that are formed word-initially (§5.5.1)
2. consonant + glide sequences that are formed word-medially but with variable resyllabification (§5.5.2)
3. consonant-consonant clusters that typically get a schwa [CəCV] but can lose the schwa in casual speech from schwa elision [CCV] (§5.5.3)

Another category are consonant + sonorant sequences from non-nativized loanwords. For example, the name <k'lara> [klɑɾɑ] 'Clara' Քլարա or the word <gram> [gəɾɑm] 'gram' կրամ. However, such loanwords can be nativized by adding a schwa: [kəlɑɾɑ, gəɾɑm]. We don't discuss such loanwords further because they are quite limited.

5.5.1 Complex onsets of consonant + glide

Grammars often report that the only acceptable 'normal' complex onset is a consonant + glide /j/ combination, such as [gjɑŋk] 'life' կեանք. However, although such complex codas exist, they are significantly restricted in their distribution. These restrictions involve the following asymmetries:

1. Few word-initial /Cj/ sequences.
2. /Cj/ is restricted to being almost always before /ɑ//

3. Variation in how word-medial /Cj/ clusters are syllabified.

First, word-initially, such [Cj] clusters are limited to a handful of roots (Table 5.70). The Kouyoumdjian dictionary listed 12 roots that start with a [CjV] sequence. We list these roots below.

Table 5.70: Word-initial [Cj] complex onsets

ˈgjɑl	'to exist'	կեալ	ˈʁjɑg	'rudder'	ղեակ
ˈsjɑv	'black'	սեաւ	ˈkjɑɾ	'necklace'	քեառ
ˈljɑɾt	'liver'	լեարդ	ˈgjɑŋk	'life'	կեանք
ˈnjɑɾt	'fibre'	նեարդ	ˈdjɑɾk	'gentlemen'	տեարք
ˈsjɑmk	'threshold'	սեամք	ˈsjɑmk	'threshold'	սեամք
ˈljɑɾən	'mountain'	լեառն	ˈdjɑɾən	'of the Lord'	տեառն
ˈzjɑn	'pain'	զեան			

5 of these roots are then used to form 39 derivatives, such as [njɑɾt-ɑvoɾ] 'fibrilous' նեարդաւոր. The total number of [CjV] words ends up as just 49 words (12 roots and 39 derivatives).

Second, as is visible above, these word-initial [CjV] clusters are limited to cases where the vowel is /ɑ/. For the other vowels, there are some variable cases of [Cju] clusters. Some roots like [kʏʁ] 'village' գիւղ have a front-round vowel [ʏ] that can optionally be pronounced as [ju], as in [kjuʁ]. This is discussed in Section §4.2.3.[7]

Another exceptional case is the archaic word մեօքակերպ <meōk'agerb> 'similar to us'. As an orthographic rule, the sequence եօ <ōk> is supposed to be pronounced as [jo]. For this word, the default pronunciation is thus [mjokɑgeɾb] with a [Cj] complex onset. But HD feels that because this word is low-frequency (and previously unknown to him), then this word can also be pronounced as [mejokɑgeɾb] with glide epenthesis.

In fact, it seems that in Western Armenian, the [Cj] complex onset must always precede either a /ɑ/ (in the usual case) or a /u, o/ (in rare cases). We examined Armenian Wiktionary, and we confirmed this generalization. Armenian Wiktionary did report a handful of words that with [Cj] and other vowels. But these counter-examples are either obvious loanwords like [gjoɾliŋk] (WA), [kjoɾliŋg] (EA) 'curling' կյորլինգ, or dialectal words that don't genuinely exist in Western Armenian.

[7] However, Armenian dialects do vary in how often they have [Cj] complex onsets. The Western Armenian [ʏ] vowel corresponds to a [ju] sequence in Eastern Armenian. Does a Western word like [ɑχpʏɾ] 'fountain' աղբիւր is pronounced as [ɑχpjuɾ] in Eastern Armenian. Eastern Armenian thus has significantly more cases of [Cj] complex onsets than Western Armenian.

Third, even though [Cj] complex onsets are permitted in Western Armenian, these clusters are dispreferred word-medially (Table 5.71). Specifically, if a /Cj/ cluster is intervocalic, then the /Cj/ cluster can be pronounced either as a complex onset [V.CjV] or into separate syllables [VC.jV]. For example, most word-medial cases of [Cj] are due to adding one of the following derivational suffixes: /jɑn, jɑg, jɑl, jɑ/. Each of these suffixes allow resyllabification. This variation is discussed in more depth in Section §5.5.2.

Table 5.71: Variation in re-syllabification for /j/-initial suffixes

/jɑn/	ɑ.tɑ.mɑn.ˈ**tj**ɑn ɑ.tɑ.mɑn**t.j**ɑn	‘diamond-encrusted’ ադամանդեան	cf. ɑ.tɑ.mɑnt	‘diamond’ ադամանդ
/jɑl/	ʃeʃ.ˈ**tj**ɑl ʃeʃ**t.**ˈ**j**ɑl	‘accented’ շեշտեալ	cf. ˈʃeʃt	‘stress’ արբած
/jɑ/	pɑj.ˈ**dj**ɑ pɑj**d.**ˈ**j**ɑ	‘wooden’ փայտեայ	cf. ˈpɑjd	‘wood’ փայտ
/jɑg/	de.ˈ**ʁj**ɑg de**ʁ.**ˈ**j**ɑg	‘well-informed’ տեղեակ	cf. ˈdeʁ	‘place’ տեղ

In sum, even though [Cj] complex onsets are allowed, they seem to be either a marginal or restricted part of the language, such that the language has strategies to avoid pronouncing the [Cj] onsets.

5.5.2 Variation in syllabifying word-medial consonant-glide clusters

Armenian generally avoids all complex onsets except for /Cj/ sequences. However, there is significant free variation in how easily or how often [Cj] complex onsets are formed.

Word-initially, a [Cj] complex onset generally cannot alternate with other forms. Thus a word like [**gj**ɑŋk] ‘life’ կեանք is pronounced only with a [Cj] complex onset. But there a few words which show free variation such that [Cj] is replaced by [Cij]: [**lj**ɑɾt ~ **lij**ɑɾt] ‘liver’ լեարդ.

Word-medially, we find much more free variation. This variation is correlated with the following parameters:

- Whether the /Cj/ follows a vowel /VCjV/ or consonant /VCCjV/.
- If the pre-/j/ consonants in /VCCjV/ form a falling-sonority complex coda.
- If the pre-/j/ consonants is two consonants or more /V(C)CCjV/.
- Orthographic rules for pronouncing [Cj] for different words.
- Preferences for or against [Cj] by certain suffixes.

We go through the various parameters, summarized in Table 5.72.

Table 5.72: Variation in syllabifying word-medially /Cj/ clusters

[V.CjV~VC.jV]	gə.ˈ**ɾj**ɑ he.ˈ**kj**ɑt	gə**ɾ**.ˈ**j**ɑ he**k**.ˈ**j**ɑt	‘turtle’ ‘fable’	կրիայ հէքեաթ
[VC.CjV~VCC.jV]	pər.ˈ**tj**ɑ uχ.ˈ**tj**ɑl	pə**ɾt**.ˈ**j**ɑ uχ**t**.ˈ**j**ɑl	‘woolen’ ‘votive’	բրդեայ ուխտեալ
[VC.CjV, *VCC.jV]	əs.toɾ.ɑk.ˈ**ɾj**ɑl jot.ˈ**nj**ɑg	*əs.toɾ.ɑk**ɾ**.ˈ**j**ɑl *jo**tn**.**j**ɑg	‘undersigned’ ‘septet’	ստորագրեալ եօթնեակ
[VCC.CjV, *VCCC.jV]	vosp.ˈ**nj**ɑ jeɾg.goʁm.ˈ**nj**ɑ	*vosp**n**.**j**ɑ *jeɾg.goʁm**n**.**j**ɑ	‘freckled’ ‘bilateral’	ոսպնեայ երկկողմնեայ
[V(C)C.ɾjV, *V(C)Cɾ.jV] or [V(C).Cəɾ.jV]	sɑnd.ˈ**ɾj**ɑ sɑn.dəɾ.ˈ**j**ɑ vosk.ˈ**ɾj**ɑ vos.kəɾ.ˈ**j**ɑ meʁ.ˈ**ɾj**ɑl me.ʁəɾ.ˈ**j**ɑ	*sɑnd**ɾ**.**j**ɑ *vosk**ɾ**.**j**ɑ *meʁ**ɾ**.**j**ɑ	‘pectinal’ ‘bony’ ‘honeyed’	սանտրեայ ոսկրեայ մեղրեալ

First, consider the intervocalic parameter. In /VCjV/ sequences, the /Cj/ can form either a complex onset or be in different syllables: [se**n.j**ɑg] or [se.**nj**ɑg] ‘room’ սենեակ. The choice of syllabification is partially just free variation, but there is some correlation with dialect. In our experience, Eastern Armenian speakers are more likely to use [V.CjV] than [VC.jV], while Western Armenian speakers are more likely to use [V.CjV].

Second, if the /Cj/ precedes a consonant, then resyllabification is possible if the preceding consonant cluster can form a complex coda: [bɑn.**dj**ɑl ~ bɑn**d.j**ɑ] ‘prisoner’ բանտեալ. But if the preceding consonant cluster can’t form complex codas in the language, then we don’t have resyllabification: [lus.**nj**ɑg] but not **lus**n.j**ag* ‘lunatic’ լուսնեակ.

Note that the preceding cluster can even be complex coda that arguably includes an extrasyllabic appendix (§5.1.4). For example, word-final [ks] clusters are attested and they’re arguably comprised of an appendix *-s*. These structures can be formed via /Cj/ resyllabification: [me.dɑk.**sj**ɑ ~ me.dɑk**s.j**ɑ] ‘silken’ մետաքսեայ

Third, when the /j/ follows three consonants, then the default pronunciation is to create the [Cj] complex onset: [hɑɾs.**nj**ɑg] ‘chrysalis’ հարսնեակ. The three consonants cannot form a complex coda because 3-consonant codas are generally banned in Armenian: **hɑɾs**n.j**ag* (§5.1.5).

An interesting type of variation occurs when the glide /j/ follows a /(C)Cɾ/ cluster. Here, the prescriptive pronunciation is to create a complex onset: [ɑ.ʁek.sɑnd.**ɾj**ɑn] ‘Alexandrine’ աղէքսանդրեան. Resyllabification with a 3-consonant coda is at

best odd to hear: **ɑ.ʁek.sandɾ.jɑ*. However, in casual speech, HD has observed that some speakers (including himself) can epenthesize a schwa before the /ɾ/ for at least some of these: [ɑ.ʁek.sɑn.dəɾ.ˈjɑn].

This sub-pattern of pre-rhotic schwas can interact with vowel reduction. Consider the root [əndiɾ] 'chosen' ընտիր. Adding a suffix /-jɑl/ causes the vowel to reduce to zero in the prescriptive form: [ənd.**ɾ**jɑl] 'elected' ընտրեալ. Resyllabification is of course impossible: **əndɾ.jal*. However in colloquial speech, a schwa can occur before the /ɾ/: [ən.də**ɾ**jɑl]. Other examples include deriving [t͡ʃɑ.mit͡ʃ] 'raisin' չամիչ to [t͡ʃɑm.**t͡ʃ**jɑ, t͡ʃɑ.mə**t͡ʃ**.jɑ] 'plum cake' չամչեայ but not **t͡ʃɑmt͡ʃ.jɑ* It's unclear if these schwas here are epenthetic or a reduced form of the root high vowel.

Another sub-variation is that in colloquial speech, it is possible to sometimes change a /CCjV/ form into [CCijV]. For example, [dɑs.njɑg] 'decade' տասնեակ can be pronounced as [dɑs.ni.jɑg]. Data is too limited to make any concrete generalizations on such variation. But HD suspects that it's relatively common when the /j/ is part of a suffix that follows a rising-sonority consonant sequence.

Fourth, the traditional orthography for Armenian has rather opaque rules on how to spell [Cj] clusters. The orthographic cluster իա <ia> is found inside various roots and suffixes. When this vowel sequence is after the word-initial consonant, the sequence is pronounced with glide epenthesis: <d**ia**r> [d**ij**ɑɾ] 'mister' տիար. But after a word-medial consonant, some cases of <ia> are pronounced strictly as [jɑ] like <mariam> [mɑɾ.jɑm] 'Mariam' Մարիամ,[8] some can show variation between [jɑ] and [ijɑ]: <tasdiarag> [tɑst**ij**ɑɾɑg, tɑst**j**ɑɾɑg] 'educator' դաստիարակ, and some always take [ija]: <badriark> [bɑdɾ**ij**ɑɾk] պատրիարք. Thus, certain roots seem to idiosyncratically ban or allow [Cj].

Fifth, some suffixes seem to have individual preferences for whether they allow [Cj] complex onsets or not. Consider the nominalizer suffix /-utɤn/ -ութիւն. This suffix has many possible pronunciations, such with replacing /ɤ/ with a glide-vowel sequence: [-utjun] (§4.5.1). One pronunciation is to create a [Cj] sequence as a complex onset: [uɾɑχ-u.ˈtjun] 'happiness' ուրախութիւն. But it's also possible to make the /Cj/ sequence be in separate syllables: [uɾɑχ-ut.ˈjun]. In HD's judgment, the complex onset [Cj] form feels more natural than [C.j].

This morpheme-specific behavior can interact with the other parameters for /j/-resyllabification. Consider the country-naming suffix <ia> իա. After a single consonant, this suffix is often pronounced as just [-jɑ]: [i.dɑ**l.j**ɑ] 'Italy' Իտալիա. Resyllabification is possible [i.dɑ.**lj**ɑ], but HD feels this is substantially less common. After two consonants that can be a complex coda, HD reports possible

[8] Eastern Armenian seems to allow the form [mɑ.**ɾj**ɑm] but this sounds odd to HD's Western ears.

resyllabification: [tu**ɾk.j**ɑ, tu**ɾ.kj**ɑ] ‘Turkey’ Թուրքիա. But if the consonant cluster can’t form a complex coda, then the [ijɑ] form feels more preferred than [jɑ]: [ɑŋk.li.jɑ] instead of [ɑ**ŋk.lj**ɑ], and not **a**ŋkl.j**a* ‘England’ Անգլիա.

Another case of morpheme-specific variation is the family-name suffix (patronymic suffix) [-jɑn] -եան. In principle, this suffix can form [Cj] complex onsets: [dov.le.**tj**ɑn ~ dov.le**t.j**ɑn] ‘Deovletian’ Տէօվլէթեան. However, in HD’s judgment, it is much more typical to break the /Cj/ sequence into separate syllables. Unfortunately, dictionary data is too restricted to determine the exact rates or preferences for this suffix.

Thus, although [Cj] complex onsets are possible, it seems that the language tries to remove such complex onsets if possible. Based on how /V(C)(C)CjV/ clusters are syllabified, it seems that creating [Cj] acts as a last resort.

5.5.3 Complex onsets created from schwa elision

write

TODO: for some reason i dont see ⊠⊠⊠⊠⊠⊠ as deriving ⊠⊠⊠⊠⊠⊠⊠⊠ in my dictionary even though its on nayiri. did i forget to find cases of ⊠⊠ with glide deletion?

TODO: ⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ – some roots have a final /a/ allomorph because foreign so treat it as allomorphy

5.6 Vowel-vowel sequences or vowel hiatus repair

Vowel hiatus is when the morphology brings together two vowels, creating a hiatus or break from one syllable to another. A sequence of vowels must be pronounced in some way, meaning that the vowel hiatus must be repaired. In Armenian, possible repairs include inserting a glide /j/ (glide epenthesis), inserting a glottal stop /ʔ/, changing one of the vowels to a consonant, deleting one of the vowels, or merging the two vowels into one vowel.[9]

This section goes over how vowel hiatus is repaired in diverse morphological contexts. The choice of rule varies by the type of first vowel and by the morphological identity of the second vowel. The following subsections go through these morphological categories and then phonological subcategories. Furthermore, some vowel-vowel sequences can undergo diverse repair rules without a clear preference, while some sequences can undergo mainly one rule with few exceptions.

[9] Unfortunately, there are no words that end in a vowel /ɤ/ (§4.2.3), and there are virtually no words that end in /ə/ and that can take derivational suffixes.

- In roots, vowel hiatus is largely diachronic or limited to loanwords (§5.6.1).
- Before derivational suffixes, various rules are possible, and the choice varies by vowel and sometimes by the word (§5.6.2).
- In compounds and after prefixoids, some words use various repair rules (similar to derivational suffixation), while some use just glottal stop epenthesis (§5.6.3).
- Before regular inflectional suffixes, glide epenthesis is the main strategy (§5.6.4).
- Before irregular inflectional suffixes, different strategies are possible (§5.6.5).
- Before clitics, glide epenthesis is the main strategy (§5.6.6).

Although vowel hiatus repair rules are naturally limited to only vowel-vowel sequences, there are some consonant-initial suffixes that use rules as a type of paradigmatically-induced overapplication (§5.6.7). There are also some minor problems in diachrony (§5.6.8).

5.6.1 Vowel hiatus in underived words and roots

In this chapter, we almost exclusively focus on how vowel hiatus or vowel-vowel sequences are handled in derived forms, not underived forms. This is because underived forms don't give solid evidence on whether what we see is actually a vowel-vowel sequence, or something else.

To illustrate, consider the word <gr**ia**y> 'turtle' կրիայ. Although the orthography shows a vowel sequence <ia>, this word is pronounced as [gəɾ**jɑ**]. The mismatch between the orthographic <ia> and the pronounced [jɑ] is because of diachrony. In Classical Armenian, such orthographic vowel sequences likely reflected some type of diphthong. Over time, this diphthong turned into a Modern [jɑ] sequence. Synchronically however, even though the [jɑ] is spelled as two vowels <ia>, there is no synchronic evidence that the pronounced [j] is derived from an underlying /i/. Assuming the schwa is epenthetic, the Armenian child has no reason to think that [gəɾjɑ] is underlyingly /gɾiɑ/ instead of just /gɾjɑ/. Similar orthography-phonology mismatches include words like <**eō**tə> that is pronounced as [**jo**tə] 'seven' եօթը. See Section §4.1.7 for discussion on the phonemic status of /j/.

A large category of such mismatches involves the digraph <ow> ու that is read as [v] before vowels, but [u] elsewhere. In a root before a vowel, the digraph is read as [v]: <aɣ**owē**s> [ɑʁ**ve**s] 'fox' աղուէս. If the [v] is part of an unsyllabiable consonant cluster, then we get an extra schwa: <n**owa**z> [n**əvɑ**z] 'less' նուազ. The reason why such mismatches exist is again because of Classical Armenian where clusters like <owē, owa> were likely read as some diphthong [ʊe̯,ʊɑ̯]. But

in the modern language, this diphthong was replaced by a [vɑ] sequence. Because such roots don't show any alternations, the child has no reason to treat a word like [ɑʁ**ve**s] as derived from an underlying /ɑʁues/ instead of just /ɑʁves/.

For such underived words, we only found two corners of the grammar where it is likely that such an orthographic cluster does reflect an underlying vowel-vowel sequence. Both corners involve free variation. One such corner is for the word <hr**ea**y> 'Jewish' հրեայ where the cluster is variably pronounced as [hər**eja**] (more archaic) or [hər**ja**] (more modern). However, such examples likely reflect a change in the underlying representation from /hɾejɑ/ (or /hɾeɑ/) to /hɾjɑ/.

The other corner involves non-nativized loanwords. Consider a word like <k'**ao**s> [k**ɑʔo**s] 'chaos' քաոս or <bant'**eo**n> [bɑnteʔon] 'pantheon' պանթէոն. Such words are obviously loanwords, and their vowel sequence is borrowed in tact, and their hiatus is resolved via a glottal stop. When such words get more frequent or common, their pronunciation is simplified. For example, consider <madt'**eo**s> 'Matthew' Մատթէոս, which prescriptively can be pronounced as [mɑt**eʔo**s], but is more commonly pronounced as [mɑt**jo**s].

Such cases of free variation indicate that vowel sequences must be repaired in some way. But they don't tell us what are productive means of handling vowel-vowel sequences that are created from the morphology, i.e., derived forms. At best, such underived forms tell us what are possible rules for reading orthographic clusters (orthography-phonology mismatches: Section §3.4) and how a word's pronunciation can get simplified over time as it gets nativized or more frequent.

5.6.2 Vowel hiatus repair between roots and derivational suffixes

When studying vowel hiatus in Armenian, the most common contexts in derivational morphology are /V+ɑ/ and /V+u/. For derivational morphology, the majority of derivational suffixes start with /ɑ/. Compounds are also typically formed by combining two stems with the linker /ɑ/. For the other vowels, a common /u/-initial derivational suffix is the nominalizer /-utjɤn/ -ութիւն. There are some derivational suffixes for /e/, /o/, and other /u/ suffixes.

It is rather difficult to study how some phonological process works in derivational morphology for various reasons. The first is that different dictionaries have different lists of words. Thus a word that we found in one major dictionary like Kouyoumdjian might not exist in another major dictionary. Second, creating new words is a very creative process. So we can't say that some vowel-vowel sequence is always repaired in some specific way, simply because some dictionary out there may list such a sequence with an alternative repair. The third reason

is variability. For a given root, we can find some derivatives that utilize one rule, and other set of derivatives that utilize another rule.

The above factors ultimately mean that any study on vowel hiatus repair that is using a dictionary has limitations. However, because the dictionary we use is rather large, we hope that the following descriptive patterns that we provide can reflect the tendencies in the language.

- /ɑ/ + vowel: repaired via glide epenthesis for most words, but /ɑ/ deletion for some words (§5.6.2.1).
- /e/ + vowel: repaired via glide epenthesis for most words, but /e/ deletion for some words,).
- /i/ + vowel: repaired via glide epenthesis, deletion, or coalescence/merger (/i-ɑ/→[e]), depending on the word (§5.6.2.3).
- /ə/ + vowel: unattested.
- /o/ + vowel: repaired via glide epenthesis (§5.6.2.4).
- /u/ + vowel: repaired via de-vocalization for most words (/u-ɑ/→[vɑ], /u-i/→[vi], etc.), and glide epenthesis for some (§5.6.2.5).

Note that in derivational morphology, glide epenthesis can often be replaced with a glottal stop epenthesis, with unclear sociolinguistic effects or factors.

5.6.2.1 Stem final /ɑ/

When a /ɑ/-final stem precedes the compound linker /ɑ/ or a vowel-initial derivational suffix, there are two attested repair rules: deleting the stem /ɑ/ or glide epenthesis. The choice between the two rules correlates with spelling. If the stem /ɑ/ is spelled as with a final <a> ա, then deletion is the default rule. But if the stem /ɑ/ is spelled as <ay> այ, then glide epenthesis is the default rule. However, there are a handful of stems which either display both rules or display the opposite rule.

First let us consider stems with a final <ay>. Glide epenthesis is the default rule before a derivational suffix that starts with an /ɑ/ (/-ɑX/), /u/ (in /-utjɤn/), /u/ in some other suffix (/-uX/), /i/ (-/iX/, /e/ (/-eX/), and /o/ (/-oX/). Glide epenthesis is also the rule before the compound linker /-ɑ-/.

In Table 5.73 and onward, we list the number of roots that we found in Kouyoumdjian such that a) this root had this specific phonology-orthographic structure, b) this root was listed as an entry in the dictionary, and c) the dictionary listed derivatives for this root, and d) the derivatives displayed only this vowel hiatus rule in this vowel-vowel sequence. For illustration, we also show the underlying form of the suffix. We show a simplified segmentation for verbal suffixes, meaning /-il/ instead of /-i-l/ (TH-INF).

Table 5.73: Glide epenthesis in derivation for stems with final [ɑ] <ay>

+/-ɑX/	<hsg**ay**>	həsˈk**ɑ**	‘giant’	հսկայ	n=45
+/-ɑgɑn/	<hsg**ay**agan>	həsk**ɑj-ɑ**ˈgɑn	‘giant-like’	հսկայական	
+/-ɑ-/	<kowrb**ay**>	kuɾˈb**ɑ**	‘hose’	գուրպայ	n=35
	<kowrb**ay**akord͡z>	kuɾb**ɑj-ɑ**-ˈkoɾd͡z	‘hosier’	գուրպայագործ	
	cf <kord͡z>	koɾd͡z	‘work’	գործ	
+/-utjɤn/	<xapep**ay**>	χɑpeˈp**ɑ**	‘deceitful’	խաբեբայ	n=59
	<xapep**ay**owt’iwn>	χɑpep**ɑj-u**ˈtjɤn	‘deception’	խաբեբայութիւն	
+/-uX/	<k’ahan**ay**>	kɑhɑˈn**ɑ**	‘priest’	քահանայ	n=7
+/-uhi/	<k’ahan**ay**owhi>	kɑhɑn**ɑj-u**ˈhi	‘priestess’	քահանայուհի	
+/-iX/	<ʃɾt͡ʃ**ay**>	ʃəɾt͡ʃɑˈk**ɑ**	‘roamer’	շրջագայ	n=7
+/-il/	<k’ʃɾt͡ʃ**ay**il>	ʃəɾt͡ʃɑk**ɑ**ˈj-il	‘to stroll’	շրջագայիլ	
+/-eX/	<p’ilisop’**ay**>	pilisop**ɑ**	‘philosopher’	փիլիսոփայ	n=17
+/-el/	<p’ilisop’**ay**el>	pilisop**ɑ**ˈj-el	‘to philosophize’	փիլիսոփայել	
+/-oX/	<xn**ay**>	χən**ɑ**	‘caution’	խնայ	n=11
+/-oʁ/	<xn**ay**oɣ>	χən**ɑ**ˈj-oʁ	‘thrifty’	խնայող	

In contrast, for stems that end in <a>, the default behavior is deleting the stem <a> (Table 5.74).

Table 5.74: Stem-vowel deletion in derivation for stems with final [ɑ] <a>

+/-ɑX/	<asi**a**>	ɑsˈj**ɑ**	‘Asia’	Ասիա	n=18
+/-ɑgɑn/	<asi**a**gan>	ɑsj-**ɑ**ˈgɑn	‘Asian’	ասիական	
+/-ɑ-/	<fizik’**a**>	fiziˈk**ɑ**	‘physics’	ֆիզիքա	n=10
	<fizik’**a**kēd>	fizik-**ɑ**-ˈked	‘physicist’	ֆիզիքագէտ	
	cf. <kēd>	ˈked	‘learned (archaic)’	գէտ	
+/-utjɤn/	<k’aɣte**a**>	kɑχtej**ɑ**	‘Chaldea’	Քաղդէա	n=1
	<k’aɣte**ow**t’iwn>	kɑχtej-**u**ˈtjɤn	‘astrology’	քաղդէութիւն	
+/-uX/	<frans**a**>	fəɾɑns**ɑ**	‘France’	Ֆրանսա	n=1
+/-uhi/	<frans**ow**hi>	fəɾɑns-**u**ˈhi	‘French woman’	ֆրանսուհի	
+/-eX/	<kaɣɣi**a**>	kɑʁʁij**ɑ**	‘Gaul’	Գաղղիա	n=1
+/-eɾen/	<kaɣɣi**e**rēn>	kɑʁʁij-**e**ˈɾen	‘Gallic language’	գաղղիերէն	

Based on the data so far, it is clear that glide epenthesis is the norm for stems spelled with <ay>, while deletion is the norm for words spelled with <a>. But, there is some degree of variation (Table 5.75). In the Kouyoumdjian dictionary, we found some stems that are spelled with <ay> but display deletion before some derivational suffixes.

Table 5.75: Glide epenthesis in some stems with final [ɑ] <a>

[ɑ]+/-eni/	<nowma>	nuˈm**ɑ**	‘Mandarin’	Նումա
	<nowm**ay**eni>	num**ɑj**-enˈi	‘Mandarin orange’	Նումայենի
[ɑ]+/-ɑ-/	<delta>	delˈt**ɑ**	‘delta’	տէլդա
	<delt**ay**agerb>	delt**ɑj**-**ɑ**-ˈgɛrb	‘deltoid’	տէլդայակերպ
	cf. <gerb>	ˈgɛrb	‘manner’	կերպ

Besides the above exceptions, we found a handful more words that are spelled with <ay> but display vowel deletion in their attested derivatives (Table 5.76). For some of these words like [ɑgɾɑ-pɛɾɑd] ‘toothless’, it is possible that these words underlyingly never had the linking vowel /-ɑ-/ in the first place /ɑgɾɑ + pɛɾɑd/, thus there is no vowel hiatus to repair. See Section §5.6.3.2 for such compounds.

Table 5.76: Vowel deletion in some stems with final [ɑ] <ay>

[ɑ]+/-ɑnɑl/	<momi**ay**>	momˈj**ɑ**	‘mummy’	մումիայ
	<momi**a**nal>	momj-**ɑ**ˈ**n**ɑl	‘to get mummified’	մումիանալ
[ɑ]+/-ɑ-/	<amir**ay**>	ɑmiˈɾ**ɑ**	‘lord’	ամիրայ
	<amir**a**bed>	ɑmiɾ-**ɑ**-ˈbɛd	‘caliph’	ամիրապէտ
	cf. <bed>	ˈbɛd	‘leader’	պէտ
[ɑ]+/-ɑ-/	<agṙ**ay**>	ɑgɾ**ɑ**	‘tooth’	ակռայ
	<agṙ**a**p’eṙad>	ɑgɾ-**ɑ**-peˈɾɑd	‘gap-toothed’	ակռափեռատ
	cf. <p’eṙad>	peˈɾɑd	‘toothless’	փեռատ

What is more revealing is that some words are spelled with <ay>, display glide epenthesis in some derivatives, but vowel deletion in other derivatives (Table 5.77).

Table 5.77: Vowel deletion or glide deletion in some stems with final [ɑ] <ay>

	<sadan**ay**>	sɑdɑˈn**ɑ**	‘devil’	սատանայ
+/-utjʏn/	<sadan**ayow**t’iwn>	sɑdɑn**ɑj**-**u**ˈtjʏn	‘devilry’	սատանայութիւն
+/-ig/	<sadan**ayig**>	sɑdɑˈn-**ig**	‘devilet’	սատանիկ
	<aṙarg**ay**>	ɑɾɑɾˈg**ɑ**	‘object’	առարկայ
+/-ɑgɑn/	<aṙarg**ayag**an>	ɑɾɑɾg**ɑj**-**ɑ**ˈgɑn	‘objective’	առարկայական
+/-el/	<aṙarg**e**l>	ɑɾɑɾˈg-**e**l	‘to object’	առարկել
	<mek’en**ay**>	mɛkɛn**ɑ**	‘machine’	մեքենայ
+/-ɑpɑɾ/	<mek’en**ay**apar>	mɛkɛˈn**ɑj**-**ɑ**pɑɾ	‘mechanically’	մեքենայաբար
+/-el/	<mek’en**a**par>	mɛˈkɛn-**ɑ**pɑɾ	‘mechanically’	մեքենաբար

Based on the above main patterns and exceptions, we conclude that glide epenthesis is the norm for words spelled with <ay>, while vowel deletion is the norm for words spelled with <a>. Exceptions exist for both classes of words.

However, we do not think that the mental grammar of Armenian directly uses these orthographic rules to know when to do glide epenthesis. Such orthographic rules are instead diachronic accidents, discussed more in Section §5.6.8.

Instead, the correlation with orthography is an indirect correlation with morphological structure. The <a>-spelled words tend to end with a country-naming suffix, such as /-ɑ/ in [fəɾɑns-ɑ] 'France' Ֆրանսա, /-jɑ/ in [ɑs-jɑ] 'Asia' Ասիա. Other cases are obvious loanwords like [fizikɑ] 'physics' Ֆիզիքա. In contrast, the <ay> spelling are mostly simple native words. There are of course a few exceptions for this native-loanword generalizations, for example the word [lɑmɑ] 'Lama' is an obvious loanword but it is spelled with <ay> լամայ, and it gets glide epenthesis: [lɑmɑj-ɑ-bed] 'chief Lama' լամայապէտ.

Further, the <ay> spelling is drastically more common than <a>, simply because <ay> is used for native words. For example, in the Kouyoumdjian dictionary, we found at least 100 words with a final <ay> and that have vocalic derivatives, compared with only 24 words with <a>.

Thus we argue that the actual rule for vowel hiatus repair is that native words with final [ɑ] get glide epenthesis, while loanwords or country-names get vowel deletion. Exceptions are limited to the items discussed above.

5.6.2.2 Stem final /e/

It is rather rare to find a word that ends with /e/. When a stem-final /e/ precedes a vowel-initial derivational suffix, the most common repair rule is glide epenthesis (Table 5.78). But there are some cases where either the /e/ or the following vowel is deleted. We first show cases with glide epenthesis.

Table 5.78: Glide epenthesis in derivation for stems with final [e]

+/-ɑX/	<hiwlē> <hiwlēagan>	hʏˈl**e** hʏl**ej-ɑ**ˈɡɑn	'atom' 'atomic'	հիւլէ հիւլէական	n=19
+/-ɑ-/	<osdrē> <osdrēavad͡ʒaṙ> cf. <vad͡ʒaṙ>	vostˈɾ**e** vostr**ej-ɑ**-vɑˈd͡ʒɑɾ vɑˈd͡ʒɑɾ	'oyster' 'oyster-selleter' 'sale'	ոստրէ ոստրէավաճառ վաճառ	n=10
+/-utjʏn/	<gatʼoɣigē> <gatʼoɣigēowtʼiwn>	gatoʁiˈg**e** gatoʁig**ej-u**ˈtjʏn	'cathedral' 'Catholicism'	կաթողիկէ կաթողիկէութիւն	n=6
+/-uX/ +/-uhi/	<markarē> <markarēowhi>	mɑɾkɑˈɾ**e** mɑɾkɑɾ**ej-u**ˈhi	'prophet' 'prophetess'	մարգարէ մարգարէուհի	n=1
+/-iX/ +/-ig/	<baxrē> <baxrēig>	bɑχˈɾ**e** bɑχɾ**e**ˈ**j-ig**	'money' 'small money'	պախրէ պախրէիկ	n=1

There are of course exceptions (Table 5.79). We found words which can a) delete the stem /e/, b) merge the the stem and suffix /e/ into one vowel, c) delete the vowel of the suffix or linker, or d) use glide epenthesis. Most of these exceptional roots would use one strategy in one derivative, but another strategy in another

Table 5.79: Glide epenthesis vowel deletion in derivation for some stems with final [e]

	<ap'roditē>	ɑpɾodiˈ**te**	'Aphrodite'	Ափրոդիտէ
+/-ɑgɑn/	<ap'roditagan>	ɑpɾodiˈt-**ɑ**ˈgɑn	'venereal'	ափրոդիտական
	<xahowē>	χɑhˈv**e**	'coffee'	խահուէ
+/-eni/	<xahowēni>	χɑhv-**e**ˈni	'coffee-tree'	խահուենի
+/-ɑrɑɾ/	< xahow**ea**rar>	χɑhv**ej-ɑ**ˈrɑɾ	'coffeehouse keeper'	խահուէարար
	<pazē>	pɑˈz**e**	'falcon'	բազէ
+/-ɑno͡ts/	<pazēno͡ts'>	pɑz**e**-ˈno͡ts	'hawking-pouch'	բազէնոց
+/-ɑ-giɾt/	<pazēagirt'>	pɑz**ej-ɑ**-giɾt	'falconer'	բազէակիրթ

As a last note, for words like [hɤlej-ɑgɑn] 'atomic' հիւլէական or [ej-utjɤn] 'existence' էութիւն (from [e] 'is' է), we transcribe the glide as a fully pronounced glide [j]. However, in HD's judgment, in careful speech this glide can be considerably weakened to either a transient glide [hɤle^{j}-ɑgɑn] or even a glottal stop [hɤleʔ-ɑgɑn]. He also reports that the preference of a glottal stop feels more salient if the root sounds more clearly like a loanword: [seɾofpe] 'seraph' սերովբէ and [seɾofpeʔ-ɑgɑn] 'seraphic' սերովբէական.

In HD's judgements, the full glide form is quite common in Western Armenian. In contrast, VP informs us that Eastern Armenian uses a transient glide or glottal stop more often. In Armenian philology, the transcription of <ē.a> է.ա can be variably pronounced as [ejɑ], [e^{j}ɑ], or [eʔɑ]. This makes it difficult to get reliable statistical data on this free variation.

5.6.2.3 Stem final /i/

When a stem-final /i/ precedes a vowel-initial derivational suffix, we find various possible repair strategies: glide epenthesis, deleting the /i/, or merging the /i/ and the next vowel /ɑ/ into [e] (coalescence). The choice of strategy sometimes correlates with the morphological identity of the /i/, but also seems random.

First, there is productive derivational suffix *-i* that can be added after virtually any verb (ending in /el/ or /ɑl/) to create an adjective (Table 5.80). This suffix can then undergo further derivational suffixation with either the nominalizer /-

utjɤn/ or the adverbalizer /-oɾen/. Vowel hiatus between this suffix /i/ and the next vowel is repaired by glide epenthesis.

Table 5.80: Glide epenthesis in derivation for stems with deverbal suffix [-i]

+/-utjɤn/	<badʒel>	bɑdˈʒel	'to punish'	պատժել	n=51
	<badʒel**i**>	bɑdʒeˈl-**i**	'punishable'	պատժելի	
	<badʒel**iow**t'iwn>	bɑdʒel-**ij-u**ˈtjɤn	'penalization'	պատժելիութիւն	
+/-oɾen/	<xɣd͡ʒal>	χəʁˈd͡ʒɑl	'to pity'	խղճալ	n=6
	<xɣd͡ʒal**i**>	χəʁd͡ʒɑˈl-**i**	'pitiful'	խղճալ	
	<xɣd͡ʒal**iō**ren>	χəʁd͡ʒɑˈl-**ij-o**ɾen	'pitifully'	խղճալիօրէն	

Thus, glide epenthesis is the norm for this suffix /i/ and its derivatives. However, in HD's judgment, careful speech allows replacing this inserted glide [j] in [bɑdʒel-**ij-u**tjɤn] with either a transient glide [ʲ] or a glottal stop: [bɑdʒel-**iʲ-u**tjɤn, bɑdʒel-**iʔ-u**tjɤn]. Based on judging the Eastern Armenian entries on English Wiktionary, the weak glide or glottal stop form seems to be more common in casual speech in Eastern Armenian.

Note that there are many words with a final /i/ that is a) ambiguously a derivational suffix and b) is absent in some related forms. We set aside these words from discussion. For example, [ɑɾpi] and [ɑɾp] both mean 'sun' արփ, արփի. Thus in a form like [ɑɾp-ɑvoɾ] 'luminous' արփաւոր, we have no way of knowing if this word was derived from [ɑɾp] or from [ɑɾpi] with deletion.

Glide epenthesis is also attested in some roots (Table 5.81). There are five monosyllabic words that end in [i]. When a vowel suffix is added, we see glide epenthesis.

Table 5.81: Glide epenthesis in derivation for monosyllabic stems with final [i]

+/-ɑX/	<t͡s**i**>	ˈt͡s**i**	‘horse’	ձի
+/-ɑvoɾ/	<t͡s**ia**wor>	t͡s**ij-ɑ**ˈvoɾ	‘horseman’	ձիաւոր
+/-ɑX/	<t**i**>	ˈt**i**	‘corpse’	դի
+/-ɑg/	<t**i**ag>	t**i**ˈ**j-ɑ**g	‘corpse’	դիակ
+/-ɑ-/	<t’**i**>	ˈt**i**	‘shovel’	թի
+/-ɑ-/	<t’**ia**t͡sowg>	t**ij-ɑ**-ˈt͡sug	‘paddle fish’	թիաձուկ
	cf. <t͡sowg>	ˈt͡sug	‘fish’	ձուկ
+/-utjɤn/	<m**i**>	ˈm**i**	‘one’	մի
	<m**iow**t’iwn>	m**ij-u**ˈtjɤn	‘unity’	միութիւն
+/-oX/	<l**i**>	ˈl**i**	‘full’	լի
+/-ov/	<l**io**v>	l**i**ˈ**j-o**v	‘fully’	լիով

As before, some of these tokens could be pronounced with a glottal stop in more formal speech, such as [mij-utjɤn] or [miʔ-utjɤn] ‘unity’.

There is likewise a monosyllabic root [di] տի which seems to be a bound root, because it is almost always found as part of a compound with the word [jezeɾk] ‘edge’, as [dij-ezeɾk] ‘cosmos’ տիեզերք.

Although glide epenthesis is the norm for the above words, there are morphemes that prefer vowel deletion (Table 5.82). The suffix /-eni/ that used to derive plant names and other words. This suffix tends to delete before other vowel-initial derivational suffixes.

Table 5.82: Vowel deletion in derivation for stems with suffix [-eni]

+/-ɑX/	<t’t’**e**ni>	tət**e**n**i**	‘mulberry tree’	թթենի	n=7
+/-ɑgɑn/	<t’t’**e**nagan>	tət**e**n-**ɑ**ˈgɑn	‘moric’	թթենական	
+/-ɑ-/	<ayd͡z**e**ni>	ɑjd͡zeˈn**i**	‘goat’s hair’	այծենի	n=4
	<ayd͡z**e**nakord͡z>	ɑjd͡zen-**ɑ**-ˈkord͡z	‘chamois dresser’	այծենագործ	
	cf. <kord͡z>	ˈkord͡z	‘work’	գործ	
+/-utjɤn/	<vayr**e**ni>	vɑjɾeˈn**i**	‘savage’	վայրենի	n=1
	<vayren**ow**t’iwn>	vɑjɾen-**u**ˈtjɤn	‘savageness’	վայրենութիւն	
+/-oX/	<epen**i**>	jepeˈn**i**	‘ebony tree’	եբենի	n=2
+/-os/	<epen**o**s>	jepeˈn-**o**s	‘ebony’	եբենոս	

Although deletion is the norm for this suffix, we have found some instances where the vowel of [-eni] is optionally deleted: [kɑχkeni] ‘townsman’ քաղքենի can derive the word ‘middle class’ with either deletion [kɑχken-utjɤn] քաղքենութիւն or glide epenthesis [kɑχkenij-utjɤn] քաղքենիութիւն.

Besides the above easily organized categories, most cases of final /i/ seem to behave randomly (Table 5.83). For example, some words with final /i/ always used a glide in their derivatives in Kouyoumdjian.

Table 5.83: Other stems with final /i/ that always use glide epenthesis in Kouyoumdjian

+/-ɑX/ +/-ɑgɑn/	<naxaɣ**i**> <naxaɣ**ia**gan>	nɑχɑˈʁ**i** nɑχɑʁ**ij-ɑ**ˈgɑn	‘duodenum’ ‘duodenal’	նախաղի նախաղիական	n=11
+/-ɑ-/	<ar**i**> <ar**ia**sird> cf. <sird>	ɑˈɾ**i** ɑɾ**ij-ɑ**-ˈsɪɾd ˈsɪɾd	‘brave’ ‘courageous’ ‘heart’	արի արիասիրտ սիրտ	n=7
+/-utjɤn/	<amowr**i**> <amowr**iow**t’iwn>	ɑmuˈɾ**i** ɑmuɾ**ij-u**ˈtjɤn	‘bachelor’ ‘single life’	ամուրի ամուրիութիւն	n=17
+/-uX/ +/-uhi/	<k’aɣak’at͡s’**i**> <k’aɣak’at͡s’**iow**hi>	kɑʁɑkɑˈt͡s**i** kɑʁɑkɑt͡s**ij-u**ˈhi	‘citizen’ ‘citizeness’	քաղաքացի քաղաքացիուհի	n=1
+/-eX/ +/-el/	<sdnt**i**> <sdnt**ie**l>	əstənˈt**i** əstənt**iˈj-e**l	‘fosterling’ ‘to suckle’	ստնդի ստնդիել	n=1
+/-oX/ +/-on/	<k’an**i**> <k’an**iō**n>	kɑˈn**i** kɑn**iˈj-o**n	‘how many’ ‘how many’	քանի քանիօն	n=2

For the above set of words, it’s possible that some of these include a derivational suffix *i*, and it is this suffix which prefers glide epenthesis. For example, the word [ɑjɾi] ‘widow’ այրի could be historically derived from the word [ɑjɾ] ‘man’ այր. This word takes glide epenthesis in its derivatives: [ɑjɾij-ɑnɑl] ‘to become a widow’ այրիանալ.

In another set of words, the vowel /i/ is deleted in all the derivatives that are reported in Kouyoumdjian (Table 5.84).

Table 5.84: Other stems with final /i/ that always deletes in Kouyoumdjian

+/-ɑX/	<tapni> <tapnayin>	tɑpˈni tɑpn-**ɑ**ˈjin	‘laurel’ ‘of laurel’	դաբնի դաբնային	n=29
+/-ɑ-/	<kaɣdni> <kaɣdnnabah> cf. <bah>	kɑχtˈni kɑχtn-**ɑ**-ˈbɑh ˈbɑh	‘secret’ ‘discreet’ ‘keeper’	գաղտնի գաղտնապահ պահ	n=25
+/-utjɤn/	<amehi> <amehowt’iwn>	ɑmeˈhii ɑmeh-**u**ˈtjɤn	‘wild’ ‘wildness’	ամեհի ամեհութիւն	n=28
+/-uX/ +/-uhi/	<barmani> <barmanowhi>	bɑrmɑˈni bɑrmɑn-**u**ˈhi	‘young man’ ‘young woman’	պարմանի պարմանուհի	n=5
+/-iX/ +/-il/	<kadaɣi> <kadaɣil>	gɑdɑˈʁi gɑdɑˈʁ-il	‘mad’ ‘to go mad’	կատաղի կատաղիլ	n=1
+/-eX/ +/-eɾen/	<vrat͡s’i> <vrat͡s’erēn>	vəɾɑˈt͡si vəɾɑt͡s-eˈɾen	‘Georgian’ ‘Georgian language’	վրացի վրացերէն	n=10
+/-oX/ +/-ot͡s/	<asɣani> <asɣanot͡s’>	ɑsχɑˈni ɑsχɑˈn-ot͡s	‘thread’ ‘needle-case’	ասղանի ասղանոց	n=4

Another set of words shows a special rule of fusing or merging /i/ with /ɑ/ to form [e] (Table 5.85). We found 8 words which only had derivatives before /ɑ/, and here the /i/ and /ɑ/ merged to [e].

Table 5.85: Other stems with final /i/ where the /i/ coalesces with /ɑ/ to form [i]

+/-ɑvoɾ/	<aki> <akewor> cf. <kord͡zawor>	ɑˈki ɑk-**e**ˈvoɾ koɾd͡z-**ɑ**ˈvoɾ	‘tail’ ‘tailed’ ‘worker’	ագի ագեւոր գործաւոր
+/-ɑ-/	<dari> <darekirk’> cf.<kirk’>	dɑˈɾi dɑɾ-**e**ˈkiɾk ˈkiɾk	‘year’ ‘annal’ ‘book’	տարի տարեգիրք գիրք

Much more common are roots which seem to randomly pick one out of the three possible repair rules (Table 5.86: glide epenthesis, deletion, /e/ coalescence). Although the coalescence rule is restricted to before /ɑ/, these roots can also randomly use either glide epenthesis or vowel deletion in this same phonological context or in other phonological contexts. We found 35 such roots, many of which are quite high-frequency.

Table 5.86: Other stems with final /i/ that randomly show coalescence, glide epenthesis, or deletion

	<koti>	koˈti	‘leper’	գոդի
/i-ɑnot͡s/→[e]	<kotenot͡s’>	kote-ˈnot͡s	‘Lazar-house’	գոդենոց
	cf. <aɣpanot͡s’>	ɑχp-**ɑ**ˈnot͡s	‘sewer’	աղբանոց
/i-utjɤn/→[u]	<kot**ow**t’iwn>	kot-**u**ˈtjɤn	‘leprosy’	գոդութիւն
	<abag**i**>	ɑbɑˈg**i**	‘glass’	ապակի
/i-ɑ-/→[e]	<abag**e**-kord͡z>	ɑbɑg-**e**ˈkoɾd͡z	‘glass-maker’	ապակեգործ
	cf. <kord͡z>	ˈkoɾd͡z	‘work’	գործ
/i-ɑ-/→[ɑ]	<abag**a**vad͡ʒaṙ>	ɑbɑg-**ɑ**-vɑˈd͡ʒɑɾ	‘glass-seller’	ապակավաճառ
	cf. <vad͡ʒaṙ>	vɑˈd͡ʒɑɾ	‘sale’	վաճառ
	<hok**i**>	hoˈk**i**	‘soul’	հոգի
/i-ɑvoɾ/→[e]	<hok**e**woṙ>	hok-**e**vɑˈd͡ʒɑɾ	‘spiritual’	հոգեւոր
	cf. <t’ak**a**wor>	tɑk-**ɑ**ˈvoɾ	‘king’	թագաւոր
/i-ɑnɑl/→[ijɑ]	<hok**i**anal>	hok**ij**-**ɑ**ˈnɑl	‘to revive’	հոգիանալ

In sum, a stem-final /i/ can display a quite random assortment of possible changes before a vowel-initial derivational suffix. Although some morphological categories show consistent behavior, many words seem to just randomly pick one of three possible repairs. The rule is used in some derivatives, but not others, and there is no clear semantic, morphological, or phonological rationale behind this variation.

5.6.2.4 Stem final /o/

Words with a final /o/ are rare. Thus, it is even rarer to find a /o/-final stem that takes a derivational suffix (Table 5.87). The only cases we found in Kouyoumdjian were stems where the /o/ was spelled as <oy>. The vowel sequence undergoes glide epenthesis.

Table 5.87: Glide epenthesis in derivation for stems with final [o]

+/-ɑX/	<martahad͡ʒ**oy**>	mɑɾtahoˈd͡ʒ**o**	‘flatterer’	մարդահաճոյ	n=1
+/-ɑnɑl/	<martahad͡ʒ**oy**anal	mɑɾtahad͡ʒ**oj**-**ɑ**ˈɑ>	‘to flatter’	մարդահաճոյանալ	
+/-ɑ-/	<tʃx**oy**>	təʃˈχ**o**	‘queen’	դշխոյ	n=1
	<tʃx**oy**ahaw>	təʃχ**oj**-**ɑ**ˈhɑv	‘fat pullet’	դշխոյահաւ	
	cf. <haw>	ˈhɑv	‘chicken’	հաւ	
+/-utjɤn/	<vadapar**oy**>	vɑdɑpɑˈɾ**o**	‘coward’	վատաբարոյ	n=12
	<vadapar**oy**ow**t**’iwn>	vɑdɑpɑɾ**oj**-**u**ˈtjɤn	‘cowardice’	վատաբարոյութիւն	
+/-eX/	<gog**oy**>	goˈg**o**	‘cocoa’	կոկոյ	n=1
+/-eni/	<gog**oy**eni>	gog**oj**-**e**ˈni	‘coco tree’	կոկոյենի	

5.6.2.5 Stem final /u/

For words with final /u/, the most common vowel hiatus repair rule is turn the /u/ into [v] (Table 5.88). We call this process /u/-devocalization. There many common words which show this process before virtually all types of vowels.

Table 5.88: /u/-devocalization in derivation for stems with final [o]

+/-ɑX/ +/-ɑgɑn/	<owr**ow**> <owr**owa**gan>	uˈɾ**u** uɾ**v-ɑ**ˈgɑn	‘ghost’ ‘ghost’	ուրու ուրուական	n=11
+/-ɑ-/	<t͡s**ow**> <t͡s**owa**gat’>	t͡sˈ**u** t͡s**ə-v-ɑ**-ˈgɑt	‘egg’ ‘egg-nog’	ձու ձուակաթ	n=14
+/-utjɤn/	<erglez**ow**> <erglez**owowt**’iwn>	jɛɾgleˈz**u** jɛɾglez**v-u**ˈtjɤn	‘bilingual’ ‘duplicity’	երկլեզու երկլեզուութիւն	n=2
+/-iX/ +/-ig/	<lez**ow**> <lez**owi**g>	leˈz**u** lez**v-i**g	‘tongue’ ‘little tongue’	լեզու լեզուիկ	n=4
+/-eX/ +/-el/	<t͡ʃ’**ow**> <t͡ʃ’**owe**l>	t͡ʃˈ**u** t͡ʃə**v-e**l	‘travel’ ‘to migrate’	չու չուել	n=7
+/oX/ +/-od/	<t’t’**ow**> <t’t’**owo**d>	təˈ**tu** tətˈ**v-o**d	‘sour’ ‘sourish’	թթու թթուոտ	n=1

Note that [v] can trigger schwa epenthesis to syllabify the consonant cluster. The [v] is variably devoiced to [f] after voiceless sounds; this is discussed in Section §4.3.1.5.

In the traditional orthography, both the /u/ and the devocalized [v] are spelled the same as <ow> ու. But in the reformed orthography as used for Eastern Armenian, the devocalized [v] is spelled as [v] վ.

The rule of /u/-devocalization is the default rule for handling /u/ before a vowel-initial derivational suffix. There are limited exceptions (Table 5.89). The word for ‘two’ [jɛɾgu] and its derivatives like ‘twelve’ [dɑsnəjɛɾgu] generally resist /u/-devocalization in their derivatives. Instead we get glide epenthesis. Although some of these derivatives can be pronounced with a [v] like [jɛɾgv-ɑgɑn], the glide-based derivatives like [jɛɾguj-ɑgɑn] are significantly more commonly heard. The word ‘pilot’ [otɑt͡ʃu] ‘pilot’ behaves the same.

Table 5.89: Words that take glide epenthesis for stems with final [o]

	<erg**ow**>	jeɾˈg**u**	‘two’	երկու
	<erg**owa**gan>	jeɾg**uj-ɑ**ˈgɑn	‘dual’	երկուական
	<erg**owow**t’iwn>	jeɾg**uj-u**ˈtjɤn	‘bilocation’	երկուութիւն
but	<erg**owo**reag>	jeɾg**v-o**ɾˈjɑg	‘twin’	երկուորեակ
	<ōtat͡ʃ’**ow**>	otɑˈt͡ʃ**u**	‘pilot’	օդաչու
	<ōtat͡ʃ’**owa**gan>	otɑt͡ʃ**uj-ɑ**ˈgɑn	‘aeronautic’	օդաչուական
or		otɑt͡ʃ**v-ɑ**ˈgɑn		

Besides the above words, there are some morphological categories of words with final /u/. Some of these groups tend to show devocalization, while others arguably use deletion.

There are many words that end in the morpheme [-du], such as [hɑmɑɾ-ɑ-d**u**] ‘accountable’ համարատու. Here, we know this [-du] is some sort of compound-like suffix (a suffixoid) because there’s a word [hɑmɑɾ] ‘account’. This suffix/-compound [-du] is used with the general meaning of ‘I give X or have X’, and is related to the verb ‘to give’ [dɑl] տալ. In general, these morpheme undergoes /u/-devocalization before the suffix /-utjɤn/: [hɑmɑɾ-ɑ-də**v-u**tjɤn] ‘report’ համարատուութիւն.

Though we have found some cases of variation. For the word [t͡sɑjn-ɑ-du] ‘sonorous’ ձայնատու, we found a form with devocalization in [t͡sɑjn-ɑ-də**v-u**tjɤn] ‘resonance’ ձայնատուութիւն , but also a form with deletion in [t͡sɑjn-ɑ-d-**u**tjɤn] ‘aphonia’ ձայնատուութիւն.

On a last note, there are many words that end in a suffix /-u/, especially as part of some suffix sequence /-ɑɾ-u/ where the /-ɑɾ/ is related to the verb [ɑrnel] ‘to take’: [poɾt͡s] ‘attempt’ փորձ derives [poɾt͡s-ɑɾ-u] ‘experienced’ փորձառու. But for such suffixes, it is hard to determine how they behave with vowel hiatus. When we come across a derivative like [poɾt͡s-ɑɾ-ɑgɑn] ‘experimental’ փորձառական, it is difficult to know if the suffix /-u/ was deleted, or if it was just never added in the first place.

In sum, when a vowel-initial derivational suffix is added after an /u/, the default rule is devocalizing the /u/ to [v], but with some variation.

5.6.3 Vowel hiatus repair in compounds and prefixoids

write Compound formation is a derivational process. Most ‘typical’ compounds are formed by combining two stems with a vowel /-ɑ-/. This vowel triggers the same vowel hiatus rules as /ɑ/-initial derivational suffixes (§5.6.3.1). This vowel

is generally deleted before vowel-initial roots. But in some ‘atypical’ compounds and with prefixoids (§5.6.3.2), the /-ɑ-/ is present before a root and triggers glottal stop epenthesis.

5.6.3.1 Vowel hiatus repair in typical compounds

The most typical way to form a compound is to combine two stems with a linking vowel /-ɑ-/. If the first stem is V-final, then the vowel sequence between that stem and /-ɑ-/ is typically modified in some way, such as glide epenthesis. The various strategies for /V-ɑ-/ sequences is discussed in Section §5.6.2. But if the second stem starts with a vowel, then the linking vowel /-ɑ-/ is typically omitted (Table 5.90).

Table 5.90: Vowel hiatus repair in typical compound constructions

XC + CX →XC-ɑ-CX	ɑŋˈgɤ**n** + ˈ**k**i**d͡z** ɑŋgɤ**n-ɑ-**ˈ**k**i**d͡z**	‘angle + line’ ‘diagonal’	անկիւն, գիծ անկիւնագիծ
Xɑ + CX →Xɑj-ɑ-CX	vəˈg**ɑ** + ˈ**t**uχt vəg**ɑj-ɑ-**ˈ**t**uχt	‘witness + paper’ ‘certificate’	վկայ, թուղթ վկայաթուղթ
XC + VC →XC-VX	ˈmeɾ**t͡s** +**i**ˈmɑst meɾ**t͡s-i**ˈmɑst	‘close + meaning’ ‘rough meaning	մերձ, իմաստ ’ մերձիմաստ
XC + jeX → XC-eX	ˈvoχ**p** + ˈ**je**ɾk voχˈ**p-e**ɾk	‘lamentation + song’ ‘tragic ballad’	ողբ, երգ ողբերգ
XC + voX → XC-oX	ˈkɑj**l** + ˈ**vo**ɾs kɑjˈ**l-o**ɾs	‘wolf + hunt’ ‘wolf-hunter’	գայլ,որս գայլորս

If the second stem’s first two segments are [je] ե or [vo] ո, then the prescriptive rule is that that the second stem is treated as if it’s vowel-initial. The [j] and [v] are absent. Adding the [j] or [v] however is attested in some special contexts (§5.6.3.2).

Very rarely, a compound is formed where a) the second stem starts with a consonant, but b) there is no linking vowel (Table 5.91). The first stem can either end in a consonant or vowel. It seems that the final vowel of the first stem does not undergo alternations. We use the qualifier ‘seems’ because the relevant data is quite few, thus making it hard to form concrete generalizations.

Table 5.91: Lack of vowel hiatus repair when there is no subsequent vowel in compounds

Xɑ + CX →Xɑ-CX	ʃɑpʏˈʁ**ɑ** + ˈ**t͡s**ev ʃɑpʏʁ-**ɑ**ˈ**t͡s**ev	‘sapphire + shape’ ‘sapphire-shaped’	շափիւղայ, ձեւ շափիւղաձեւ
Xe + CX →Xe-CX	pɑˈz**e** + ˈ**d͡ʒ**ɑnd͡ʒ pɑz**e**-ˈ**d͡ʒ**ɑnd͡ʒ	‘hawk + fly’ ‘hawk fly’	բազէ, ճանճ բազէճանճ
Xi + CX →Xi-CX	keˈʁ**i** + ˈ**d**un keʁ**i**-ˈ**d**un	‘rudder + house’ ‘wheelhouse’	քեղի, տուն քեղիտուն

But if the first stem ends in a vowel, while the second stem starts with a vowel, then the linking vowel is absent (Table 5.92). Vowel hiatus is usually repaired as if the second stem was a suffix. For example, /u/ generally becomes [v] before both suffix-vowels and root-vowels. /i/ tends to either delete or cause glide epenthesis, depending on the root. /ɑ/ can trigger a glide or delete. And /e/ tends to take glide epenthesis as well.

Table 5.92: Vowel hiatus repair in compounds where first stem ends with vowel, and second one starts with vowel

Xɑ + ɑX →Xɑj-ɑX	ɑɾˈk**ɑ** + **ɑ**ŋˈkəʁ ɑɾk**ɑj-ɑ**ŋˈkəʁ	‘king + vulture’ ‘king vulture’	արքայ, անգղ արքայանգղ
Xɑ + əX →X-əX	peˈs**ɑ** + **ə**ŋˈgeɾ pes-**ə**ŋˈgeɾ	‘groom + friend’ ‘groomsman’	փեսայ, ընկեր փեսընկեր
Xu + əX →Xv-əX	**t͡su** + **ə**ŋgɑˈl-it͡ʃ t͡s**ə**v-**ə**ŋˈgɑl	‘egg + receiver’ ‘ovary’	ձու, ընկալիչ ձուընկալ
Xi + oX →X-oX	jegeʁe**t͡si** + ˈ**o**ɾhˈnek jegeʁe**t͡so**ɾhˈnek	‘church + blessing’ ‘consecration’	եկեղեցի, օրհնէք եկեղեցօրհնէք
Xi + ɑX →Xij-ɑX	**t͡si** + ˈ**ɑ**ɾt͡sɑg **t͡sij-ɑ**ɾt͡sɑg	‘horse + untied’ ‘consecration’	ձի, արձակ ձիարձակ
Xi + oX →Xij-oX	ˈk**i** + **o**ˈʁi k**ij-o**ˈʁi	‘juniper + gin’ ‘gin’	գի, օղի գիօղի
Xe + ɑX →Xej-ɑX	χɑhˈv**e** + **ɑ**ˈmɑn χɑhv**ej**-ˈ**ɑ**ˈmɑn	‘coffee + pot’ ‘coffee-pot’	խահուէ, ամաn խահուէաման

For some these cases, HD feels its possible to just use a glottal stop instead of glide epenthesis: [χɑhveʔ-ɑmɑn] ‘coffee-pot’. He likewise reports that using a glottal stop before a root ‘feels’ more common than using a glottal stop before a suffix.

And as for the second stem, if the stem starts with [je,vo], then it is again treated as starting with [e,o] (Table 5.93). Vowel hiatus repair rules apply, as if the the second stem was a suffix. For example, /i/ and /ɑ/ can delete or trigger glide epenthesis. Unfortunately, it is hard to know if this glide was epenthetic because of hiatus, vs. just retained from the second stem.

Table 5.93: Vowel hiatus repair in compounds where first stem ends with vowel, and second one starts with [vo,je]

Xi + voX	keˈɾ**i** + **vo**ɾˈti	‘uncle + son’	քեռի, որդի
→X-oX	keɾ-**o**ɾˈti	‘king vulture’	քեռորդի
Xɑ + voX	ɑɾˈk**ɑ** + **vo**ɾˈti	‘king + son’	արքայ, որդի
→Xɑj-oX	ɑɾk**ɑj-o**ɾˈti	‘prince’	արքայորդի
Xi + jeX	ˈm**i** + **je**ˈɾɑŋk	‘one + color’	մի, երանգ
→Xij-eX	m**ij-e**ˈɾɑŋk	‘monochromous’	միերանգ
Xi + jeX	vosˈk**i** + **je**χ͡ʃʏɾ	‘gold + horn’	ոսկի, եղջիւր
→X-eX	vosk-**e**χ͡ʃʏɾ	‘golden horn’	ոսկեղջիւր
Xɑ + jeX	ɑɾˈk**ɑ** + **je**ˈɾɑg	‘king + vein’	արքայ, երակ
→Xɑj-eX	ɑɾk**ɑj-e**ˈɾɑg	‘basilisk’	արքայերակ

Thus, when a compound has both a V-final first stem and a V-initial second stem, then it seems that the default pattern is to resolve the vowel hiatus by using the same rules as if the second stems is a derivational suffix.

However, we have found some bizarre cases of vowel hiatus repair in compounds that are hard to explain (Table 5.94). There are some compounds where the first stem ends in /i/, and this vowel becomes [e] before a vowel-initial stem. It seems that for such words, the linking vowel /-ɑ-/ was temporarily added creating an underlying three vowel-cluster /i-ɑ-V/. The /i-ɑ/ then merged or coalesced into /e/, and then the /e-V/ triggers a glottal stop. Surprisingly, a glide sounds quite bizarre here in HD’s judgments.

Table 5.94: Vowel hiatus repair in compounds where first stem ends with a derived [e], and second one starts with a vowel

Xi + VX	kɑˈɾ**i** + **ɑ**ˈlʏɾ	‘barley + flour’	գարի, ալիւր
	kɑɾ**eʔ-ɑ**ˈlʏɾ	‘barley flour’	գարէալիւր
	vosˈk**i** + ˈ**o**ʁ	‘gold + ring’	ոսկի, օղ
→Xeʔ-VX	vosk**e**ˈ**ʔ-o**ʁ	‘gold ring’	ոսկէօղ
	vosˈk**i** + ˈ**ɑ**gən	‘gold + gem’	ոսկի, ակն
→Xeʔ-VX	vosk**e**ˈ**ʔ-ɑ**gən	‘gold ring’	ոսկեակն

5.6.3.2 Vowel hiatus repair in prefixoids and atypical compounds

The previous section discussed 'typical' compounds. However, there are rare cases where a compound's second stem starts with a vowel, but a vowel linker /-ɑ-/ is inserted. Here, the linking vowel is phonologically unneeded but it is arbitrarily used because of the morphology. The hiatus between the linker and the second stem is repaired by a glottal stop. We discuss three such cases: numerals, prefixoids, and normal compounds.

One case comes from complex numerals where the linker is [ə]. Dialects and registers differ in whether this linker is pronounced before V-initial roots (Table 5.95). This is discussed in numeral morphology. HD's Western dialect prefers using this schwa even before a vowel, thus triggering a glottal stop.

Table 5.95: Numerals with a glottal stop after the linker

	ˈdɑs + ˈ**u**tə	'ten + eight'	տաս, ութը
→	dɑsn-**əˈʔ-u**tə	'eighteen'	տասնըութը
	ˈdɑs + ˈ**i**nə	'ten + nine'	տաս, ինը
	dɑsn-**əˈʔ-i**nə	'nineteen'	տասնըինը

Another common case are compounds that use prefixoids like [hɑm-ɑ-] 'pan-', [hɑg-ɑ] 'anti-' or [ɑmen-ɑ-] 'most' (Table 5.96). As explained in prefixoid chapter, the prefixoid is made up of a root-like prefix and a linking vowel /-ɑ-/. For some prefixoids, the linking vowel is usually absent before vowels. But there are arbitrary cases where the linking vowel is present, triggering a glottal stop. Some prefixoids like [ɑmen-ɑ-] exceptionally always take the linking vowel.

Table 5.96: Prefixoids with a glottal stop after the linker

	[hɑm-ɑ] ‘pan-’, [hɑg-ɑ] ‘anti-’		[ɑmen-ɑ] ‘most’	
/ɑ/	**ɑ**ɾɑpɑˈgɑn ‘Arabian’ արաբական	hɑm-**ɑʔ-ɑ**ɾɑpɑgɑn-uˈtjɤn ‘Pan-Arabism’ համաարաբականություն	**ɑ**nuˈnov ‘renowned’ անունով	ɑmen-**ɑʔ-ɑ**nuˈnov ‘most renowned’ ամենաանունով
/e/	**e**juˈtjɤn ‘existence’ էութիւն	hɑm-**ɑʔ-e**juˈtjɤn ‘consubstantiality’ համաէութիւն	**e**jɑˈgɑn ‘essential’ էական	ɑmen-**ɑʔ-e**jɑˈgɑn ‘most essential’ ամենաէական
/i/	**i**sˈlɑm ‘Islam’ իսլամ	hɑm-**ɑʔ-i**slɑˈm-izm ‘Pan-Islamism’ համաիսլամիզմ	**i**mɑsˈtun ‘wise’ իմաստուն	ɑmen-**ɑʔ-i**mɑsˈtun ‘wisest’ ամենաիմաստուն
/ə/	**ə**ŋgeɾɑˈjin ‘social’ ընկերային	hɑg-**ɑʔ-ə**ŋgeɾɑˈjin ‘anti-social’ հակաընկերային	**ə**ntuˈnɑg ‘capable’ ընդունակ	ɑmen-**ɑʔ-ə**ntuˈnɑg ‘most capable’ ամենաընդունակ
/o/	**o**ɾinɑˈgɑn ‘legitimate’ օրինական	hɑm-**ɑʔ-o**ɾinɑˈgɑn ‘illegitmate’ հակաօրինական	**o**ɾtˈnjɑl ‘blessed’ օրհնեալ	ɑmen-**ɑʔ-o**ɾtˈnjɑl ‘Most Blessed’ ամենաօրհնեալ
/u/	**u**doˈbjɑ ‘utopia’ ուտոպիա	hɑg-**ɑʔ-ə**doˈbjɑ ‘anti-utopia’ հակաուտոպիա	**u**ˈʒeʁ ‘strong’ ուժեղ	ɑmen-**ɑʔ-u**ˈʒeʁ ‘strongest’ ամենաուժեղ

Similarly, in a typical compound, the linking vowel is absent before a vowel. But in some arbitrary cases (Table 5.97), especially neologisms or technically vocabulary, the linking vowel is present. The vowel hiatus caused by the linking vowel and root triggers a glottal stop. Note that we know that these compounds are single words because a) they written as one word, b) they have final stress, and c) the first stem undergoes morphophonological alternations like vowel reduction (last two columns in Table 5.97).

Table 5.97: Compounds with a glottal stop after the linker

	Without reduction		With reduction	
/ɑ/	əstɑˈmoks + **ɑ**ˈʁik ‘stomach + intestine’ ստամոքս, աղիք	əstɑmoks-**ɑʔ-ɑʁ**ik-ɑˈjin ‘gastrointestinal’ ստամոքսաաղիքային	jeɾɑˈʒiʃt + **ɑ**ˈlik ‘musician + wave’ երաժիշտ, ալիք	jeɾɑʒəʃt-**ɑʔ-ɑ**lik ‘music channel’ երաժշտաալիք
/e/	kəɾisˈtos + **e**t͡ʃ- ‘Christ + √descend’ Քրիստոս, էջ	kəɾistos-**ɑˈʔ-e**t͡ʃ ‘Christ descent’ Քրիստոսաէջ	ˈlujs + ˈ**e**t͡ʃ ‘light + page’ լույս, էջ	lus-**ɑˈʔ-e**t͡ʃ ‘illuminated page’ լուսաէջ
/i/	iˈmɑst + ˈ**i**χts ‘meaning + wish’ իմաստ, իղձ	imɑst-**ɑˈʔ-i**χts ‘sensible’ իմաստաիղձ	həɴˈtig + **i**ˈɾɑn ‘Indian + Iran’ հնդիկ, Իրան	həntk-**ɑʔ-i**ɾɑn-ˈjɑn ‘Indo-Iranic’ հնդկաիրանեան
/ə/	ɑʃˈχɑɾ + **ə**mpəɾˈnum ‘world + perception’ աշխարհ, ըմբռնում	ɑʃχɑɾ-**ɑʔ-ə**mpəɾˈnum ‘worldview’ աշխարհաըմբռնում	həɴˈtig + **ə**ŋˈgujz ‘Indian + walnut’ հնդիկ, ընկոյզ	həntk-**ɑʔ-ə**ŋˈgujz ‘coconut’ հնդկընկոյզ
/o/	ˈzɑɾt + ˈ**o**ʁ ‘decoration + ear-ring’ զարդ, օղ	zɑɾt-**ɑˈʔ-o**ʁ ‘decorative ear-ring’ զարդաօղ	jeɾˈgin + ˈ**o**t ‘heaven (archaic) + air’ երկին, օդ	jeɾgn-**ɑʔ-o**tɑˈjin ‘airborne’ երկնաօդային
/u/	ˈot + **u**ˈʁi ‘air + way’ օդ, ուղի	ot-**ɑʔ-u**ˈʁi ‘airway’ օդաուղի	ˈmiʃt + **u**ˈɾɑχ ‘always + happy’ միշտ, ուրախ	məʃt-**ɑʔ-u**ˈɾɑχ ‘ever-happy’ մշտաուրախ

When the prefixoid or compound uses the linker /ɑ/ and when the second stem starts [vo,je], then the [vo,je] surfaces in the compound (Table 5.98).

Table 5.98: Compounds with a linker before [je,vo]

	/-ɑ/ + [je]		/-ɑ/ + [vo]	
Prefixoid	**je**ɾˈgiɾ ‘Earth’ երկիր	hɑg-**ɑ-je**ɾgˈɾ-jɑ ‘antichthon’ հակաերկրեայ	**vo**lɔɾdɑˈgɑn ‘horizontal’ ոլորտական	hɑg-**ɑ-vo**lɔɾdɑˈgɑn ‘antiperistaltic’ հակաոլորտական
Prefixoid	**je**ɾˈgɑɾ ‘longest’ երկար	ɑmen-**ɑ-je**ɾˈgɑɾ ‘longest’ ամենաերկար	**vo**t͡ʃənt͡ʃɑˈt͡sum ‘annihilation’ ոչնչացում	ɑmen-**ɑ-vo**t͡ʃənt͡ʃɑˈt͡sum ‘annihilation of all’ ամենաոչնչացում
Compound	kəˈluχ + **je**ˈdev ‘head + back’ գլուխ, ետեւ	kəlχ-**ɑ-je**ˈdev ‘back of head’ գլխաետև	ɑʁˈves + **vo**ɾˈsoɾt ‘fox + hunter’ աղուէս, որսորդ	ɑʁves-**ɑ-vo**ɾˈsoɾt ‘fox hunter’ աղուէսաորսորդ

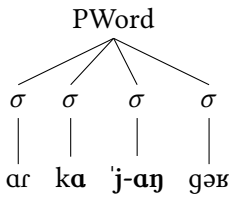
For these ‘atypical’ compounds, one can argue that the reason why there’s a glottal stop is because speakers want to treat such as compounds as some grey area between a phrase and a word. In terms of phonological domains or prosody, one could argue that the glottal stop signifies a weak word boundary before the second stem.

Cross-linguistically, such phenomena have been modeled using recursive prosodic structure cite, like greeks. For a typical compound like [ɑɾk**ɑj-ɑ**ŋgəʁ] ‘king vulture’ with glide epenthesis and a deleted linker (Table 5.91), the entire word is just

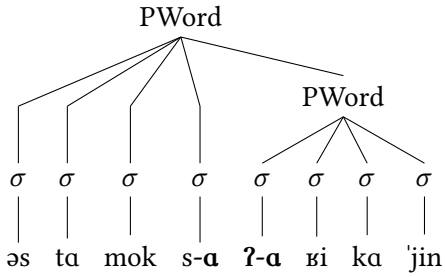
one phonological word. See Section §?? for discussion on phonological words. But for an atypical compound with an unneeded linker and a glottal stop like [əstɑmoks-**ɑʔ-ɑʁ**ik-ɑˈjin] ‘gastrointestinal’ (Table 5.97), the entire word would act as a phonological word, while the second stem would act as an extra internal phonological word. The first stem could also be treated as its own word (Representation 2).

Representation 2. Prosodic structure of a typical compound vs. an atypical compound

One prosodic word in [ɑɾk**ɑˈj-ɑ**ŋgəʁ] ‘king vulture’



Layered prosodic words in [əstɑmoks-**ɑʔ-ɑʁ**ikɑˈjin] ‘gastrointestinal’



But it is an open question if such recursive prosodic structure is truly applicable to atypical Armenian compounds. The main problem is that typical compounds show more refined word-internal domains, discussed in compound prosody chapter. Furthermore, the first stem stills shows morphophonological alternations (like vowel reduction), and these changes are arguably stem-level changes instead of word-level changes. Thus whatever prosodic boundary exists before the glottal stop, this boundary is not strong enough to prevent all stem-internal changes.

5.6.4 Vowel hiatus repair before regular inflectional suffixes

There are few V-initial inflectional suffixes. Some of these suffixes are productive and can be easily added after any word: *-i* (G/D), *-e* (ABL), *-ov* (INS). One suffix is productive but can only take monosyllabic roots: *-eɾ* (PL). Some suffixes are productive but can only follow C-final words: *-ə* DEF, *-əs* (POSS.1SG), *-ət* (POSS.2SG)

Besides these productive suffixes, there is one suffix that is unproductive and can follow only a handful of roots *-u* (G/D). There are also some verbal inflectional suffixes like past *-a,-i* which we discuss in Section §5.6.5.

Based on these suffixes, we can easily examine how vowel hiatus is handled between any type of V1 and a V2 that belongs to the set {e, i, o}. We find glide

epenthesis is the norm. Unfortunately, we can't extensively examine how vowel hiatus is handled before an inflectional /ɑ,u/ because the data is too limited by the morphology.

5.6.4.1 Stem-final /ɑ/

First, consider words that end in /ɑ/ (Table 5.99). Before all V-initial inflection, glide epenthesis is the regular rule. Irregularities are limited (§5.6.5).

Table 5.99: Glide epenthesis between /ɑ/ and V-initial inflection

	Words ends in orthographic <ay> այ			Words ends in orthographic <a> ա		
	'king'	'rosy'	'monk'	'Messiah'	'Europe'	'chemistry'
	արքայ	վարդեայ	աբեղայ	Մեսիա	Եւրոպա	քիմիա
	<ark'ay>	<varteay>	<apeɣay>	<Mesia>	<Ewroba>	<k'imia>
	ɑɾ'k**ɑ**	vɑɾt'j**ɑ**	ɑpe'ʁ**ɑ**	mes'j**ɑ**	jevɾo'b**ɑ**	kim'j**ɑ**
G/D *-i*	ɑɾk**ɑ**'**ji**	vɑɾtj**ɑ**'**ji**	ɑpeʁ**ɑ**'**ji**	mesj**ɑ**'**ji**	jevɾob**ɑ**'**ji**	kimj**ɑ**'**ji**
ABL *-e*	ɑɾk**ɑ**'**je**	vɑɾtj**ɑ**'**je**	ɑpeʁ**ɑ**'**je**	mesj**ɑ**'**je**	jevɾob**ɑ**'**je**	kimj**ɑ**'**je**
INS *-ov*	ɑɾk**ɑ**'**jov**	vɑɾtj**ɑ**'**jov**	ɑpeʁ**ɑ**'**jov**	mesj**ɑ**'**jov**	jevɾob**ɑ**'**jov**	kimj**ɑ**'**jov**

As is clear, glide epenthesis is the norm for repair vowel sequence of /ɑ/ plus essentially any type of inflectional vowel. This process applies after both roots like [ɑɾkɑ] 'king', and after derivational suffixes like *-ja* in [vɑɾt.jɑ] 'rosy', derived from [vɑɾt] 'rose' վարդ.

5.6.4.2 Stem-final /e/

Given a words that end in /e/ like [ɾo.be] 'second', if a V-initial inflectional suffix like instrumental [-ov] is added, then we get glide epenthesis: [ɾo.be.jov]. Glide epenthesis is the rule before all V-initial inflectional suffixes (Table 5.100).

Table 5.100: Glide epenthesis between /e/ and V-initial inflection

	'second'	'prophet'	'made of bricks'	'Cairo'
	որպէ	մարգարէ	աղիւսէ	Գահիրէ
	ɾo.'**be**	mɑɾ.kɑ.'ɾ**e**	ɑ.ʁʏ.'s**e**	kɑ.hi.'ɾ**e**
G/D *-i*	ɾo.b**e**.'**ji**	mɑɾ.kɑ.ɾ**e**.'**ji**	ɑ.ʁʏ.s**e**.'**ji**	kɑ.hi.ɾ**e**.'**ji**
ABL *-e*	ɾo.b**e**.'**je**	mɑɾ.kɑ.ɾ**e**.'**je**	ɑ.ʁʏ.s**e**.**j**'**e**	kɑ.hi.ɾ**e**.'**je**
INS *-ov*	ɾo.b**e**.'**jov**	mɑɾ.kɑ.ɾ**e**.'**jov**	ɑ.ʁʏ.s**e**.'**jov**	kɑ.hi.ɾ**e**.'**jov**

All words that end in /e/ take a glide /j/ before V-initial inflection. This rule applies across native words like [ɾobe] ‘second’ and to loanwords like [kɑhiɾe] ‘Cairo’. Derivational suffixes also obey this rule, such as the derivational suffix *-e* in [ɑʁʏse] ‘made of bricks’, derived from [ɑʁʏs] ‘brick’ աղիւս.

5.6.4.3 Stem-final /i/

When /i/ is before V-initial inflection, the norm is to get glide epenthesis in Western Armenian (Table 5.101).

Table 5.101: Glide epenthesis between /i/ and V-initial inflection

	Roots with final /i/			Suffixes with final /i/		
	‘island’	‘son’	‘pigeon’	‘queen’	‘venerable’	‘apple-tree’
	կղզի	որդի	աղաւնի	թագուհի	յարգելի	խնձորենի
	gəʁˈzi	voɾˈti	ɑʁɑvˈni	tɑkuˈhi	hɑɾkeˈli	χənt͡soɾeˈni
G/D *-i*	gəʁziˈji	voɾtiˈji	ɑʁɑvniˈji	tɑkuhiˈji	hɑɾkeliˈji	χənt͡soɾeniˈji
ABL *-e*	gəʁziˈje	voɾtiˈje	ɑʁɑvniˈje	tɑkuhiˈje	hɑɾkeliˈje	χənt͡soɾeniˈje
INS *-ov*	gəʁziˈjov	voɾtiˈjov	ɑʁɑvniˈjov	tɑkuhiˈjov	hɑɾkeliˈjov	χənt͡soɾeniˈjov

Glide epenthesis applies after both roots and after all derivational suffixes. For example, glide epenthesis applies after: a) The feminine nominalizer *-uhi* as in [tɑk-uhi] ‘queen’ derived from [tɑk] ‘crown’ թագ. b) The deverbal adjectivizer *-i* as in [hɑɾkel-i] ‘venerable’ derived from [hɑɾkel] ‘to respect յարգել. And c) the tree-naming suffix *-eni* in [χənt͡soɾ-eni] ‘apple tree’ derived from [χənt͡soɾ] խնձոր.

There is likewise an irregular word [t͡si] ‘horse’ ձի which takes the irregular dative *-u*. Here again we get glide epenthesis: [t͡si-ju].

We emphasize that glide epenthesis is the norm in Western Armenian (Table 5.102). In contrast in Eastern Armenian, stem-final /i/ usually deletes before V-initial inflection. In Eastern, the regular dative and ablative are /-i, it͡s/. After /i/-final stems, the Eastern dative and ablative are /-u, ut͡s/; Eastern likewise has a locative *-um*. Many but not all *i*-final words can lose their /i/ before these Eastern suffixes, while Western keeps the /i/.

Table 5.102: Dialectal variation in how /i/ appears before V-initial inflection

	‘soul’	հոգի	‘Azeri’	ազերի
	Western	Eastern	Western	Eastern
	ho'**ki**	ho'**ki**	ɑze'ɾ**i**	ɑze'ɾ**i**
G/D *-i* (WA), *-u* (EA)	hoki'**ji**	ho'k**u**	ɑzeɾ**i**'**ji**	ɑzeɾ**i**'**ji**
ABL *-e* (WA), *-i͡ts, -u͡ts* (EA)	hoki'**je**	ho'k**u͡ts**	ɑzeɾ**i**'**je**	ɑzeɾ**i**'**ji͡ts**
INS *-ov*	hoki'**jov**	ho'k**ov**	ɑzeɾ**i**'**jov**	ɑzeɾ**i**'**jov**
		hoki'**jov**		
LOC *-um*		ho'k**um**		ɑzeɾ**i**'**jum**
		hoki'**jum**		

5.6.4.4 Stem-final /ə/

There are relatively few words that end in /ə/ and that can take inflection. One category of such words is the names for sounds, like [pə] բը to mean the sound ‘p’ or the letters բ,փ <p,p’>. Before V-initial inflection, we get glide epenthesis (Table 5.103).

Table 5.103: Glide epenthesis between /ə/ and V-initial inflection

	բը	պը	մը	հը
	'pə	'bə	'mə	'hə
G/D *-i*	pə'**ji**	bə'**ji**	mə'**ji**	hə'**ji**
ABL *-e*	pə'**je**	bə'**je**	mə'**je**	hə'**je**
INS *-ov*	pə'**jov**	bə'**jov**	mə'**jov**	hə'**jov**

5.6.4.5 Stem-final /o/

There are relatively few words that end in [o]. Before V-initial inflection, there is glide epenthesis (Table 5.104).

Table 5.104: Glide epenthesis between /o/ and V-initial inflection

	Words ends in orthographic <ō> ο		Words ends in orthographic <oy> այ	
	'zero'	'tango'	'collection'	'agreeable'
	զէրο	թանկο	հաւաքածոյ	հաճոյ
	<zērō>	<t'angō>	<hawak'ad͡zoy>	<had͡ʒoy>
	zeˈɾ**o**	taŋˈg**o**	havakaˈd͡z**o**	haˈd͡ʒ**o**
G/D *-i*	zeɾ**o**ˈ**ji**	taŋg**o**ˈ**ji**	havakad͡z**o**ˈ**ji**	had͡ʒ**o**ˈ**ji**
ABL *-e*	zeɾ**o**ˈ**je**	taŋg**o**ˈ**je**	havakad͡z**o**ˈ**je**	had͡ʒ**o**ˈ**je**
INS *-ov*	zeɾ**o**ˈ**jov**	taŋg**o**ˈ**jov**	havakad͡z**o**ˈ**jov**	had͡ʒ**o**ˈ**jov**

Glide epenthesis applies both after roots like [zeɾo] 'zero', and after derivational suffixes like the nominalizer *-o* in [havavad͡z-o] 'collection' that is added onto resultative participles like [havakad͡z] 'collected' հաւաքած.

Orthographically, loanwords with final [o] tend to be spelled as <ō> ο, while native words are spelled with final <oy> ոյ. The reason is diachronic, as explained in Section §5.6.8.

5.6.4.6 Stem-final /u/

When a stem-final /u/ is before a V-initial inflectional suffix, the most common repair in Western Armenian is glide epenthesis (Table 5.105).

Table 5.105: Glide epenthesis between /u/ and V-initial inflection

	'owl'	'bee'	'sour'	'male'
	բու	մեղու	թթու	արու
	ˈ**pu**	meˈʁ**u**	təˈ**tu**	aˈɾ**u**
G/D *-i*	pu**ˈji**	meʁu**ˈji**	tətu**ˈji**	aɾu**ˈji**
ABL *-e*	pu**ˈje**	meʁu**ˈje**	tətu**ˈje**	aɾu**ˈje**
INS *-ov*	pu**ˈjov**	meʁu**ˈjov**	tətu**ˈjov**	aɾu**ˈjov**

There is however dialectal variation and some degree of lexical variation (Table 5.106). In Eastern Armenian, the more typical rule is that /u/ becomes [v] (*u*-devocalization) before V-initial inflection. The [v] can then trigger schwa epenthesis. However even in Eastern Armenian, there is lexical variation in that some words use devocalization, some use glide epenthesis, and some can do both. Data is from English Wiktionary via VP. In contrast in Western Armenian, glide epenthesis is the norm.

Table 5.106: Dialectal variation in how /u/ appears before V-initial inflection

	'flea'	լու	'pilot'	օդաչու	'cat'	կատու
	Western	Eastern	Western	Eastern	Western	Eastern
	ˈ**lu**	ˈ**lu**	otaˈ**t͡ʃu**	otaˈ**t͡ʃu**	gɑˈ**du**	kɑˈ**tu**
G/D *-i*	**luˈji**	**ləˈvi**	otat͡ʃ**uˈji**	otat͡ʃ**uˈi**	gɑ**duˈji**	kɑˈ**tuˈi**
						kɑtˈ**vi**
ABL *-e* (WA), *-its͡* (EA)	**luˈje**	**ləˈvit͡s**	otat͡ʃ**uˈjit͡s**	otat͡ʃ**uˈit͡s**	gɑ**duˈje**	kɑˈ**tuˈit͡s**
						kɑtˈ**vit͡s**
INS *-ov*	**luˈjov**	**ləˈvov**	otat͡ʃ**uˈjov**	otat͡ʃ**uˈov**	gɑ**duˈjov**	kɑˈ**tuˈov**
						kɑtˈ**vov**

Note for the above Eastern words [otat͡ʃu] and [kɑtu], we transcribe the inflected forms with a glide [j]. However, VP informs us that the glide in the above examples feels weaker than a full glide, thus potentially a transient glide.

In Western, the *u*-devocalization process is common when a stem-final /u/ precedes a derivational suffix (§5.6.2.5), but uncommon when it precedes an inflectional suffix. For example, [gɑdu] 'cat' is inflected as dative [gɑdu-ji] but is derived to [gɑdv-ɑgɑn] 'feline' կատուական.

However, there are some words like the word 'tongue' [lezu] that show optionality in whether use devocalization or glide epenthesis (Table 5.107). In Western Armenian at least, the devocalized version has a more archaic connotation.

Table 5.107: Lexical variation in how /u/ appears before V-initial inflection

	'tongue/language'		լեզու	
	Western		Eastern	
	leˈ**zu**		leˈ**zu**	
G/D *-i*	lez**uˈji**	lezˈ**vi**	lez**uˈji**	lezˈ**vi**
ABL *-e* (WA), *-its͡* (EA)	lez**uˈje**	lezˈ**ve**	lez**uˈjit͡s**	lezˈ**vit͡s**
INS *-ov*	lez**uˈjov**	lezˈ**vov**	lez**uˈjov**	lezˈ**vov**

The devoicalized form is impressionistically more common in high-frequency collocations or sayings or proverbs (2).

(2) **lezv-i-t** dɑg pɑn-mə g-ɑ-∅
tongue-GEN-POSS.2SG under thing-INDF exist-TH-3SG
Literally: 'There is something under your thing'
Meaning: 'There is something that you want to say but you're having trouble it.'

Լեզուիդ տակ բան մը կայ:

5.6.5 Vowel hiatus repair in irregular inflection and verbal inflection

When a vowel-final stem gets a vowel-initial regular inflectional suffix, then the normal way to resolve the vowel sequence is to use glide epenthesis (§5.6.4). However, irregular inflection uses different strategies.

There are a handful of irregular words that end in /ɑ/: [dəʁɑ] 'boy' տղայ and [d͡ʒɑmpɑ] 'road' ճամբայ. In standard speech, the vowel is irregularly deleted before irregular inflectional suffixes: [dəʁ-o͡ts] 'boy-PL.GEN' տղոց and [d͡ʒɑmp-u] 'road-GEN' ճամբու. But in colloquial speech, we can use regular inflectional suffix and potentially glide epenthesis: [dəʁɑ-neɾ-u] 'boy-PL-GEN' տղաներու and [d͡ʒɑmpɑj-i] 'road-GEN' ճամբայի. Fuller declension classes for these irregular words are found in cite chapter declesion irregular.

For the irregular word [d͡ʒɑmpɑ] 'road', it likewise has derived words where the [ɑ] is deleted: [d͡ʒɑmp-oɾt] 'traveller' ճամբորդ or [d͡ʒɑmp-el] 'to send away' ճամբել.

The country-naming suffix *-ja* -իա follows its own declension class: [əspɑn-jɑ] 'Spain' Սպանիա but [əs]. In standard speech, the [ɑ] is deleted and replaced by a dative-genitive suffix *-o*: [əspɑn-jo] 'Spain-GEN' Սպանիոյ. In colloquial speech, we can use the regular suffix *-i* and take a glide: [əspɑn-jɑj-i] Սպանիայի. This declension class is discussed more in cite chapter declesion irregular.

Some words of time take an irregular dative-genitive suffix *-va*. This suffix deletes the stem-final vowel or changes it to a schwa. For example, [ɑɾdu] 'morning' արտու but [ɑɾdə-vɑ] արտուայ, and [dɑɾi] 'year' տարի but [dɑɾ-vɑ] տարուայ. This is discussed more in cite chapter declesion irregular.

Among verbs, we find very few cases of clear vowel hiatus repair. The clearest is when the past suffix *-i* is added after the theme vowels [-e-, -ɑ-] and triggers a glide. For example for [-e-], [jeɾk-e-n] 'sing-TH-3PL' meaning 'they sing' երգեն, and its past form [jeɾk-ej-i-n] 'sing-TH-PST-3PL' meaning 'if they were singing' երգէին. Similarly for the theme vowel [ɑ]: [gɑɾt-ɑ-n] 'read-TH-3PL' meaning 'they read' կարդան, and its past form [gɑɾt-ɑj-i-n] 'read-TH-PST-3PL' meaning 'if they were reading' կարդային.

An unclear case involves irregular past inflection. For irregular words in the past perfective, the past markers [ɑ,i] never follow a theme. For example, [peɾ-e-n] 'bring-TH-3PL' meaning 'they bring' բերեն but [pəɾ-i-n] 'bring-PST-3PL' meaning 'they brought' բերին. It is unclear if this is because the [-i] suffix deletes theme vowels, or if the morphology just never added the theme vowel here. Irregular verbal past marking is discussed more in cite chapter verb irregular.

5.6.6 Vowel hiatus repair before clitics

Armenian has few V-initial clitics (Table 5.108). These are the 'also' clitic [ɑl] ալ, and the copula [e] է that can take on various inflected forms like present 1SG [em] եմ, past 1SG [eji] էի, and so on. When these clitics are added after a V-final word, we get glide epenthesis.

Table 5.108: Glide epenthesis between before V-initial clitics

		+/ɑl/ 'also' ալ	+/e/ 'is' է		
/ɑ/	vəˈg**ɑ**	vəˈg**ɑjɑl**	vəˈg**ɑje**	'witness'	վկայ
/e/	kəˈv**e**	kəˈv**ejɑl**	kəˈv**eje**	'vote'	քուէ
/i/	leˈʁ**i**	leˈʁ**ijɑl**	leˈʁ**ije**	'bitter'	լեղի
/ə/	ˈv**e**	ˈv**əjɑl**	ˈv**əje**	'/v/ sound'	վը
/o/	kiˈl**o**	kiˈl**ojɑl**	kiˈl**oje**	'kilogram'	քիլոյ
/o/	gəˈd͡z**u**	gəˈd͡z**ujɑl**	gəˈd͡z**uje**	'spicy'	կծու

One exception is the indefinite suffix [mə] մը. Before a clitic, it changes its form to [mən]. See indefinite floating But in more archaic speech, the [mə] is reduced to [m]. find reference from an old grammar bcz impossible to find online [me] forms

(3) kit͡ʃ-**mən=ɑl** ɑnuʃeʁen (modern: HD)
kit͡ʃ-**m=ɑl** ɑnuʃeʁen (archaic)
little-INDF=also sweets
'And a bit of sweets.'[10]
Քիչ մը՝ն ալ
Քիչ մ՚ալ անուշեղէն

5.6.7 Overapplication of vowel hiatus repair

We went through various rules for vowel hiatus repair like vowel deletion or glide epenthesis. In general, such rules apply when two vowels become next to each because of the morphology. But there are corners of the grammar where these rules overapply even though there's no vowel sequence. The overapplication is because of a mix of morphological and diachronic factors. The corners are passives (§5.6.7.1) and /j/-initial suffixes (§5.6.7.2).

[10] For the [m-ɑl] form, the archaic sentence was taken from the title of an article from 1900; URL: https://arar.sci.am/dlibra/publication/74562/edition/67396/.

5.6.7.1 Overapplication of glide epenthesis in passives

For passives, the passive suffix is a consonant /v/. But after /ɑ/-final roots, a glide is epenthesized and written in the orthography. For example, from the root <vga**y**> [vəgɑ] 'witness' վկայ, we can form a passive verb <vga**yow**il> [vəgɑ**j-v**-i-l] 'to be testified' վկայուիլ. The reason this glide appears is because of two reasons, one diachronic and one synchronic.

The diachronic reason is that the modern passive suffix /v/ was historically a vowel */-u-/ in Classical Armenian. Thus vowel hiatus was created in Classical Armenian, and thus we got a glide. The synchronic reason is that the change from a vocalic morpheme */-u-/ to a consonantal morpheme [-v-] was accompanied by a huge reanalysis of passive morphophonology. Briefly put, passive stems try to resemble active stems, even if that means a glide is unnecessarily added: <vga**y**el> [vəgɑ**j**-e-l] 'to testify' վկայել. The need for resemblances affects many other morphophonological changes such as vowel reduction. This passive problem is discussed in depth in passive phonology chapter.

5.6.7.2 Vowel hiatus repair before /j/-initial suffixes

For /j/-initial suffixes, these suffixes are pronounced with an initial glide, but they are spelled with an initial vowel (Table 5.109). For example, the suffix /-jɑ/ is spelled as <ea**y**> in the word <aɣiws**eay**> [ɑʁʏs-**jɑ**] 'made of brick' աղիւսեայ, derived from the word <aɣiws> [ɑʁʏs] 'brick' աղիւս. When added to a vowel-final stem, the vowel undergoes alternations as if the /j/ were a vowel. Such changes include deletion or de-vocalization.

Table 5.109: Overapplication of vowel hiatus repair before /j/-initial suffixes

/ɑ-j/	<giwt'era**y**>	gʏteˈɾ**ɑ**	'Cythera'	Կիւթերա
→[j]	<giwt'ere**a**n>	gʏteɾ-ˈ**j**ɑn	'Cytherean'	կիւթերեան
/u-j/	<brd**ow**>	bəɾˈd**u**	'payprus'	պրտու
→[vj]	<brd**owe**ay>	bəɾdˈ**v-j**ɑ	'made of papyrus'	պրտուեայ
	<pazmalez**ow**>	pɑzmɑleˈz**u**	'polyglot'	բազմալեզու
→[vj]	<pazmalez**owe**an>	pɑzmɑlezˈ**v-j**ɑn	'polyglot'	բազմալեզուեան
/i-j/	<kin**i**>	kiˈn**i**	'wine'	գինի
→[j]	<kin**e**ag>	kin-ˈ**j**ɑg	'sour wine'	գինեակ
	<əndan**i**>	əndɑˈn**i**	'familiar'	ընտանի
→[j]	<əndan**e**ōk'>	əndɑn-ˈ**j**ok	'familiarly'	ընտանեօք
/e-j/	<ap'rotid**ē**>	ɑpɾotiˈd**e**	'Aphrodite'	Ափրոդիտէ
→[j]	<ap'rotid**e**an>	ɑpɾotid-ˈ**j**ɑn	'venereal'	ափրոդիտեան

No such changes typically happen after /ɑ/ or /e/ (Table 5.110). This is because, as explained in Section §5.6.2, such vowels usually take glide epenthesis before vowel-initial suffixes. The glide of the suffix here does the job of the epenthetic glide.

Table 5.110: No application of of vowel hiatus repair between /ɑ,e/ and a /j/-initial suffix

/ɑ-j/	<ark’**ay**>	ɑɾˈk**ɑ**	‘king’	արքայ
→[ɑj]	<ark**ay**ean>	ɑɾk**ɑ**-ˈjɑn	‘royal’	արքայական
	<norəndz**ay**>	noɾənˈd͡z**ɑ**	‘novice’	նորընծայ
→[ɑj]	<norəndz**ay**eal>	noɾənd͡z**ɑ**-ˈjɑl	‘novice’	նորընծայական
	<yut**ay**>	huˈt**ɑ**	‘Judas’	Յուդայ
→[ɑj]	<yut**ay**ean>	hut**ɑ**-ˈjɑn	‘Judaical’	յուդայական
/e-j/	<hiwlē>	hɤˈl**e**	‘atom’	հիւլէ
→[ej]	<eōtn’hiwlē**e**an>	jotnəhɤl**e**-ˈjɑn	‘heptatomic’	եօթնհիւլէական

We have found many words where a /j/-initial suffix was added to a /i/-final word, and then the /i/ is deleted (Table 5.111).

Table 5.111: Deletion of /i/ before a /j/-initial suffix

<ʃoki>	ʃoˈk**i**	‘vapor’	շոգի	<badani>	bɑdɑˈn**i**	‘adolescent’	պատանի
<ʃokeag>	ʃok-ˈjɑg	‘light vapor’	շոգեակ	<badaneag>	bɑdɑn-ˈjɑg	‘adolescent’	պատանեակ
<yoṙi>	hoˈṙ**i**	‘evil’	յոռի	<dēruni>	deɾuˈn**i**	‘dominical’	տէրունի
<yoṙeag>	hoɾ-ˈjɑg	‘wicked’	յոռեակ	<dērunean>	deɾun-ˈjɑn	‘dominical’	տէրունական
<d͡ʒd͡ʒi>	d͡ʒəˈd͡ʒ**i**	‘worm’	ճճի	<madani>	mɑdɑˈn**i**	‘ring’	մատանի
<d͡ʒd͡ʒeag>	d͡ʒəd͡ʒ-ˈjɑg	‘animalcule’	ճճեակ	<madaneag>	mɑdɑn-ˈjɑg	‘small ring’	մատանեակ
<orti>	voɾˈt**i**	‘son’	որդի	<gɣzi>	gəʁˈz**i**	‘island’	կղզի
<orteag>	voɾt-ˈjɑg	‘little child’	որդեակ	<gɣzeag>	gəʁˈz-jɑg	‘islet’	կղզեակ

For /i/-final roots, many but not all of the above suffixes are [jɑg], such as [ʃokjɑg] ‘light vapor’. This suffix could be reanalyzed as just the root plus the diminutive suffix /ɑg/. The underlying form could thus be either /ʃoki-jɑg/ or /ʃoki-ɑg/. With the latter form, the /i/ becomes [j] before a vowel. Such a reanalysis however does not work for words that end in sequences like [jɑn].

As is clear, for some reason, the /j/-initial suffixes act like vowel-initial suffixes and they trigger vowel hiatus rules like /i/-deletion. But why do they do this? The answer is likely diachronic. The fact these suffixes are spelled with an <e> suggests that in Classical Armenian, these /j/-initial suffixes were pronounced as */e/-initial suffixes. They would then transparently trigger vowel hiatus repair rules. Over time, the ancient */e/ became a modern /j/. This sound change

required Armenian speakers to reanalyze these /j/-initial suffixes as arbitrarily requiring that the preceding vowel is deleted, de-vocalized, or be left unchanged.

One could perhaps argue that the data above suggests that modern speakers still treat the /j/-initial suffixes as underlyingly /e/-initial. That is, perhaps a suffix like [-jɑg] is actually /-eag/. We don't think such an analysis is realistic however as we explain below.

In the Kouyoumdjian dictionary, we found around 1430 words which end in a /j/-initial suffix. Thus these suffixes are relatively common and productive. However, we only found 27 words such that a) the word had such a suffix, and b) the word was clearly derived from a vowel-final stem. Condition (b) is important. Such derived words are all extremely-low frequency words, and they are often high-register technical words or liturgical words. The Armenian child is unlikely to be systematically exposed to such derived words in their formative years. Based on this statistical skew, we suspect that the Armenian child will at first treat the /j/-initial suffixes like [-jɑg] as truly just underlyingly /-jɑg/, without any abstraction.

In HD's impression, knowing how to morphologically form such derived words and how to pronounce them is something that the child is exposed to later in life at school or church. The child then learns that these /j/-initial suffixes for unknown reasons trigger vowel hiatus repair on the preceding vowel.

5.6.8 Diachronic problems in glide epenthesis

For vowel hiatus, glide epenthesis is a common rule. However, glide epenthesis has an intricate connection with the history of glide deletion in Armenian.

In modern Armenian, there are many native words that are pronounced with a final [ɑ] or [o], but that are spelled with a final glide: <ark'ay> [ɑɾkɑ] 'king' արքայ or <xaʃoy> [χɑʃo] 'broth' խաշոյ. Such final glides are silent letters. In contrast, relatively few words are spelled with a final <a> ա, <ō> օ, or <o> ո. For example, <fransa> [fəɾɑnsɑ] 'France' Ֆրանսա, [d͡zo] <d͡zo> 'an interjection' ծo, <vet'o> [veto] 'veto' վեթo. Such words are usually either loanwords, interjections, or have a country-naming suffix *-ja*.

The reason for this state of affairs is because in Classical Armenian, these silent letters were pronounced. In the change from Classical Armenian to Modern Armenian, the final glide was deleted from polysyllabic words, while it was kept in monosyllabic words, or in compounds where the second stem was monosyllabic. In Table 5.112, we transcribe the Classical Words using the 'pronunciation rules' that modern Western Armenian speakers would use to read Classical texts. We ultimately don't know how Classical Armenian was exactly pronounced.

Table 5.112: Diachronic loss of final glides for polysyllabic words

	'king' <ark'ay>	'broth' <xaʃoy>	'Armenian' <hay>	'Latinized Armenian' <ladinahay>	'ram' <xoy>
Classical	[ɑɾˈkɑj]	[χɑˈʃoj]	[ˈhɑj]	[lɑdin-ɑ-ˈhɑj]	[ˈχoj]
Modern	[ɑɾˈkɑ]	[χɑˈʃo]	[ˈhɑj]	[lɑdin-ɑ-ˈhɑj]	[ˈχoj]
	արքայ	խաշոյ	հայ	լատինահայ	խոյ

In contrast, Classical Armenian seems to not have many words that are spelled with a final <a> or <ō>. For <a>, one of the few examples we found on the Calfa dictionary was <alk'imia> [ɑlkimjɑ] 'alchemy' ալքիմիա, an obvious loanword.[11]. For <ō>, this letter did not exist in Classical Armenian. The final <o> ո is also pretty rare in Classical: <ayo> [ɑjo] 'yes' այո.

When a modern Western speaker uses a word like [ɑɾkɑ] 'king', the final silent glide <y> is present in the spelling <ark'ay> արքայ but it not present in the lexical representation /ɑɾkɑ/. When a vowel-initial suffix is added, a glide is inserted: [ɑɾkɑj-i] 'king-GEN' արքայի. This complicated diachronic sequence of deleting a final glide and then inserting an epenthetic glide constitutes a type of rule inversion (Vennemann 1972), which is common in historical linguistics.

Given these historical facts, some argue that words like [ɑɾkɑ] <ark'ay> 'king' are underlyingly glide-final /ɑɾkɑj/ in order to explain why the glide surfaces before vowels: [ɑɾkɑji] (G/D) cite vaux. But this psychologically unrealistic and unneeded for various reasons.

First, words that are spelled without a glide like [jevɾobɑ] <ewroba> 'Europe' show the same epenthetic glide in inflection: [jevɾobɑji] (G/D). They may differ before derivational morphology, but such variation is for independent reasons (§5.6.2.1).

Second, the Kouyoumdjian dictionary lists 294 words that end in an orthography <a> ա and pronounced as [ɑ], and 708 words that end in an orthographic <ay> այ and pronounced as [ɑ].

Third, the reformed orthography does not spell these final deleted glides: reformed spelling <ark'a> արքա vs. traditional <ark'ay> արքայ. The orthographic reform was likely guided by the reformers intuition that these silent glides are phonologically absent.

A related fourth point is that some loanwords are variably spelled with a glide (like a native word), or without a glide (like a typical loanword). For example, [zeɾo] 'zero' can be <zēroy> զէրոյ or <zērō> զէրo. There's no reason to suspect that these alternating spellings mean there's any difference in their phonology.

[11] https://dictionary.calfa.fr/entries/

Fifth, this unwritten glide is completely invisible to other inflectional suffixes. The definite suffix is /-ə/ after consonants, but /-n/ after vowels. Words like [ɑɾkɑ] are transparently treated as vowel-final and take the definite form [ɑɾkɑ-n]. The orthography deletes this glide before consonant-initial inflectional suffixes like the definite <ark'an> արքան and the plural [ɑɾkɑ-neɾ] <ark'aner> արքաներ.

Thus, the Armenian child really has no reason to treat this glide as anything other than an inserted glide. The have no reason to postulate that most cases of final [ɑ] are underlyingly /ɑj/ with a rule of deleting final glides.

Note that this diachronic-synchronic problem has an interesting interaction with morphology. Consider the word <bargay> [bɑɾgɑ] 'Fatal Sister' Պարկայ, a Greek mythological character. The plural marker in Classical was *-k* -ք. The modern language uses this suffix as a nominalizer instead. When these nominalizer is added after a silent glide, speakers recognize the 'learned' or 'archaic' nature of this rule, and would pronounce the glide because they're reading the silent glide: <bargayk'> [bɑɾgɑjk] 'The Fate sisters' Պարկայք.

6 note

readings here https://en.wiktionary.org/wiki/Talk:%D5%B8%D5%B2%D5%A2%D5%A1%D5%AC#c-Vahagn_Petrosyan-20250219194100-Nikosev-20250218214900

jk vowel hiatus syncope pseudoredup schwa elision

References

Adjarian, Hratchia. 1899. Les explosives de l'ancien arménien étudiées dans les dialectes modernes. *La parole ou revue internationale de rhinologie, otologie, laryngologie et phonétique expérimentale*. 119–127. https://gallica.bnf.fr/ark:/12148/bpt6k57626142/f121.item.

Barkaï, Malachi & Julia Horvath. 1978. Voicing assimilation and the sonority hierarchy: Evidence from Russian, Hebrew and Hungarian. *Linguistics* 16(212). 77–88. DOI: 10.1515/ling.1978.16.212.77.

Beguš, Gašper. 2019. Post-nasal devoicing and the blurring process. *Journal of Linguistics* 55(4). 689–753. DOI: 10.1017/S002222671800049X.

Braun, Angelika. 2013. An early case of "VOT". In *Interspeech 2013*, 119–122. ISCA. DOI: 10.21437/Interspeech.2013-50.

Dolatian, Hossep. 2021. Cyclicity and prosodic misalignment in Armenian stems: Interaction of morphological and prosodic cophonologies. *Natural Language & Linguistic Theory* 39(3). 843–886. DOI: 10.1007/s11049-020-09487-7.

Downing, Laura J. 1999. Prosodic Stem ≠ Prosodic Word in Bantu. In T. Alan Hall & Ursula Kleinhenz (eds.), *Studies on the phonological word*, vol. 174, 73–98. Amsterdam: John Benjamins. DOI: 10.1075/cilt.174.05dow.

Dum-Tragut, Jasmine. 2009. *Armenian: Modern Eastern Armenian* (London Oriental and African Language Library 14). Amsterdam: John Benjamins. DOI: 10.1075/loall.14.

Fairbanks, Gordon H. 1948. *Phonology and morphology of modern spoken West Armenian*. University of Wisconsin-Madison. (Doctoral dissertation).

Hall, Daniel Currie. 2004. A formal approach to /v/: Evidence from Czech and Slovak. In Olga Arnaudova, Wayles Browne, María Luisa Rivero & Danijela Stojanović (eds.), *Formal approaches to Slavic linguistics 12: The Ottawa Meeting*, 187–205. Ann Arbor: Michigan Slavic Publications.

Hossep, Dolatian. 2024. *Adjarian's Armenian dialectology (1911): Translation and commentary* (Languages of the Caucasus 4). Berlin: Language Science Press. DOI: 10.5281/zenodo.14008766.

Kelly, Niamh E. & Lara Keshishian. 2019. The voicing contrast in stops and affricates in the Western Armenian of Lebanon. In *Interspeech 2019*, 1721–1725. ISCA. DOI: 10.21437/Interspeech.2019-2529.

Kelly, Niamh E. & Lara Keshishian. 2021. Voicing patterns in stops among heritage speakers of Western Armenian in Lebanon and the US. *Nordic Journal of Linguistics* 44(2). 103–129. DOI: 10.1017/S0332586521000093.

Kouyoumdjian, Mesrob G. 1970. *A comprehensive dictionary, Armenian-English*. Beirut: Atlas Press.

Macak, Martin. 2017. The phonology of Classical Armenian. In Jared Klein, Brian Joseph & Matthias Fritz (eds.), *Handbook of comparative and historical Indo-European linguistics*, 1037–1079. Berlin/Munich/Boston: Walter de Gruyter. DOI: 10.1515/9783110523874-016.

Macak, Martin Jakub. 2016. *Studies in Classical and Modern Armenian phonology*. University of Georgia. (Doctoral dissertation).

Minassian, Martiros. 1980. *Grammaire d'arménien oriental*. Delmar, NY: Caravan Books.

Padgett, Jaye. 2002. Russian voicing assimilation, final devoicing, and the problem of [v]. Unpublished manuscript. http://roa.rutgers.edu/article/view/538.

Pater, Joe. 1999. Austronesian nasal substitution and other NC effects. In René Kager, Harry van der Hulst & Wim Zonneveld (eds.), *The prosody-morphology interface*, 310–343. Cambridge University Press. DOI: 10.1017/CBO9780511627729.009.

Rubach, Jerzy. 1996. Nonsyllabic analysis of voice assimilation in Polish. *Linguistic Inquiry* 27(1). 69–110.

Rubach, Jerzy. 1997. Extrasyllabic consonants in Polish: Derivational Optimality Theory. In Iggy Roca (ed.), *Derivations and constraints in phonology*, 551–581. Oxford: Clarendon Press.

Rubach, Jerzy & Geert E. Booij. 1990. Edge of constituent effects in Polish. *Natural Language & Linguistic Theory* 8(3). 427–463. DOI: 10.1007/BF00135620.

Sanjian, Avedis K. 1996. The Armenian alphabet. In Peter T. Daniels & William Bright (eds.), *The world's writing systems*, 356–363. New York & Oxford: Oxford University Press.

Seyfarth, Scott, Hossep Dolatian, Peter Guekguezian, Niamh Kelly & Tabita Toparlak. 2024. Armenian (Yerevan Eastern and Beirut Western varieties). *Journal of the International Phonetic Association* 54(1). 445–478. DOI: 10.1017/S0025100323000130.

Tahtadjian, Talia. 2020. *Western Armenian rhotics: A differential phonetic study*. Carleton University. (Bachelor's thesis).

Tahtadjian, Talia Lisa. 2021. *Phonetic interference in the production of stops by Western Armenian bilinguals*. University of Toronto. (MA thesis).

Toparlak, Tabita. 2019. *Etudes phonétiques en arménien*. Université Paris 3 - Sorbonne Nouvelle. (MA thesis).

Vaux, Bert. 1998. *The phonology of Armenian*. Oxford: Clarendon Press.

Vaux, Bert & Andrew Wolfe. 2009. The appendix. In Eric Raimy & Charles E. Cairns (eds.), *Contemporary views on architecture and representations in phonology*, 101–143. Cambridge, MA: The MIT Press. DOI: 10.7551/mitpress/9780262182706.003.0005.

Vennemann, Theo. 1972. Rule inversion. *Lingua* 29(3-4). 209–242. DOI: 10.1016/0024-3841(72)90025-3.

Աճառեան, Հրաչեայ. 1902. Թուրքերէնէ փոխառեալ բառերը Պոլսի հայ ժողովրդական լեզուին մէջ համեմատութեամբ Վանի, Ղարաբաղի եւ Նոր-Նախիջեւանի բարբառներուն *[Turkish loanwords in the colloquial language of Istanbul Armenians, in comparison with the dialects of Van, Karabakh, and New Nakhichevan]* (Էմինեան ազգագրական ժողովածու 3). Մոսկուա-Վաղարշապատ: Լազարեան Ճեմարան Արեւելեան Լեզուաց.

Աճառեան, Հրաչեայ. 1911. Հայ Բարբառագիտութիւն *[Armenian dialectology]*. Մոսկուա-Նոր-Նախիջեւան: Լազարեան Ճեմարան Արեւելեան Լեզուաց.

Աճառեան, Հրաչեայ. 1926. Հայերէն արմատական բառարան *[Armenian etymological dictionary]*. Երեւան: Երեւանի Համալսարանի Հրատարակչութիւն.

Ասատրյան, Մանվել. 1976. Արևելահայ գրական լեզվի ձայնեղ պայթական բաղաձայների արտասանությունը [Pronunciation of voiced plosives in the literary Eastern Armenian language]. Բանբեր Երևանի Համալսարանի 29(2). 32–47.

Ավետիսյան, Յուրի. 2005. Արտասանական տեղաշարժեր ժամանակակից հայերենի ուղղախոսության մեջ [Pronunciation shifts in modern Armenian orthoepy]. Բանբեր Երևանի Համալսարանի 116(2). 93–104.

Ավետիսյան, Յուրի Սրապիոնի. 2007. Արևելահայերենի և արևմտահայերենի զուգադրական քերականություն *[Comparative grammar of Eastern and Western Armenian]*. Երևան: Երևանի Պետական Համալսարանի Հրատարակչություն.

Ավետիսյան, Յուրի Սրապիոնի. 2011. Արևելահայերենի և արևմտահայերենի զուգադրական հնչյունաբանություն *[Comparative phonology of Eastern and Western Armenian]*. Երևան: Հայաստանի սփյուռքի նախարարություն.

Ավետյան, Սարգիս. 2015. Գրական արևմտահայերենում հին հայերենի իւ երկբարբառի արտացոլման հարցի շուրջ [On the question of the manifestation of the Old Armenian diphthong իւ in literary Western Armenian]. In Վ. Լ. Կատվալյան (ed.), Հայերենագիտական միջազգային տասներորդ գիտաժողովի զեկուցումների ժողովածու, 17–26. Երևան: ՀՀ ԳԱԱ Հրաչյա Աճառյանի անվան լեզվի ինստիտուտ.

Եզեկյան, Լևոն. 2007. Հայոց լեզու *[Armenian language]*. Երևան: Երևանի Պետական Համալսարանի Հրատարակչություն.

Խաչատրյան, Ամալյա. 1988. Ժամանակակից հայերենի հնչույթաբանություն *[Phonetics of contemporary Armenian]*. Երևան: Հայկական ՍՍՀ Գիտությունների Ակադեմիա Հրատարակչություն.

Հովհաննիսյան, Հասմիկ Ռազմիկի. 2014. Վանկի դերը հայերենի հնչույթաբանության մեջ *[The role of the syllable in Armenian phonetics]*. Երևան: Երևանի Պետական Համալսարանի Հրատարակչություն.

Ղարագյուլյան, Թերեզա. 1974. Ժամանակակից հայերենի ուղղախոսությունը *[Modern Armenian orthoepy]*. Երեւան: Հայկական ՍՍՀ Գիտությունների Ակադեմիա Հրատարակչություն.

Մարգարյան, Ալեքսանդր. 1997. Ժամանակակից հայոց լեզու. Հնչյունաբանություն *[Contemporary Armenian language: Phonology]*. Երևան: Երեւանի Պետական Համալսարանի Հրատարակչություն.

Սաքապետոյեան, Ռուբէն Կարապետի. 2011. Արեւմտահայերէն-արեւելահայերէն նոր բառարան *[New dictionary of Western Armenian-Eastern Armenian]*. Երեւան: «Տիգրան Մեծ» հրատարակչություն.

Սևակ, Գուրգեն. 2009. Ժամանակակից հայոց լեզվի դասընթաց *[Course in Modern Armenian]*. Երևան: ԵՊՀ Հրատարակչություն.

Սուքիասյան, Աշոտ Մուրադի. 2004. Ժամանակակից հայոց լեզու (հնչյունաբանություն, բառագիտություն, բառակազմություն*) [Modern Armenian phonology, lexicology, and word-formation]*. Երևան: Երևանի Պետական Համալսարանի Հրատարակչություն.

Grammar of Western Armenian

Set blurb on back with \BackBody{my blurb}